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Introduction

Rydberg atoms, atoms in states of high principal quantum number, n , are atoms with exaggerated properties. While they have only been studied intensely since the nineteen seventies, they have played a role in atomic physics since the beginning of quantitative atomic spectroscopy. Their role in the early days of atomic spectroscopy is described by White.¹

The first appearance of Rydberg atoms is in the Balmer series of atomic H. Balmer's formula, from 1885, for the wavelengths of the visible series of atomic H, is given by¹

$$\lambda = \frac{bn^2}{n^2 - 4}, \quad (1.1)$$

where $b = 3645.6 \text{ \AA}$. We now recognize Eq. (1.1) as giving the wavelengths of the Balmer series of transitions from the $n = 2$ states to higher lying levels.

While the H atom was the first atom to be understood in a quantitative way, other atoms played a crucial role in unravelling the mysteries of atomic spectroscopy. For example, Liveing and Dewar² made the important observation that the observed spectral lines of Na could be grouped into different series. Specifically they observed the Na $ns-3p$ and $nd-3p$ emissions from an arc. Up to the 9s and 8d states they observed doublets due to the fine structure splitting of the 3p state, but for the 10s and 9d states they were unable to resolve the 3p doublet. The $ns-3p$ doublets appeared as sharp lines while the $nd-3p$ doublets appeared as diffuse lines. Most important, they recognized that these sharp and diffuse doublets were members of two series of related lines, although they were unsuccessful in their attempts to relate the wavelengths of the observed transitions.

Knowing, as we do today, that the Na d states are almost degenerate with the higher angular momentum states of the same n , we are not surprised that the series of transitions to the d Rydberg series is diffuse. They are much more likely to suffer pressure broadening. In contrast to Na, Liveing and Dewar noted that in the K spectrum the s and d series were not particularly different, both being "more or less diffuse".² We now realize that the difference between Na and K is due to the fact that both the s and the d Rydberg states of K are energetically well removed from the high angular momentum states.

A crucial advance was made by Hartley, who in his studies of the spectra of Mg, Zn, and Cd, first realized the significance of the frequency of a transition as opposed to the experimentally measured wavelength.^{1,3} Hartley observed that

the splittings of series of multiplets always had the same wavenumber splitting, irrespective of the wavelengths of the transitions. The wavenumber, ν , is the inverse of the wavelength in vacuum. The importance of this realization is hard to overstate and is immediately apparent if we rewrite Balmer's formula in terms of the wavenumber of the observed lines instead of the wavelength,

$$\nu = \left(\frac{1}{4b}\right)\left(\frac{1}{4} - \frac{1}{n^2}\right). \quad (1.2)$$

It is evident that it simply reflects the energy difference between the $n = 2$ and higher lying states.

Following the precedent of Liveing and Dewar, Rydberg began to classify the spectra of other atoms, notably alkali atoms, into sharp, principal, and diffuse series of lines.⁴ Each series of lines has a common lower level and a series of ns , np , or nd levels, respectively, as the upper levels of the sharp, principal, and diffuse series. He realized that the wavenumbers of the series members were related and that the wavenumbers of the observed lines could be expressed as^{1,4}

$$\begin{aligned} \nu_s &= \nu_{\infty s} - \frac{Ry}{(n - \delta_s)^2}, \\ \nu_p &= \nu_{\infty p} - \frac{Ry}{(n - \delta_p)^2}, \\ \nu_d &= \nu_{\infty d} - \frac{Ry}{(n - \delta_d)^2}, \end{aligned} \quad (1.3)$$

where the constants $\nu_{\infty s}$, $\nu_{\infty p}$, and $\nu_{\infty d}$ and δ_s , δ_p , and δ_d are the series limits and quantum defects of the sharp, principal, and diffuse series. The constant Ry is a universal constant, which can be used to describe the wavenumbers of the transitions, not only for different series of the same atom, but for different atoms as well. This constant, Ry , is now known as the Rydberg constant. Defining the wavenumber as the inverse of the wavelength in air, Rydberg⁴ assigned Ry the value 109721.6 cm^{-1} . Recognition of the generality of the Rydberg constant is one of Rydberg's two major accomplishments. The other was to show that all the spectral lines from an atom are related. Specifically, he realized that $\nu_{\infty s}$ and $\nu_{\infty d}$ were the same to within experimental error. For example, for the series originating in the Na $3p_{3/2}$ and $3p_{1/2}$ levels he assigned the values 24485.9 and 24500.5 cm^{-1} for $\nu_{\infty s}$, and 24481.8 and 24496.4 cm^{-1} for $\nu_{\infty d}$. In addition, he observed that $\nu_{\infty p} - \nu_{\infty s}$ was equal to the wavenumber of the first member of the principal series. He reasoned that it should be possible to see other series which did not have the first term of the principal series in common, but some other term instead. This notion led him to a general expression for the wavenumbers of lines connecting different series. The wavenumbers of lines connecting the s and p series, for example, are given by⁴

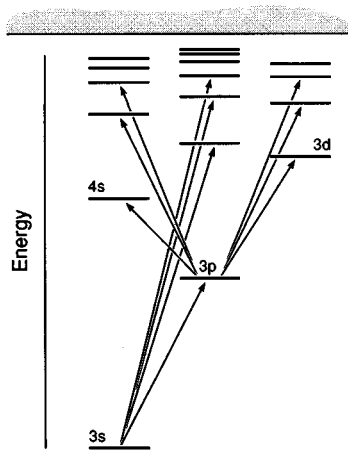


Fig. 1.1 Energy level diagram for Na showing the sharp 3p–ns, principal 3s–np, and diffuse 3p–nd series.

$$\pm \nu = \frac{Ry}{(m - \delta_s)^2} - \frac{Ry}{(n - \delta_p)^2}. \quad (1.4)$$

In Eq. (1.4) the + sign and constant n describe a sharp series of s states and the minus sign and a constant m describe a principal series of p states. If we consider the special case $\delta_s = \delta_p = 0$ and $m = 2$ we arrive at Balmer's formula for the H transitions from $n=2$.

From the early spectroscopic work it is possible to construct an energy level diagram for Na, as shown in Fig. 1.1. From this figure it is apparent that the difference in the principal and sharp series limits is the wavenumber of the 3s–3p transition. It is also apparent that the Rydberg states, states of high principal quantum number n , lie close to the series limit.

The physical significance of high n did not become clear until Bohr proposed his model of the H atom in 1913.¹ In spite of its shortcomings, the Bohr atom contains the properties of Rydberg atoms which make them interesting. The basis of the Bohr atom is an electron moving classically in a circular orbit around an ionic core. To the currently accepted ideas of classical physics Bohr added two notions; that the angular momentum was quantized in integral units of $\hbar$, Planck's constant divided by 2π , and that the electron did not radiate continuously in a classical manner, but gave off radiation only in making transitions between states of well defined energies. The first was an assumption, and the second was forced upon him by the experimental observations.

The Bohr theory can be summarized as follows. An electron of charge $-e$ and mass m in a circular orbit of radius r about an infinitely heavy positive charge of Ze obeys Newton's law for uniform circular motion⁵

$$\frac{mv^2}{r} = \frac{kZe^2}{r^2}, \quad (1.5)$$

where $k = 1/4\pi\epsilon_0$, ϵ_0 being the permittivity of free space. Adding Bohr's requirement of the quantization of angular momentum yields

$$mvr = n\hbar. \quad (1.6)$$

Combining Eqs. (1.5) and (1.6) leads immediately to an expression for the radius r of the orbit

$$r = \frac{n^2\hbar^2}{Ze^2mk}. \quad (1.7)$$

In other words, the size of the orbit increases as the square of the principal quantum number, and states of high n have very large orbits. The energy, W , of a state is obtained by adding its kinetic and potential energies,

$$W = \frac{mv^2}{2} - \frac{kZe^2}{r} = \frac{-k^2Z^2e^4m}{2n^2\hbar^2}. \quad (1.8)$$

The energies are negative, i.e. the electron is bound to the proton, and the binding energy decreases as $1/n^2$. The allowed transition frequencies are the differences in the energies. Explicitly

$$W_2 - W_1 = \frac{k^2Z^2e^4m}{2\hbar^2} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right). \quad (1.9)$$

Comparing this expression to the general form of Rydberg's formula we can see that

$$Ry = \frac{k^2Z^2e^4m}{2\hbar^2}. \quad (1.10)$$

Historically, the most important aspect of the Bohr atom was that it related the spectroscopic Rydberg constant to the mass and charge of the electron. From our present point of view, the Bohr atom physically defines Rydberg atoms and shows us why they are interesting. In high n , or Rydberg, states the valence electrons have binding energies which decrease as $1/n^2$ and orbital radii which increase as n^2 . In other words, in a Rydberg atom the valence electron is in a large, loosely bound orbit. Although Rydberg atoms of principal quantum number well over 100 have been studied in the laboratory, it is instructive to consider a relatively low Rydberg state, $n = 10$, and compare it to a ground state atom. The ground state of H is bound by 1 Ry, 13.6 eV, and has an orbital radius of $1a_0$ and a geometric cross section of a_0^2 . In contrast, the $n = 10$ state has a binding energy of 0.01 Ry, an

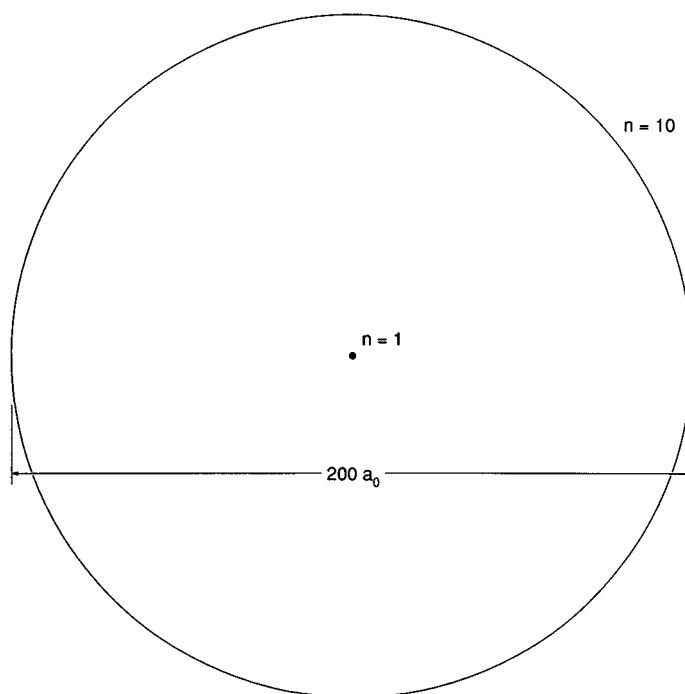


Fig. 1.2 Bohr orbits of $n = 1$ and $n = 10$. The solid dark circle at the center is the $n = 1$ orbit of radius $1a_0$. The $n = 10$ orbit has radius $100a_0$. $1a_0 = 0.53 \text{ \AA}$.

orbital radius of $100a_0$, and a geometric cross section of $10^4 a_0^2$. Fig. 1.2 is a drawing of the Bohr orbits for $n = 1$ and $n = 10$, which demonstrates graphically the difference between a Rydberg atom and a ground state atom. The binding energy of the $n = 10$ atom is comparable to thermal energies, and the geometric cross section is orders of magnitude larger than gas kinetic cross sections.

Once the Bohr theory was formulated it was apparent that Rydberg atoms should have bizarre properties. Why then were Rydberg atoms not studied more extensively until the nineteen seventies? There are two reasons. First, the most pressing problem at the time was not the properties of atoms near the series limit, but the development of the quantum theory. Second, there was simply no means of making Rydberg atoms efficiently enough to study them in the detail that is now possible. Nonetheless, the most sophisticated tool available at the time, high resolution absorption spectroscopy, was used in the first experiments to demonstrate the bizarre properties of Rydberg atoms. By examining the fine details of the spectra, the splittings, shifts, and broadenings, it was possible to learn a great deal about Rydberg states.

One of the first properties to be explored was the sensitivity of Rydberg atoms to external electric fields, the Stark effect. While ground state atoms are nearly immune to electric fields, relatively modest electric fields not only perturb the Rydberg energy levels, but even ionize Rydberg atoms, as was shown in early

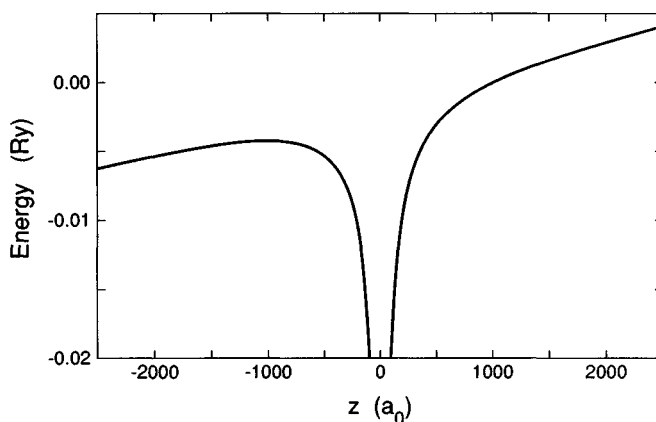


Fig. 1.3 Combined Coulomb-Stark potential along the z axis for a field of 5.14 kV/cm in the z direction.

studies of the Balmer series in electric fields of up to 10^6 V/cm.^{6,7} The Balmer lines exhibit a splitting which is approximately linear in the electric field, and the components actually disappear at well defined values of the electric field due to field ionization of the Rydberg state.

While the Bohr atom is of no help in understanding the splittings of the Balmer lines, using it we can calculate the field at which a state is ionized by an electric field. Consider a H atom with its nucleus at the origin in the presence of an electric field in the z direction. The potential experienced by an electron moving along the z axis is given by

$$V = -\frac{k}{|z|} + Ez \quad (1.11)$$

and is shown in Fig. 1.3. Only electrons with energies lower than the local maximum of the potential, at $z = -1000a_0$, are classically bound. In the real three dimensional potential the local maximum at $z = -1000a_0$ is a saddle point, and electrons with energies above the saddle point of the potential are ionized by the field. The potential at the saddle point is given by

$$V_s = -2\sqrt{kE}. \quad (1.12)$$

If this energy is equated to the energy of the Rydberg state, $-Ry/n^2$, we find the classical field for ionization of a state of principal quantum number n

$$E_c = \frac{Ry^2}{4kn^4}. \quad (1.13)$$

This formula is usually encountered in atomic units, in which it reads

$$E_c = 1/16n^4. \quad (1.14)$$

In the next chapter we shall introduce atomic units. Although we have derived Eq. (1.13) with no regard for tunneling or the shifts of the energy levels in an electric

field, it is quite a useful expression, as it is, in a practical sense, nearly exact for nonhydrogenic atoms.

The size of a Rydberg atom is large, the geometric cross section scaling as n^4 , and, not surprisingly, early experiments were focused on this aspect of Rydberg atoms. A classic experiment was undertaken by Amaldi and Segre to observe the energy shifts of the high lying members of the Rydberg series of K in the presence of high pressures of rare gas atoms.⁸ Specifically, they observed the shifts of the $4s\text{--}np$ absorption lines due to the added rare gas. The expectation at the time was that a red shift of the lines would be observed due to the fact that the space between the Rydberg electron and the K^+ ion is filled with a polarizable dielectric, the rare gas atoms, instead of vacuum. While the pressure shifts for Ar and He were to the red, the pressure shift for Ne was to the blue, an observation completely incompatible with the dielectric model. Shortly after the experiment Fermi explained the pressure shifts as arising predominantly from the short range scattering of the Rydberg electron from the rare gas atom, not from the anticipated dielectric effect.⁹ This experiment was perhaps the first of many in which the result was diametrically opposed to expectations.

Due to their large size, Rydberg atoms also exhibit large diamagnetic energy shifts. Since the Zeeman and Paschen–Back effects are proportional to the angular momentum, they are not particularly striking in the optically accessible low ℓ Rydberg states. The diamagnetic shifts on the other hand depend on the area of the orbit and thus scale as n^4 . The diamagnetic shifts are very difficult to see in low n states, but in high n states they are quite evident. The diamagnetic shifts were first observed by Jenkins and Segre in the absorption spectrum of K taken in the 27 kG field of the Berkeley cyclotron.¹⁰ As n increases the field first mixes states of different ℓ , then mixes states of different n . Only within the last 20 years have experiments progressed substantially beyond these first experiments of Jenkins and Segre. Ironically, the study of Rydberg atoms in magnetic fields, one of the first topics studied, is still a topic of active research 50 years later.

The interest in Rydberg atoms waned until it became apparent that they might play an important role in real physical systems. In particular it became apparent that they were important astrophysically.¹¹ The radio recombination lines detected by radio astronomers are the emissions from transitions between Rydberg states subsequent to radiative recombination. A particularly favorable frequency regime in which to search for radio signals is 2.4 GHz, which corresponds to the $n = 109$ to $n = 108$ transition. While it is not surprising that Rydberg atoms might be found in interstellar space, where the density is very low, they also play an important role in astrophysical and laboratory plasmas.^{12,13} Radiative recombination of low energy electrons and ions results in Rydberg atoms, and dielectronic recombination of higher energy electrons and ions often proceeds by the capture of the electrons into autoionizing Rydberg states and the subsequent radiative decay to bound Rydberg states. Furthermore, the microscopic electric fields in a plasma cut off the Rydberg series at or near the classical ionization limit,

as described by Eq. (1.4). Precisely how the Rydberg series is terminated is a matter of some importance, for it determines the thermodynamic properties of the plasma.¹²

The single event which probably contributed the most to the resurgence of interest in Rydberg atoms was the development of the tunable dye laser.^{14,15} With a tunable dye laser and very simple apparatus it became possible to excite large numbers of atoms to a single, well defined Rydberg state. This ability allowed one to study a wide range of properties in astonishing detail. Many properties familiar from low lying excited states, such as the radiative lifetimes, energy level spacings, and collision cross sections were measured systematically, and in many, but not all, cases the properties followed the expected extrapolations from the low lying states. Perhaps more interesting were the investigations of the properties which are in a practical sense peculiar to Rydberg atoms. A good example of such properties is the effect of an electric field. While the gross effects of electric fields on Rydberg atoms had been observed long before in optical spectra, and worked out theoretically for some cases, they had not been understood in a systematic way, mainly because there appeared to be no pressing reason to do so. Not only do electric fields produce dramatic effects on Rydberg atoms, but they also afford us novel techniques for the detection and manipulation of Rydberg atoms. To exploit fully these techniques has required a systematic study of the static and dynamic effects of electric fields.

The most fascinating opportunities presented by Rydberg atoms are those which take advantage of their exaggerated properties to carry out experiments that would be impossible in other systems. One example of this is the interaction of atoms with radiation fields. In the low intensity limit, Rydberg atoms provide an ideal system with which to test the interaction of an atom with the vacuum, particularly with the structured vacuum produced by a resonant cavity. Outstanding examples are the one atom maser¹⁶ and the two photon maser.¹⁷ Rydberg atoms have two properties which make them ideal for such investigations. First, the characteristic frequencies of Rydberg atom transitions are low, and the wavelengths correspondingly long, making the construction of resonant or near resonant cavities a straightforward matter. Second, the dipole moments of Rydberg atoms are so large that they have usable radiative decay rates in spite of the low transition frequencies. At the other intensity extreme, advances in laser technology have made atoms in strong radiation fields, fields comparable in magnitude to the coulomb field, a topic of both practical and fundamental interest.¹⁸⁻²⁰ While it is apparent that any practical applications will involve intense lasers and ground state atoms and molecules, a Rydberg atom in a microwave field is an ideal system in which to carry out quantitative measurements of the properties of atoms in strong radiation fields.²¹

All the properties we have discussed thus far are those of atoms with one Rydberg electron. Double Rydberg atoms, atoms in which there are two highly excited electrons, exhibit pronounced correlation between the motions of the two

electrons and have properties which differ in a qualitative way from those of normal Rydberg atoms.²² While some experiments have been done, they have only scratched the surface, and much is yet to be learned about these exotic atoms.

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Oscillator strengths and lifetimes

In Chapter 3 we considered briefly the photoexcitation of Rydberg atoms, paying particular attention to the continuity of cross sections at the ionization limit. In this chapter we consider optical excitation in more detail. While the general behavior is similar in H and the alkali atoms, there are striking differences in the optical absorption cross sections and in the radiative decay rates. These differences can be traced to the variation in the radial matrix elements produced by nonzero quantum defects. The radiative properties of H are well known, and the radiative properties of alkali atoms can be calculated using quantum defect theory.

Oscillator strengths

A convenient way of expressing the strengths of transitions is to use the oscillator strength. The oscillator strength $f_{n'\ell'm',n\ell m}$ from level $n\ell m$ to level $n'\ell'm'$ is defined by¹

$$f_{n'\ell'm',n\ell m} = 2 \frac{m}{\hbar} \omega_{n'\ell',n\ell} |\langle n'\ell'm' | x | n\ell m \rangle|^2, \quad (4.1)$$

where $\omega_{n'\ell',n\ell} = (W_{n'\ell'} - W_{n\ell})/\hbar$. We could equally well have used the matrix elements of y or z in Eq. (4.1), but in any case, $f_{n'\ell'm',n\ell m}$ is m dependent. Since it is evident that the radiative decay of an atom in free space cannot depend upon m , it is useful to introduce the average oscillator strength $\bar{f}_{n'\ell',n\ell}$ which does not depend upon m . It may be written as¹

$$\bar{f}_{n'\ell',n\ell} = \frac{2}{3} \omega_{n'\ell',n\ell} \frac{\ell_{\max}}{2\ell+1} |\langle n'\ell' | r | n\ell \rangle|^2, \quad (4.2)$$

where $\ell_{\max}$ is the larger of ℓ and ℓ' . If we reverse the roles of ℓ and ℓ' , it is straightforward to show that

$$\bar{f}_{n'\ell',n\ell} = -\frac{2\ell'+1}{2\ell+1} \bar{f}_{n\ell,n'\ell'}. \quad (4.3)$$

The usefulness of the oscillator strengths stems in part from the fact that they satisfy several sum rules. The Thomas–Reiche–Kuhn sum rule is given by¹

$$\sum_{n'\ell'm'} f_{n'\ell'm',n\ell m} = Z, \quad (4.4)$$

where Z is the number of electrons in the atom and the sum implicitly includes the accessible continua. When applied to Na, this sum rule leads to $\sum f_{n'\ell'm',n\ell m} = 11$,

which tells us nothing about the oscillator strength of transitions of the outer electron alone. More useful sum rules, valid for one electron in a central potential, are¹

$$\sum_{n'} \bar{f}_{n'\ell-1,n\ell} = -\frac{1}{3} \frac{\ell(2\ell-1)}{2\ell+1} \quad (4.5)$$

and

$$\sum_n \bar{f}_{n'\ell+1,n\ell} = \frac{1}{3} \frac{(\ell+1)(2\ell+3)}{2\ell+1}, \quad (4.6)$$

from which it is apparent that

$$\sum_{n'\ell'} \bar{f}_{n'\ell',n\ell} = 1, \quad (4.7)$$

where $\ell' = \ell \pm 1$. Eqs. (4.5)–(4.7) apply to, for example, the transitions of the valence electron of Na, in spectral regions where the transitions of the Na^+ core are unimportant. Since the lowest lying excited states of Na^+ lie approximately 25 eV above the ground state of Na^+ , and 30 eV above the ground state of Na, Eqs. (4.5)–(4.7) can be used to describe the transitions of the valence electron of Na due to photons of energy less than 10 eV.

The attractions of the oscillator strengths are that they are dimensionless and they sum to 1. Thus the determination of one oscillator strength is often enough to give a good indication of the overall distribution of oscillator strengths. We shall see that the effect of nonzero quantum defects is to redistribute the alkali oscillator strengths relative to those of H. If we return to Eqs. (4.5) and (4.6) we can see that the strongest $\ell \rightarrow \ell-1$ transitions are to lower energies and that the strongest $\ell \rightarrow \ell+1$ transitions are to higher energies.

Finally, we introduce the Einstein A coefficient, $A_{n'\ell',n\ell}$, which defines the spontaneous decay rate of the $n\ell$ state to the lower lying $n'\ell'$ state. Explicitly,¹

$$A_{n'\ell',n\ell} = \frac{4e^2\omega_{n\ell,n'\ell'}^3}{3\hbar c^3} \frac{\ell_{\max}}{2\ell+1} |\langle n'\ell'|r|n\ell\rangle|^2. \quad (4.8)$$

In terms of the average oscillator strength we can write the A coefficient as

$$A_{n'\ell',n\ell} = \frac{-2e^2\omega_{n\ell,n'\ell'}^2}{\hbar c^3} \bar{f}_{n'\ell',n\ell}. \quad (4.9)$$

The minus sign comes from the fact that $\bar{f}_{n'\ell',n\ell} < 0$. The radiative lifetime, $\tau_{n\ell}$, of the $n\ell$ state is simply the inverse of the total radiative decay rate, which is obtained by summing $A_{n'\ell',n\ell}$ over all lower lying $n'\ell'$ states. Explicitly

$$\tau_{n\ell} = \left[\sum_{n'\ell'} A_{n'\ell',n\ell} \right]^{-1}. \quad (4.10)$$

From Eq. (4.8) we see that the A values contain a factor of ω^3 , which generally means that the transition with the highest frequency contributes the most to the radiative decay rate and therefore dominates the overall dependence on n .

For all Rydberg $n\ell$ states save the s states the highest frequency decay transition is to the lowest lying state of $\ell - 1$. For an s state it is the transition to the lowest p state. In either case, for a high $n\ell$ state, as $n \rightarrow \infty$ the frequency of the highest frequency transition approaches a constant. Thus, in the limit of high n , the A value depends only on the radial matrix element between the Rydberg state and the low lying state. Only the part of the Rydberg state wavefunction which spatially overlaps the wavefunction of the low lying state contributes to the matrix element in Eq. (4.8). As a result, the squared radial matrix element of Eq. (4.8) exhibits an n^{-3} scaling due to the normalization of the Rydberg state wavefunction. Correspondingly,

$$\tau_{n\ell} \propto n^3. \quad (4.11)$$

For $\ell \approx n$ states the same reasoning does not apply. For the $\ell = n - 1$ state the only allowed transition is to the $n' = n - 1$, $\ell' = n - 2$ state. In this case the frequency of the transition is not constant but is given by $1/n^3$. When cubed, this frequency contributes an n^{-9} scaling to the decay rate. Offsetting this scaling is the fact that the radial matrix element represents the size of the atom in the n and $n - 1$ states. Since $\langle n - 1 \ n - 2 | r | n \ n - 1 \rangle \sim n^2$ the lifetimes of the highest ℓ states scale as n^5 , i.e.

$$\tau_{n\ell=n-1} \propto n^5. \quad (4.12)$$

The average lifetime for a statistical mixture of ℓm states of the same n is given by^{1,2}

$$\bar{\tau}_n \propto n^{4.5}. \quad (4.13)$$

Finally, it is useful to extend the definition of oscillator strength above the ionization limit using

$$\frac{d\bar{f}_{\varepsilon'\ell',n\ell}}{dW} = \frac{2}{3} \omega_{\varepsilon'\ell',n\ell} \frac{\ell_{\max}}{2\ell+1} |\langle \varepsilon\ell' | r | n\ell \rangle|^2, \quad (4.14)$$

where the continuum wavefunction is normalized per unit energy.

The photoionization cross section of a statistical mixture of m levels of the $n\ell$ state to the $\varepsilon'\ell'$ continuum is then given by

$$\sigma = \frac{2\pi^2}{c} \cdot \frac{d\bar{f}_{\varepsilon'\ell',n\ell}}{cdW}. \quad (4.15)$$

It is interesting to examine the dependence on n and ℓ of the oscillator strength and lifetimes for the H and alkali atoms. First we show in Fig. 4.1 a plot of the oscillator strengths of the H $1s \rightarrow np$ transitions using the format of Fano and Cooper.³ Above the ionization limit $d\bar{f}/dW$ is plotted directly. Below the ionization limit the oscillator strength to the np level is plotted as a rectangular block of area $\bar{f}_{np,1s}$ from $W = -1/2(n - 1/2)^2$ to $W = -1/2(n + 1/2)^2$, i.e. it is approximately $1/n^3$ wide, $\bar{f}_{np,1s} n^3$ high, and centered at energy $W = -1/2n^2$. Thus the area of each block corresponds to the oscillator strength for the excitation of one np state. Plotting the oscillator strength in this fashion brings out clearly the fact that

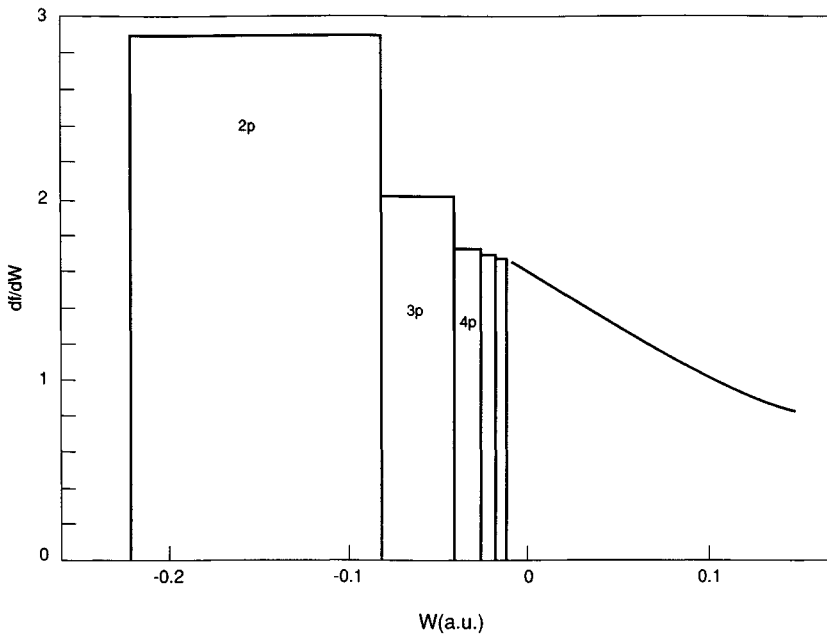


Fig. 4.1 Oscillator strength distribution from the H 1s state to the excited p states. Below the ionization limit the area of each block corresponds to the average oscillator strength from the 1s state to an np state.

oscillator strength per unit energy passes smoothly from the bound to the continuous region of the spectrum. From Fig. 4.1 it is also apparent that at the limit the oscillator strength is slowly varying, and thus $\bar{f}_{np,1s} \propto n^{-3}$ as expected from the normalization of the np wavefunctions near the origin. Fig. 4.1 and direct calculation show that a reasonable fraction, about one half, of the oscillator strength lies above the ionization limit.

In the excited states of H the lifetimes exhibit a predictable behavior. For states of a given n the p states decay most rapidly, followed by the d, f, g, s, h . . . states in that order.⁴ Putting aside the s states, the dominant decay mechanism for each $n\ell$ series is to the lowest $n'\ell - 1$ state. The frequency dependence of this transition upon ℓ leads to the observed ℓ ordering of the lifetimes. The ns states have long lifetimes due to the fact that the radial matrix elements to the lower lying p states are small. For the high ns states of H the first two lobes of the ns radial wavefunction overlap the 2p wavefunction, and their contributions cancel to some extent. In contrast, only one lobe, the innermost, of the nd radial wavefunction overlaps the 2p radial wavefunction. The nd wavefunction also has a larger amplitude than the ns wavefunction at the position of the 2p wavefunction because of the centrifugal potential, which ensures that the nd atom has less kinetic energy in radial motion. In general, matrix elements involving decreases in n accompanied by a decrease in ℓ by one are larger than matrix elements involving

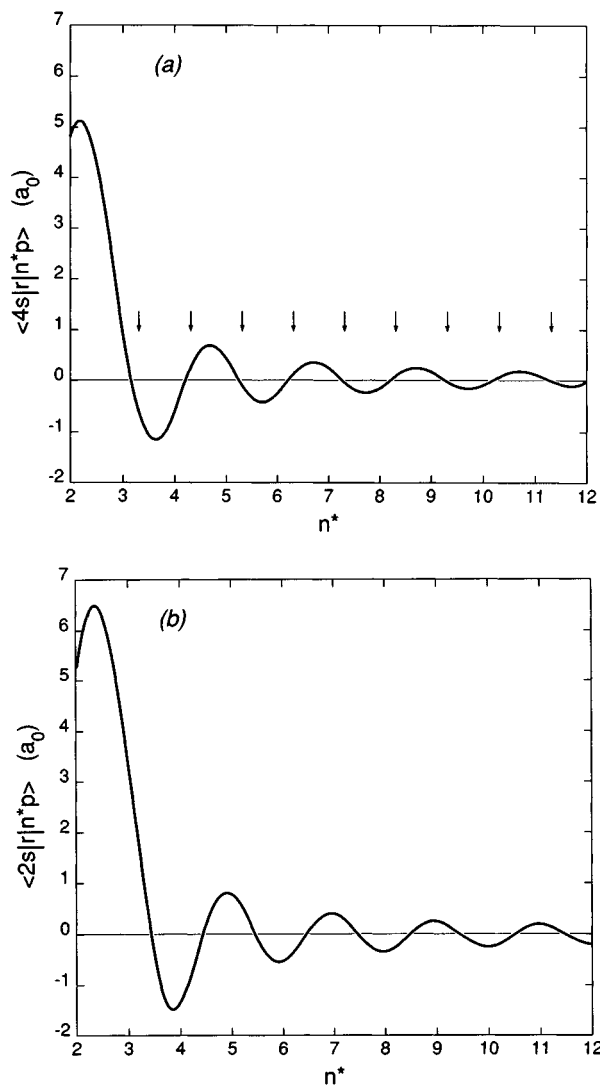


Fig. 4.2 (a) Radial matrix element of the K $4s \rightarrow n^*p$ transition with n^* a continuous variable. The quantum defect of the $4s$ state is 2.23. The quantum defects of the np states are 1.71, so the np states fall at the locations shown by the arrows near where the matrix element crosses zero. (b) H $2s \rightarrow n^*p$ radial matrix element as a function of n^* . Note that the maximum amplitudes of the matrix element occur at integer values of n^* .

decreases in n with an increase of ℓ by one. With the exception of the ns states, all states decay predominantly by $\Delta\ell = -1$ transitions.^{1,4}

Now let us consider alkali atoms. We begin by considering the K $4s \rightarrow np$ excitations analogous to the excitation from the ground state of H. In K the situation is in fact quite different, as shown by Fig. 4.2(a), a plot of the radial

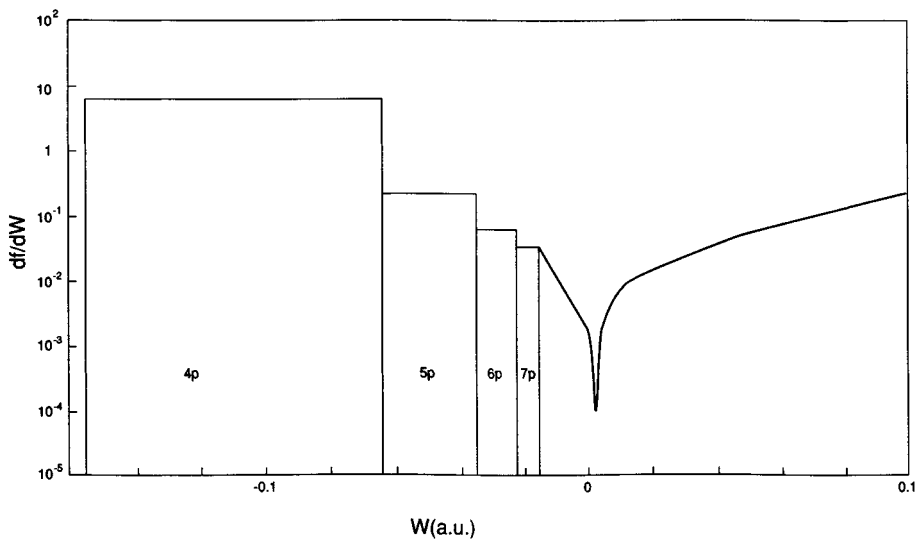


Fig. 4.3 Oscillator strength distribution from the K 4s state to the np states and ϵp continuum. Note that the vertical scale is logarithmic, not linear as in Fig. 4.1. The oscillator strengths to the high np states are orders of magnitude smaller than in H.

matrix element $\langle n^*p|r|4s \rangle$ for K with n^* a continuous variable. The K 4s state has a quantum defect of 2.23, and the high lying p states have quantum defects of 1.71, so that $n^* = 0.29(\text{mod } 1)$. In Fig. 4.2 we have shown the locations of the np states. It is quite apparent that the $4s \rightarrow 4p$ transition has a large matrix element, but the matrix elements to higher n states fall near the zero crossings in the matrix element and are therefore very small. The analogous plot of the matrix element between the H 2s and np states is shown in Fig. 4.2(b). The H $\langle 2p|r|2s \rangle$ matrix element is larger than the K $4s \rightarrow 4p$ matrix element due to the smaller binding energy of the H 2s state, and all the matrix elements between the 2s and the higher np states are further away from the zero crossings and hence larger than the K $\langle np|r|4s \rangle$ matrix elements.

If we convert the squares of the matrix elements shown in Fig. 4.2 to average oscillator strengths, by multiplying by $2\omega/3$, we find that in H the $2s \rightarrow 2p$ transition has vanishing oscillator strength due to the $2s \rightarrow 2p$ degeneracy. All the oscillator strength must come from the $n \geq 3$ $2s \rightarrow np$ and $2s \rightarrow \epsilon p$ transitions. On the other hand, in K the $4s \rightarrow 4p$ transition has a significant oscillator strength since the 4p state lies halfway between the 4s state and the ionization limit. In fact, it has an oscillator strength of very nearly 1, consuming nearly all of the available K $4s \rightarrow np$ oscillator strength. In Fig. 4.3 we show a plot of $d\bar{f}/dW$ for K. The values for the bound oscillator strengths and for the continuum more than 0.03 au above the limit are taken from Marr and Creek.^{5,6} The values of $d\bar{f}/dW$ near the minimum just above the limit are taken from the relative measurements of Sandner *et al.*⁷ normalized to the bound state oscillator strengths given by Marr

and Creek.⁵ In K the variation in $d\bar{f}/dW$ is so great that Fig. 4.3 is plotted on a logarithmic scale. As shown, $d\bar{f}/dW$ declines throughout the bound spectrum reaching a minimum just above the ionization limit. This minimum, usually termed the Cooper minimum, is found in the continua of all the alkalis save Li.

The origin of the minimum is most easily understood by considering a spinless K atom. In a spinless K atom the $4s$ – np radial matrix elements decrease with increasing energy, as implied by Fig. 4.3. The decrease with increasing energy of the np state occurs because the nodes of the bound np , or continuum ϵp , wavefunction become increasingly close together with increasing energy. Stated another way, the $4s$ – np or $4s$ – ϵp radial matrix element is an integral of the product of a fixed function, $r^3 R_{4s}(r)$, and a function the wavelength of which decreases with energy, $R_{np}(r)$ or $R_{\epsilon p}(r)$. In general the resulting integral oscillates about zero as a function of energy. In K the first change in sign is just above the ionization limit, leading to $d\bar{f}/dW = 0$ and a vanishing photoionization cross section. Having neglected spin we predict a zero cross section instead of the nonzero minimum of Fig. 4.3. The energy at which the minimum occurs is determined by the radial phases of the ground state and ϵp wavefunctions. In Na, K, Rb, and Cs these phases differ by $\sim \pi/2$, since the ground state and np quantum defects differ by $1/2$. For these atoms the Cooper minimum occurs in the continuum not far above the ionization limit. In Li the quantum defects of the ground state and the np states differ by $1/4$, and $d\bar{f}/dW$ has its minimum below the ionization limit.

As shown by Fig. 4.3, the cross section does not actually vanish, but only decreases to a nonzero minimum. As pointed out by Seaton,⁸ when the spin orbit interaction is taken into account there are two ϵp continuum waves, $\epsilon p_{1/2}$ and $\epsilon p_{3/2}$, which have slightly different radial phases corresponding to the slightly different quantum defects of the bound $np_{1/2}$ and $np_{3/2}$ series. As a result of having different phases the $\epsilon p_{1/2}$ and $\epsilon p_{3/2}$ radial matrix elements cross zero at slightly different energies. Each of these continuum channels has its own oscillator strength, and the total cross section is never zero. While the location of the minimum is a function of the difference in the quantum defects of the ground state and the excited np states, the minimum cross section and the width of the minimum are proportional to the np fine structure splitting, or the difference in the $np_{1/2}$ and $np_{3/2}$ quantum defects. Consequently the minimum cross section and the width of the Cooper minimum increase from Na to Cs.

Since the radial matrix elements from the ground state to the $np_{1/2}$ and $np_{3/2}$ states are not the same, due to the differing radial phases, the oscillator strengths and intensity ratios of the transitions to the $np_{1/2}$ and $np_{3/2}$ states cannot be in the 1:2 ratio predicted by the angular factors.⁸ The largest deviations occur in Cs, for the Cs $6s_{1/2}$ – $\epsilon p_{1/2}$ matrix element vanishes just above the ionization limit.⁹ As a result, just below the ionization limit large intensity ratios have been measured.^{10–13} For example, at $n=30$ Raimond *et al.* measured a ratio of 1:1170.¹³ The measurements agree with calculations of Norcross¹⁴ which predict that the

intensity ratio at $n=\infty$ should be roughly 1:12,000. While the Cooper minimum in K falls much closer to the ionization limit than it does in Cs, both the $4s\text{--}\epsilon p_{1/2}$ and $4s\text{--}\epsilon p_{3/2}$ matrix elements cross zero at nearly the same energy, and in the bound states the maximum $np_{1/2}\text{:}np_{3/2}$ intensity ratio is only 1:4.¹⁵

Radiative lifetimes

Both the absorption cross sections and the lifetimes of low ℓ states of alkali atoms are different from the hydrogenic values. The most glaring examples occur the alkali np states. In H the np states have very short radiative lifetimes, but in an alkali atom they have very long lifetimes, due to their small oscillator strengths to the ground state. For example, as shown in Figs. 4.1 and 4.3, the oscillator strengths from the ground states of H and K to the high, $n \approx 10$, np states differ by a factor of 100. This factor, when multiplied by the ratio of the squares of the frequencies, 16, leads to K $np \rightarrow 4s$ Einstein A coefficients roughly 1000 times smaller than their H $np \rightarrow 1s$ counterparts. The K np lifetimes are not a thousand times longer than the H np lifetimes because decay to states other than the ground state plays a significant role in the total decay rate.

The radiative lifetimes of many excited states of Li,¹⁶ Na,^{17–21} K,²² Rb,^{23–25} and Cs,^{26–28} have been measured using a variety of techniques, the most common being time resolved laser induced fluorescence, which is typically carried out using a cell, as shown in Fig. 4.4. In all cases, the observed lifetimes are in reasonable agreement with values calculated in the coulomb approximation, corrected for the decrease due to black body radiation.²⁹ In the following chapter we show that if the 0 K lifetime of a state is τ that its lifetime at a finite temperature T is given by^{24,30}

$$\frac{1}{\tau^T} = \frac{1}{\tau} + \frac{1}{\tau^{bb}}. \quad (4.16)$$

where $1/\tau^{bb}$ is the black body radiation induced decay rate.

Systematic lifetime measurements have been made for Rb^{23–25} and Na.^{17–21} In both cases the lifetimes of the s, p, d, and f states have been measured, and we discuss the Na lifetimes as a representative example. The lifetimes of Na $n < 15$ s, p, d, and f states have been measured using the time and wavelength resolved laser induced fluorescence arrangement shown in Fig. 4.4. Na atoms at a pressure of 10^{-6} Torr in a pyrex cell are excited from ground state to the 3p state and then to the ns or nd states using two dye laser pulses of 5 ns duration. The ns and nd lifetimes are easily measured by observing the time resolved $ns \rightarrow 3p$ or $nd \rightarrow 3p$ fluorescence with a photomultiplier^{17,18}. These fluorescent decays have large branching ratios and are easily detected in spite of the fact that they are at the

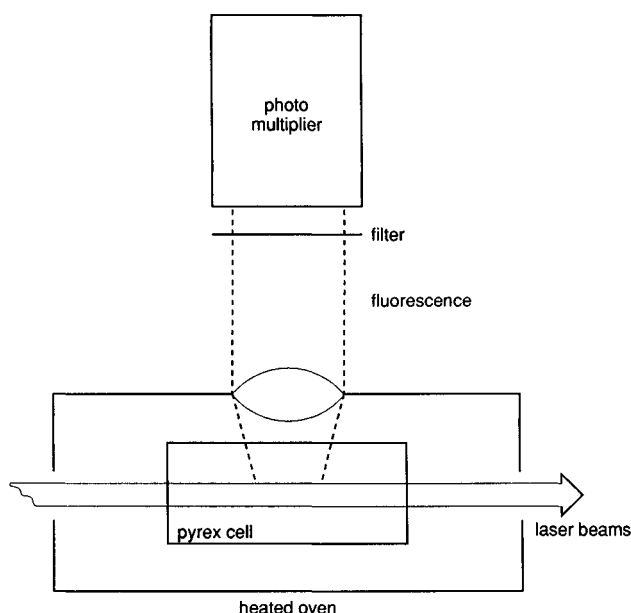


Fig. 4.4 Schematic diagram of a typical arrangement for laser induced fluorescence measurements of lifetimes. A pulsed laser beam (or beams) passes through a heated glass cell containing alkali vapor and the time and wavelength resolved fluorescence is detected at a right angle.

same wavelength as the second exciting laser. For rapidly decaying low lying states the fluorescence is typically stronger than the scattered laser light, and for slowly decaying high lying states the phototube can be gated on after the laser pulse. The longer lived Na np lifetimes can be observed by exciting the nd state which quickly decays, leaving an observable fraction, $\sim 10\%$, of the initial nd population in the much longer lived np state.¹⁹ The time resolved $4p \rightarrow 3s$ fluorescence can be directly detected, but for higher np states the $np \rightarrow 3s$ fluorescence has too short a wavelength to pass through a glass cell. Instead, the time resolved $ns \rightarrow 3p$ fluorescence, the second step of the $np \rightarrow ns \rightarrow 3p$ cascade is observed. The long time behavior of the observed signal reflects the lifetime of the np state, which is much longer than the lifetime of either the ns or nd states. Finally, the lifetimes of several Na nf states have been measured using a resonant microwave field to equalize the populations in the nd and nf levels.²⁰ Observing the time resolved fluorescence from either the nf or nd state gives a decay rate indistinguishable from the nd state decay rate, indicating that the nf lifetime is very nearly the same as the nd lifetime. Had the nf lifetimes been much longer than the nd lifetimes, this method would have been relatively insensitive to the nf lifetimes.

The fluorescence methods described above work well for states of principal quantum numbers less than 15. For higher principal quantum numbers they do not work so well for three reasons. First, the amplitude of the fluorescence signal

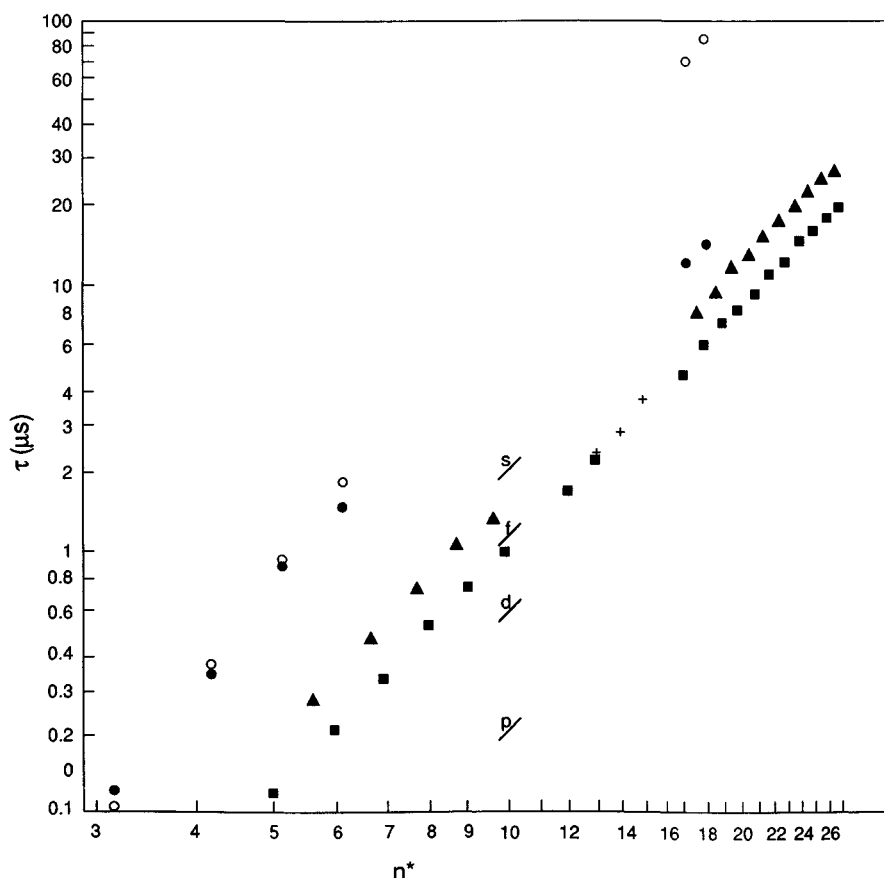


Fig. 4.5 Na radiative lifetimes vs n^* . Experimental values for ns ($\blacktriangle$), np ($\circ$), nd ($\blacksquare$) and nf ($+$) states are shown. The lifetimes states below $n = 15$ have been measured by fluorescence techniques, at temperatures of approximately 400 K. The lifetimes of $n > 15$ states have been measured by field ionization, the ns and nd states at 30 K and the np states at 300 K. The theoretical 0 K np lifetimes are also shown ($\circ$). They are far above the measured values at high n due to black body radiation. Finally, the hydrogenic lifetimes are shown by the line segments ($—$) (from refs. 4, 18–22, 30).

decreases as n^{-6} , second, the atoms pass out of the field of view due to their longer lifetimes, and, third, black body radiation can appreciably alter the lifetimes.⁸ The time delayed field ionization method employed by Spencer *et al.*²¹ to measure the Na ns and nd lifetimes circumvents all of these problems. They passed a well collimated Na atomic beam through an interaction region cooled to 30 K. The atoms were excited by two pulsed dye laser beams collinear with the atomic beam and tuned to the $3s \rightarrow 3p$ and $3p \rightarrow ns$ or $3p \rightarrow nd$ transitions. At a variable time after the laser excitation a field ionization pulse was applied to atoms in the interaction region and the resulting ions detected. The atoms which passed out of the interaction region between the laser pulse and the field ionization pulse were

Table 4.1. *Lifetime parameters for alkali atoms.*

		s	p	d	f
Li ^a	τ_0 (ns)	1.39	5.69	0.59	0.96
	α	2.80	2.78	2.92	3.06
Na ^b	τ_0 (ns)	1.38	8.35	0.96	1.13
	α	3.00	3.11	2.99	2.96
K ^b	τ_0 (ns)	1.32	6.78	5.94	0.83
	α	3.00	2.78	2.82	2.95
Rb ^b	τ_0 (ns)	1.43	2.76	2.09	0.76
	α	2.94	3.02	2.85	2.95
Cs ^b	τ_0 (ns)	1.43	4.42	0.96	0.69
	α	2.96	2.94	2.93	2.94

^a(computed from values in ref. 29)^b(from ref. 31)

replaced by atoms which had been excited further upstream by the laser. Using this approach they were able to make measurements at delay times of up to 25 μ s, corresponding to about 2 cm of travel of the atoms. It is useful to note that if the interaction region is cooled the only criterion for the field ionization pulse is that it be strong enough to ionize the Rydberg state in question. If significant black body redistribution occurs, as it does at 300 K, the field ionization pulse must be chosen with more care, as described in Chapter 5.

In Fig. 4.5 we show measured values of Na *ns*, *np*, *nd*, and *nf* lifetimes vs n^{*3} on logarithmic scales. The scales differ by a factor of 3, so an n^{*3} dependence appears as a 45° line. As shown by Fig. 4.5 the n^{*3} dependence suggested by Eq. (4.11) is obtained. The values for $n \leq 15$ have all been obtained using fluorescence detection from atoms in a cell at about 425 K. For all but the *np* states the black body radiation is a negligible effect. For the *np* states we also show the calculated 0 K values of Theodosiou, and for the 7*p* state, the measured value is about 30% below the calculated 0 K value and consistent with the calculated 425 K value of Theodosiou.²⁹ For $n > 15$ all values have been obtained using field ionization. The *ns* and *nd* values are those obtained by Spencer *et al.*²¹ at 30 K and are clearly in reasonably good agreement with an extrapolation of the values obtained at lower n using fluorescence detection. For the 18*p* and 19*p* lifetimes measured at 300 K, however, black body radiation reduces the lifetimes far below an n^{*3} extrapolation of the lifetimes of the lower *np* states.³⁰

We also show in Fig. 4.5 short 45° lines at $n = 10$ corresponding to the lifetimes of the hydrogenic *ns*, *np*, *nd*, and *nf* series. The Na *np* and *nd* lifetimes are respectively factors of 50 and 2 longer than their hydrogenic counterparts, while the Na *ns* lifetimes are about 30% shorter than the H *ns* lifetimes. The Na and H *nf* lifetimes are the same, which is not surprising as the Na *nf* states and the Na *nd* states to which they predominantly decay all have quantum defects of nearly zero.

The lifetimes of alkali Rydberg states can be calculated accurately, as shown by Theodosiou,²⁹ who has made extensive comparisons between observed lifetimes and calculated values. From this work it is also apparent that the black body radiation correction of Eq. (4.16) is important for high lying states. In light of these two points, it seems that the most compact and consistent way of presenting the alkali lifetimes is by a fit of the calculated 0 K values to the form³¹

$$\tau = \tau_0(n^*)^\alpha \quad . \quad (4.17)$$

The values of τ_0 and α are presented in Table 4.1. As shown by Table 4.1, the values of α are all near 3, as expected from Eq. (4.11). The biggest discrepancies occur in the Li and K np series which have their Cooper minima nearest the ionization limit. It is also apparent that the np lifetimes are the longest, as expected from Figs. 4.1 and 4.2. In using the values of Table 4.1 it is worth remembering that for high n states the black body decay rate is ℓ independent, so the lifetimes of the longest lived states are the most affected by black body radiation.³²

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Black body radiation

Lying ≥ 4 eV above the ground state, Rydberg states are not populated thermally, except at very high temperatures. Accordingly, it is natural to assume that thermal effects are negligible in dealing with Rydberg atoms. However, Rydberg atoms are strongly affected by black body radiation, even at room temperature. The dramatic effect of thermal radiation is due to two facts. First, the energy spacings ΔW between Rydberg levels are small, so that $\Delta W < kT$ at 300 K. Second, the dipole matrix elements of transitions between Rydberg states are large, providing excellent coupling of the atoms to the thermal radiation. The result of the strong coupling between Rydberg atoms and the thermal radiation is that population initially put into one state, by laser excitation for example, rapidly diffuses to other energetically nearby states by black body radiation induced dipole transitions.^{1–3} Both the redistribution of population and the implicit increase in the radiative decay rates are readily observed. Although the above mentioned effects on level populations are the most obvious effects, the fact that a Rydberg atom is immersed in the thermal radiation field increases its energy by a small amount, 2 kHz at 300 K. While the radiation intensity is vastly different in the two cases, this effect is the same as the ponderomotive shift of the ionization limit in high intensity laser experiments.

Black body radiation

The most familiar way of representing black body radiation is to use the Planck radiation law for the energy density, or the square of the electric field, $\rho(\nu)$. Explicitly⁴

$$\rho(\nu) d\nu = \frac{8\pi h \nu^3}{c^3 (e^{h\nu/kT} - 1)}, \quad (5.1)$$

where k is the Boltzmann constant, h is Planck's constant, ν is the frequency of the black body radiation, and T is the temperature.

In Fig. 5.1 we show $\rho(\nu)$ vs ν at 300 K using both frequency and wavenumber as abscissae. A typical optical transition from the ground state of an atom has frequency $\nu = 3 \times 10^{14}$ Hz, and a transition between two Rydberg states has frequency $\nu \sim 3 \times 10^{11}$ Hz. Thus, it is apparent from Fig. 5.1 that to a ground state atom the black body radiation appears as a slowly varying, nearly static field, whereas to a Rydberg atom it appears to be a rapidly varying field.

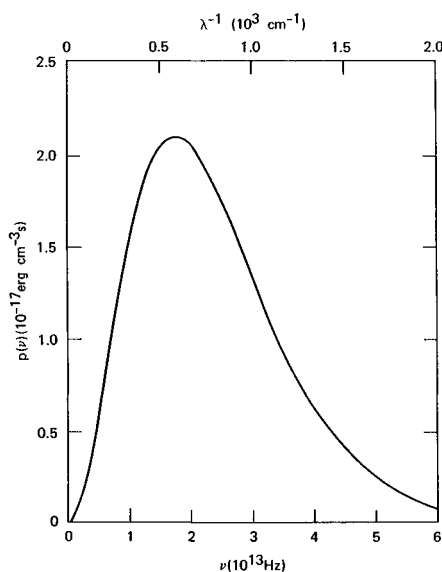


Fig. 5.1 Energy density $\rho(\nu)$ as a function of frequency ν at 300 K (from ref. 5).

While the representation of black body radiation given in Eq. (5.1) and Fig. 5.1 is the most familiar, it is not the most useful for the most important effect of black body radiation on Rydberg atoms, inducing transitions between neighboring levels.

To calculate the rates of these transitions it is more convenient to express the radiation field in terms of the number of photons per mode of the radiation field, i.e. the photon occupation number $\bar{n}$ of each mode. The photon occupation number $\bar{n}$ is given by⁴

$$\bar{n} = \frac{1}{e^{h\nu/kT} - 1}. \quad (5.2)$$

At low frequencies, for which $h\nu \ll kT$, Eq. (5.2) reduces to

$$\bar{n} \approx \frac{kT}{h\nu} \quad (5.3)$$

Fig. 5.2 shows the dependence of $\bar{n}$ on ν for $T = 300$ K, again with both frequency (Hz) and wavenumber (cm^{-1}) used as abscissae. It is useful to recall that at 300 K, $kT/h \approx 6 \times 10^{12}$ Hz or $kT/hc \approx 200$ cm^{-1} . This point is apparent in Fig. 5.2, since $\bar{n} = 1$ at $\lambda^{-1} \approx 200$ cm^{-1} . Since the vacuum fluctuations, which lead to spontaneous emission, are given by $\bar{n} = 1/2$, black body radiation at frequencies greater than $kT/h\bar{n}$, where $\bar{n} \ll 1$, does not lead to significant effects. For an atom in its ground state with transitions at $\sim 10^4$ cm^{-1} , black body induced transitions are unimportant, since $\bar{n} \ll 1$. However, for a Rydberg state with transitions at 10 cm^{-1} , where $\bar{n} \sim 10$, the black body induced transition rates can

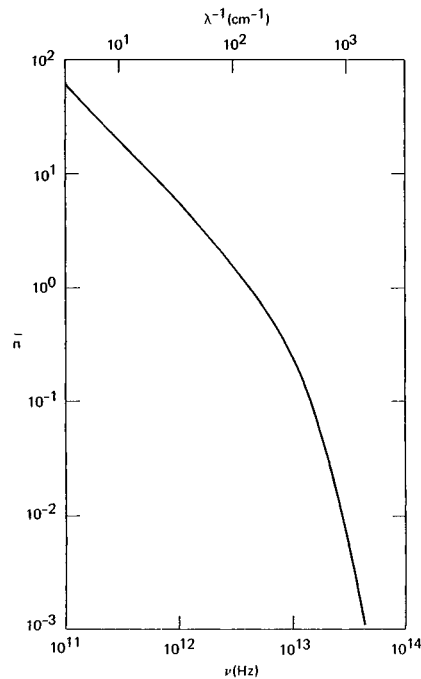


Fig. 5.2 Occupation number $\bar{n}$ as a function of frequency ν at 300 K (from ref. 5).

be an order of magnitude larger than the spontaneous emission rates, and the effect of black body radiation is significant.

Eqs. (5.1)–(5.3) present black body radiation in a familiar form. Both to conform to the general use of atomic units in this book and to simplify the calculation of the Rydberg atoms' response to the radiation we reexpress Eqs. (5.1)–(5.3) in atomic units.⁵

We reexpress Eq. (5.1) as

$$\rho(\omega)d\omega = \frac{2\alpha^3\omega^3d\omega}{\pi(e^{\omega/kT} - 1)}, \quad (5.4)$$

where α is the fine structure constant and ω is the energy in au. Similarly, we may reexpress Eq. (5.2) as

$$\bar{n} = \frac{1}{e^{\omega/kT} - 1} \quad (5.5)$$

and Eq. (5.3) for $\omega \ll kT$ as

$$\bar{n} \approx \frac{kT}{\omega}. \quad (5.6)$$

We now consider the transition rates for absorption of and stimulated emission induced by the black body radiation and compare these rates to the spontaneous

emission rates. This comparison provides a reasonable indication of the importance of the effects of black body radiation induced transitions for Rydberg atoms.⁶

Black body induced transitions

As shown in the previous chapter, the spontaneous decay rate of the $n\ell$ state to the lower lying $n'\ell'$ state is given by the Einstein A coefficient $A_{n'\ell',n\ell}$.⁷ In the presence of thermal radiation the stimulated emission rate $K_{n'\ell',n\ell}$ is simply $\bar{n}$ times as large as the spontaneous rate. Explicitly,

$$K_{n'\ell',n\ell} = \bar{n}A_{n'\ell',n\ell}, \quad (5.7)$$

where $\bar{n}$ is implicitly evaluated at the frequency $\omega_{n'\ell',n\ell}$. It is convenient to reexpress Eq. (5.7) in terms of the average oscillator strength,

$$K_{n'\ell',n\ell} = -2\bar{n}\alpha^3\omega_{n'\ell',n\ell}^2\bar{f}_{n'\ell',n\ell}, \quad (5.8)$$

where α is the fine structure constant, $\omega_{n'\ell',n\ell}$ is the energy difference $W_{n'\ell'} - W_{n\ell}$, and

$$\bar{f}_{n'\ell',n\ell} = \frac{2}{3}\omega_{n'\ell',n\ell}\frac{\ell_{\max}}{2\ell+1}|\langle n'\ell'|r|n\ell\rangle|^2, \quad (5.9)$$

where $\ell_{\max}$ is the larger of ℓ and ℓ' .

The black body photons can also be absorbed as the atoms in the $n\ell$ state make the transition to a higher lying $n'\ell'$ state. Both the stimulated emission and absorption rates are given by

$$K_{n'\ell',n\ell} = 2\bar{n}\alpha^3\omega_{n'\ell',n\ell}^2|\bar{f}_{n'\ell',n\ell}|. \quad (5.10)$$

Because of the variation of $\bar{n}$ with frequency, the frequency dependence of $K_{n'\ell',n\ell}$ is quite different from that of $A_{n'\ell',n\ell}$, and these two processes favor different final states. This point is made graphically in Fig. 5.3, which is a plot of $A_{n'p,18s}$ and $K_{n'p,18s}$ vs n' for the Na 18s state and $T = 300$ K. As shown in Fig. 5.3, black body radiation favors transitions to nearby states and spontaneous emission favors transitions to the lowest lying states.

In addition to driving transitions to discrete states the black body radiation can also photoionize the Rydberg atoms. The photoionization rate, $1/\tau_{n\ell}^P$ is given by⁸

$$\frac{1}{\tau_{n\ell}^P} = \frac{2\pi}{3} \int_{1/2n^2}^{\infty} \left\{ \frac{\ell}{2\ell+1} |\langle n\ell|r|\varepsilon\ell-1\rangle|^2 + \frac{(\ell+1)}{2\ell+1} |\langle n\ell|r|\varepsilon\ell+1\rangle|^2 \right\} \rho(\omega) d\omega. \quad (5.11)$$

The matrix elements are the radial matrix elements of r between the initial $n\ell$ state and continuum $\ell-1$ and $\ell+1$ waves which are normalized per unit energy.

Just as we previously summed the spontaneous emission rates over the possible final states to obtain the total spontaneous decay rate $1/\tau_{n\ell}$, we can sum the black

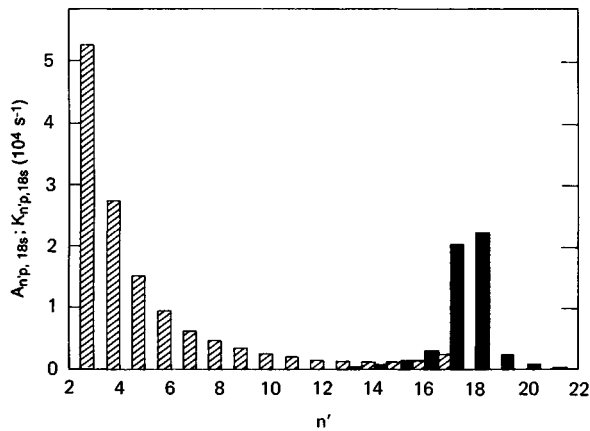


Fig. 5.3 Spontaneous emission rates of the 18s state to lower lying $n'p$ states, $A_{n'p, 18s}$, ($\square$) and the 300 K black body transition rates of the 18s state to higher and lower lying $n'p$ states, $K_{n'p, 18s}$, ($\blacksquare$) as a function of n' (from ref. 5).

body radiation transition rates to compute the total black body radiation induced decay rate $1/\tau_{n\ell}^{\text{bb}}$. Explicitly,

$$\frac{1}{\tau_{n\ell}^{\text{bb}}} = 2\alpha^3 \sum_{n'} \bar{n} \omega_{n'e', n\ell}^2 |f_{n'e', n\ell}| + \frac{1}{\tau_{n\ell}^{\text{P}}}, \quad (5.12)$$

and the total decay rate $1/\tau_{n\ell}^{\text{T}}$ is given by

$$\frac{1}{\tau_{n\ell}^{\text{T}}} = \frac{1}{\tau_{n\ell}} + \frac{1}{\tau_{n\ell}^{\text{bb}}}. \quad (5.13)$$

From Fig. 5.3 it is apparent that the most important contributions to the black body radiation decay rate are from transitions to nearby states for which $|\omega_{n'e', n\ell}| < kT$. In this case we may substitute Eq. (5.6) for $\bar{n}$ and write $1/\tau_{n\ell}^{\text{bb}}$ as

$$\frac{1}{\tau_{n\ell}^{\text{bb}}} = 2\alpha^3 kT \sum_{n'} \omega_{n'e', n\ell} \bar{f}_{n'e', n\ell}. \quad (5.14)$$

In Eq. (5.14) the summation over n' implicitly includes the continuum. For a one-electron atom we may use the sum rule⁷

$$\sum \omega_{n'e', n\ell} \bar{f}_{n'e', n\ell} = \frac{2}{3n^2}, \quad (5.15)$$

which we may use to write

$$\frac{1}{\tau_{n\ell}^{\text{bb}}} = \frac{4\alpha^3 kT}{3n^2}. \quad (5.16)$$

For $T = 300 \text{ K}$, Eq. (5.16) is accurate to 30% for $n > 15$, and as n is increased Eq. (5.16) becomes an increasingly good approximation. More important, it brings

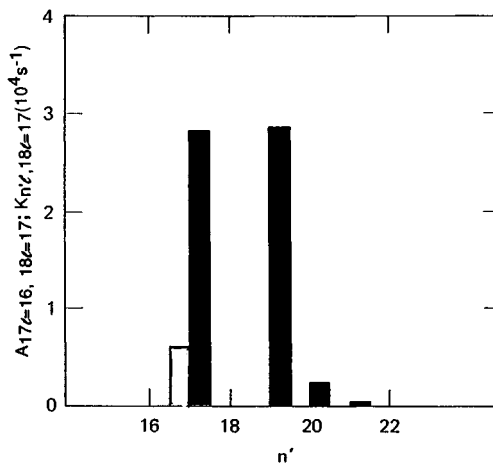


Fig. 5.4 Spontaneous emission rate of the $n = 18, \ell = 17$ state to the $n = 17, \ell = 16$ state, $A_{17\ell=16, 18\ell=17}$, ($\square$) and the 300 K black body transition rates of the $n = 18, \ell = 17$ state to $n'\ell'$ states of $\ell' = \ell \pm 1$, $K_{n'\ell', 18\ell=17}$, ($\blacksquare$) as a function of n' (from ref. 5).

out two interesting points. First, the total black body decay rate $1/\tau_{n\ell}^{\text{bb}}$ does not depend on ℓ , only on n . Since the radiative decay rate decreases rapidly as ℓ is increased,⁹ for high ℓ states the black body induced decay rate is often much larger than the spontaneous emission rate. This point is shown graphically in Fig. 5.4 in which the spontaneous emission rates and black body radiation emission and absorption rate are shown for the $n = 18, \ell = 17$ state. In contrast to the 18s state, for which $1/\tau^{\text{bb}} \sim 0.2/\tau$, the $n = 18, \ell = 17$ state has a black body decay rate which is ten times larger than its spontaneous decay rate. Second, because of the $1/n^2$ scaling of the black body radiation decay rate, it is evident that even for low ℓ states that at high enough n the black body radiation rate exceeds the spontaneous emission rate, which scales as $1/n^3$. Even at $n = 20$, the black body radiation rate is at least 20% of the spontaneous emission rate for all ℓ states, and in many experiments it is a significant source of population redistribution.

Black body energy shifts

That part of the black body spectrum coincident with the atomic transition frequencies leads to the transitions which redistribute the population. In contrast, all the energy of the black body radiation contributes to the shift of the energy levels. The energy shift is a second order ac Stark shift, and for state n the shift ΔW_b is given by¹⁰

$$\Delta W_{n\ell} = \sum_{n'\ell'} \int_0^\infty \frac{|\langle n'\ell' | r | n\ell \rangle|^2 E_{\omega_b}^2 \omega_{n\ell, n'\ell'}}{2(\omega_{n\ell, n'\ell'}^2 - \omega_b^2)} d\omega_b \quad (5.17)$$

where E_{ω_b} is the electric field of the black body radiation in a bandwidth $d\omega_b$ at frequency ω_b . Since $E_{\omega_b}^2$ is proportional to the energy density, examining Fig. 5.2 we can see that at room temperature the peak of the black body radiation energy density is at $\sim 500 \text{ cm}^{-1}$, far above the strong $n \sim 20$ transitions which have wavenumbers of $\sim 20 \text{ cm}^{-1}$. In fact, for atoms in Rydberg states it is often true that the frequencies of the strong transitions are much lower than the frequency of the black body radiation, i.e. $\omega_{n\ell, n'\ell'} \ll \omega_b$, in which case we can ignore $\omega_{n\ell, n'\ell'}$ in the denominator of Eq. (5.17) and rewrite $\Delta W_{n\ell}$ as

$$\Delta W_{n\ell} = \left(\sum_{n'\ell'} \bar{f}_{n'\ell', n\ell} \right) \left(\int_0^\infty \frac{E_{\omega_b}^2 d\omega_b}{4\omega_b^2} \right). \quad (5.18)$$

Using the oscillator strength sum rule⁷

$$\sum_{n'\ell'} \bar{f}_{n'\ell', n\ell} = 1 \quad (5.19)$$

and integrating over ω_b in Eq. (5.18), we may express Eq. (5.18) as

$$\Delta W_{n\ell} = \frac{\pi}{3} \alpha^3 (kT)^2. \quad (5.20)$$

Eq. (5.20) is accurate to $\sim 10\%$ for $n > 15$ at $T = 300 \text{ K}$. Eq. (5.20) indicates that all Rydberg states experience the same energy or frequency shift, which is $+2.2 \text{ kHz}$ at 300 K . Eq. (5.20) corresponds to the energy shift of a free electron in an oscillating electric field. This point is more apparent if we simply use the oscillator strength sum of Eq. (5.19) and apply it to the case of a monochromatic field of angular frequency ω_b for which $\int_0^\infty E_b^2 d\omega_b = E^2$. In this case Eq. (5.18) can be written

$$\Delta W_{n\ell} = \frac{E^2}{4\omega^2}, \quad (5.21)$$

which is the average kinetic energy of a free electron in a field $E \cos \omega t$. The equality of the energy shift of a free electron and an electron in a Rydberg state is not surprising since the physical effect of the ac black body field, at a frequency high compared to the orbital frequency of the electron, is to superimpose a fast, $\sim \omega_b$, wiggle on the orbital motion of the electron. The energy of the wiggle motion is independent of the much slower orbital velocity.

While the black body radiation energy is mostly at frequencies high compared to the Rydberg state frequencies, it is low compared to the frequencies of transitions of low lying states. Explicitly, $\omega_{n'\ell', n\ell} \gg \omega_b$, and for low lying states we can ignore ω_b in the denominator of Eq. (5.18). In this case the Stark shift is equal to that produced by a static field and can be expressed as

$$\Delta W_{n\ell} = \sum_{n'\ell'} - \frac{\bar{f}_{n'\ell', n\ell}}{\omega_{n'\ell', n\ell}^2} \int_0^\infty \frac{E_{\omega_b}^2 d\omega_b}{4}. \quad (5.22)$$

For a one-electron Rydberg atom we may use the hydrogenic sum rule¹¹

$$\sum_{n'} \frac{\bar{f}_{n'\ell',n\ell}}{\omega_{n'\ell',n\ell}^2} = \frac{9}{2} \quad (5.23)$$

so that we may write Eq. (5.22) as

$$\Delta W_n = - \int \frac{9}{8} E_{\omega_b}^2 d\omega_b = - \frac{3}{5} (\alpha\pi)^3 (kT)^4. \quad (5.24)$$

Evaluation of Eq. (5.24) gives a shift of -0.036 Hz at 300 K, which is negligible for all practical purposes.

Finally, we must remember that for some states it will be true that $\omega_{n'\ell',n\ell} \sim \omega_b$, in which case neither of the above approximations is valid and Eq. (5.17) must be evaluated explicitly as has been done by Farley and Wing.¹² At 300 K $\omega_{n\ell,(n+1)\ell'} \sim \omega_b$ for $n \sim 8$.

If we confine our attention to Rydberg states of $n > 15$, for which Eq. (5.20) is valid, it is apparent that all the Rydberg states and the ionization limit are shifted up in energy by an amount which is proportional to the square of the temperature. If the atoms are exposed to a monochromatic radiation field satisfying the same frequency criteria, the Rydberg state energies and the ionization limit are shifted according to Eq. (5.21), which is often termed the ponderomotive energy shift.¹³

In this chapter we have implicitly assumed the Rydberg atom to be a one electron-atom. In the perturbed Rydberg series of, for example, alkaline earth atoms, Rydberg states can have mixed valence–Rydberg character. In such states the black body effects are reduced by a factor equal to the fractional Rydberg character.¹⁴

Initial verification

The experimental observations which called attention to the effects of room temperature black body radiation on Rydberg atoms were made using Rydberg states of Xe and Na.^{1–3} However, Pimbert had earlier noted the effect of higher temperatures.¹⁵ Two kinds of experiments were performed, measurements of population redistribution and increased decay rates. Both of the effects measured could conceivably arise from collisions, and this possibility was systematically ruled out in the experiments. In the Xe experiment atoms in a metastable beam were excited to the 26f state by a pulsed laser and exposed to the thermal radiation for periods of 1.5, 7.5, and 15.5 μ s, after which a ramped field was applied to the atoms. This procedure leads to the field ionization signals shown in Fig. 5.5.¹ For each time delay there is a clear progression of peaks in the field ionization spectra which are assigned the n values shown. At longer delay times the intensities of the

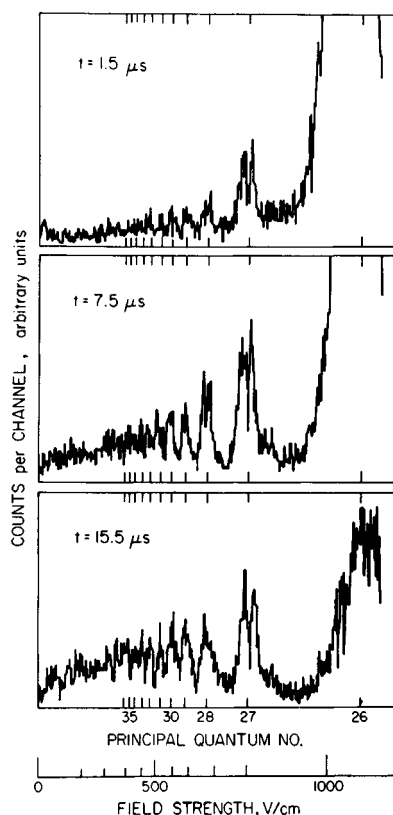


Fig. 5.5 Field ionization spectra at different times after the laser pulse excites Xe atoms to the 26f level showing the black body radiation induced transfer to higher levels (from ref. 1).

$n > 26$ peaks grow, as would be expected for progressively longer exposure times to the thermal radiation. In addition, due to the fact that the black body radiation induced transitions are dipole transitions, only d and g final states are expected. The fact that the field ionization spectra exhibit a sharp peak for each n is consistent with only one or two, not all, ℓ states being populated by the thermal radiation.¹ If all the ℓ states were populated there would be no sharp peaks in Fig. 5.5.

In an analogous experiment in which the initial population was placed in the Na 18s state, the transferred population was found to be in the higher lying p states, as expected for dipole transitions. Furthermore, the fraction of the atoms which had undergone transitions to each of the higher p states after 5 μ s was in good agreement with values calculated on the basis of 300 K black body radiation induced transitions, as shown by Table 5.1.²

The other initial experiment was the observation of the increase in the decay rate due to 300 K black body radiation, shown by Eq. (5.13).³ The decay rate of

Table 5.1. *Calculated and observed populations in higher lying np states 5 μ s after the initial population of the Na 18s state expressed in terms of the initial 18s population.*^a

Final state	Calculated yield	Observed yield
18p	4.2%	5.0%
19p, 20p	1.6%	1.6%
>20p, ϵ p	1.0%	1.2%
Total	6.8%	7.8%

^a(see ref. 2)

Table 5.2. *Lifetimes of the Na 17p and 18p states.*^a

State	τ (μ s)	τ^{bb} (μ s)	τ^{T} (calc) (μ s)	τ^{T} (obs) (μ s)
17p	48.4 ^b	22.7	15.5	11.4 ^{+5.6} _{-1.4}
18p	58.4 ^b	25.6	17.9	13.9 ^{+8.8} _{-2.9}

^a(from ref. 3)

^b(from ref. 17)

the Na 18s state at 300 K differs from its 0 K rate by about 20%, which is not a large enough difference that a measurement at 300 K alone is entirely convincing. The Na np states of $n \sim 20$, on the other hand, have long lifetimes^{16,17} and offer a reasonable prospect for a convincing measurement, even if made only at 300 K. The Na 17p and 18p lifetimes were measured by exciting Na atoms in a beam to the 17p and 18p states and observing the populations in these states as a function of time after the laser pulse. The results of the experiment are given in Table 5.2, which shows clearly that the experimentally observed values are in good agreement with the lifetimes calculated with the black body radiation, but in stark disagreement with the 0 K lifetimes.

Temperature dependent measurements

Subsequent to the initial 300 K observations, measurements were done as a function of temperature. The first measurements, by Koch *et al.*,¹⁸ were done at

elevated temperatures, and later measurements were done at temperatures below room temperature. A critical issue in such measurements is knowing the temperature the atoms are experiencing. Typically an enclosure cooled to a temperature from 6 K to 300 K surrounds the atoms, shielding them from the 300 K radiation from the walls of the vacuum chamber. However, simply making an enclosure in which 90% of the solid angle seen by the atoms is cold is not adequate, because at far infrared wavelengths many materials, and virtually all metal surfaces, are excellent reflectors. Thus a 10% aperture to the warm world outside is adequate to raise substantially the temperature the atoms in the enclosure experience.

Two approaches have been used to ensure that the temperature experienced by the atoms is close to the temperature of the enclosure. Hildebrandt *et al.* lined the inside of the enclosure with graphite coated Cu wool, which they verified separately to be an effective way of absorbing far infrared radiation.¹⁹ An alternative approach was followed by Spencer *et al.*²⁰ They covered the apertures through which laser beams propagated with glass, which is opaque at far infrared wavelengths, and those through which the atomic beam propagated with a fine, 70 μm , mesh, which blocks all radiation with $\lambda > 140 \mu\text{m}$.

Experiments with Xe analogous to the previously described Xe experiment were carried out at both 90 K and 300 K. Specifically the time dependences of the populations in higher nd and ng states were measured subsequent to population of an nf state by a pulsed laser.¹⁹ The time dependences of the observed populations were fit to a model which yielded the radiative transfer rates from the initially excited state to the final states. Not surprisingly, as the temperature was reduced from 300 K to 90 K, not only was the overall radiative transfer reduced, but the transfer to the highest lying states was most sharply reduced, as expected from the dependence of the photon occupation number on frequency and temperature. For example the transfer from the Xe 25f to 26d,g states was reduced by a factor of 3 while the 25f–27d,g transfer was reduced by a factor closer to 5.

Spencer *et al.* populated the Na 19s state and observed the sum of the populations in the 19s, 19p, and 18p states, at times from 2–32 μs after the laser excitation, to determine the sum of the radiative transition rates from the 19s to 18p and 19p states.²⁰ The measurements were made at temperatures from 6 K to 210 K with the results shown in Fig. 5.6. As shown by Fig. 5.6, the observed rates agree well with the calculated rates, shown by the line. At 0 K the rate is entirely due to the spontaneous 19s $\rightarrow$ 18p transition, and the increase above this rate is due to thermal radiative transfer to both the 18p and 19p states. It is interesting to note that above ~ 30 K the transfer rate increases linearly with T as expected from Eq. (5.16). The linear behavior only occurs above 30 K since only then is $kT > \omega$ for these two transitions.

While the Na 19s $\rightarrow$ 18p, 19p transfer rate is an excellent illustration of the validity of Eq. (5.16) for $\omega < kT$, the photoionization of Rydberg atoms of $n \approx 20$ by black body radiation is a test of the regime $\omega \sim kT$, and this regime has been explored by Spencer *et al.*²¹ Specifically, they measured the relative rates for

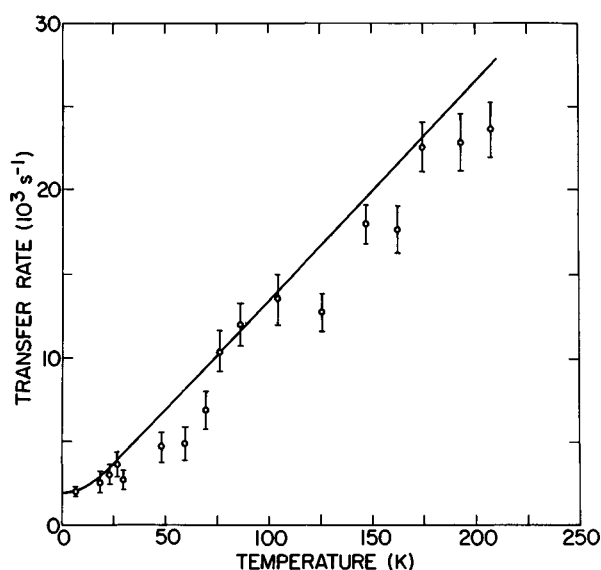


Fig. 5.6 Experimental and theoretical transfer rate of the Na 19s state to the 18p and 19p states vs temperature. The solid line is calculated with no adjustable parameters (from ref. 20).

photoionization of the Na 17d state by thermal radiation at temperatures of 90–300 K. The experiments were done by exciting the Na 17d state, allowing photoionization to occur for 500 ns, a time ten times shorter than the 17d lifetime, and then collecting the photoions by applying a small field pulse of 8 V/cm. A field this small was verified to produce negligible field ionization of bound states excited by black body radiation. When the relative rates are normalized to the calculated rate at 300 K, they are found to be in excellent agreement with the calculated rates.

Since the 17d state is bound by 380 cm^{-1} , which exceeds kT , the photoionization rate varies rapidly with temperature, roughly a factor of 100 between 90 K and 300 K, as shown by Fig. 5.7. The rapid, nearly exponential, dependence of the photoionization rate on temperature shown in Fig. 5.7 is to be contrasted with the linear temperature dependence of the $19s \rightarrow 19p$, $18p$ transfer rate shown in Fig. 5.6.

Suppression and enhancement of transition rates

We have described the effects of black body radiation in free space. In a closed cavity the radiation is confined to the allowed modes of the cavity. In essence all the thermal radiation is forced into the cavity modes, raising the intensity at the

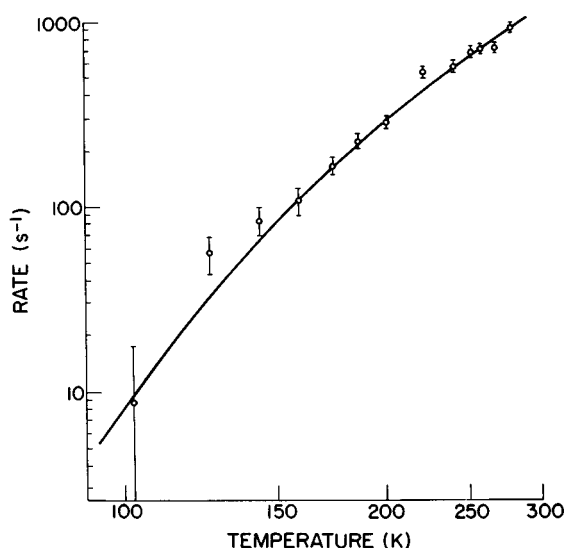


Fig. 5.7 Black body induced photoionization rate vs temperature for the 17d state in Na. Scales are logarithmic. The solid line represents the calculated values. Experimental points are normalized to the calculated value at 300 K (from ref. 21).

frequencies of the modes by a factor of Q relative to the free space value.²² Here Q is the quality factor of the cavity, the frequency of a mode divided by its linewidth (FWHM), i.e. $Q = \nu/\Delta\nu$. Similarly, between the cavity modes the intensity is substantially depressed. At a cavity mode resonance the radiation transfer rate is increased, by a factor of Q , and away from a resonance, the transfer rate is suppressed. As pointed out by Kleppner, Rydberg atoms, with their long wavelength transitions, provide a nearly ideal system for the study of such notions.²³

The first experimental realization of this notion was by Vaidyanathan *et al.* who excited Na atoms to nd states between two parallel plates 0.337 cm apart.²⁴ The region between these plates is cutoff for propagation of radiation of wavelengths longer than 0.674 cm polarized parallel to the plate surfaces, and such radiation is not present between the plates. Stated another way, frequencies less than the cutoff frequency $\nu_c = 1.48 \text{ cm}^{-1}$ do not propagate. Radiation polarized perpendicular to the surfaces can propagate freely between the plates. The zero static field 29d–30p interval is just under 1.48 cm^{-1} . However, the application of a small static field increases the separation of the levels, and a spacing of 1.48 cm^{-1} occurs at a static field of 2.4 V/cm. Thus as the static field is increased from zero to beyond 2.4 V/cm the 29d $\rightarrow$ 30p transition frequency passes from below to above cutoff, and the black body radiation transition rate increases accordingly.

In the experiment Na atoms are excited by two lasers from the ground state to the nd state in an environment cooled to 180 K at which point $\bar{n} = 86$ for the 29d $\rightarrow$ 30p transition. The atoms are exposed to the thermal radiation for 20 μs ,

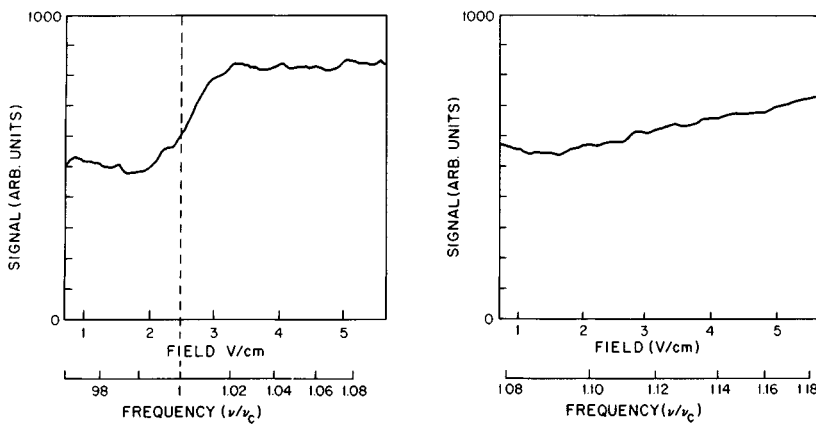


Fig. 5.8 Black body radiative transfer signals in Na located between parallel conducting plates for $29d \rightarrow 30p$ (left-hand side) and $28d \rightarrow 29p$ (right-hand side) as a function of the absorption frequency. The cutoff frequency is $\nu_c = 1/2d = 1.48 \text{ cm}^{-1}$, where d is the plate separation. The increase in the transfer rate at $\nu = \nu_c$ (left-hand side) is due to the “switching on” of the radiation polarized parallel to the plates (from ref. 24).

after which they are exposed to a field ionization pulse to ionize atoms in the $(n + 1)p$ state selectively. In Fig. 5.8 we show the results obtained when the $28d$ and $29d$ states are excited as functions of static field over the range $0\text{--}5 \text{ V/cm}$. As shown in Fig. 5.8, the $29d\text{--}30p$ interval is tuned through the cutoff frequency of the plates in this field range. Correspondingly, there is a sharp increase in the population found in the $30p$ state when the field tunes the $29d\text{--}30p$ interval through the cutoff frequency. In contrast, the $28d\text{--}29p$ interval always exceeds the cutoff frequency, and no sharp increase in the $29p$ population is observed as the field is tuned.

Just as the black body transitions can be suppressed, they can also be enhanced, by tuning to a cavity resonance. Raimond *et al.*²⁵ have passed a thermal beam of Na atoms excited to the $30s_{1/2}$ state through a 300 K Fabry–Perot cavity, which can be tuned to the $30s_{1/2}\text{--}30p_{1/2}$ resonance at 134 GHz , and detected the atoms downstream by field ionization. With the cavity tuned off resonance 5% of the atoms make the $30s_{1/2} \rightarrow 30p_{1/2}$ transition, but when it is tuned to resonance an additional 5% make the transition, in agreement with the calculated value.²⁵

In any low angular momentum state the radiative decay rate is usually dominated by the high frequency transitions to low lying states, and as a result it is impossible to control completely the decay rate using a millimeter wave cavity. In a circular $\ell = m = n - 1$ state the only decay is the far infrared transition to the $n - 1$ level, and Hulet *et al.* have observed the suppression of the decay of this level.²⁶ They produced a beam of Cs atoms in the circular $n = 22$, $\ell = m = 21$ state by pulsed laser excitation and an adiabatic rapid passage technique.²⁷ The beam of circular state atoms then passed between a pair of plates 6.4 cm wide, 12.7 cm long, spaced by $230.1 \mu\text{m}$, and held at 6 K . The 0 K radiative lifetime is $460 \mu\text{s}$, and

the transit time of the atoms through the plates to the detector is approximately one lifetime at the peak of the velocity distribution.

In zero static field the wavelength λ of the $n = 22, \ell = 21 \rightarrow n = 21, \ell = 20$ transition is $450 \mu\text{m}$, and $\lambda = 0.98(2d)$ where d is the plate spacing. However, using the plates to apply a static field it is possible to reduce continuously the energy separation between the levels by up to 4%, increasing the wavelength of the transition to $1.02(2d)$. In other words it is possible to increase the wavelength of the transition from below cutoff to above cutoff by varying the tuning field. At this point it is worth recalling that in the static field the $\ell = 21, m = 21$ circular $n = 22$ state must interact with radiation polarized parallel to the field plates to decay to the $n = 21, \ell = m = 20$ state. The radiation polarized perpendicular to the plates cannot drive this transition. When the wavelength is longer than the cutoff wavelength, i.e. $\lambda > 2d$, radiative decay is suppressed, i.e. $\tau_{\text{cav}} = \infty$, but when the wavelength is less than the cutoff wavelength $\lambda < 2d$ the radiative decay rate is increased by 50%.²⁶

They recorded time of flight spectra of atoms remaining in the circular state as a function of the spacing of the plates and the tuning field applied, and from these spectra determined the decay rates. Decay rates were measured and calculated for three conditions. First the plate spacing was increased by a factor of 30 to observe the free space decay. They observed a lifetime of $450(10) \mu\text{s}$, in good agreement with the calculated 6 K value of $451 \mu\text{s}$. With the plate spacing returned to $230.1 \mu\text{m}$ the decay rates were measured with the electric field set so as to tune the wavelength both below and above the cutoff wavelength. With a low electric field, for which $\lambda < 2d$, the measured decay rate was 50% larger than the free space value, to within 5%. When the field was increased so that $\lambda > 2d$, roughly twice the number of atoms was detected and the decay rate was consistent with zero, to within 5% of the free space decay rate.

Level shifts

Rydberg states which are pure Rydberg states, not containing admixtures of valence states, all have the same positive energy shift in a thermal radiation field. To measure this shift requires careful measurement of a transition from a long lived low lying atomic state. Holberg and Hall have measured the thermal shift of Rb atoms exposed to a thermal source which ranged in temperature from 350 K to 1000 K using two photon, Doppler free spectroscopy between the ground state and the Rydberg states.²⁸ In their experiment a Rb beam passes through the two arms of a folded optical cavity, so the atoms interact with the optical radiation twice, providing a Ramsey interference pattern. An attraction of the Ramsey method is that the Ramsey interference pattern is located at the optical field free location of the $5s \rightarrow ns$ transition. It is thus unaffected by laser

light shifts. The central Ramsey interference fringe is 40 kHz wide, the inverse of the transit time of the atoms between the two arms of the optical cavity. The radiation from the heated black body source is focused on the atomic beam between its intersections with the two laser beams. Due to the small solid angle subtended by the heated black body source at the intersection of the atomic and laser beams, it is only $\sim 10\%$ as effective as a complete enclosure at the same temperature. The radiation from the heated black body source is chopped and the position of the central fringe of the Ramsey pattern observed with and without the thermal radiation. In spite of the reduced efficiency of the source in producing a shift, blue shifts of up to 1.4 kHz were observed, at 876 K. The blue shift increases with temperature as predicted by Eq. (5.20), and while the results do not provide a stringent test, they are clearly consistent with Eq. (5.20).

The black body shifts are not confined to Rydberg atoms, but also alter the frequency of atomic clock transitions. Itano *et al.* have shown that the Cs 9 GHz ground state hyperfine interval, the definition of the second, is increased by one part in 10^{14} when the temperature is raised from 0 K to 300 K.²⁹

Experimental manifestations and uses

As we have already pointed out, the presence of black body radiation invariably decreases the lifetime of a Rydberg state, and the decreased lifetimes have been reported in many systems.^{3,30–32} What is most important, though, is the transfer of population to nearby states, and this effect must be taken into account in making measurements. To illustrate this point we consider the measurement of the 300 K lifetime of a Rydberg state using selective field ionization, an apparently straightforward measurement.⁶ The 18s state of Na is calculated to have a 0 K radiative lifetime of 6.37 μs , which is reduced to 4.87 μs by the 300 K black body radiation. Na atoms were excited to the 18s state at time $t = 0$, and the population remaining at later times was determined by applying an electric field pulse high enough to ionize easily the atoms in the 18s state. This measurement yielded an apparent lifetime of 7.8 μs , a value longer than the 0 K radiative lifetime. The problem was that the field also ionized all the long lived, np states of $n \geq 17$ to which population was transferred by black body radiation. Only when the signal from higher lying states was detected separately was it possible to correct the observed 18s signal for the presence of the long lived np states. The final value was 4.78 μs , in excellent, and probably fortuitous, agreement with the calculated value of 4.87 μs .

As shown by Table 5.2, at 300 K the $n = 17$ and $n = 18$ black body decay rates are $\approx 5 \times 10^4 \text{ s}^{-1}$, which might seem tolerable. However, in even slightly dense

samples thermal radiation can trigger superradiance, which depopulates an initially populated state in 10 ns.^{33,34} For superradiance to occur between two levels (crudely speaking, laser action without mirrors), a necessary condition is that the gain, G , be 1 along the length of the sample. This criterion may be expressed as³³

$$G = NL\sigma \geq 1, \quad (5.25)$$

where N is the difference in population densities of the upper and lower levels, L is the length of the sample, and σ , the optical cross section, is given by³⁵

$$\sigma = \frac{g_\ell \lambda^2 A_{u\ell}}{8\pi g_u \Delta}, \quad (5.26)$$

where g_ℓ and g_u are the degeneracies of the upper and lower states. $A_{u\ell}$ is the Einstein coefficient for the transition, and Δ is the linewidth of the transition, to which the main contributions are usually radiative and Doppler broadening. Because of the long wavelength of the transitions between Rydberg states, the above gain criterion for superradiance can be met with a very low density of atoms, hundreds of atoms in a volume of 10^{-3} cm^{-3} , and is easily met even in a low density atomic beam. The mere presence of adequate gain does not, however, ensure superradiance. Some initial photons are required to trigger the superradiance, and due to the low spontaneous transition rates of the long wavelength transitions between Rydberg states, they are unlikely to come from spontaneous emission. On the other hand, the omnipresent black body radiation easily triggers superradiance.

Gounand *et al.* have used superradiant cascades initiated by black body radiation to populate efficiently states which would normally be inaccessible using optical excitation.³⁴ In vapor cell experiments they observed the population in Rb Rydberg states to undergo several very rapid superradiant cascades to populate lower lying levels. The population is transferred in times far shorter than would be possible by simple radiative decay.³⁴ Fig. 5.9 shows the population in various Rb Rydberg states after the initial population of the 12s state. Normal radiative decay would require $\sim 3 \mu\text{s}$ to populate the 7f state, but as shown in Fig. 5.9, only 20 ns is required when superradiant cascades occur.

Stoneman and Gallagher have used black body radiation to make precise measurements of the avoided crossings between the Kn s and $n - 2$ Stark manifold states in electric fields,³⁶ and these measurements are described in Chapter 6.

Far infrared detection

An interesting potential application of Rydberg atoms is as a far infrared (or microwave) detector, a notion first suggested by Kleppner and Ducas.³⁷ The basic

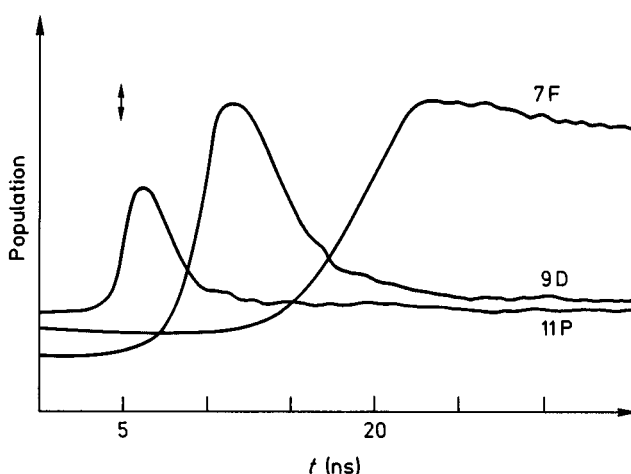


Fig. 5.9 Time dependence of the population in Rb Rydberg states after the initial population of the 12s state showing the rapid superradiant cascades (from ref. 35).

idea of the most straightforward type of device is to use a laser to make a target of Rydberg state atoms in one state, *A*. This target is exposed to the radiation to be detected at the infrared frequency ω_{AB} , which is equal to the frequency of the atomic transition from *A* to a higher energy state *B*. If the density of atoms in state *A* in the target is high enough that the target is optically thick to the radiation at frequency ω_{AB} , a photon at this frequency will be absorbed with nearly unit probability so that one atom will undergo the transition to state *B*. The fact that the atom has undergone the transition can be detected in either of two ways. The first is selective field ionization of atoms in state *B* but not those in state *A*, and the second is to detect the optical fluorescence from state *B* but not from state *A*.³⁸ In either case each incoming photon is converted into an ion and an electron or a visible photon, all of which are easily detected. Several variants of this approach have been examined.^{2,39–41}

With optical detection it is unimportant whether state *B* lies above or below state *A*. If *B* lies below *A*, a second approach, triggered superradiance, is possible, and the technique has been carefully examined by Goy *et al.*⁴² To detect the same radiation at ω_{AB} requires an optically thick target of atoms in the upper state *A*, which in a cooled environment would not become superradiant, since spontaneous emission is too slow to initiate the process. However, an incoming photon at frequency ω_{AB} would trigger a superradiant avalanche, converting one incoming photon into $\sim 10^4$ outgoing photons and leaving 10^4 atoms in state *B*, which would be trivial to detect. As shown by the sensitivity given in Table 5.3, this approach is equally as sensitive as the more straightforward approach described above.

While Rydberg atoms have been used to detect infrared radiation with wavelengths as short as $2.34 \mu\text{m}$, they are most effective as detectors of far infrared

Table 5.3. *Far infrared detection sensitivity in Rydberg atom experiments.*

Method	Source	Wavenumber cm^{-1}	NEP $\text{W}/\text{cm}^2\text{Hz}^{1/2}$
Absorption, ^a field ionization	300 K black body	22	10^{-14}
Absorption, ^b field ionization	Far infrared laser	20	5×10^{-15}
Absorption, ^c field ionization	Variable temperature black body	3.3	10^{-17}
Triggered ^d superradiance	Klystron harmonic	3.6	3×10^{-17}

^a(see ref. 2)^b(see ref. 41)^c(see ref. 40)^d(see ref. 42)

radiation, and in Table 5.3 we present measured sensitivities for Rydberg atom detection of far infrared radiation.

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6

Electric fields

The effect of an electric field on Rydberg atoms, the Stark effect, provides an interesting example of how the deviation from the coulomb potential at small r in, for example an alkali atom, significantly alters its behavior in a field from hydrogenic behavior. The difference is somewhat surprising since field effects are fundamentally long range effects, and this difference alone would make the Stark effect worthy of study. However, in addition to its intrinsic interest, the Stark effect is of great practical importance for the study of Rydberg atoms.

Hydrogen

We begin by considering the behavior of the H atom in a static field, ignoring the spin of the electron. We start with the familiar zero field $n\ell m$ angular momentum eigenstates with the immediate goal of showing that important qualitative features of the Stark effect are easily understood using familiar notions. If the applied field E is in the z direction, the potential seen by the electron is given by

$$V = -\frac{1}{r} + Ez. \quad (6.1)$$

Using the zero field $n\ell m$ states we calculate the matrix elements $\langle n\ell m | Ez | n'\ell' m' \rangle$ of the Stark perturbation to the zero field Hamiltonian. Writing the matrix element in spherical coordinates and choosing the z axis as the axis of quantization,

$$\langle n\ell m | Ez | n'\ell' m' \rangle = E \langle n\ell m | r \cos \theta | n'\ell' m' \rangle. \quad (6.2)$$

From the properties of the spherical harmonic angular functions we know that the Ez matrix elements are nonvanishing if $m' = m$ and $\ell' = \ell \pm 1$. The most obvious effect of the field is that it lifts the degeneracy of the ℓm states of a particular n . If we neglect the electric dipole coupling to other n states, the Hamiltonian matrix has the same entry, $-1/2n^2$, for all the diagonal elements and a set of off diagonal $\langle \ell | Ez | \ell \pm 1 \rangle$ matrix elements, all of which are proportional to E . If we subtract the common energy $-1/2n^2$ we find that all entries of the matrix are now proportional to E . Therefore, when the matrix is diagonalized to find its eigenvalues, they must

all be proportional to E , i.e. the eigenvalues are given by $\lambda_i E$. Stated another way, the H atom exhibits a linear Stark shift. Since the eigenvalues are all proportional to E , the eigenvectors are independent of E , and the Stark states are thus field independent linear combinations of zero field ℓ states of the same n and m . The fact that the Stark states have linear Stark shifts means that they have permanent electric dipole moments. Stark states which have positive Stark shifts have the electron localized on the upfield side of the atom and Stark states which have negative Stark shifts have the electron localized on the downfield side of the atom.

Since m is a good quantum number, each set of m states is independent of the others. We consider first the circular $|m| = n - 1$ states. They have no first order Stark shift since there are no other states of the same m and n . There are, however, two $m = n - 2$ states, which exhibit small linear Stark shifts. The Stark shifts are not large since the radial matrix element is small,¹

$$\langle n\ell | r | n\ell + 1 \rangle = \frac{-3n \sqrt{n^2 - \ell^2}}{2}. \quad (6.3)$$

The minus sign is for radial wavefunctions $R_{n\ell}(r) \sim +r^\ell$ as $r \rightarrow 0$. On the other hand, the $m = 0$ states also include the low ℓ states for which the radial matrix element is large, and the extreme $m = 0$ states have Stark shifts of approximately $\pm 3n^2 E/2$.

The most straightforward way to treat the Stark effect is to use parabolic coordinates, for in parabolic coordinates the problem remains separable even with the electric field.¹⁻³ The parabolic coordinates are defined in terms of the familiar Cartesian and spherical coordinates by

$$\begin{aligned} \xi &= r + z = r(1 + \cos \theta) \\ \eta &= r - z = r(1 - \cos \theta) \\ \phi &= \tan^{-1} y/x, \end{aligned} \quad (6.4)$$

and

$$\begin{aligned} x &= \sqrt{\xi\eta} \cos \phi \\ y &= \sqrt{\xi\eta} \sin \phi \\ z &= (\xi - \eta)/2 \\ r &= (\xi + \eta)/2. \end{aligned} \quad (6.5)$$

Surfaces of constant ξ or η are paraboloids of revolution about the z axis. From Eqs. (6.4) it is apparent that $\xi = 0$ corresponds to the $-z$ axis, and $\xi = \infty$ corresponds to $r \rightarrow \infty$, $\theta \neq 0$. Correspondingly, $\eta = 0$ corresponds to the $+z$ axis and $\eta \rightarrow \infty$ corresponds to $r \rightarrow \infty$, $\theta \neq 0$. When the field is pointing in the z direction, corresponding to Eq. (6.1), the electron escapes to $\eta = \infty$. Equivalently, the motion in the ξ direction is bound, but the motion in the η direction is not.

In parabolic coordinates the Schroedinger equation for an electron orbiting a singly charged ion with an external field in the z direction, i.e. in the potential given by Eq. (6.1), is written using Eqs. (6.5) as¹⁻³

$$\left[-\frac{\nabla^2}{2} - \frac{2}{\xi + \eta} + \frac{E(\xi - \eta)}{2} \right] \psi = W\psi, \quad (6.6)$$

where

$$\nabla^2 = \frac{4}{\xi + \eta} \frac{\partial}{\partial \xi} \left(\xi \frac{\partial}{\partial \xi} \right) + \frac{4}{\xi + \eta} \frac{\partial}{\partial \eta} \left(\eta \frac{\partial}{\partial \eta} \right) + \frac{1}{\xi \eta} \frac{\partial^2}{\partial \phi^2},$$

and W is the energy.

If we assume that solution can be written as the product,¹⁻³

$$\Psi(\xi, \eta, \phi) = u_1(\xi)u_2(\eta)e^{im\phi}, \quad (6.7)$$

with m an integer or zero, and substitute Eq. (6.7) into Eq. (6.6), we find two independent equations for $u_1(\xi)$ and $u_2(\eta)$

$$\frac{d}{d\xi} \left(\xi \frac{du_1}{d\xi} \right) + \left(\frac{W\xi}{2} + Z_1 - \frac{m^2}{4\xi} - \frac{E\xi^2}{4} \right) u_1 = 0 \quad (6.8a)$$

and

$$\frac{\partial}{\partial \eta} \left(\eta \frac{du_2}{d\eta} \right) + \left(\frac{W\eta}{2} + Z_2 - \frac{m^2}{4\eta} + \frac{E\eta^2}{4} \right) u_2 = 0. \quad (6.8b)$$

From Eqs. (6.8a) and (6.8b) it is apparent that the sign of m is unimportant, and that the wavefunctions for $\pm m$ are degenerate, as might be expected for this cylindrically symmetric problem. In Eqs. (6.8a) and (6.8b) the separation constants Z_1 and Z_2 are related by^{1,2}

$$Z_1 + Z_2 = 1. \quad (6.9)$$

Z_1 and Z_2 may be thought of as the positive charges binding the electron in the ξ and η coordinates. This point will become more apparent shortly. Note that the two differences between Eqs. (6.8a) and (6.8b) are that the field enters with different signs in the two equations and that Z_1 and Z_2 may have different values.

The classic way of determining the energies of hydrogenic levels in a field is to solve the zero field problem in parabolic coordinates and calculate the effect of the field using perturbation theory. The zero field parabolic wavefunctions obtained by solving Eqs. (6.8a) and (6.8b) have, in addition to the quantum numbers n and $|m|$, the parabolic quantum numbers n_1 and n_2 , which are nonnegative integers.¹ n_1 and n_2 are the numbers of nodes in the u_1 and u_2 wavefunctions and are related to n and $|m|$ by

$$n = n_1 + n_2 + |m| + 1. \quad (6.10)$$

They are in addition related to the effective charges Z_1 and Z_2 by

$$Z_1 = \frac{1}{n} \left(n_1 + \frac{|m| + 1}{2} \right) \quad (6.11a)$$

and

$$Z_2 = \frac{1}{n} \left(n_2 + \frac{|m| + 1}{2} \right). \quad (6.11b)$$

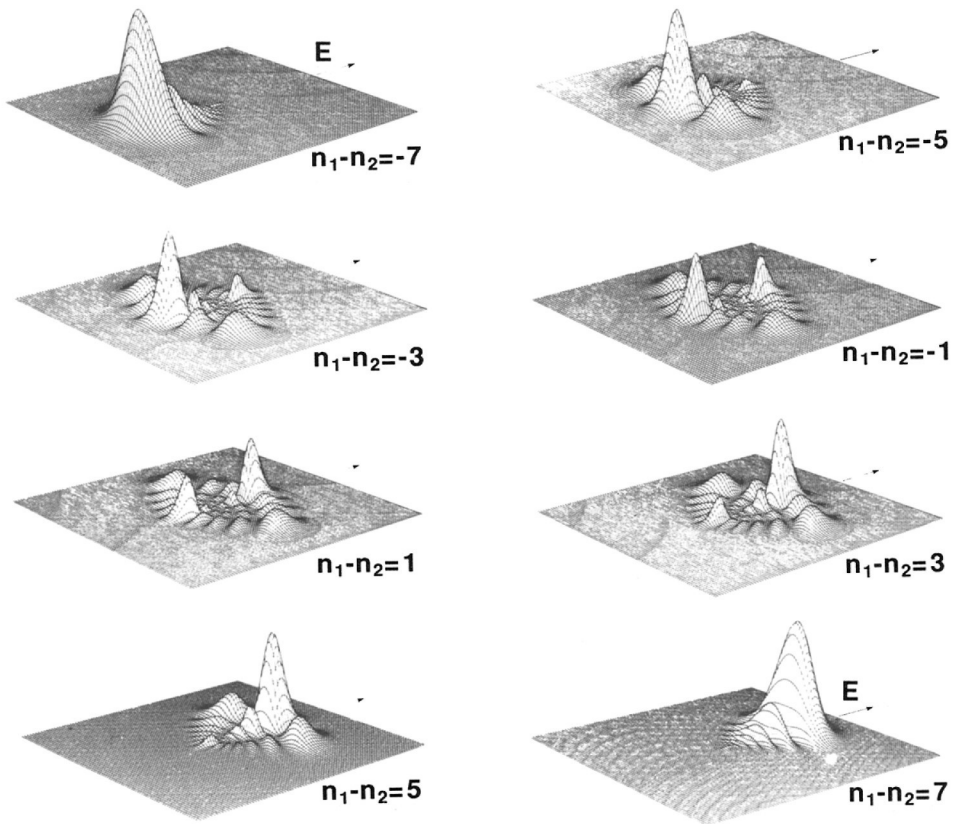


Fig. 6.1 Charge distribution for H, for “parabolic” eigenstates: $n = 8$, $m = 0$, $n_1 - n_2 = -7$ to 7. The dipole moments that give rise to the first order Stark effect are conspicuous (from ref. 4).

The wavefunctions are given in terms of associated Laguerre polynomials, and using the approximate form of the Laguerre polynomials for large arguments, we can write an approximate, unnormalized wavefunction as¹

$$\psi_{nn_1n_2m} \propto e^{im\phi} \xi^{n_1+|m|/2} \eta^{n_2+|m|/2} e^{-(\xi+\eta)/2n}. \quad (6.12)$$

Taking the squared absolute value of Eq. (6.12) and using the definition of the parabolic coordinates given in Eq. (6.4), we can write an expression for the electron probability distribution in spherical coordinates,¹

$$|\psi_{nn_1n_2m}|^2 = r^{2n-2} (1 + \cos \theta)^{2n_1+|m|} (1 - \cos \theta)^{2n_2+|m|} e^{-2r/n}. \quad (6.13)$$

Eq. (6.13) shows explicitly that the $n_1 - n_2 \approx n$ wavefunctions are localized along the $+z$ axis while the $n_2 - n_1 \approx n$ wavefunctions are localized along the $-z$ axis. Similarly, $n_1 - n_2 \approx 0$ wavefunctions are localized near the $z = 0$ plane. In Fig. 6.1 we show the $n = 8$, $m = 0$, $n_1 - n_2 = -7$ to 7 wavefunctions to illustrate the polarization along the field direction.⁴ Careful inspection also reveals that the

nodes lie on parabolas. Using the zero field wavefunctions the first order energies are¹

$$W_{nn_1n_2m} = \frac{-1}{2n^2} + \frac{3}{2} E(n_1 - n_2)n. \quad (6.14)$$

While there is no explicit m dependence, there is an implicit m dependence due to the fact that $n_1 + n_2 + m + 1 = n$. For $m = 0$, allowed values of $n_1 - n_2$ are $n - 1, n - 3 \dots, -n + 1$ while for $m = 1$ they are $n - 2, n - 4 \dots -n + 2$. The even and odd m levels are interleaved. Furthermore, for the quantum numbers n and m there are $n - |m|$ Stark levels. Note that for the circular states, $|m| = n - 1, n_1 = n_2 = 0$, and the first order Stark shift vanishes, as we have already seen.

By taking into account the matrix elements off diagonal in n , the second and higher order contributions to the Stark effect can be calculated. If the calculation is carried through second order, the energies are given by¹

$$W_{nn_1n_2m} = \frac{-1}{2n^2} + \frac{3En}{2}(n_1 - n_2) - \frac{E^2}{16}n^4[17n^2 - 3(n_1 - n_2)^2 - 9m^2 + 19]. \quad (6.15)$$

The second order shift is always to lower energy, as expected from oscillator strength sum rules.⁵ Equally important, the second order shift breaks the m degeneracy. The energies can be calculated to higher order,^{6,7} but for many applications the first and second order shifts are adequate. This point is made explicitly by Fig. 6.2, a plot of the H $|m| = 1$ energy levels from $n = 8$ to $n = 14$ in electric fields from 0 to 2×10^{-5} (0 to 10^5 V/cm). Fig. 6.2 makes several important points about the Stark effect in H. First, the levels exhibit apparently linear Stark shifts from zero field to the point at which field ionization occurs at significant rates, shown by the broken lines in Fig. 6.2. Only when looking along the diagram from the side of the page is the curvature of the levels evident. Second, the Stark levels of adjacent n cross; there is no coupling, at least at this level of resolution. Finally, the red, or down shifted, Stark states ionize near the classical ionization limit, but the higher energy blue, or upshifted, states ionize only at higher fields.

As shown by Fig. 6.2, the Stark shifts are quite linear, except at the highest fields shown, and the first order energies of Eq. (6.14) are adequate for many purposes. On the other hand, even the second order energies of Eq. (6.15) are not adequate for comparisons to precise measurements of the energy levels in field. For example, using a fast H beam Koch⁸ observed the $(10,8,0,1) \rightarrow (25,21,2,1)$ and $(10,0,9,0) \rightarrow (30,0,29,0)$ transitions in fields of 2.514 and 0.689 kV/cm respectively using the $10 \mu\text{m}$ R24 and P24 CO_2 laser lines. Here the states are denoted $(n, n_1, n_2, |m|)$. From his results it is apparent that the energy of the red shifted $(30,0,29,0)$ state is given accurately by perturbation theory results at about tenth order. In contrast, the blue shifted $(25,21,2,1)$ state is never given to the same accuracy, and, in fact, the perturbation theory result does not converge but oscillates about the experimental result with the minimum amplitude of the oscillations at about tenth order.⁸

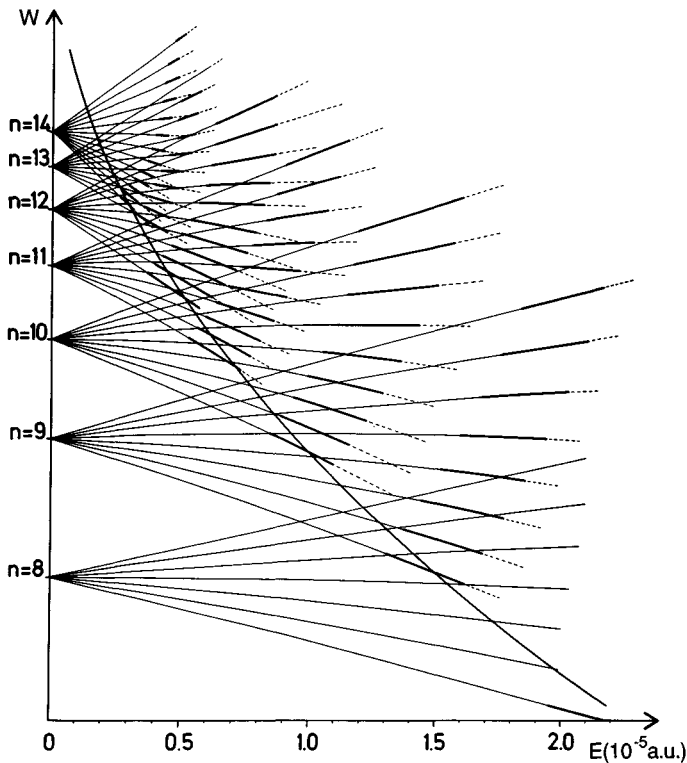


Fig. 6.2 Stark structure and field ionization properties of the $|m| = 1$ states of the H atom. The zero field manifolds are characterized by the principal quantum number n . Quasi-discrete states with lifetime $\tau > 10^{-6}$ s (solid line), field broadened states 5×10^{-10} s $< \tau < 5 \times 10^{-6}$ s (bold line), and field ionized states $\tau < 5 \times 10^{-10}$ s (broken line). Field broadened Stark states appear approximately only for $W > W_c$. The saddle point limit $W_c = -2\sqrt{E}$ is shown by a heavy curve (from ref. 3).

The second point made by Fig. 6.2, although not with very high resolution, is that the levels of n and $n + 1$ cross. The extreme $m = 0$ red and blue states have Stark shifts of approximately $\pm 3n^2 E/2$, which combined with the $n - (n + 1)$ energy spacing of $1/n^3$, yields a crossing field of

$$E = 1/3n^5. \quad (6.16)$$

This field is associated with the Inglis–Teller limit, where Stark broadening in a plasma causes levels of adjacent n to become unresolvable.⁹ The fact that the red $n + 1$ and blue n levels actually cross, in spite of the fact that they have the same m , is a result of the fact that the use of the parabolic coordinates diagonalizes the Runge–Lenz vector, which has a different eigenvalue in the two crossing states. The fact that both the Runge–Lenz vector and angular momentum are conserved in H in zero field was exploited by Park¹⁰ to show in an elegant way that the transformation coefficient between the parabolic nn_1n_2m states and the spherical

$n\ell m$ states was simply a Clebsch–Gordon coefficient. If we expand the Stark state $|nn_1n_2m\rangle$ as

$$|nn_1n_2m\rangle = \sum_{\ell} |n\ell m\rangle \langle n\ell m | nn_1n_2m \rangle, \quad (6.17)$$

then the transformation coefficient is given in terms of Wigner 3J symbols as

$$\begin{aligned} \langle nn_1n_2m | n\ell m \rangle &= (-1)^{(1-n+m+n_1-n_2)/2 + \ell} \\ &\times \sqrt{2\ell + 1} \begin{pmatrix} \frac{n-1}{2} & \frac{n-1}{2} & \ell \\ \frac{m+n_1-n_2}{2} & \frac{m-n_1+n_2}{2} & -m \end{pmatrix}. \end{aligned} \quad (6.18)$$

An equivalent form is given by Englefield.¹¹ It is possible to find quite a variety of phases for the transformation coefficients of Eq. (6.18).^{10–13} The phase depends on the phase conventions established for the spherical and parabolic states. The choice of phase in Eq. (6.18) is for spherical functions with an r^ℓ , as opposed to $(-r)^\ell$, dependence at the origin and the spherical harmonic functions of Bethe and Salpeter. A few examples of the spherical harmonics are given in Table 2.2. The parabolic functions are assumed to have an $(\xi n)^{|m|/2}$ behavior at the origin and an $e^{im\phi}$ angular dependence. This convention means, for example, that for all Stark states with the quantum number m , the transformation coefficient $\langle nn_1n_2m | nmm \rangle$ is positive. To the extent that the Stark effect is linear, i.e. to the extent that the wavefunctions are the zero field parabolic wavefunctions, the transformation of Eqs. (6.17) and (6.18) allows us to decompose a parabolic Stark state in a field into its zero field components, or vice versa.

Harmin¹⁴ derived an approximate form of the transformation of Eq. (6.18) which is particularly useful. The information contained in the quantum numbers n_1 and n_2 always appears as $\pm(n_1 - n_2)$ and can also be represented by Z_1 and Z_2 , the separation constants, a notion which easily passes into the regime in which n_2 is not a well defined quantum number. (In a field n_1 is always a good quantum number.) Explicitly, Harmin showed that Eq. (6.18) can also be written as¹⁴

$$\langle nn_1n_2m | n\ell m \rangle = \sqrt{\frac{2}{n}} (-1)^\ell P_{\ell m}(Z_1 - Z_2), \quad (6.19)$$

where $P_{\ell m}$ is a normalized associated Legendre polynomial of Bethe and Salpeter.¹ It is defined for $m < 0$, and $\int_{-1}^1 P_{\ell m}^2(x) dx = 1$.¹ Eq. (6.19) is only defined over the region $-1 \leq Z_1 - Z_2 \leq 1$, which is, however, a less restrictive requirement than that both n_1 and n_2 quantum numbers have integral values. We may also write Eq. (6.19) in terms of the parabolic n_1 and n_2 quantum numbers using $Z_1 - Z_2 = (n_1 - n_2)/n$ from Eq. (6.11). From Eq. (6.19) it is easy to see how the parabolic states are composed of the $n\ell m$ states. From either the properties of the 3J symbol or the fact that $P_{00}(x) = 1/\sqrt{2}$ it is apparent that the ns state is evenly

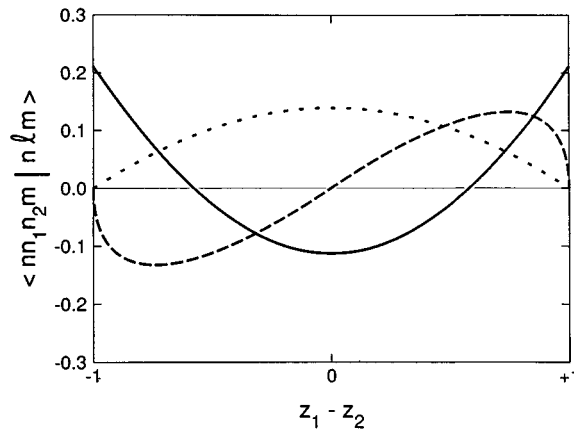


Fig. 6.3 Plots of the projection of the $n = 100$ Stark states of $m = 0, 1$, and 2 on the $n = 100$ zero field d states, $\langle 100 n_1 n_2 m | 100 2 m \rangle$ for $m = 0, 1$, and 2 : $m = 0$ (—), $m = 1$ (---) $m = 2$ (· · ·). The values are obtained from Eqs. (6.11) and (6.19).

spread over the nn_1n_20 states. The $\ell \geq 1$ states are more interesting. In Fig. 6.3 we plot Eq. (6.19) for $\ell = 2$ $|m| = 0, 1$, and 2 . From Fig. 6.3 it is clear that the $m = 0$ states at the edge of the Stark manifold, i.e. those for which $|n_1 - n_2| \sim n$, have much more d character than those in the center. On the other hand, the $|m| = 2$ states at the center of the Stark manifold have substantial d character while the states at the edges do not. For $n = 10$ and $|m| = 2$ the largest value of $|Z_1 - Z_2| = |n_1 - n_2|/n$ is 0.7 , but at higher n it approaches 1 , at which point the $|m| = 2$ Stark states at the edge of a Stark manifold have vanishingly small d character.

Working out the parabolic wavefunctions in terms of the Laguerre polynomials is useful in the analytic calculation of the Stark effect using perturbation theory. However, it is not useful in very strong electric fields. Here we outline a more general procedure which is valid in strong fields and lends itself to numerical computations. If we replace $u_1(\xi)$ and $u_2(\eta)$ in Eqs. (6.8a) and (6.8b) by^{1-3,15}

$$u_1(\xi) = \frac{\chi_1(\xi)}{\sqrt{\xi}} \quad (6.20a)$$

$$u_2(\eta) = \frac{\chi_2(\eta)}{\sqrt{\eta}} \quad (6.20b)$$

and introduce a normalizing factor of $\sqrt{2\pi}$ to conform to common usage, the wavefunction is given by

$$\psi(\xi, \eta, m) = \frac{\chi_1(\xi) \chi_2(\eta) e^{im\phi}}{\sqrt{2\pi\epsilon\eta}}, \quad (6.21)$$

which leads to the separated equations written as

$$\frac{d^2\chi_1}{d\xi^2} + \frac{\chi_1}{2} [W - V(\xi)] = 0 \quad (6.22a)$$

and

$$\frac{d^2 \chi_2}{d\eta^2} + \frac{\chi_2}{2} [W - V(\eta)] = 0, \quad (6.22b)$$

where

$$V(\xi) = 2 \left(-\frac{Z_1}{\xi} + \frac{m^2 - 1}{4\xi^2} + \frac{E\xi}{4} \right), \quad (6.23a)$$

and

$$V(\eta) = 2 \left(-\frac{Z_2}{\eta} + \frac{m^2 - 1}{4\eta^2} - \frac{E\eta}{4} \right) \quad (6.23b)$$

The χ_1 and χ_2 wavefunctions describe the motion of particles of total energy $W/4$ in the potentials $V(\xi)/4$ or $V(\eta)/4$. It is also interesting to note that for the case $E=0$ the wave equations of Eqs. (6.22a) and (6.22b) are similar to the radial equation for the coulomb potential in spherical coordinates. Explicitly, making the substitutions

$$\begin{aligned} 2W &\rightarrow W/2 \\ \ell(\ell + 1) &\rightarrow m^2 - 1 \\ Z &\rightarrow Z_1, Z_2 \\ r &\rightarrow \xi, \eta \end{aligned} \quad (6.24)$$

in Eq. (2.10) for $\rho(r) = rR(r)$ yields Eqs. (6.22a) and (6.22b).

When $E = 0$, the potential $V(\eta)$ with Z_2 substituted for Z_1 is the same as $V(\xi)$. It is apparent that for small values of Z_1 and Z_2 the potentials $V(\xi)$ and $V(\eta)$ are not as deep as for the higher values of Z_1 and Z_2 . Thus at the same energy W the number of nodes which occur in the wavefunction increases with Z_1 or Z_2 . Inspecting Eq. (6.24), we can see that when $E \neq 0$ that $V(\xi)$ and $V(\eta)$ are no longer identical for the same values of Z_1 and Z_2 . The motion in the ξ direction is bounded for all energies since $V(\xi) \rightarrow E\xi$ as $\xi \rightarrow \infty$. In contrast, the motion in the η direction is unbounded, for any energy, since $V(\eta) \rightarrow -E\eta$ as $\eta \rightarrow \infty$. The different behaviors of $V(\xi)$ and $V(\eta)$ as ξ and $\eta \rightarrow \infty$ lead to qualitatively different wavefunctions. The motion in the ξ direction has a well defined integer quantum number, n_1 , and the motion in the η direction is, in principle, a continuum, containing resonances. In practice, at energies far below the saddle point in $V(\eta)$ the motion in the η direction also has a good quantum number, n_2 . In Fig. 6.4 we show the potentials $V(\xi)$ and $V(\eta)$ for $E=10^{-6}$, and $|m| = 1$. The potentials are shown for Z_1 and $Z_2 = 0.1, 0.5$, and 0.9 , corresponding to a blue shifted Stark state, a middle state, and a red shifted state. As shown by Fig. 6.4(a), the ξ motion is always bounded. In Fig. 6.4(b) it is apparent that the saddle point in $V(\eta)$ occurs at lower energy for the red state ($Z_2 = 0.9$) than for the blue state. Had we drawn the potential $V(\eta)$ for $m=0$, it would be deeper at $\eta = 0$ and would lie generally lower than the $|m| = 1$ potential. For $|m| > 1$ the potential has a $1/\eta^2$ centrifugal barrier at small η and lies generally above the $m = 1$ potentials of Fig. 6.4.

To solve Eqs. (6.22a) and (6.22b) for $E \neq 0$ we take advantage of the fact that the motion in ξ is bounded. This motion always has a bound wavefunction and a

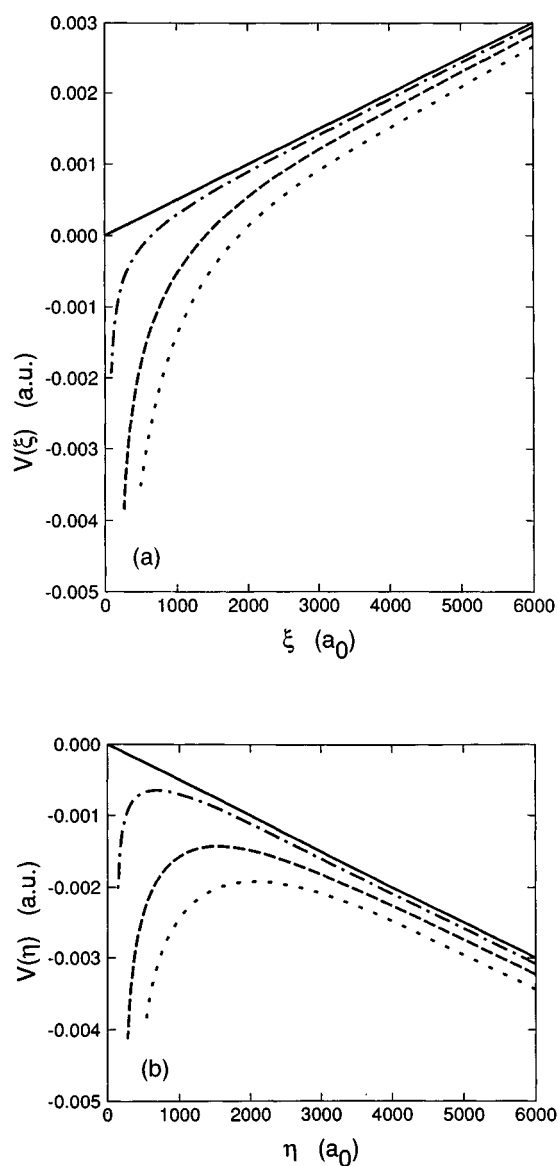


Fig. 6.4 Plots of the potentials $V(\xi)$ and $V(\eta)$ in a field $E = 10^{-6}$ for $|m| = 1$. (a) $V(\xi)$ for $Z_1 = 0$ (—), $Z_1 = 0.1$ (---), $Z_1 = 0.5$ (— — —), and $Z_1 = 0.9$ (· · ·) (b) $V(\eta)$ for $Z_2 = 0$ (—), $Z_2 = 0.1$ (---), $Z_2 = 0.5$ (— — —), and $Z_2 = 0.9$ (· · ·). Note that the saddle point in $V(\eta)$ occurs at lower energy for larger values of Z_2 . As $\eta \rightarrow \infty$ the Stark effect dominates both potentials, so that the ξ motion is always bounded.

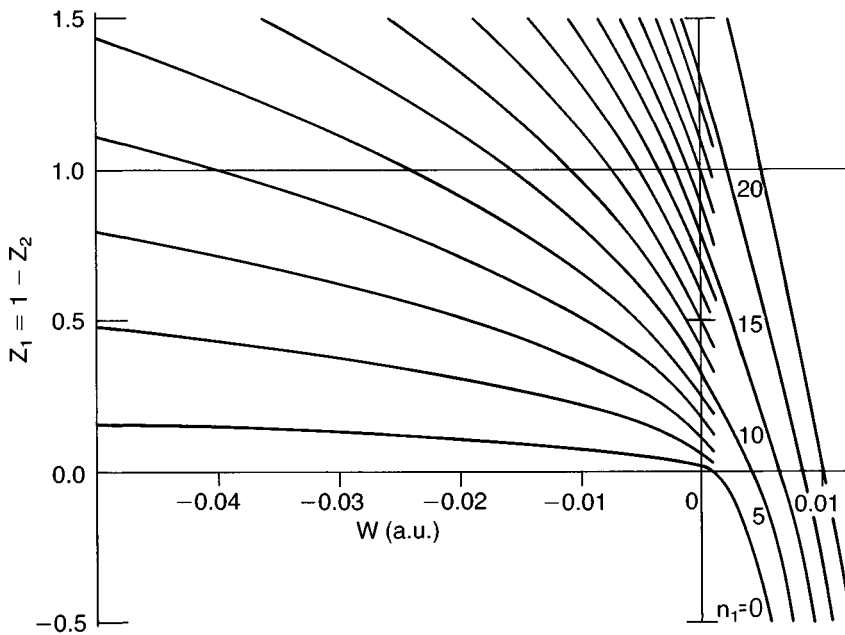


Fig. 6.5 Eigenvalues $Z_1 = 1 - Z_2$ as a function of energy for $m = 0$, $E = 1.5 \times 10^{-5}$ au = 77 kV/cm, and $n_1 = 0-20$ (from ref. 15).

well defined, integral quantum number, n_1 , which is simply the number of nodes in the $\chi_1(\xi)$ wavefunction. For any energy W and any value of n_1 we can find the separation constant, or eigenvalue, Z_1 . A straightforward way of finding the eigenvalues of Z_1 is to use the unnormalized WKB approximation i.e. ^{1,3,15}

$$\chi_1(\xi) = \frac{1}{[T(\xi)]^{1/4}} \cos \left[\int_{\xi_1}^{\xi} \sqrt{T(x)} dx - \frac{\pi}{4} \right], \quad (6.25)$$

where $T(\xi) = [W - V(\xi)]/2$, and ξ_1 is the inner turning point. The WKB wavefunction of Eq. (6.25) satisfies the quantization condition

$$\int_{\xi_1}^{\xi_2} \sqrt{T(\xi)} d\xi = \left(n_1 + \frac{1}{2} \right) \pi, \quad (6.26)$$

where n_1 is a positive integer and ξ_1 and ξ_2 are the inner and outer classical turning points, where $W = V(\xi)$. An alternative approach is use a Numerov algorithm to find the allowed wavefunctions, as discussed by Luc-Koenig and Bachelier.³ In either case it is important to remember that for the same values of W , m , and E there exist many solutions to Eq. (6.26) corresponding to different values of the quantum number n_1 and the eigenvalue Z_1 . These solutions are orthogonal, which is why the H Stark energy levels of the same m cross, as shown by Fig. 6.2.

In Fig. 6.5 we show the eigenvalues Z_1 associated with several values of the quantum number n_1 as a function of the energy W . These values of Z_1 are obtained by means of the WKB quantization condition of Eq. (6.26). For a constant value

of n_1 , the value of Z_1 decreases with increasing energy. Keeping n_1 constant means the WKB integral of Eq. (6.26) must remain a constant. The WKB integral is an integral of the momentum, or the square root of twice the kinetic energy, over the classically allowed spatial region. Increasing the energy increases both the spatial range covered by the integral and the kinetic energy. To keep the same value of n_1 as W is increased, the depth of the small ξ part of the potential is reduced, by reducing Z_1 .

For fixed W , m , and E there exists a series of values of n_1 and Z_1 implying a series of allowed values of $Z_2 = 1 - Z_1$. The known value of Z_2 can now be used to solve Eq. (6.22b). Since $V(\eta) \rightarrow -E\eta/2$ as $\eta \rightarrow \infty$, in principle, the solution to Eq. (6.22b) is a continuum wave for all values of the energy. Accordingly we normalize the $\chi_2(\eta)$ wavefunction per unit energy as $\eta \rightarrow \infty$. The asymptotic form of the χ_2 wavefunction is easily obtained using the WKB approximation as³

$$\chi_2(\eta) = \frac{N}{(E\eta + 2W)^{1/4}} \sin \left[\frac{1}{3E} (E\eta + 2W)^{3/2} + \alpha \right], \quad (6.27)$$

where α is a phase. In Eq. (6.27) we see the expected oscillatory behavior and an amplitude decreasing as the square root of the momentum. Normalization per unit energy is often obtained by requiring that the inward or outward flux $I = 1/2\pi$. An approach consistent with the one given in Chapter 2 is to require

$$N^2 \int \psi^* d\tau \int_{W-\Delta W}^{W+\Delta W} \psi dW' = 1. \quad (6.28)$$

In parabolic coordinates the volume element $d\tau = (\xi + \eta)d\xi d\eta d\phi/4$. Using the general form of the wavefunction given in Eq. (6.21) and carrying out the angular integration, leads to

$$N^2 \int_0^\infty \int_0^\infty \chi_1^*(\xi) \chi_2^*(\eta) \left(\frac{1}{\eta} + \frac{1}{\xi} \right) \frac{d\xi d\eta}{4} \int_{W-\Delta W}^{W+\Delta W} \chi_1(\xi) \chi_2(\eta) dW'. \quad (6.29)$$

The major contribution to the integral of Eq. (6.29) comes from $\eta \rightarrow \infty$, in which case the $1/\eta$ term may be ignored, leaving

$$N^2 \int_0^\infty \frac{(\chi_1^*(\xi) \chi_1(\xi))}{\xi} d\xi \int_0^\infty \chi_2^*(\eta) d\eta \int_{W-\Delta W}^{W+\Delta W} \chi_2(\eta) dW' = 1. \quad (6.30)$$

Requiring that the bound $\chi_1(\xi)$ wavefunction be normalized according to

$$\int_0^\infty \frac{\chi_1^2(\xi)}{\xi} d\xi = 1 \quad (6.31)$$

yields

$$N = \frac{1}{\sqrt{2\pi}}. \quad (6.32)$$

Thus, the amplitude of the continuum $\chi_2(\eta)$ wave varies insignificantly with energy as $r \rightarrow \infty$.

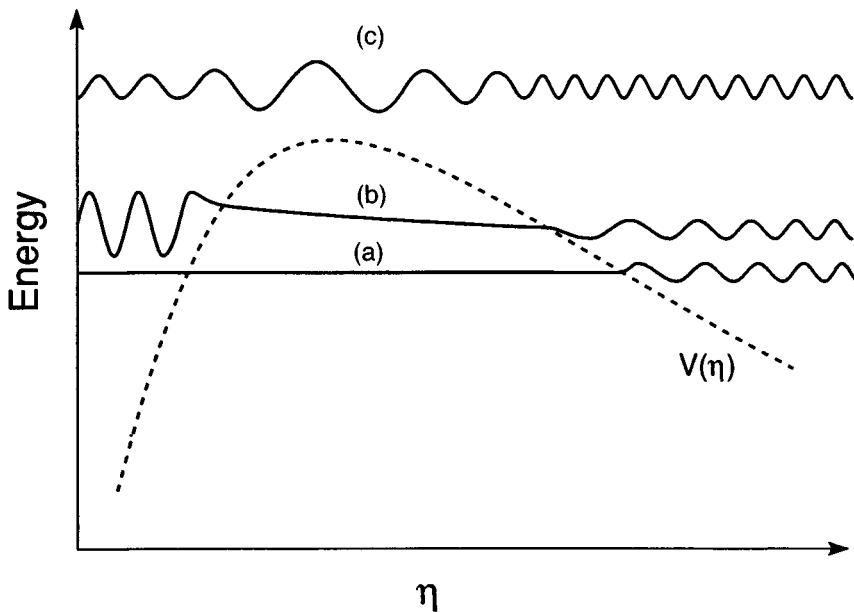


Fig. 6.6 Schematic wavefunctions $\chi_2(\eta)$ for several choices of energy W . (a) An energy below V_b the peak of the potential $V(\eta)$ but away from a resonance of the inner well of the potential. The wavefunction is vanishingly small in the inner well of the potential. (b) An energy below V_b and equal to one of the resonances of the inner well. (c) An energy above V_b .

We are not so much interested in the wavefunction at $r = \infty$ as we are in the wavefunction in the inner potential well of $V(\eta)$. Examining plots of the potential $V(\eta)$, it is apparent that there is, in general, a peak V_b in the potential for $\eta > 0$. If the energy W is above the peak of the potential, $W > V_b$, the entire region is classically accessible. If the energy is less than the peak of the potential barrier, V_b , there are two classically allowed regions, which are connected by a classically forbidden region through which tunnelling occurs. The ratio of the amplitude of the wavefunction in the inner well to that in the outer well is the quantity of interest.

Consider for a moment the energy dependence of $\chi_2(\eta)$ shown qualitatively in Fig. 6.6. Outside the barrier it has the standing wave form of Eq. (6.27) with virtually the same amplitude irrespective of energy. At an energy below the peak of the barrier the small transmission through the barrier allows some of the exterior wavefunction to penetrate into the inner well. Due to the small transmission, in general the wavefunction in the inner well of $V(\eta)$ is negligible, as shown by Fig. 6.6(a). However, at a resonance of the inner well, when an integral number, n_2 , of nodes fit, the amplitude of the wavefunction in the inner well becomes very large, as shown in Fig. 6.6(b). This situation is no different from a microwave cavity tenuously coupled to a transmission line.

Far below the barrier in $V(\eta)$ the transmission through the barrier is vanishingly small and the resonances are so sharp as to be bound states for all practical purposes, i.e. n_2 is, for all intents and purposes, a good quantum number. As the energy is increased the transmission through the barrier increases, increasing the width of the resonances and decreasing the amplitude of $\chi_2(\eta)$ in the inner well. Finally, when the energy exceeds the top of the barrier, the wavefunction has an amplitude which is spatially smoothly varying, displaying minimal energy dependence as shown by Fig. 6.6(c). While Fig. 6.6 shows the variation of the amplitude of the wavefunction in the inner well of $V(\eta)$ in a qualitative way, we need a quantitative measure. A useful one takes advantage of the fact that for small values of ξ and η the regular solutions of $\chi_1(\xi)/\sqrt{\xi}$ and $\chi_2(\eta)/\sqrt{\eta}$ behave as $\xi^{|m|/2}$ and $\eta^{|m|/2}$. A useful approach is to define a small ξ, η wavefunction as³

$$\psi = \frac{\sqrt{C_{n_1}^m}}{\sqrt{2\pi}} (\xi\eta)^{|m|/2} e^{im\phi}, \quad (6.33)$$

where $C_{n_1}^m$, the density of states, is a measure of the probability of finding the electron near the origin, which is of practical interest for photoexcitation from the ground state and for field ionization of nonhydrogenic atoms. $C_{n_1}^m$ is easily obtained from the normalized $\chi_1(\xi)$ and $\chi_2(\eta)$ wavefunctions.

Figure 6.7 is a plot of $\sqrt{C_{n_1}^m}$, the square root of the density of states for $m = 0$, $n_1 = 7$, and $E = 1.5 \times 10^{-5}$ au. In this case the maximum in the potential barrier of $V(\eta)$ is at $V_b = -0.00372$. Above V_b there is, in essence, a continuum. Below V_b there is a series of sharp states which each have the approximate quantum numbers n_2 shown, corresponding to n_2 nodes of the $\chi_2(\eta)$ wavefunction in the inner potential well of $V(\eta)$. For any m , each value of the quantum number n_1 has a spectrum of $\sqrt{C_{n_1}^m}$ analogous to the one shown in Fig. 6.7.

Field ionization

Field ionization, which is of great practical importance in the study of Rydberg atoms, has been introduced implicitly in Figs. 6.2, 6.6, and 6.7. We now consider it explicitly. The order of magnitude of the field required for ionization can be estimated using the method outlined in Chapter 1. The combined coulomb–Stark potential,

$$V = -1/r + Ez, \quad (6.34)$$

has its saddle point on the z axis at $z = -1/\sqrt{E}$ where the potential has the value $V = -2\sqrt{E}$. Thus if an atom is in a state of $m = 0$, so that there is no additional centrifugal potential, and if the electron is bound by an energy W , a field given by

$$E = \frac{W^2}{4} \quad (6.35)$$

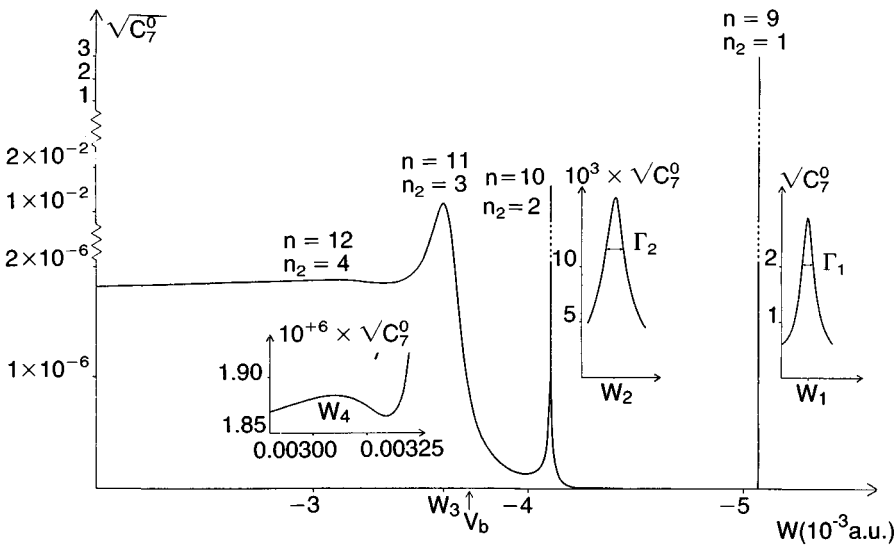


Fig. 6.7 Typical energy dependence of the square root of the density of states $\sqrt{C_{n_1}^m}$ for the states $n_1 = 7$, $|m| = 0$ in the field $E = 1.5 \times 10^{-5}$ au. V_b is the energy of the peak in the potential $V(\eta)$. Each resonance is characterized by the parabolic quantum number n_2 and the width Γ_{n_2} which varies rapidly with n_2 . For $W < V_b$ the spectrum exhibits quasi-discrete resonances; for $W > V_b$ the continuum exhibits oscillations which damp out with increasing n_2 . The energies and widths of the prominent resonances are.

$W_1 = -0.005066$	$\Gamma_1 = 3.7 \times 10^{-16}$ au
$W_2 = -0.004097$	$\Gamma_2 = 6.368 \times 10^{-8}$ au
$W_3 = -0.003601$	$\Gamma_3 = 1.353 \times 10^{-4}$ au
$W_4 = -0.003100$	$\Gamma_4 = 1.637 \times 10^{-3}$ au

(from ref. 3).

is adequate for ionization to occur classically. This field is usually termed the classical field for ionization. If we ignore the Stark shift of a Rydberg state of principal quantum number n and write the binding energy in terms of n we obtain the familiar result for the classical field for ionization in terms of the principal quantum number,

$$E = \frac{1}{16n^4}. \quad (6.36)$$

These results, Eqs. (6.35) and (6.36), are only valid for $m = 0$ states. In higher $|m|$ states there is a $1/(x^2 + y^2)$ centrifugal potential keeping the electron away from the z axis, and the centrifugal barrier raises the threshold field of $m \neq 0$ states.¹⁶ Specifically, for $m \neq 0$ states the fractional increase in the field required for ionization, compared to an $m = 0$ state of the same energy is¹⁶

$$\frac{\Delta E}{E} = \frac{|m|\sqrt{W}}{\sqrt{2}} = \frac{|m|}{2n}. \quad (6.37)$$

Actually, the classical picture outlined above has two serious defects. First, Eq. (6.36) ignores the Stark shifts, as does Eq. (6.37) if we choose to use the $|m|/2n$ form. Eq. (6.35) and the $|m|\sqrt{W}/\sqrt{2}$ form of Eq. (6.37) do not suffer from this defect. Second, the classical approach ignores the spatial distribution of the wavefunctions. As shown by Fig. 6.2 for H states of the same n and $|m|$, the higher energy, or blue, $n_1 - n_2 \sim n$, Stark states require higher fields to ionize than the lower energy, or red, $n_2 - n_1 \sim n$ states. This point is made quite graphically by Bethe and Salpeter¹ using the data of Rausch von Traubenberg.¹⁷ A simple physical argument for why blue states are harder to ionize than red states is that the electron is located on the side of the atom away from the saddle point in the potential in the blue states, whereas the electron is adjacent to the saddle point in the red states.

In parabolic coordinates the motion in the ξ direction is bounded. Thus for ionization to occur the electron must escape to $\eta = \infty$. Classically, ionization only occurs for energies above the peak V_b of the potential barrier in $V(\eta)$. If we ignore the short range $1/\eta^2$ term in Eq. (6.23b), an approximation good for low m , and set $W = V_b$ we find the required field for ionization to be

$$E = \frac{W^2}{4Z_2}. \quad (6.38)$$

Eq. (6.38) differs from Eq. (6.35) by the factor of Z_2 , the effective charge. Consider three low $|m|$ states of the same energy. A blue state of $n_1 - n_2 \approx n$ has $Z_2 \approx 1/n$, a central state has $n_1 = n_2 \sim n/2$ and $Z_2 \approx 1/2$, and a red state has $n_2 - n_1 \sim n$ and $Z_2 \sim 1$. If the three states have the same energy the red one will be most easily ionized.

For the extreme red Stark state of high n Eq. (6.38) reduces to Eq. (6.35) since $Z_2 \approx 1$. For this Stark state the Stark shift increases the binding energy, and for an $m = 0$ state the energy is adequately given using the linear Stark effect as

$$W = -\frac{1}{2n^2} - \frac{3n^2 E}{2}. \quad (6.39)$$

Using this energy we find a threshold field

$$E = \frac{1}{9n^4}. \quad (6.40)$$

The numerical factor of $1/9$ instead of $1/16$ is due to the Stark shift of the level.

For the blue states it is not possible to estimate simply the threshold field. However, blue and red states of the same n and $m=0$ often have threshold fields differing by a factor of 2.

We have thus far discussed field ionization as occurring only when it is classically allowed, when $E > W^2/4Z_2$, i.e. when the energy is above V_b , the peak of the barrier in $V(\eta)$, as shown in Fig. 6.6. At lower energies tunnelling occurs, and accurate calculations of ionization rates have been made.^{2,18-20} As an example, we show in Fig. 6.8 the ionization rates calculated by Bailey *et al.*²⁰ using

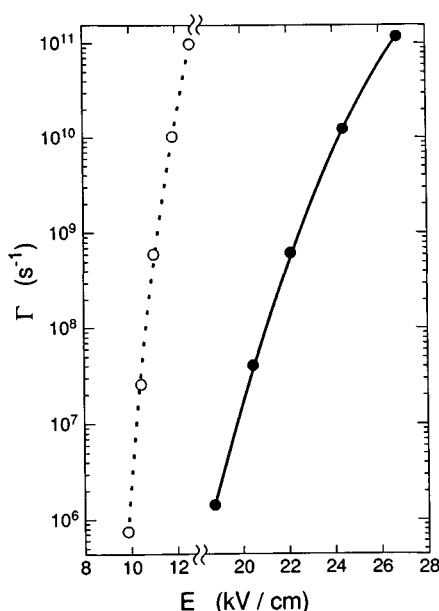


Fig. 6.8 Calculated ionization rates of the (n, n_1, n_2, m) states as a function of field using the values of ref. 20: the $(15, 0, 14, 0)$ red state (○) and the $(15, 14, 0, 0)$ blue state (●).

the WKB method of Rice and Good¹⁸ for the $H n=15$ extreme red and blue $m=0$ Stark states over the range of ionization rates from 10^6 s^{-1} to 10^{11} s^{-1} . From Fig. 6.8 it is apparent that the blue states ionize at higher fields than red states of the same n . It is also useful to note that the ionization rates increase rapidly with field, implying that simply computing when ionization is classically allowed is often quite adequate for practical applications.

Using a fast (8 keV) beam of H Koch and Mariani²¹ have made precise measurements of the ionization rates of state selected $|m|=0$ and 1, $n=30\text{--}40$ blue Stark states of H. As can be seen in Fig. 6.9, the measured rates are in quite good agreement with, but typically offset from, the analytic results obtained by Damburg and Kolosov,²² using the approximation that the field is weak enough that n_2 is very nearly a good quantum number. However, as shown, the measured rates are in excellent agreement with the exact numerical calculations of Damburg and Kolosov,²³ which are in good agreement with the ionization rates calculated by Bailey *et al.*²⁰ The excellent agreement between theory and experiment suggests that a nonrelativistic treatment is quite adequate. However, Bergeman has calculated that it should be possible to see the effects of spin orbit coupling in field ionization.²⁴

The ionization measurements of Koch and Mariani²⁰ are for blue states of $n_1 \approx n$. Rottke and Welge²⁵ carried out measurements on the relatively red $|m|=0$ and 1 states of $n=18$ and blue states of $n=19$ verifying the rapid decrease in ionization rate with n_1 shown theoretically in Fig. 6.9. In their measurements

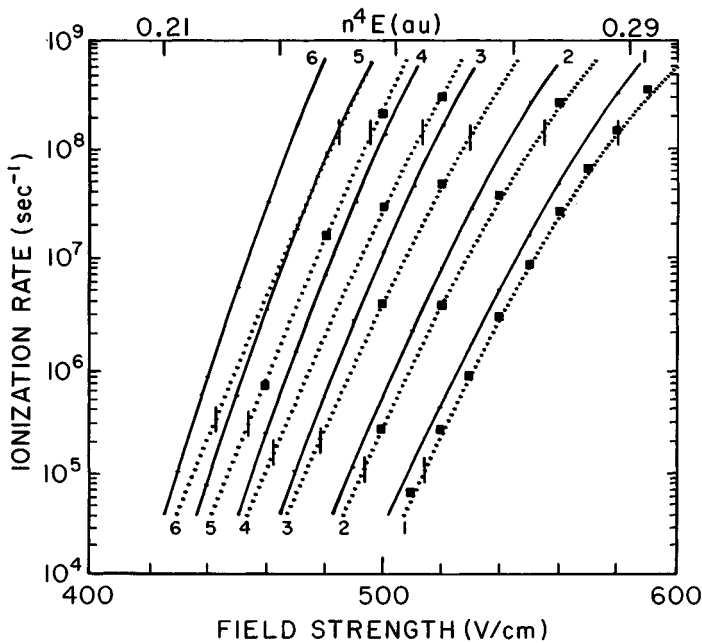


Fig. 6.9 Dotted lines, experimental ionization rate curves for the $(n, n_1, n_2, |m|)$ states 1: (40,39,0,0); 2: (40,38,0,1); 3: (40,38,1,0); 4: (40,37,1,1); 5: (40,37,2,0); 6: (40,36,2,1). The tick marks represent the range of validity of the experiment. Solid line, theoretical curves calculated with Eq. (6) of ref. 22. Squares, numerical theoretical results from ref. 23 (from ref. 21).

they used pulsed laser excitation of a beam of H atoms in a field of 5714 V/cm followed by time resolved detection of the electrons from the ionizing H atoms.

Most of the discussion of field ionization thus far has been focused on states of low $|m|$. For states of high $|m|$ we may no longer neglect the centrifugal potential, and it is not possible to derive a simple expression such as Eq. (6.38). Nonetheless, since the centrifugal term raises the barrier in the potential $V(\eta)$ it is clear that the ionizing field must be higher for high m states. A graphic illustration of this point is provided by Fig. 6.10, a simulation of the ionizing fields of all the $|m| \geq 3$ states of $n = 31$.²⁶ Measurements of the field ionization profiles of collisionally induced mixtures of the degenerate high $\ell, |m|$ states of the same n are in substantial agreement with the profile predicted in Fig. 6.10.²⁶

Nonhydrogenic atoms

Atoms other than H have characteristics essentially similar to those of the H atom in an electric field, but there are important differences due to the presence of the finite sized ionic core. In zero field the presence of the core simply depresses

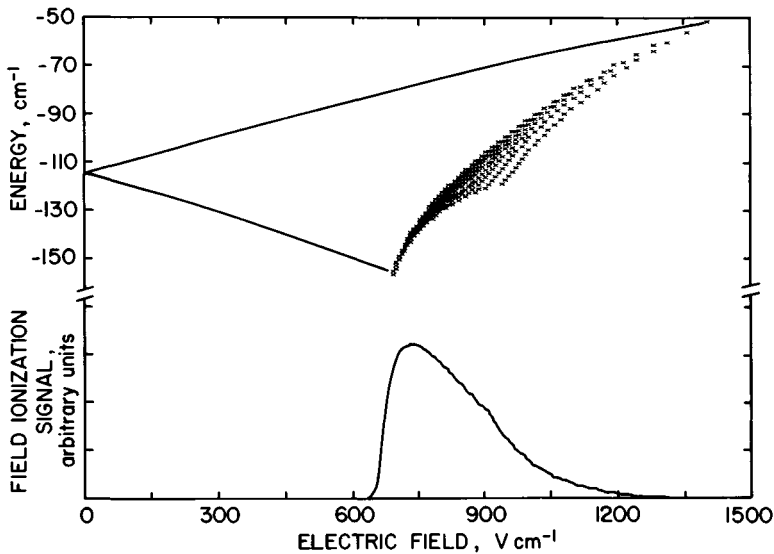


Fig. 6.10 Calculated SFI profile for diabatic ionization of the H like $|m| \geq 3$ states. Top, extreme members of the $n = 31$, $|m| = 3$ Stark manifold. The crosses represent the points at which each $|m| \geq 3$ Stark state achieves an ionization rate of 10^9 s^{-1} . Bottom, calculated SFI profile for diabatic ionization of a mixture containing equal numbers of atoms in each $|m| \geq 3$ Stark level for $n = 31$ at a slow rate of 10^9 V/cm s . (from ref. 26).

the energies, particularly those of the lowest ℓ states. As long as the core is spherically symmetric, it does not alter the spherical symmetry of the problem, and the effect is relatively minor. However, with a finite sized ionic core the wavefunction is no longer separable in parabolic coordinates. As a result the parabolic quantum number n_1 , which is a good quantum number in H, is no longer good.

The most important implication of n_1 not being a good quantum number is that blue and red states are coupled by their slight overlap at the core. In the region below the classical ionization limit blue and red states of adjacent n do not cross as they do in H, but exhibit avoided crossings as a result of their being coupled. Above the classical ionization limit blue states, which would be perfectly stable in H, are coupled to degenerate red states, which are unbound, and ionization occurs rapidly compared to radiative decay. It is really an autoionization process in which the blue state is coupled to the red continuum state at the ionic core.

In this chapter we shall treat the Stark effect in nonhydrogenic atoms by methods which, while not the most powerful, provide physical insight. We begin by calculating the energy levels of Na, as an example, in the regime where the field ionization rates are negligible. Ignoring the spin of the Rydberg electron, the Hamiltonian is given by

$$H = -\frac{\nabla^2}{2} + \frac{1}{r} + V_d(r) + Ez, \quad (6.41)$$

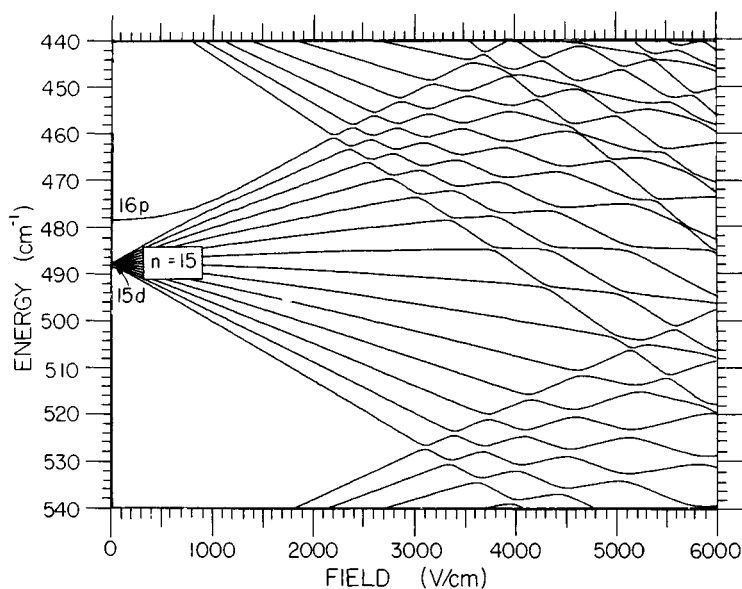


Fig. 6.11 Calculated Na $|m| = 1$ energy levels (from ref. 27).

where $V_d(r)$ is the difference between the Na potential and the coulomb potential, $-1/r$. Consequently, $V_d(r)$ is only nonzero near $r = 0$. A straightforward and effective method of calculating the energies and wavefunctions is direct diagonalization of the Hamiltonian matrix. If we use the Na $n\ell m$ spherical states as basis functions, the Hamiltonian has diagonal elements given by the known energies of the zero field $n\ell m$ states, i.e.

$$\langle n\ell m | H | n\ell m \rangle = \frac{1}{2(n - \delta_\ell)^2}, \quad (6.42)$$

and off diagonal matrix elements $\langle n\ell m | E_z | n'\ell \pm 1m \rangle$ which may be calculated using the Numerov method outlined in Chapter 2. If several n manifolds of ℓ states are included, the eigenvalues of the matrix give the energy levels of the Na atom in the field, and the eigenvectors give the Stark states in terms of the zero field $n\ell m$ states.

In Figs. 6.11 and 6.12 we show the energy levels for the Na states of $n \sim 15$ for $|m| = 1$ and 0, respectively, obtained by direct diagonalization of the Hamiltonian matrix.²⁷ In zero field the quantum defects 1.35 and 0.85, of the Na s and p states displace them from the high ℓ states and they only exhibit large Stark shifts when they intersect the manifold of Stark states. A second obvious difference between Figs. 6.11 and 6.12 and Fig. 6.2 is that the energy levels of different n do not cross as they do in H. There are avoided crossings, which are clearly visible for the $|m| = 1$ states of Fig. 6.11 and overwhelming for the $m = 0$ states of Fig. 6.12. The energy level diagram for the Na $|m| = 2$ states would not be observably different from the analogous diagram for H on the scale of Figs. 6.11 and 6.12. At zero field

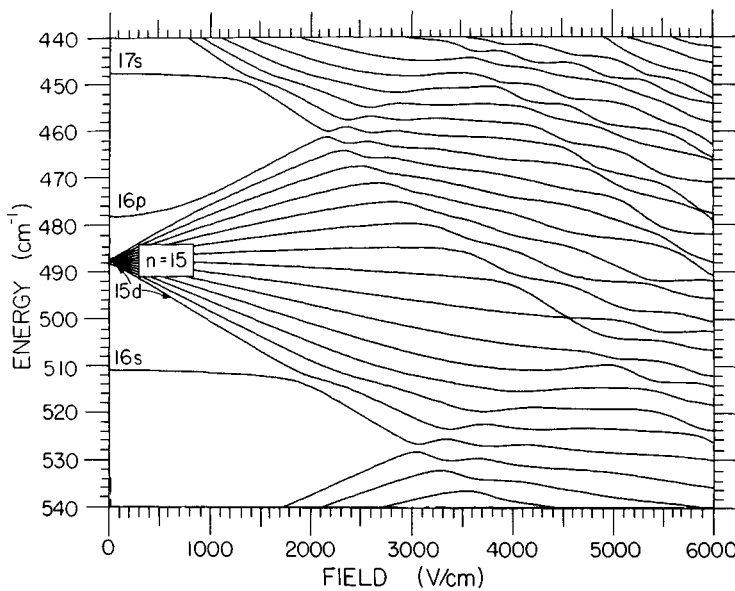


Fig. 6.12 Calculated Na $m = 0$ energy levels (from ref. 27).

the levels would be degenerate, and for $E > 1/3n^5$ the blue and red levels of adjacent n would cross.

A subtle point which is not immediately apparent when inspecting Figs. 6.11 and 6.12 is that the $|m| = 0$ and 1 levels are not interleaved as they are in H. Note that in Fig. 6.12 the 16s state joins the manifold at a field of 2000 V/cm. At fields below 1500 V/cm the $|m| = 0$ and $|m| = 1$ levels of the lower half of the $n = 15$ Stark manifold lie practically on top of one another. This point was demonstrated explicitly by Fabre *et al.*²⁸ who used a diode laser to excite Na atoms from the 3d state to states of the $n = 29$ Stark manifold with high resolution. In Fig. 6.13 we show the result of scanning the wavelength of the diode laser across three high lying pairs of $|m| = 0$ and 1 levels of the $n = 29$ Stark manifold in a field of 20.5 V/cm. The $n = 29$ and $n = 30$ Stark manifolds intersect at a field of 83 V/cm, so the field of 20.5 V/cm corresponds roughly to a field of 550 V/cm in Figs. 6.11 and 6.12, i.e. the 30p state has not yet joined the $n = 29$ Stark manifold. The $|m| = 0$ and 1 states are < 200 MHz apart while the adjacent pairs of Stark levels are ~ 2 GHz apart. In H the $|m| = 0$ and 1 levels would be 1 GHz apart. The $|m| = 0$ levels lie above the $|m| = 1$ levels, i.e. they are further displaced from the center of the manifold, or the zero field $n = 29$ high ℓ levels.

Fabre *et al.*²⁸ used a projection operator technique to describe the Stark shifts at fields below where low ℓ states of large quantum defects join the manifold. A less formal explanation is as follows. If, for example, the s and p states are excluded, as in Fig. 6.13 below 800 V/cm, effectively only the nearly degenerate $\ell \geq 2$ states are coupled by the electric field. The only differences among the $|m| = 0, 1$, and 2 manifolds occur in the angular parts of the matrix element, i.e.¹

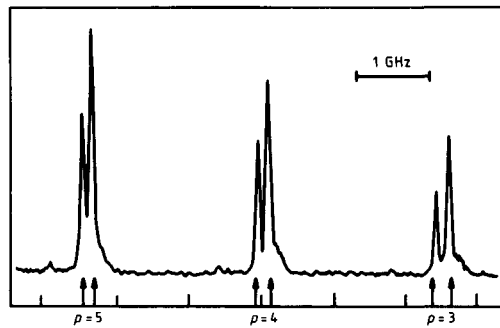


Fig. 6.13 Part of the excitation spectrum of the Na $n = 29$ Stark levels from the 3d state in an electrostatic field of 20.5 V/cm corresponding to n_1 values of $n - 3$, $n - 4$, and $n - 5$. The energy splitting between $m = 0$ (highest energy line in the doublets) and $|m| = 1$ states is of the order of 180 MHz. The arrows indicate theoretical positions of energy levels obtained by a numerical diagonalization of the Stark Hamiltonian (from ref. 28).

$$\langle n\ell m | \cos \theta | n\ell + 1m \rangle = \frac{\sqrt{(\ell + 1)^2 - m^2}}{\sqrt{(2\ell + 3)(2\ell + 1)}}. \quad (6.43)$$

Only in a few low ℓ states is the m dependence significant, and as a result the $|m| = 0, 1$, and 2 energy levels are almost degenerate. However, due to the low ℓ states, the $|m| = 0$ levels are displaced more than the $|m| = 1$ levels, which are displaced insignificantly more than the $|m| = 2$ levels.

Reexamining Figs. 6.11 and 6.12, the avoided crossings of the blue and red states of adjacent n are evident. When the avoided crossings are large they can be measured optically as was done by Zimmerman *et al.*²⁷ They excited Li atoms in a beam from the 2s state to the 2p state and then to the 3s state by two pulsed dye lasers at fixed wavelengths. The wavelength of a third laser was scanned to drive atoms from the 3s state to Rydberg Stark states in the presence of a static field. The transition to the Rydberg state was detected by applying a high field pulse to the atoms subsequent to the laser pulses. The ions were detected with a particle multiplier as the wavelength of the third laser was scanned. In Fig. 6.14 we show their observation of the avoided crossing of the (18,16,0,1) and (19,1,16,1) states. The experimental picture of Fig. 6.14 is built up by scanning the third laser over the energy of the avoided crossing for several fields near the avoided crossing field of 943 V/cm. As shown by Fig. 6.14, the observed energies of the levels match those calculated by matrix diagonalization. It is also interesting to note that the oscillator strength to the upper state vanishes at the anticrossing. At the anticrossing the eigenstates have the form $(1/\sqrt{2})(\Psi_A \pm \Psi_B)$ where Ψ_A and Ψ_B are the eigenstates away from the crossing. As shown by Fig. 6.14, away from the avoided crossing both states apparently have similar amounts of p character, and at the crossing the upper state loses it all to the lower state.

The method used by Zimmerman *et al.*²⁷ to measure the Li avoided crossing shown in Fig. 6.14 requires that the resolution of the laser be finer than the size of

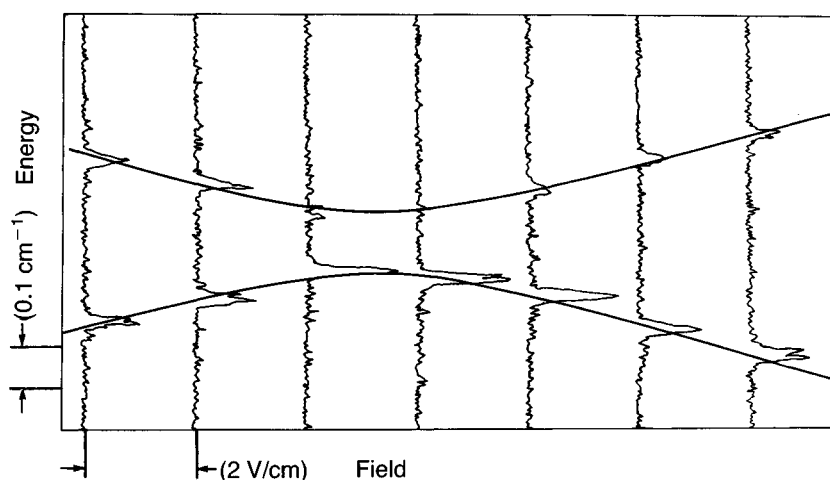


Fig. 6.14 Level anticrossing in Li. Intersection of the $(n, n_1, n_2, |m|)$ states $(18, 16, 0, 1)$ and $(19, 1, 16, 1)$ at 321.5 cm^{-1} and 943 V/cm . The calculated level structure is superimposed on the data. Note the vanishing oscillator strength to the upper level at the avoided crossing (from ref. 27).

the avoided crossing. An alternative approach used by Stoneman *et al.*²⁹ bypasses this requirement. They observed the narrow anticrossings between the $K (n+2)s$ states, which have a quantum defect of 2.18, and the low lying members of the n Stark manifold, which is composed of $\ell \geq 3$ states. They excited atoms in a beam from the $4s$ state to the $4p$ state and then to the ns state with two pulsed dye lasers with linewidths of 1 cm^{-1} . Two to three microseconds after the laser excitation the atoms were selectively field ionized, and only atoms which had undergone black body radiation induced transitions to the higher lying $(n+2)p$ states were detected. The $(n+2)p$ signal was recorded as the static field was slowly scanned through the avoided crossing field over many shots of the laser. Away from the avoided crossing only the $(n+2)s$ state was excited, and it could undergo black body radiation induced transitions to the $(n+2)p$ state. The n Stark state, composed of $\ell \geq 3$ states, cannot be excited from the $4p$ state or to the $(n+2)p$ state. At the avoided crossing the eigenstates are 50–50 mixtures of the two states, and both are excited from the $4p$ state, with the result that the same number of Rydberg atoms is always excited. However, each of these two states has half the $(n+2)s \rightarrow (n+2)p$ black body radiation induced transition rate, and the observed $(n+2)p$ signal decreases. An example of their signals, the avoided crossing signals of the $K 20s$ state with the lowest lying $|m| = 0$ and 1 states of the $n = 18$ Stark manifold, is shown in Fig. 6.15.²⁹ First we note the decrease in the $20p$ ion signal at each of the avoided crossings, as described above. Second we note that the $|m| = 0$ and 1 Stark states are nearly degenerate. If they were interleaved as in H, the two avoided crossings would be separated by 50 V/cm not 5 V/cm as they are in Fig. 6.15. We have again encountered the noninterleaving of the energy

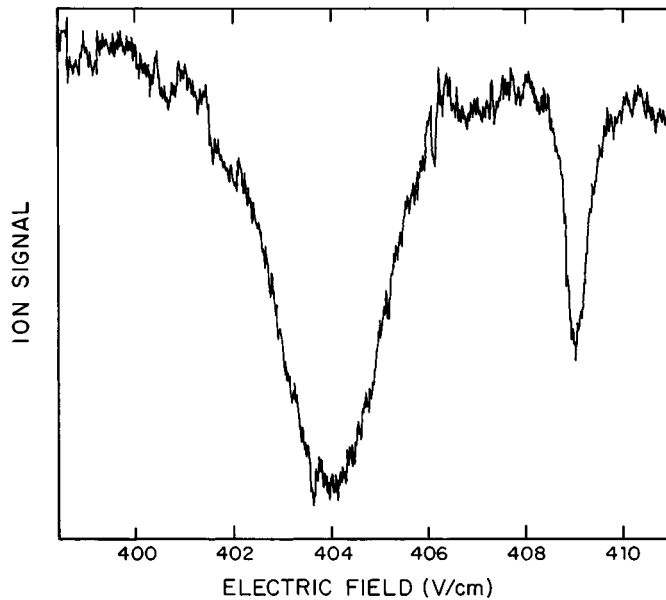


Fig. 6.15 Anticrossing signal from the avoided crossing of the K 20s state with the lowest energy $n = 18$ Stark manifold state. The left-hand peak corresponds to the $m = 0$ anticrossing, and the right-hand peak to the $|m| = 1$ anticrossing (from ref. 29).

levels of different m , as shown in Fig. 6.13. The spin orbit interaction, which we have ignored, is responsible for the avoided crossing between the 20s state and the $|m| = 1$ Stark state. Diagonalizing the Hamiltonian matrix, including the spin orbit interaction, yields 450 and 47 MHz for the widths of the $|m| = 0$ and 1 anticrossings of Fig. 6.16, in reasonably good agreement with the measured values of 510 MHz and 80 MHz.²⁹ In the heavier alkali atoms the spin orbit interaction plays an even more dominant role.²⁷

When the quantum defects are large and $|m|$ is low enough that several states have large quantum defects, matrix diagonalization using the $n\ell m$ states as basis functions is probably the most efficient approach. However, if the contributing quantum defects are small, a different approach to Eq. (6.41), suggested by Komarov *et al.*³⁰ is useful. They suggested using the hydrogenic parabolic states as a basis instead of the zero field $n\ell m$ states, in which case the Hamiltonian of Eq. (6.41) has diagonal matrix elements

$$\langle nn_1n_2m|H|nn_1n_2m\rangle = -\frac{1}{2n^2} + \frac{3}{2}n(n_1 - n_2)E + O(E^2) + \langle nn_1n_2m|V_d(r)|nn_1n_2m\rangle, \quad (6.44)$$

and nonzero off diagonal elements,

$$\langle nn_1n_2m|H|n'n'_1n'_2m\rangle = \langle nn_1n_2m|V_d(r)|n'n'_1n'_2m\rangle. \quad (6.45)$$

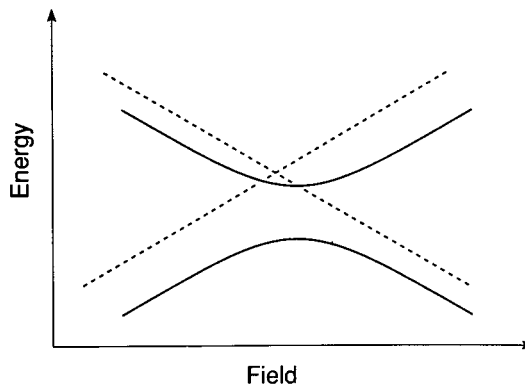


Fig. 6.16 Hydrogenic Stark energy levels (---) before considering the effect of the core perturbation $V_d(r)$. The alkali levels after incorporating the diagonal and off diagonal matrix elements of $V_d(r)$ (—). The diagonal elements shift the levels and the off diagonal elements lift the degeneracy at the level crossing. Note that the different shifts of the levels move the avoided crossing from the field of the hydrogenic crossing.

The diagonal elements are the hydrogenic energies plus a small energy shift, and the off diagonal elements are the couplings between levels. In Fig. 6.16 we show two hydrogenic levels as dotted lines and the non hydrogenic levels as solid lines. The shifts of the nonhydrogenic levels from their hydrogenic counterparts are due to the diagonal matrix elements of $V_d(r)$ and the avoided crossings to the off diagonal matrix elements of $V_d(r)$.

It is straightforward to evaluate the matrix elements of $V_d(r)$ using the known transformation between parabolic and spherical states given in Eqs. (6.18) and (6.19). If we expand the parabolic states in terms of hydrogenic spherical states, the matrix elements of $V_d(r)$ may be written as³⁰

$$\langle nn_1n_2m|V_d(r)|n'n'_1n'_2m\rangle = \sum_{\ell} \langle nn_1n_2m|n\ell m\rangle \langle n\ell m|V_d(r)|n'\ell m\rangle \langle n'\ell m|n'n'_1n'_2m\rangle, \quad (6.46)$$

where we have taken advantage of the spherical symmetry of $V_d(r)$ to require the same ℓ on both sides of the matrix element. We recall that the core interaction causes the depression of the energy levels below their H analogues. Thus,

$$\langle n\ell m|V_d(r)|n\ell m\rangle = \frac{-\delta_{\ell}}{n^3}. \quad (6.47)$$

This expression may be generalized by realizing that $V_d(r)$ is only nonzero for small r , where the dependence on n is through the normalization factor $n^{-3/2}$. Accordingly Eq. (6.47) can be generalized to³⁰

$$\langle n\ell m|V_d(r)|n'\ell m\rangle = \frac{-\delta_{\ell}}{\sqrt{n^3 n'^3}}. \quad (6.48)$$

Thus,

$$\langle nn_1n_2m|V_d|n'n_1'n_2'm\rangle = \sum_{\ell} \langle nn_1n_2m|n\ell m\rangle \frac{-\delta_{\ell}}{\sqrt{n^3n'^3}} \langle n'\ell m|n'n_1'n_2'm\rangle. \quad (6.49)$$

Since $|\langle nn_1n_2m|n\ell m\rangle|^2$ is the $n\ell m$ fraction of the nn_1n_2m state, it is evident that the depression of the nonhydrogenic Stark levels below the hydrogenic value, given by the diagonal matrix elements of $V_d(r)$, is simply the amount of each low ℓ state it possesses multiplied by its quantum defect. The nonhydrogenic energy must be lower than the hydrogenic energy since $\delta_{\ell} > 0$ for all ℓ .

At the field where the diagonal energies of two levels, given by Eq. (6.43), are equal, the off diagonal matrix element lifts the degeneracy. At the avoided crossing the energy levels are split by

$$\Delta W = 2|\langle nn_1n_2m|V_d(r)|n'n_1'n_2'm\rangle|, \quad (6.50)$$

and the eigenstates are symmetric and antisymmetric combinations of the two Stark states. Eq. (6.49) shows how the avoided crossings scale with n . Averaged over ℓ , $\langle nn_1n_2m|n\ell m\rangle^2$ must be equal to $1/(n-m)$. Thus, for low $|m| \ll n$ and $n \sim n'$, the matrix element of Eq. (6.49) is proportional to $1/n^4$, and the avoided crossings exhibit a $1/n^4$ scaling.

Ionization of nonhydrogenic atoms

A nonhydrogenic atom, Na for example, differs from H not only in its energy level spectrum but also in how it is ionized by a field. In particular, in a nonhydrogenic atom there are two forms of field ionization. The first is exactly like H. A state of approximate quantum number n_1 can itself ionize, keeping the same value of n_1 , in a field E if its energy W is enough for the electron to surmount the potential barrier V_b in $V(\eta)$. Rates for this form of ionization increase rapidly in the vicinity of the threshold for classical ionization, just as in H. In this form of ionization the blue states require higher fields for ionization than red states of the same energy.

The second form of ionization is similar to autoionization.³¹ In a nonhydrogenic atom n_1 is not a good quantum number and bound states of high n_1 are coupled to Stark continua of low n_1 . This form of ionization applies to those states other than the reddest Stark states. The extreme red Stark states have $n_1 = 0$ and ionize, as do the red H states, at the classical ionization limit given by Eq. (6.35), modified by Eq. (6.37) for $m \neq 0$ states. This point has been demonstrated explicitly by Littman *et al.*,³² who measured the time resolved ionization of Na $|m| = 2$ states subsequent to pulsed laser excitation. Their measured rates are in good agreement with the rates obtained by extrapolation of the rates of Bailey *et al.*²⁰ shown in Fig. 6.8.

In H it is possible to have $n_1 \geq 1$ states which are quite stable against ionization even though energetically degenerate states of lower n_1 have been converted to continua. In other atoms these blue $n_1 \geq 1$ states are coupled by the finite core to the red continua and autoionize. A simple physical picture is that an electron in a blue orbit comes near the core once on each orbit and has a finite probability of being scattered from the stable blue orbit into the degenerate red continuum. A significant difference between this form of ionization and hydrogenic ionization is that its rate is not exponentially dependent on the field.

The existence of the second form of ionization means that there is a large regime of field and energy in which nonhydrogenic atoms have states which decay more rapidly by ionization than by radiative decay, but are still spectroscopically narrow. In H all states have ionization rates which are exponentially dependent on the field so the analogous range is much smaller. The fact that there exist sharp but ionizing levels above the classical ionization limit but below the hydrogenic ionization limit was first observed by Jacquinet *et al.* in Rb.³³ It was most clearly shown by Littman *et al.* in Li.³¹ They excited a beam of Li atoms in static fields by two fixed frequency lasers from the ground state 2s via the 2p state to the 3s state. They then scanned the wavelength of a third laser to drive transitions from the 3s state to Rydberg states having np character. The excitation to a Rydberg state was detected by field ionization of the atom and subsequent detection of the ion with a particle multiplier. Spectra were recorded at many static fields, as shown by Fig. 6.17, which is composed of spectra recorded with the third laser polarized perpendicular to the static field so as to excite only $|m| = 1$ states. The spectra of Fig. 6.17 were recorded in two different ways. The spectra of Fig. 6.17(a) were recorded by applying a field ionization pulse $3 \mu\text{s}$ after laser excitation and only detecting the ions which result from ionization by the pulse, not spontaneous ionization. The resulting spectra are then composed of states which are stable for at least $3 \mu\text{s}$ against field ionization, or radiative decay, which is longer than $3 \mu\text{s}$ for the states of Fig. 6.17. It is quite apparent that there are no stable states above the classical ionization limit.

On the other hand, if no field ionization pulse is applied and the ions from atoms which ionize in the first $3 \mu\text{s}$ after the laser excitation are detected, the result is Fig. 6.17(c). The spectra begin at the classical limit and extend to much higher static fields. What is most important to note is that levels appear over a field range of typically a factor of 2 with no apparent change in width, a phenomenon which does not occur in H. Fig. 6.17(b) shows the calculated H $|m| = 1$ energy levels. Inspecting Fig. 6.17(c) it is easy to see that the levels disappear from the experimental spectrum when their decay rates exceed the laser linewidth by a large margin.

We may develop a more quantitative understanding of nonhydrogenic ionization by starting with the hydrogenic Stark states and adding the core perturbation $V_d(r)$, just as we calculated the sizes of the avoided level crossings. The level crossings described earlier are the result of core couplings between the bound

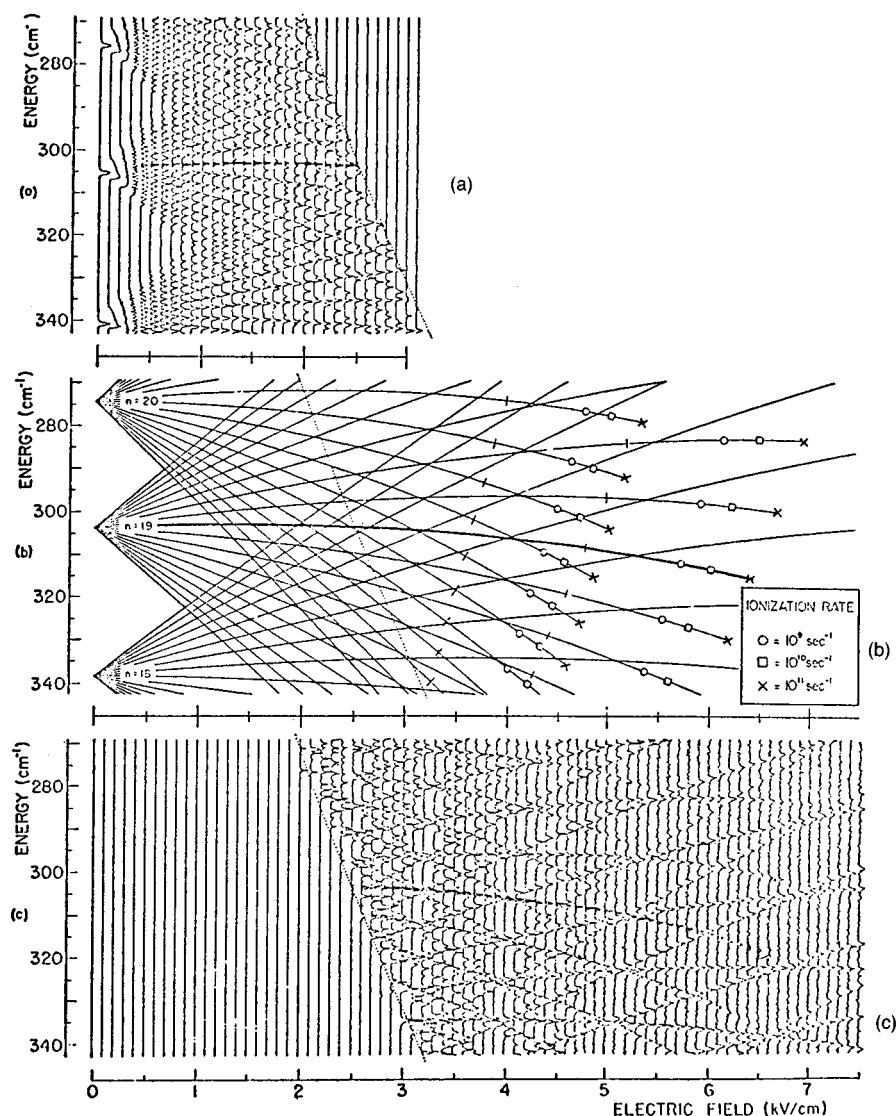


Fig. 6.17 Tunnelling and saddle point ionization in Li. (a) Experimental map of the energy levels of Li $|m| = 1$ states in a static field. The horizontal peaks arise from ions collected after laser excitation. Energy is measured relative to the one-electron ionization limit. Disappearance of a level with increasing field indicates that the ionization rates exceed $3 \times 10^5 \text{ s}^{-1}$. The dotted line is the classical ionization limit given by Eqs. (6.35) and (6.36). One state has been emphasized by shading. (b) Energy levels for H ($n = 18-20$, $|m| = 1$) according to fourth order perturbation theory. Levels from nearby terms are omitted for clarity. Symbols used to denote the ionization rate are defined in the key. The tick mark indicates the field where the ionization rate equals the spontaneous radiative rate. (c) Experimental map as in (a) except that the collection method is sensitive only to states whose ionization rate exceeds $3 \times 10^5 \text{ s}^{-1}$. At high fields, the levels broaden into the continuum in agreement with tunnelling theory for H (from ref. 32).

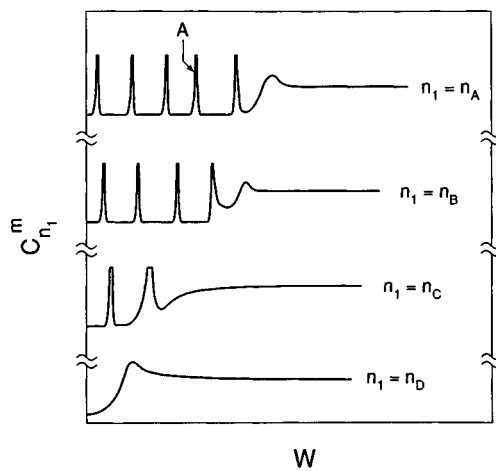


Fig. 6.18 The density of states $C_{n_1}^m$ for four different values of the quantum number n_1 : $n_A > n_B > n_C > n_D$. For each value of n_1 $C_{n_1}^m$ is composed of narrow resonances up the $W = V_b$, the energy of the saddle point in $V(\eta)$. Above $W = V_b$ $C_{n_1}^m$ is continuous as shown. The bound state A in $n_1 = n_A$ can ionize into the $n_1 = n_C$ and n_D continua in a nonhydrogenic atom. There is often a peak in $C_{n_1}^m$ at the onset of the continuous spectrum, as shown.

nn_1n_2nm and $n'n_1'n_2'm$ states, couplings between bound levels of different n_1 . The same coupling between a hydrogenically bound n_1m state and a continuum state of lower n_1 results in ionization.

Ignoring for the moment coupling between nominally bound states of different n_1 , the ionization rate of the state nn_1m to the $\epsilon n_1'm$ continua is given by Fermi's Golden Rule,

$$\Gamma = 2\pi \sum_{n_1'} |\langle nn_1m | V_d(r) | \epsilon n_1'm \rangle|^2. \quad (6.51)$$

The summation extends over all values of n_1' which are continua at the energy and field in question.

Since $V_d(r)$ is only nonzero near $r = 0$ the matrix element of Eq. (6.51) reflects the amplitude of the wavefunction of the continuum wave at $r \sim 0$. Specifically, the squared matrix element is proportional to $C_{n_1}^m$, the density of states defined earlier and plotted in Fig. 6.18. From the plots of Fig. 6.18 it is apparent that the ionization rate into a continuum substantially above threshold is energy independent. However, as shown in Fig. 6.18, there is often a peak in the density of continuum states just at the threshold for ionization, substantially increasing the ionization rate for a degenerate blue state of larger n_1 . This phenomenon has been observed experimentally by Littman *et al.*³² who observed a local increase in the ionization rate of the Na (12,6,3,2) Stark state where it crosses the 14,0,11,2 state, at a field of 15.6 kV/cm, as shown by Fig. 6.19. In this field the energy of the

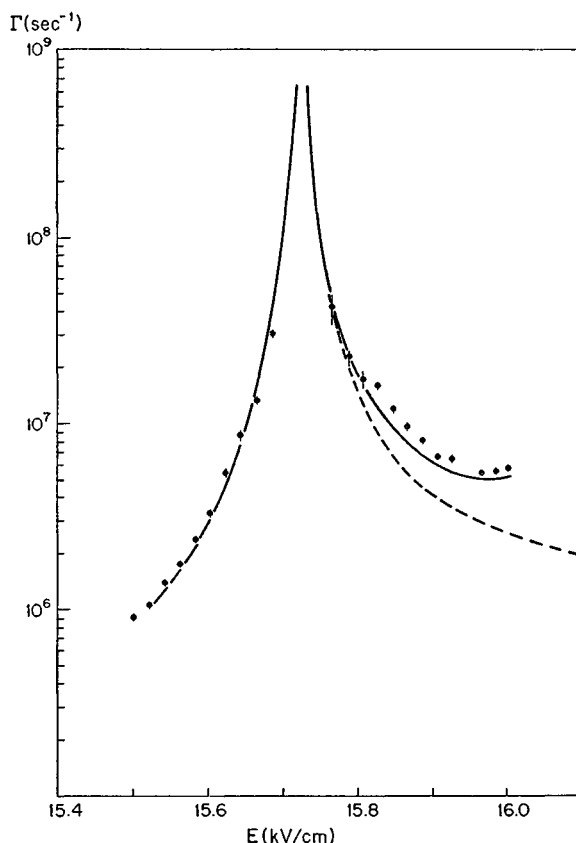


Fig. 6.19 Ionization rates for the (12,6,3,2) level of Na in the region of the crossing with the rapidly ionizing (14,0,11,2) level. The levels are specified as $(n, n_1, n_2, |m|)$. The solid line is a theoretical curve. The dashed line is calculated neglecting another avoided level crossing at 17.3 kV/cm (from ref. 31).

(14,0,11,2) state is just below the top of the barrier in $V(\eta)$ and its ionizing rate is 10^{10} s^{-1} . With this low an ionization rate the (14,0,11,2) wavefunction still has a large amplitude at the core, and the autoionization rate of blue states into it is at its highest.

Finally, it is useful to obtain an estimate of the typical ionization rates in the region between the classical ionization limit and the hydrogenic limit using Eq. (6.51). Each term in the sum of Eq. (6.51) can be written as

$$|\langle nn_1 m | V_d(r) | \epsilon n_1' m \rangle|^2 = |\langle nn_1 m | n \ell m \rangle \langle n \ell m | V_d | \epsilon \ell m \rangle \langle \epsilon \ell m | \epsilon n_1' m \rangle|^2 \quad (6.52)$$

The maximum value of n_1' in Eq. (6.51) is the value of n_1 for which $Z_2 = W^2/4E$, or $Z_1 = 1 - Z_2 = 1 - W^2/4E$. As shown by Harmin,¹⁴ a continuum form of Eq. (6.19), valid for small radii, is

$$\langle \epsilon \ell m | \epsilon n_1' m \rangle = (-1)^\ell P_{\ell m}(Z_1 - Z_2). \quad (6.53)$$

We are now in a position to evaluate Eq. (6.51) using the explicit form of Eq. (6.52) and replacing the sum over n_1 by an integral over $Z_1 - Z_2$. The first term of the matrix element of Eq. (6.52) is given by Eqs. (6.11) and (6.19), the second one by the continuum modification of Eq. (6.48), and the third by Eq. (6.53). In addition to these substitutions we convert the summation of Eq. (6.51) to an integral over $Z_1 - Z_2$. According to Eq. (6.38), the classical ionization limit occurs at $E = W^2/4Z_2$. Consequently, the contributing range of Z_2 is from $W^2/4E$ to 1. Written in terms of $Z_1 - Z_2$ the integral corresponding to the ionization rate of Eq. (6.51) is given by

$$\Gamma = 2\pi P_{\ell m}^2 \left[\left(\frac{2n_1}{n} \right) - 1 \right] \frac{\sin^2 \delta'_\ell}{n^3} \int_{-1}^{1-W^2/2E} P_{\ell m}^2(Z_1 - Z_2) d(Z_1 - Z_2), \quad (6.54)$$

where δ'_ℓ is the magnitude of the difference between δ_ℓ and the nearest integer. For example, $\delta_\ell = 0.85$ implies $\delta'_\ell = 0.15$. The first term in Eq. (6.54) reflects the amount of ℓ character in the initial nn_1m state, the second term is the coupling to the ℓ part of the continuum, and the third, the integral, reflects how much of the continuum is available having ℓ character. The value of the integral ranges from zero at energies just above the classical ionization limit to one when $W = -\sqrt{2E}$.

Approximating $\langle nn_1n_2m | n\ell m \rangle^2$ by $1/n$ and the integral of Eq. (6.54) by $1/2$ we find the approximate result

$$\Gamma \sim \pi \frac{\sin^2 \delta'_\ell}{n^4} \quad (6.55)$$

Observations to date indicate that Na $|m| = 0, 1$, and 2 Stark states of $n \approx 20$ above the classical ionization limit have widths of ~ 30 – 90 GHz, 1 – 3 GHz, and 100 MHz,³⁴ in rough accord with Eq. (6.55).

In estimating ionization rates we have ignored all interactions between bound states of high n_1 . These interactions also lead to local variations in the autoionization rates which can be quite striking and can be understood in a straightforward way. For a moment let us disregard the coupling to the continuum and focus on two nominally bound Stark states which have an avoided crossing. As we have already discussed, at the avoided crossing the eigenstates are linear combinations of the two Stark states. If away from the avoided crossing one of the two Stark levels is rapidly ionizing and the other slowly ionizing, at the crossing the ionization rates of the two eigenstates will most likely be about the same and half the rate of the rapidly ionizing state.

A more interesting case is one in which the two Stark states have equal autoionization rates. If the coupling is to the same continuum in both cases, then at the avoided crossing the coupling will vanish in one of the eigenstates and double in the other one. The phenomenon is basically the same as the vanishing of the oscillator strength to the upper level at the avoided crossing of Fig. 6.14. If the two Stark states do not have precisely the same ionization rates away from the avoided crossing, one of the ionization rates cannot vanish at the avoided

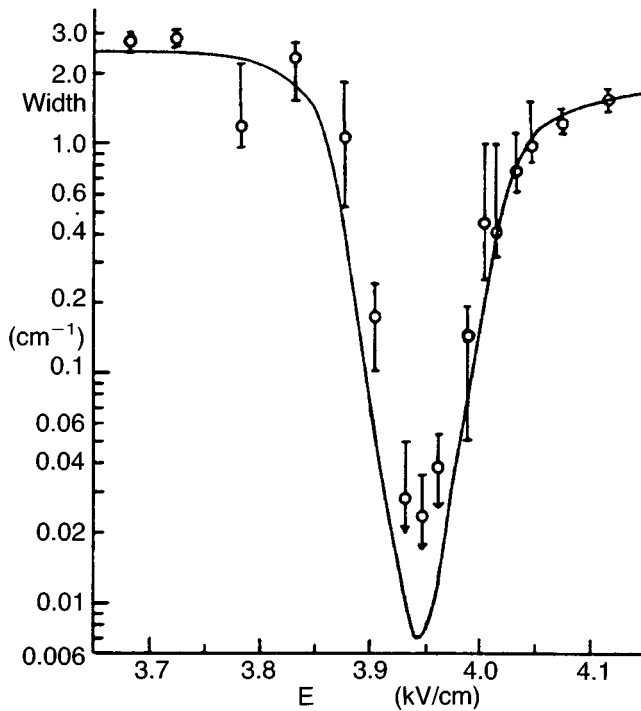


Fig. 6.20 Ionization width vs electric field for the Na (20,19,0,0) level near its crossing with the (21,17,3,0) level from experiment (data points) and from WKB-quantum defect theory (solid line). The levels are specified as $(n, n_1, n_2, |m|)$. Because the lineshapes are quite asymmetric (except for very narrow lines), the width in this figure is taken to be the FWHM of the dominant feature corresponding to the (20,19,0,0) level in the photoionization cross section. For the narrowest line, experimental widths are limited by the 0.7 GHz laser linewidth. Error limits are asymmetric because of the peculiar line shapes and because of uncertainties due to the overlapping $|m| = 1$ resonance (from ref. 37).

crossing. This phenomenon was first observed by Feneuille *et al.*³⁵ and discussed further by Luc-Koenig and LeCompte.³⁶ However, it is shown most clearly in the data of Liu *et al.*³⁷ who measured the width of the Na (20,19,0,0) state in the vicinity of its intersection with the (21,17,3,0) state at 3.95 kV/cm. As shown by Fig. 6.20, the width of the state decreases from 2 cm^{-1} away from the avoided crossing to 0.02 cm^{-1} at the avoided crossing. Correspondingly, the width of the (21,17,3,0) state approximately doubles at the avoided crossing.

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Resonant Rydberg–Rydberg collisions

Resonant energy transfer collisions, those in which one atom or molecule transfers only internal energy, as opposed to translational energy, to its collision partner require a precise match of the energy intervals in the two collision partners. Because of this energy specificity, resonant collisional energy transfer plays an important role in many laser applications, the He–Ne and CO₂ lasers being perhaps the best known examples.^{1–4} It is interesting to imagine an experiment in which we can tune the energy of the excited state of atom B through the energy of the excited state of atom A, as shown in Fig. 14.1.⁵ At resonance we would expect the cross section for collisionally transferring the energy from an excited A atom to a ground state B atom to increase sharply as shown in Fig. 14.1. In general, atomic and molecular energy levels are fixed, and the situation of Fig. 14.1 is impossible to realize. Nonetheless systematic studies of resonant energy transfer have been carried out by altering the collision partner, showing the importance of resonance in collisional energy transfer.^{6–8}

The use of atomic Rydberg states, which have series of closely spaced levels, presents a natural opportunity for the study of resonant collisional energy transfer. One of the earliest experiments was the observation of resonant rotational to electronic energy transfer from NH₃ to Xe Rydberg atoms by Smith *et al.*⁹ Resonant electronic to vibrational energy transfer has also been observed, from Rydberg states of Na to CH₄ and CD₄.¹⁰ Both of these experiments, which are described in Chapter 11, correspond to tuning in steps equal to the discrete spacing of the Rydberg levels. In an ideal study of resonant collisions it would be

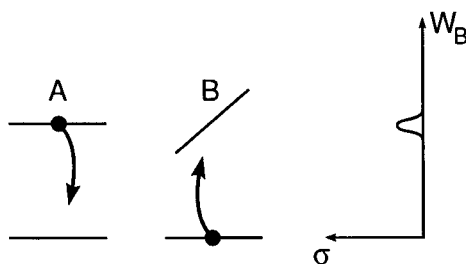


Fig. 14.1 Two atoms, A and B have energy levels as shown. Initially atom A is in its excited state, and atom B is in its ground state. If the energy of the excited state of atom B could be tuned by some means, we would expect the cross section for resonant energy transfer from atom A to atom B to increase at resonance as shown on the right (from ref. 5).

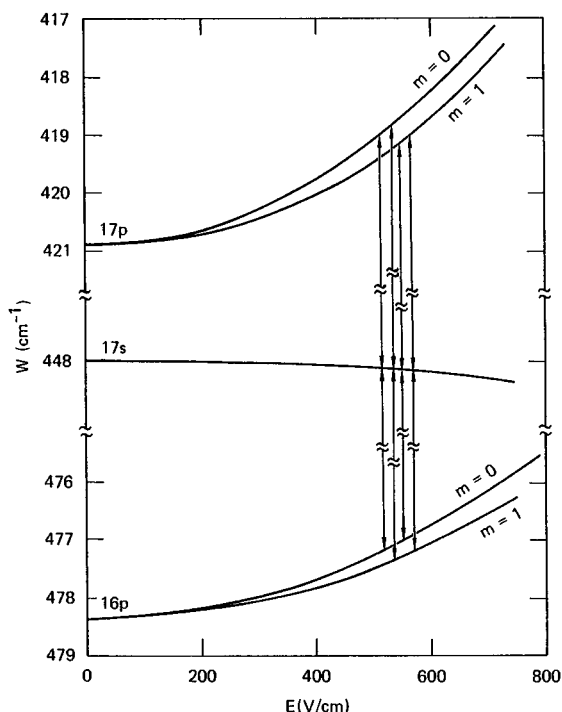
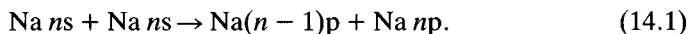


Fig. 14.2 Energy level diagram for the Na 16p, 17s, and 17p states in a static electric field. The vertical lines are drawn at the four fields where the s state is midway between the two p states and the resonant collisional transfer occurs (from ref. 12).

possible to tune continuously, as shown in Fig. 14.1, to observe the collisional resonance explicitly. In fact, due to the large Stark shifts of Rydberg atoms it is possible to tune them through a collisional resonance. A good example is the resonant energy transfer in the collision of two Na ns atoms in an electric field.¹¹ In Fig. 14.2 we show the energy levels of the Na ns , $(n-1)p$, and np states as a function of electric field. In zero field the ns state lies slightly above midway between the two p states. However, as the field is increased the p states are shifted to higher energies by their Stark shifts. As shown by Fig. 14.2, the field also lifts the degeneracy of the $|m| = 0$ and 1 levels of the p states. Here m is the azimuthal orbital angular momentum quantum number. Due to the lifting of the m degeneracy there are four fields at which the ns state lies halfway between the two p states. At these fields two ns atoms can collide to produce an $(n-1)p$ and an np atom by the process



As shown by Fig. 14.2, there are four collisional resonances for each ns state. We label them by (m_ℓ, m_u) , where m_ℓ and m_u are the $|m|$ values of the lower and upper final p states. As shown by Fig. 14.2 in order of increasing field the

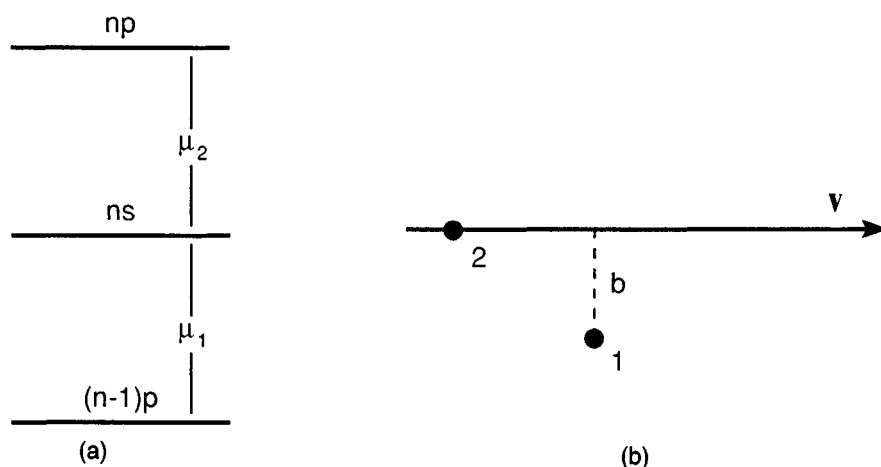


Fig. 14.3 (a) Energy levels for the Na ns , np and $(n-1)p$ states showing the dipole matrix elements coupling them. (b) Diagram of the geometry of the collision in which atom 2 passes by atom 1 with velocity v and impact parameter b (from ref. 5).

resonances are $(0,0)$, $(1,0)$, $(0,1)$, and $(1,1)$. The Na collision process described in Eq. (14.1) and depicted in Fig. 14.2 is a resonant dipole–dipole process, and a simple picture enables us to determine both the magnitude of the peak cross section and the width of the collisional resonance. We consider the collision of two Na atoms initially in the ns state. During the collision they undergo transitions to the np and $(n-1)p$ states, by means of dipole transitions with the dipole matrix elements μ_1 and μ_2 , as shown by Fig. 14.3. We assume that the atoms follow undeflected straight line trajectories described by the impact parameter b and collision velocity v , as shown in Fig. 14.3. For simplicity, we first consider the special case of the static field's being tuned to the collisional resonance, i.e. to the field at which the energy of the ns state is midway between those of the np and $(n-1)p$ states. We wish to calculate the probability of the atoms' making the transition to the np and $(n-1)p$ states during the collision. One approach is to treat one atom, atom 2, as providing an oscillating field which drives the transition in the other atom, atom 1.¹² In other words, we treat atom 2 as a classical dipole. Classically, a dipole transition matrix element becomes a dipole of strength equal to the transition matrix element oscillating at the transition frequency. Thus atom 2 can be thought of as a collection of dipoles corresponding to the $ns \rightarrow n'p$ dipole transitions allowed from the ns state. Most of these transitions are off resonant when compared to the $ns \rightarrow (n-1)p$ transition, and we ignore them. The $ns \rightarrow (n-1)p$ transition in atom 2, while degenerate can be ignored on the basis of energy conservation (both atoms cannot make transitions to lower energy states). Thus the only important transition of atom 2 is the $ns \rightarrow np$ transition, in which case the field produced by atom 2 at atom 1 is given by $E_2 \cong \mu_2/r^3$, μ_2 being the ns – np dipole matrix element, r the distance vector from atom 1 to atom 2, and

r the internuclear separation between the two atoms. This field is resonant with the $ns \rightarrow (n-1)p$ transition of atom 1 and drives the transition if the interaction is strong enough. Since the dipole field falls quickly with increasing r , it is a reasonable approximation to set

$$E_2 = \mu_2/b^3 \quad r \leq \frac{\sqrt{5}}{2} b \quad (14.2a)$$

$$E_2 = 0 \quad r > \frac{\sqrt{5}}{2} b. \quad (14.2b)$$

This choice of cutoff implies that the field from atom 2 is present for a time $\tau = b/v$. In general atom 1 undergoes the $ns \rightarrow (n-1)p$ transition if

$$E_2 \cdot \mu_1 \cdot \tau \approx 1, \quad (14.3)$$

or if

$$\frac{\mu_2 \mu_1}{b^3} \cdot \frac{b}{v} = 1. \quad (14.4)$$

In Eqs. (14.3) and (14.4) μ_1 is the $ns \rightarrow (n-1)p$ dipole matrix element. Eq. (14.3) makes it apparent that the requirement for driving the transition is that the time integrated interaction be one. Solving Eq. (14.4) for b^2 , ignoring factors of π , gives the cross section σ_R for the resonant collisional energy transfer. Explicitly,

$$\sigma_R \approx b^2 = \frac{\mu_1 \mu_2}{v}. \quad (14.5)$$

Using the impact parameter b and the velocity v we can also calculate the duration, or time, of the collision,

$$\tau = \frac{b}{v} = \frac{\sqrt{\mu_1 \mu_2}}{v^{3/2}}. \quad (14.6)$$

For the $ns \rightarrow np$ and $ns \rightarrow (n-1)p$ transitions the dipole matrix elements are $\sim n^2$. Accordingly, we insert n^2 for μ_1 and μ_2 in Eqs. (14.5) and (14.6), finding

$$\sigma_R = n^4/v \quad (14.7)$$

and

$$\tau = n^2/v^{3/2}. \quad (14.8)$$

As shown by Eq. (14.7), the cross section is given by the geometric cross section divided by the collision velocity. For thermal collisions $v \sim 10^{-4}$, and as a result, $\sigma_R \sim 10^4 n^4$. For $n=20$ this expression yields a cross section of $\sim 10^9 a_0^2$, or $\sim 2 \times 10^8 \text{ \AA}^2$, which is large by atomic standards. Perhaps more interesting than the magnitudes of the cross sections are the collision times. Evaluating Eq. (14.8) for $n=20$ and $v = 10^{-4}$ gives $\tau = 4 \times 10^8$ or $\sim 10^{-9}$ s, a time far longer than the typical atomic collision times of 10^{-12} s.

The discussion above is focused on dipole–dipole collision processes, but it may easily be extended to higher multipole processes. If atoms 1 and 2 have $2k$ pole moments of n^{2k} and $n^{2k'}$ respectively, when the atoms are separated by a distance r the interaction V due to these multipoles is

$$V = \frac{n^{2k} n^{2k'}}{r^{1+k+k'}}. \quad (14.9)$$

If we again assume that the atoms collide with an impact parameter b and velocity v , following straight line trajectories, we can again assume that V is only non zero for $r \approx b$ and a time interval $\tau = b/v$. Requiring that $V\tau = 1$ for a significant transition probability leads to

$$\sigma \cong b^2 = \frac{n^4}{v^{2/(k+k')}}. \quad (14.10)$$

For a dipole–dipole collision, $k = k' = 1$, and this result reduces to Eq. (14.5). On the other hand, as k and k' increase, the factor $v^{2/(k+k')}$ approaches 1 from below, and the cross section decreases to the geometric size of the atom. It is worth noting that the assumption of a straight line trajectory is not strictly valid in mixed multipole collisions, or in any collision in which the internal angular momentum of the atoms is not conserved. However, the amount of angular momentum in the translational motion is usually sufficiently high that small changes in it do not significantly alter the trajectories of the colliding atoms.

Two state theory

The expression of Eq. (14.7) for the cross section is adequate for a calculation of the magnitude of the cross section, but it does not take into account effects due to the orientation of the collision velocity $\mathbf{v}$ relative to the tuning field $\mathbf{E}$, nor does it allow the calculation of the lineshape of the collisional resonance. With these thoughts in mind, we present a more refined treatment of the problem which enables us to describe some of the details of the process. We shall consider the resonant collisions of two Na ns atoms to yield an np and an $(n-1)p$ atom, as indicated by Eq. (14.1). The description is, in principle, similar to the treatment of resonant rotational energy transfer between polar molecules given by Anderson.⁸ It differs, however, in that the presence of the electric field, which lowers the symmetry, must be taken into account. Treatments specific to the problem of dipole–dipole collisions in a field, such as the process depicted by Fig. 14.2, have been given by Gallagher *et al.*,¹² Fiordilino *et al.*,¹³ and Thomson *et al.*¹⁴

We assume again that the atoms follow straight line trajectories, and we calculate the transition probability, $P(b)$, from the initial to the final state in a collision with a given impact parameter, b . We then compute the cross section by integrating over impact parameter, and, if necessary, angle of $\mathbf{v}$ relative to $\mathbf{E}$ to obtain the cross section. The central problem is the calculation of the transition probability $P(b)$. The Schrodinger equation for this problem has the Hamiltonian

$$H = H_0 + V, \quad (14.11)$$

where $H_0 = H_1 + H_2$ is the Hamiltonian for two non-interacting atoms in the static field, and V is the dipole–dipole interaction

$$V = \frac{\boldsymbol{\mu}_1 \cdot \boldsymbol{\mu}_2}{r^3} - \frac{3(\boldsymbol{\mu}_1 \cdot \mathbf{r})(\boldsymbol{\mu}_2 \cdot \mathbf{r})}{r^5}. \quad (14.12)$$

We construct the molecular product states which have the spatial wavefunctions

$$\begin{aligned} \psi_A &= \psi_{1ns} \otimes \psi_{2ns} \\ \psi_B &= \psi_{1np} \otimes \psi_{2(n-1)p}, \end{aligned} \quad (14.13)$$

where $\psi_{1n\ell}$ and $\psi_{2n\ell}$ are the atomic wavefunctions of atoms 1 and 2 in the $n\ell$ state. These product states are the time independent solutions to H_0 , having energies $W_A = W_{ns} + W_{ns}$ and $W_B = W_{np} + W_{(n-1)p}$.

The dipole–dipole interaction V has a diagonal term, which slightly shifts the energies, and an off diagonal term, which induces the collisional transitions. Since V depends on r , V is time dependent, and the wavefunction is given by the solution to the time dependent Schroedinger equation

$$H\psi = i\partial\psi/\partial t. \quad (14.14)$$

When the two possible states are those given by Eq. (14.13) the general form of the solution to Eq. (14.14) is

$$\psi = C_A(t)\psi_A + C_B(t)\psi_B, \quad (14.15)$$

in which all of the time dependence resides in the coefficients $C_A(t)$ and $C_B(t)$. Using the wavefunction of Eq. (14.15) in the time dependent Schroedinger equation, Eq. (14.14), we find two coupled equations for $C_A(t)$ and $C_B(t)$,

$$W_A C_A(t) + V_{AB} C_B(t) + V_{AA} C_A(t) = i\dot{C}_A(t), \quad (14.16a)$$

$$W_B C_B(t) + V_{BA} C_A(t) + V_{BB} C_B(t) = i\dot{C}_B(t), \quad (14.16b)$$

where $V_{BA} = \langle \psi_B | V | \psi_A \rangle = \int \psi_B^* V \psi_A \, d\tau_1 \, d\tau_2$ is the matrix element of V between the two spatial wavefunctions.

In Eqs. (14.16) the matrix elements of V are time dependent, since the internuclear distance is a function of time, as are the coefficients $C_A(t)$ and $C_B(t)$. In general, Eqs. (14.16) cannot be solved analytically, but in several special cases analytic solutions can be obtained.

Box interaction strength approximation

If we set $V_{AA} = V_{BB} = 0$ and use Eq. (14.2), i.e. assume that V_{AB} is a constant, $= \mu_1 \mu_2 / b^3$, for $r < \sqrt{5}b/2$ and zero for $r > \sqrt{5}b/2$, as given in Eq (14.2), the resulting approximation is equivalent to the usual molecular beam magnetic resonance treatment.¹⁵ If we assume that $r = \sqrt{b^2 + v^2 t^2}$ and that initially both

atoms are in the ns state so that $\psi(t_0) = \psi_A$, $C_A(t_0) = 1$ and $C_B(t_0) = 0$ for $t_0 < b/2v$. After the collision, $t > b/2v$, the probability of finding the atoms in the p states, $\psi = \psi_B$, is given by

$$P(b) = |C_B(t)|^2 = \frac{|V_{BA}|^2}{\Omega^2} \sin^2\left(\frac{\Omega b}{v}\right), \quad (14.17)$$

where $\Omega = \sqrt{(W_A - W_B)^2 + 4|V_{BA}|^2}/2$. Eq. (14.17) yields Lorentzian resonances with linewidths $\sim (\sqrt{v^3/\mu_1\mu_2})$, in agreement with Eq. (14.6). While the widths are meaningful, the lineshape, which depends on the form of the interaction during the collision, is only approximate.

At resonance $W_A = W_B$, and Eq. (14.17) reduces to

$$P(b) = \sin^2\left(\frac{\mu_1\mu_2}{b^2v}\right), \quad (14.18)$$

in which we have used the explicit form $V_{AB} = \mu_1\mu_2/b^3$.

Exact resonance approximation

Another case which may be treated analytically is the case of exact resonance, $W_A = W_B$, if in addition, $V_{AA} = V_{BB} = 0$ and $V_{AB} = V_{BA}$, i.e. the coupling matrix element is real. In this case Eqs. (14.16) are readily decoupled, leading to two identical uncoupled equations. The equation for $C_A(t)$ is

$$\ddot{C}_A(t) = \frac{\dot{V}_{AB}}{V_{AB}} \dot{C}_A(t) - V_{AB}^2 C_A(t). \quad (14.19)$$

The equation for $C_B(t)$ has identical coefficients. If we again assume that initially, at $t = -\infty$, $C_A(-\infty) = 1$ and $C_B(-\infty) = 0$ the solutions to Eq. (14.19) which satisfy these boundary conditions are^{12,13}

$$C_A(t) = \cos\left(\int_{-\infty}^t V_{AB}(t') dt'\right) \quad (14.20a)$$

and

$$C_B(t) = \sin\left(\int_{-\infty}^t V_{AB}(t') dt'\right). \quad (14.20b)$$

The probability of making a transition is therefore given by

$$P(\infty) = C_B^2(\infty) = \sin^2\left(\int_{-\infty}^{\infty} V_{AB}(t') dt'\right). \quad (14.21)$$

To obtain an explicit form for the matrix element V_{AB} given in Eq. (14.12), requires that we define the geometry of the collision. While the collision velocity is normally the logical choice of quantization axis for field free collisions, the

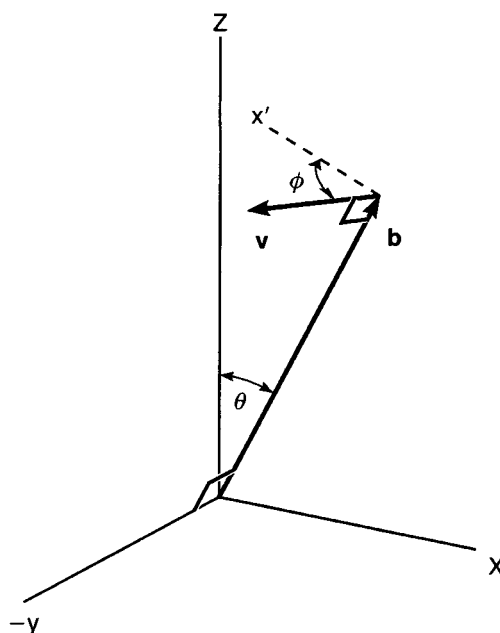


Fig. 14.4 Diagram of the geometry of the collision. The static field is in the z direction. The impact parameter vector $\mathbf{b}$ lies in the x, z plane, at an angle θ from the z axis. The velocity vector $\mathbf{v}$ lies in the plane perpendicular to $\mathbf{b}$ and makes an angle ϕ with the x' axis, which is perpendicular to $\mathbf{b}$ and in the x, z plane (from ref. 5).

presence of the tuning electric field makes the $\mathbf{E}$ field direction the logical choice, since the eigenstates of H_0 are easily described with this choice of quantization axis. If we define the field direction as the z axis, so that $\mathbf{E} \parallel \hat{z}$, then the impact parameter vector $\mathbf{b}$ makes an angle θ with the z axis as shown in Fig. 14.4. We may assume, with no loss of generality, that $\mathbf{b}$ lies in the x, z plane. The collision velocity $\mathbf{v}$, which is a constant vector due to the assumption of straight line trajectories, is normal to $\mathbf{b}$ and makes an angle ϕ with the x' axis, which is in the x, z plane and perpendicular to $\mathbf{b}$, as shown in Fig. 14.4. If the closest approach of the two atoms occurs at $t = 0$, the coordinates of atom 2 relative to atom 1 are given by

$$\begin{aligned} y &= -vt \sin \phi \\ x &= b \sin \theta - vt \cos \phi \cos \theta \\ z &= b \cos \theta + vt \cos \phi \sin \theta. \end{aligned} \quad (14.22)$$

Using these coordinate values we may now evaluate the matrix elements of Eq. (14.12) by substituting for the dipole moments the dipole matrix elements between the initial and final states. This procedure yields explicitly time dependent matrix elements $V_{AB}(t)$. It is particularly interesting to consider the (0,0) resonances, for two reasons. First, the (0,0) resonances have no further splitting due to the spin orbit interaction and are therefore good candidates for detailed experimental study. Second, since these resonances only involve the matrix

elements of μ_z , which are real, it is possible to evaluate the transition probability at resonance analytically using Eq. (14.21). The matrix element $V_{AB}(t)$ is given by

$$V_{AB}(t) = \mu_{z_1}\mu_{z_2} \left(\frac{1}{r^3} - \frac{3b^2 \cos^2 \theta}{r^5} - \frac{6bvt \cos \theta \cos \phi}{r^5} - \frac{3v^2 t^2 \cos^2 \phi \sin^2 \theta}{r^5} \right), \quad (14.23)$$

where $r = \sqrt{b^2 + v^2 t^2}$.

In Fig. 14.5 we show graphs of $V_{AB}(t)$ for two cases; $\mathbf{v} \parallel \mathbf{E}$, and one example of $\mathbf{v} \perp \mathbf{E}$, for which $\theta = 0$ and $\phi = \pi/2$. Note that the specification $\mathbf{v} \parallel \mathbf{E}$, specifies both θ and ϕ . However $\mathbf{v} \perp \mathbf{E}$, only specifies $\phi = \pi/2$; θ can take any value.

The form of $V_{AB}(t)$ given in Eq. (14.23) can be integrated analytically to find the transition probability at resonance. Carrying out the integration yields

$$\int_{-\infty}^{\infty} V_{AB}(t') dt' = \frac{2\mu_{z_1}\mu_{z_2}}{vb^2} (1 - 2\cos^2 \theta - \cos^2 \phi \sin^2 \theta). \quad (14.24)$$

Evaluating Eq. (14.24) for $\mathbf{v} \parallel \mathbf{E}$, $\theta = \pi/2$ and $\phi = 0$, reveals that the integral vanishes, as might have been anticipated from a close examination of Fig. 14.5(a). The average value of V_{AB} in Fig. 14.5(a) is clearly near zero. On the other hand, for $\mathbf{v} \perp \mathbf{E}$, $\theta = 0$ and $\phi = \pi/2$, the integral of Eq. (14.24) does not vanish, since the average value of $V_{AB}(t)$ does not vanish in this case, as can be seen in Fig. 14.5(b). Correspondingly, at resonance, the cross section must vanish for $\mathbf{v} \parallel \mathbf{E}$ but not for $\mathbf{v} \perp \mathbf{E}$.

Numerical calculations

Normally one might expect that if the transition probability vanishes on resonance it also vanishes off resonance. However, such is not the case. When the transition probability is calculated off resonance, by numerically solving Eqs. (14.16) using a Taylor expansion method, it is nonzero for both $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$.^{14,16} In Fig. 14.6 we show the transition probabilities obtained using two different approximations for $\mathbf{v} \parallel \mathbf{E}$, and $\mathbf{v} \perp \mathbf{E}$ for the 17s (0,0) collisional resonance.¹⁶ To allow direct comparison to the analytic form of Eq. (14.21) we show the transition probabilities calculated with $V_{AA} = V_{BB} = 0$. For these calculations the parameters $\mu_{z_1} = \mu_{z_2} = 156.4 \text{ ea}_0$, $b = 10^4 a_0$, and $v = 1.6 \times 10^{-4} \text{ au}$ have been used. The resulting transition probability curves are shown by the broken lines of Fig. 14.6. As shown by Fig. 14.6 these curves are symmetric about the resonance position. The $\mathbf{v} \perp \mathbf{E}$ curve of Fig. 14.6(b) has an approximately Lorentzian form, but the $\mathbf{v} \parallel \mathbf{E}$ curve of Fig. 14.6(a), while it vanishes on resonance as predicted by Eq. (14.24), has an unusual double peaked structure.

Since the transition probabilities of Fig. 14.6 are obtained numerically, it is not appreciably more difficult to include the diagonal matrix elements V_{AA} and V_{BB} arising from the permanent dipole moments of the atomic states in the field. The

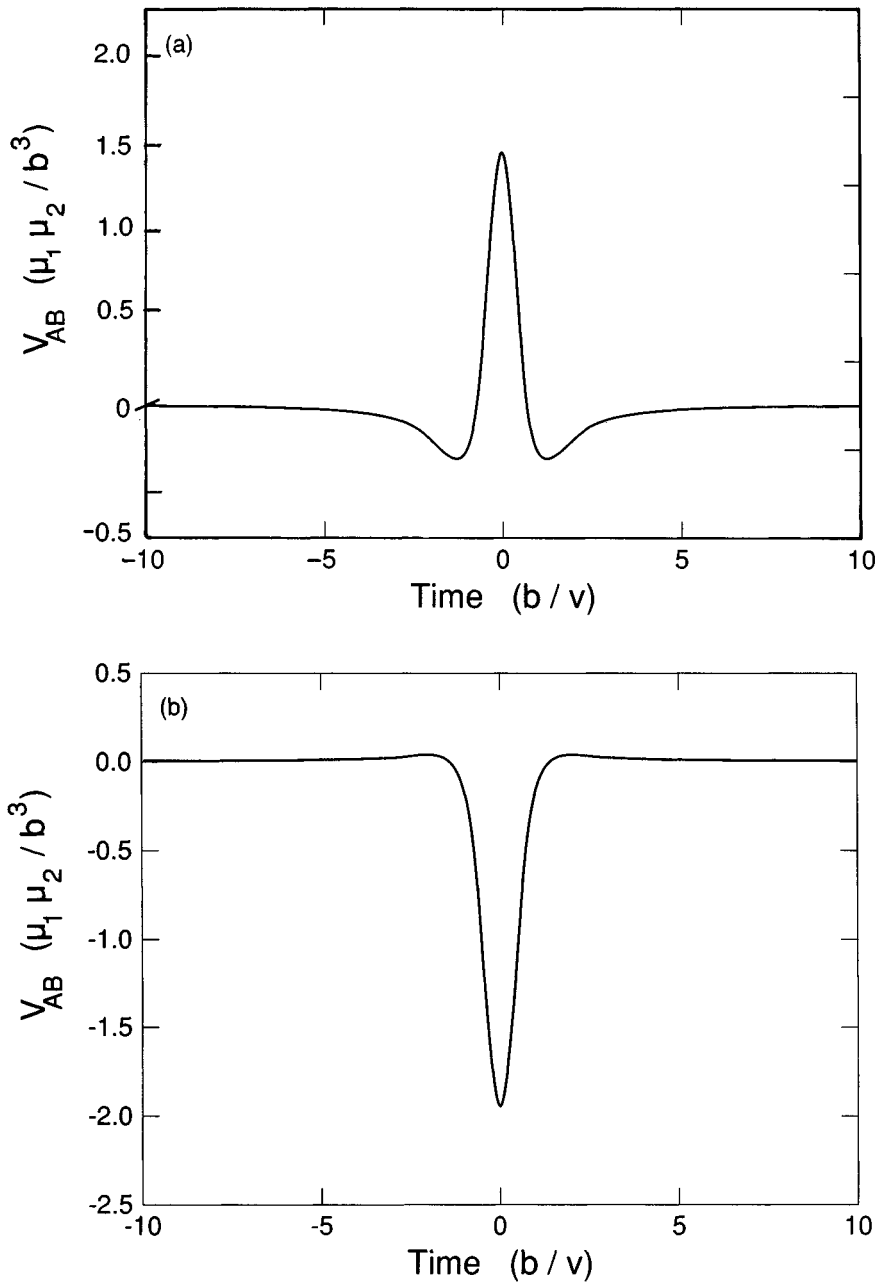


Fig. 14.5 Dipole-dipole interaction potential for (a) the case in which $\mathbf{v} \parallel \mathbf{E}$, (b) the case in which $\mathbf{v} \perp \mathbf{E}$ with $\theta = 0$ and $\phi = \pi/2$ (from ref. 14).

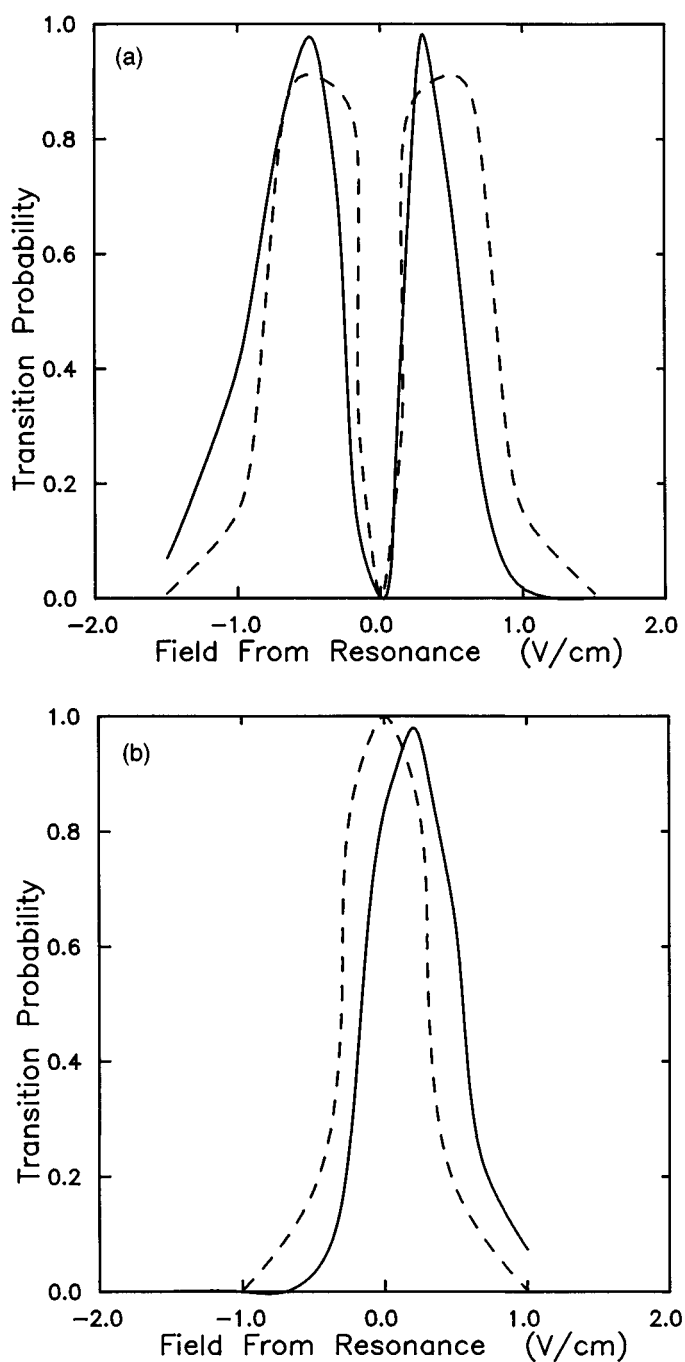


Fig. 14.6 Calculations of the transition probability for the Na $17s + 17s \rightarrow 17p + 16p$, (0,0) transition for (a) $\mathbf{v} \parallel \mathbf{E}$, and (b) $\mathbf{v} \perp \mathbf{E}$. Solid lines are for calculations which include the effects of the permanent dipole moments; dashed lines indicate that the permanent dipole moments have been neglected (from ref. 16).

solid lines of Fig. 14.6 are the transition probabilities calculated with the permanent dipole moments at the field of the collisional resonance taken into account. The permanent dipole moments of the 16p, 17p, and 17s states are $\mu_{16p} = 113.3ea_0$, $\mu_{17p} = 182.3ea_0$, and $\mu_{17s} = -16.5ea_0$, respectively. As shown by Fig. 14.6, when the permanent moments are taken into account the location of the resonance shifts slightly, by approximately a resonance linewidth, as expected from the fact that the permanent moments are nearly as large as the transition moments. In addition, the lineshapes become slightly asymmetric, due to the fact that the energy spacing of the atomic states now changes with internuclear separation.

Intracollisional interference

The unusual lineshape of Fig. 14.6(a) for $\mathbf{v} \parallel \mathbf{E}$ has approximately the same form and origin as the Ramsey separated oscillatory field pattern when there is a 180° phase shift between the two oscillatory fields.¹⁴ To see the similarity we first examine $V_{AB}(t)$, shown in Fig 14.5(a). As shown by Fig. 14.5(a), for $\mathbf{v} \parallel \mathbf{E}$ V_{AB} changes sign twice during the collision, so that $\int_{-\infty}^{\infty} V_{AB}(t') dt' = 0$. A sign change of $V_{AB}(t')$ is equivalent to a 180° phase shift between the oscillatory fields in a separated oscillatory field experiment, and the $\mathbf{v} \parallel \mathbf{E}$ collisions are thus approximately equivalent to doing an rf resonance experiment in which there are three sequential rf fields with phase changes of 180° between successive fields. At resonance, the contributions to the transition amplitude from $V_{AB}(t') < 0$ cancel those from $V_{AB}(t') > 0$ so that, integrated over the whole collision, the transition probability vanishes. Off resonance, however, the cancellation is not complete.

Calculation of cross sections

To convert the transition probabilities to cross sections we must integrate over impact parameter using

$$\sigma = \int_0^\infty P(b) 2\pi b db. \quad (14.25)$$

It is useful to consider the case of exact resonance, which may be worked out analytically. At resonance, using either Eq. (14.18) or Eq. (14.21) the transition probability is given by

$$P(b) = \sin^2\left(\frac{D}{b^2 v}\right), \quad (14.26)$$

where $D = \mu_1\mu_2$ if we use Eq. (14.18) and $D = 2\mu_{z_1}\mu_{z_2}$ ($1 - 2\cos^2\theta - \cos^2\phi\sin^2\theta$) if we use Eq. (14.21). In either case D is proportional to the product of the two dipole moments. Examining Eq. (14.26) we see that if we define b_0 such that $D/b_0^2\nu = \pi/2$, as b increases from 0 to b_0 , $P(b)$ oscillates between 0 and 1, and as b increases from b_0 to infinity $P(b)$ decreases smoothly to zero. If we insert the transition probability of Eq. (14.26) into Eq. (14.25), the integral can be done analytically, yielding

$$\sigma = \frac{\pi^3}{4} b_0^2 = \frac{\pi^2 D}{2 \nu}. \quad (14.27)$$

Recalling that $D \sim \mu_1\mu_2$, we see that Eq. (14.27) resembles Eq. (14.5) which was derived in a very simple way.

If we are interested in the cross section for $\mathbf{v} \parallel \mathbf{E}$, the angles θ and ϕ are uniquely defined, $\theta = \pi/2$ and $\phi = 0$. On the other hand, to calculate the observed $\mathbf{v} \perp \mathbf{E}$ cross section Eq. (14.27) must be averaged over θ for $\phi = \pi/2$. In either case the calculated cross sections must be averaged over the appropriate distribution of collision velocities.

In an analogous fashion the numerically obtained transition probabilities shown in Fig. 14.6 can be converted to cross sections by integrating over impact parameter and averaging over the possible collision velocities. Collisions with $\mathbf{v} \perp \mathbf{E}$ do not have a single allowed value of θ . However, since the lineshape of Fig. 14.6(b) is simple, some averaging over the possible values of θ has no appreciable effect. For $\mathbf{v} \parallel \mathbf{E}$, θ and ϕ are fixed, at 90° and 0° , so the integration over impact parameter is only over b , and the calculated $\mathbf{v} \parallel \mathbf{E}$ cross section is very similar to Fig. 14.6.

The lineshapes of Figs. 14.6(a) and 14.6(b) are very different. However, this difference is only apparent in an experiment with high resolution. At low resolution the $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ collisional resonances can be expected to exhibit the same instrumentally determined shape. Furthermore, both cross sections are approximately the same size. The calculated peak cross sections for the $\text{Na } 17s + 17s \rightarrow 16p + 17p$ collisional resonances are $6.0 \times 10^8 a_0^2$ and $1.0 \times 10^9 a_0^2$ for $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ respectively.¹⁴ When integrated over the tuning field, to allow comparison to low resolution experiments, the two integrated cross sections are $8.3 \times 10^8 a_0^2 \text{ V/cm}$ and $1.2 \times 10^9 a_0^2 \text{ V/cm}$.¹⁴

Experimental approach

The basic principle of the experimental approach is easily understood by considering the Na collisions of Eq. (14.1) and Fig. 14.2 as a concrete example. As shown by Fig. 14.7, atoms effuse from a heated oven in which the Na vapor pressure is ~ 1 Torr.¹⁴ The Na atoms are collimated into a beam by a collimator, not shown in

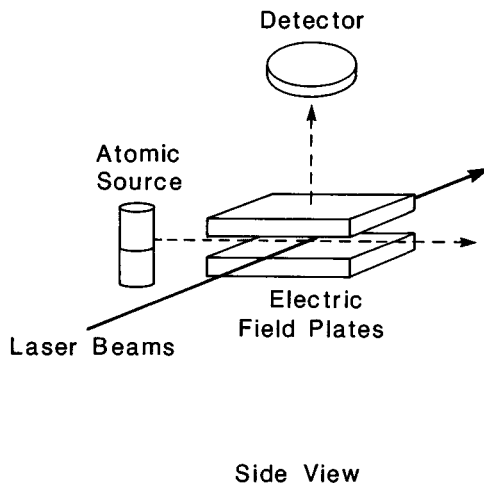


Fig. 14.7 Experimental apparatus configured with the electric field perpendicular to the collisional velocity (from ref. 14).

Fig. 14.7, and pass into the interaction region, between a pair of electric field plates, at which point the beam density can be up to 10^9 cm^{-3} . The Na atoms are excited to an ns state by two 5 ns pulsed dye lasers tuned to the Na $3s \rightarrow 3p$ and $3p \rightarrow ns$ transitions at $5890 \text{ \AA}$ and $\sim 4150 \text{ \AA}$ respectively. The ns atoms are allowed to collide with each other for a period of $\sim 1 \mu\text{s}$ during which time the slow atoms in the beam are overtaken by the fast atoms. Thus the collision velocity is approximately the width of the velocity distribution of the beam. After the $1 \mu\text{s}$ period during which the atoms collide a high voltage pulse is applied to the lower plate of the interaction region. The amplitude of the pulse is set to ionize atoms in the np state, the higher lying final state of the collision, but not atoms in the ns state, the initial state of the collision. Ions produced by field ionization are expelled from the region between the plates and impinge upon a particle multiplier above the interaction region. The signal from the particle multiplier is recorded with a gated integrator. A static tuning voltage is applied to the lower plate to tune the atomic levels into resonance. The collisional resonances are observed by monitoring the field ionization signal from atoms in the np state as the tuning voltage is slowly swept through the collisional resonances. A typical example is shown in Fig. 14.8, a recording of the field ionization signal from the Na $17p$ state when the $17s$ state is populated by laser excitation.¹²

n scaling laws

One of the most striking aspects of resonant collisional energy transfer between Rydberg atoms is the magnitude of the cross sections. Accordingly, the first

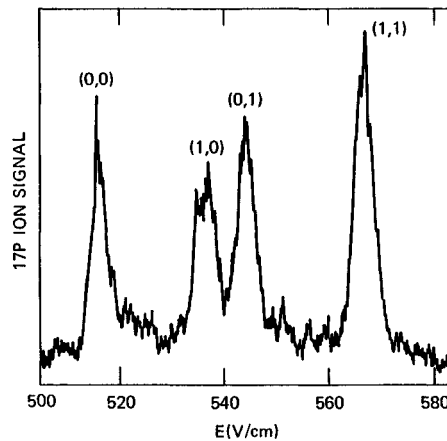


Fig. 14.8 The observed Na 17p ion signal after population of the 17s state vs dc electric field, showing the sharp collisional resonances. The resonances are labeled by the $|m|$ values of the lower and upper p states (from ref. 12).

experiments were focused on determining the magnitude of the cross sections and verifying the n scaling laws for the $\text{Na } ns + ns \rightarrow np + (n-1)p$ cross sections and collision times.

The cross sections were measured by taking advantage of the fact that the number of atoms, N_p , which are in the np state after the atoms have been allowed to collide for a time interval T is given by¹²

$$N_p = \frac{N_s^2 \sigma \bar{v} T}{V}, \quad (14.28)$$

where N_s is the number of atoms in the ns state, V is the volume of the sample of excited atoms, $\bar{v}$ is the average collision velocity, and σ is the cross section. Eq. (14.28) is valid only if the population of the ns state is not depleted significantly. If significant depletion of the population in the ns state occurs, corrections to Eq. (14.28) must be introduced. The number of atoms in the np state, N_p , is related to the signal associated with np state, N_p' , by the conversion efficiency of the detector, Γ . Specifically $N_p' = \Gamma N_p$, and $N_s' = \Gamma N_s$. Using N_p' and N_s' we can rewrite Eq. (14.28) as

$$\sigma = \frac{N_p'}{(N_s')^2} \left[\frac{\bar{v} T \Gamma}{V} \right]. \quad (14.29)$$

If the geometry of the experiment, timing gates, oven temperature, and detector gain are fixed, then the quantity in the square brackets of Eq. (14.29) is constant, and the relative cross section as a function of n is easily obtained by measuring the signals N_s' and N_p' as a function of n . The relative cross sections were put on an absolute basis by measuring the quantities in the brackets of Eq. (14.29), with the result shown in Fig. 14.9. The magnitudes and the n scaling of the

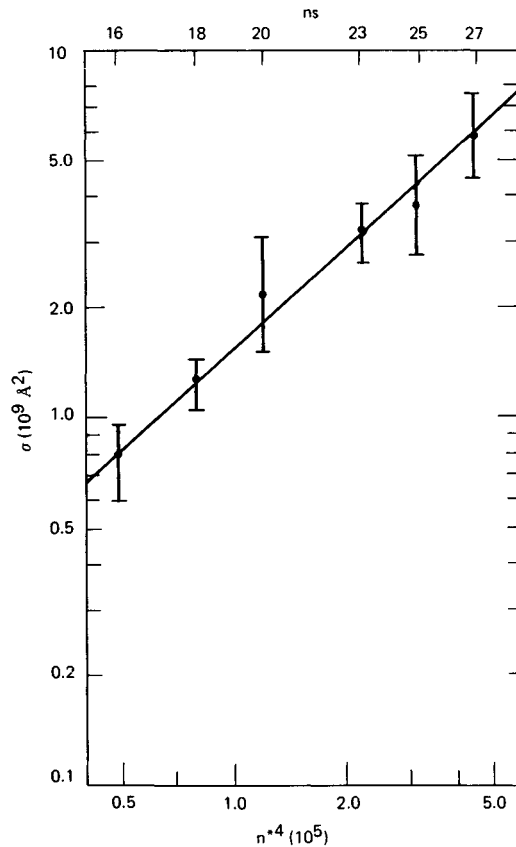


Fig. 14.9 The observed cross sections with their relative error bars (●), and the fit curve (—) (from ref. 12).

cross section as $n^{*3.7(5)}$ are in good agreement with Eq. (14.7).¹² Here n^* is the effective quantum number of the ns state, defined by $W_{ns} = -1/2n^{*2}$. The collisions resulting in the cross sections shown in Fig. 14.9 were most likely not between atoms of different velocities in a directed beam but between atoms moving in random directions. Both the excited atom sample volume and collision velocity were probably a factor of 2 larger than thought at the time, so the errors compensated to a large extent. In any event, from Fig. 14.9 it is apparent that the cross section exhibits the expected n^4 scaling and is of the correct magnitude to match the expected $\sigma = n^4/v$ behavior.

Verification that the collision time τ increases as n^2 , or that the linewidth of the collisional resonances decreases as n^{-2} , is simply a matter of measuring the widths of the collisional resonances. In Fig. 14.10 we show a plot of the width of the (0,0) resonance as a function of n . The observed widths exhibit magnitudes and an $n^{*-1.95(20)}$ dependence in agreement with Eq. (14.8).

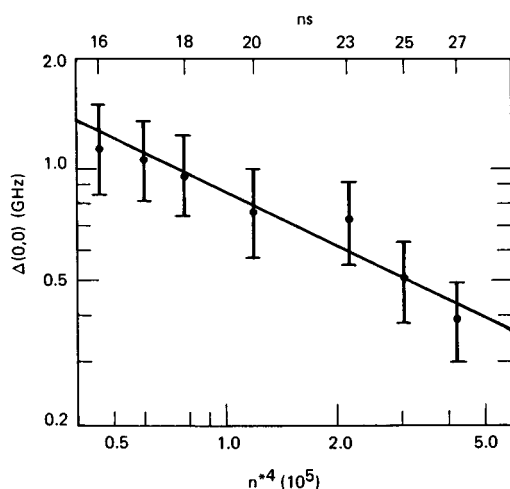
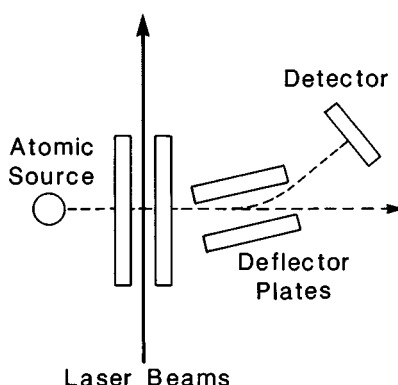


Fig. 14.10 The widths of the (0,0) resonances (●), and the fit curve (—) (from ref. 12).

Orientation dependence of $\mathbf{v}$ and $\mathbf{E}$ and intracollisional interference

One of the more interesting aspects of the collision process is the calculated dependence on the orientation of $\mathbf{v}$ relative to $\mathbf{E}$. In particular, do the lineshapes for collisions with $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ differ as dramatically as shown in Fig. 14.6? To answer this question unambiguously required two improvements upon the initial measurements. First, the Na atoms must be in a well defined beam, otherwise $\mathbf{v}$ is not well defined. This requirement is easily met by enclosing the interaction region with a liquid N_2 cooled box, which ensures that the only Na atoms in the interaction region are those in the beam. Second, the field homogeneity must be adequate, ~ 1 part in 10^4 , to resolve the intrinsic lineshape of the collisional resonances. A pair of Cu field plates 1.592(2) cm apart with 1 mm diameter holes in the top plate to allow the ions to be extracted is adequate to meet this requirement.

Making these two modifications, but keeping the basic geometry of Fig. 14.7, substantially improves the resolution so that what appear to be four collisional resonances in Fig. 14.8 are actually nine.¹⁴ All the resonances but the (0,0) resonance are split by the spin orbit splitting of the np $|m| = 1$ states. In the (0,1) and (1,0) resonances the upper and lower p states, respectively, are split, leading to doublets, while in the (1,1) resonance both upper and lower p states are split, leading to a quartet. In view of the fact that only the (0,0) resonance is a single line, it was selected for a detailed investigation of the cross section with $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$. To measure the cross section with $\mathbf{v} \perp \mathbf{E}$ the geometry of Fig. 14.8, with the liquid N_2 cooled shield, is adequate. To measure the cross section with $\mathbf{v} \parallel \mathbf{E}$ an arrangement, shown in Fig. 14.11, is used in which the atomic beam passes



Top View

Fig. 14.11 Experimental apparatus configured so that the electric field and collisional velocity are parallel (from ref. 14).

through the field plates. In both cases the collision velocity is the average of the difference in velocities of different atoms in the beam.

The observed cross sections for the 18s (0,0) collisional resonance with $\mathbf{v} \parallel \mathbf{E}$ and $\mathbf{v} \perp \mathbf{E}$ are shown in Fig. 14.12. The approximately Lorentzian shape for $\mathbf{v} \perp \mathbf{E}$ and the double peaked shape for $\mathbf{v} \parallel \mathbf{E}$ are quite evident. Given the existence of two experimental effects, field inhomogeneities and collision velocities not parallel to the field, both of which obscure the predicted zero in the $\mathbf{v} \parallel \mathbf{E}$ cross section, the observation of a clear dip in the center of the observed $\mathbf{v} \parallel \mathbf{E}$ cross section supports the theoretical description of intracollisional interference given earlier. It is also interesting to note that the observed $\mathbf{v} \parallel \mathbf{E}$ cross section of Fig. 14.12(a) is clearly asymmetric, in agreement with the transition probability calculated with the permanent electric dipole moments taken into account, as shown by Fig. 14.6.

Velocity dependence of the collisional resonances

One of the potentially most interesting aspects of the resonant collisions is that, in theory, the collision time increases and the linewidth narrows as the collision velocity is decreased. According to Eqs. (14.6) and (14.8) the collision time is proportional to $1/v^{3/2}$. Collisions between thermal atoms with temperatures of ~ 500 K lead to linewidths of the collisional resonances that are a few hundred MHz at $n = 20$. In principle, substantially smaller linewidths can be observed if the collision velocity is reduced.

The most straightforward way of reducing the collision velocity is to velocity select the atoms in an atomic beam,¹⁷ and a natural method of velocity selection is

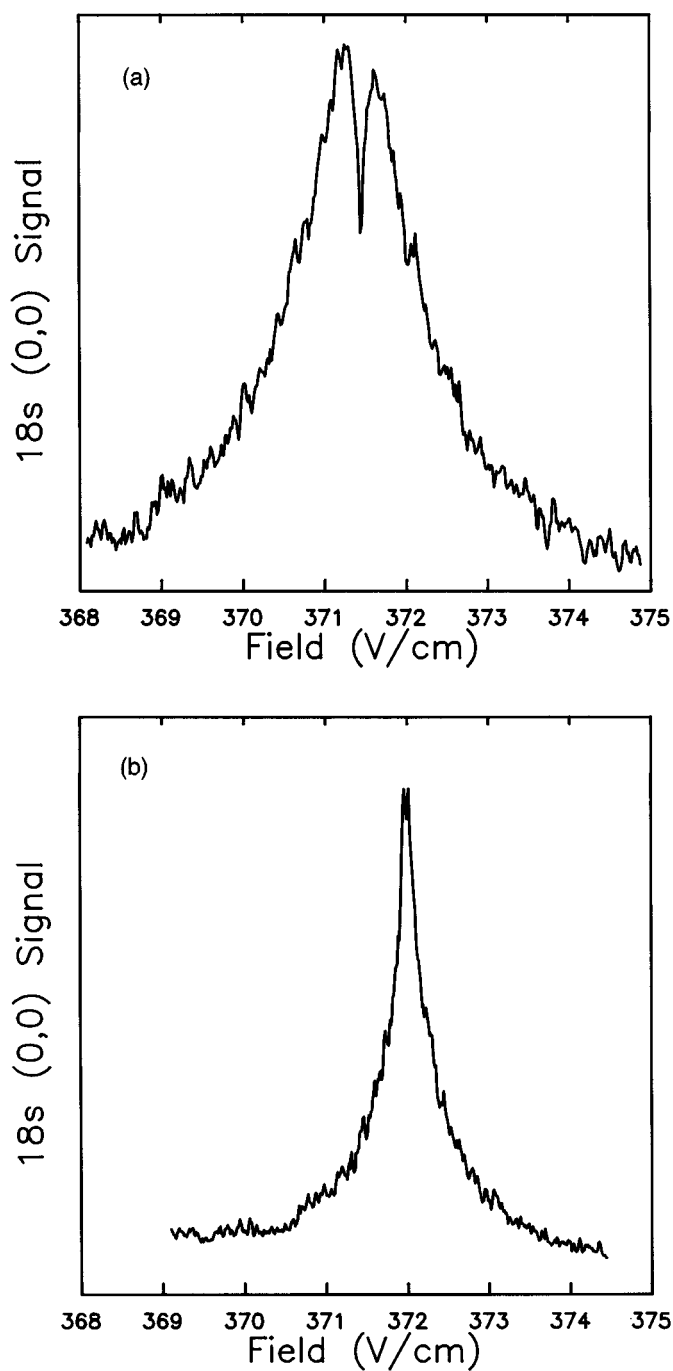


Fig. 14.12 Experimental measurements of the Na 18s (0,0) resonance for (a) $\mathbf{v} \parallel \mathbf{E}$ and (b) $\mathbf{v} \perp \mathbf{E}$. Note the definite dip near the center of the resonance in (a), as well as the slight asymmetry, both of which agree with the numerical predictions (from ref. 14).

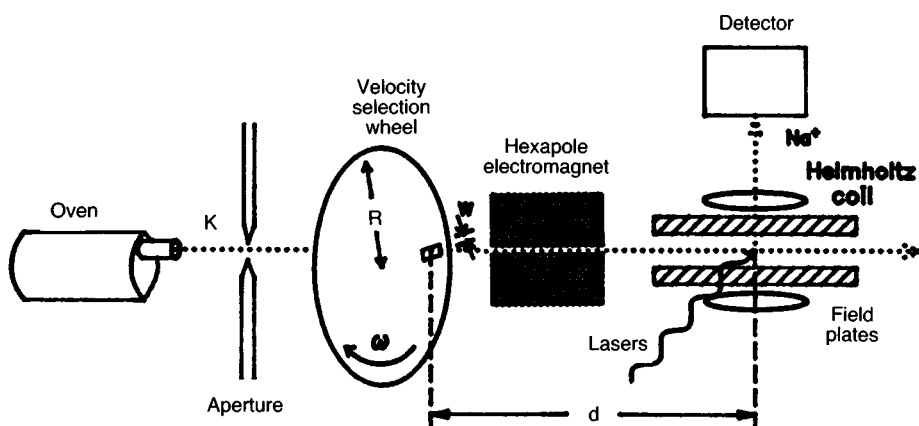


Fig. 14.13 Experimental arrangement for velocity selecting and focusing the atomic beam. The rotating slotted disc and the pulsed laser beam select atoms in a velocity group, and the hexapole magnet focuses them where they cross the laser beam (from ref. 18).

the one shown in Fig. 14.13.¹⁸ Atoms effusing from the source pass through a chopper, a rotating 9.6 cm diameter disc which has radial slits as narrow as 1.5 mm near the outer edge. The disc rotates at frequencies as high as 200 Hz, at which frequency a 1.5 mm slit lets through a $25\ \mu\text{s}$ wide pulse of atoms. The laser, which has a 5 ns pulse width, excites the atoms $250\ \mu\text{s}$ later at the interaction region, which is 10 cm from the chopper. With a 1.5 mm wide slit the velocity distribution is $\sim 10\%$ of the velocity, which is usually chosen to be the velocity of the peak of the velocity distribution of the beam. Since a single velocity group of atoms is selected, it is possible to focus them spatially using the hexapole focusing magnet, substantially increasing the number density at the interaction region.^{18,19}

The first and most obvious question is whether or not a narrower velocity distribution leads to narrower collisional resonances. In Fig. 14.14 we show the $\text{Na } 26s + \text{Na } 26s \rightarrow \text{Na } 26p + \text{Na } 25p$ resonances obtained under three different experimental conditions.²⁰ In Fig. 14.14(a) the atoms are in a thermal 670 K beam. In Figs. 14.14(b) and (c) the beam is velocity selected using the approach shown in Fig. 14.13 to collision velocities of 7.5×10^3 and 3.8×10^3 cm/s, respectively. The dramatic reduction in the linewidths of the collisional resonances is evident. The calculated linewidths are 400, 28, and 10 MHz, and the widths of the collisional resonances shown in Figs. 14.14(a)–(c) are 350, 40, and 23 MHz respectively. The widths decrease approximately as $1/v^{3/2}$ until Fig. 14.14(c), at which point the inhomogeneities of the electric field mask the intrinsic linewidth of the collisional resonance.

In the Na collision process of Eq. (14.1) the inhomogeneities in the static tuning fields required preclude the observation of very narrow collisional resonances. In

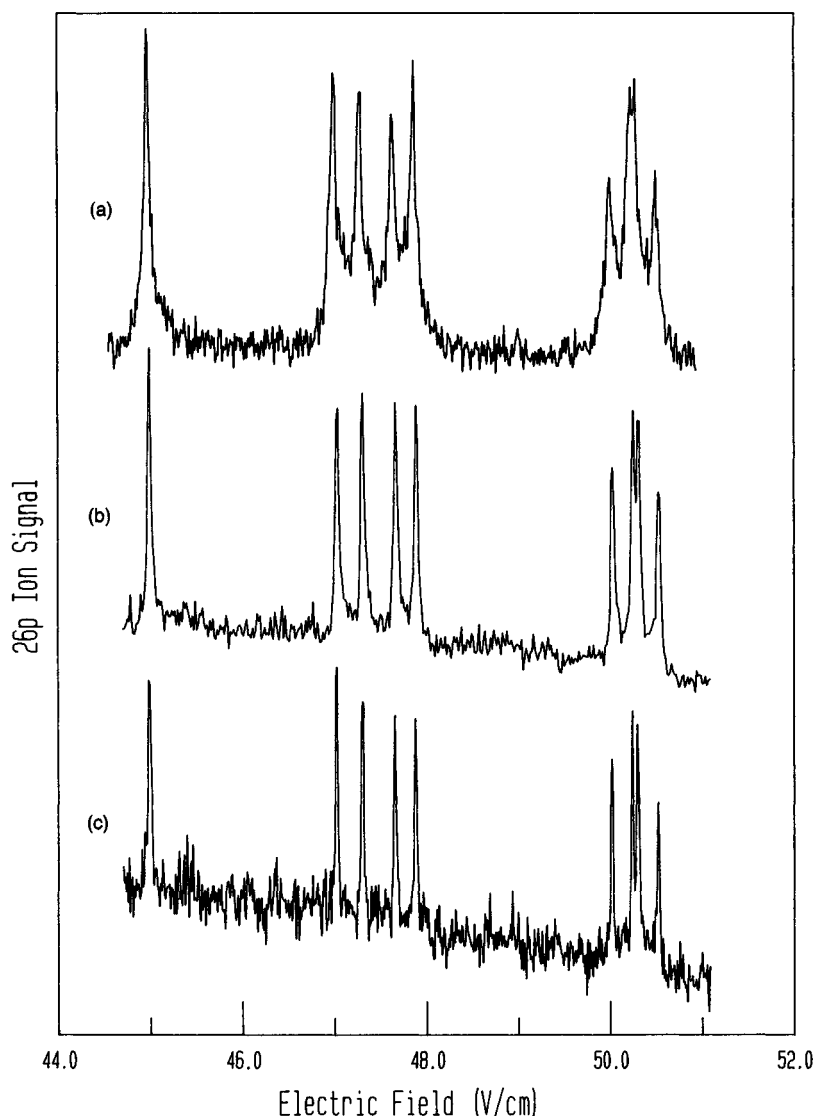
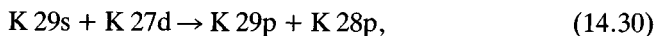


Fig. 14.14 $\text{Na } 26s + \text{Na } 26s \rightarrow \text{Na } 26p + \text{Na } 27p$ collisional resonances observed with (a) no velocity selection, collision velocity 4.6×10^4 cm/s, (b) velocity selection to a collision velocity 7.5×10^3 cm/s, (c) velocity selection to a collision velocity 3.8×10^3 cm/s (from ref. 20).

the K atom, however, dipole–dipole resonances occur near zero electric field,¹⁸ where the homogeneity of the static field is less important. A K collision process which has been studied extensively is^{17,18,21}



which occurs in relatively small fields as shown in the energy level diagram of Fig. 14.15. As shown by Fig. 14.15 and Eq. (14.30), the 29s and 27d states must both be

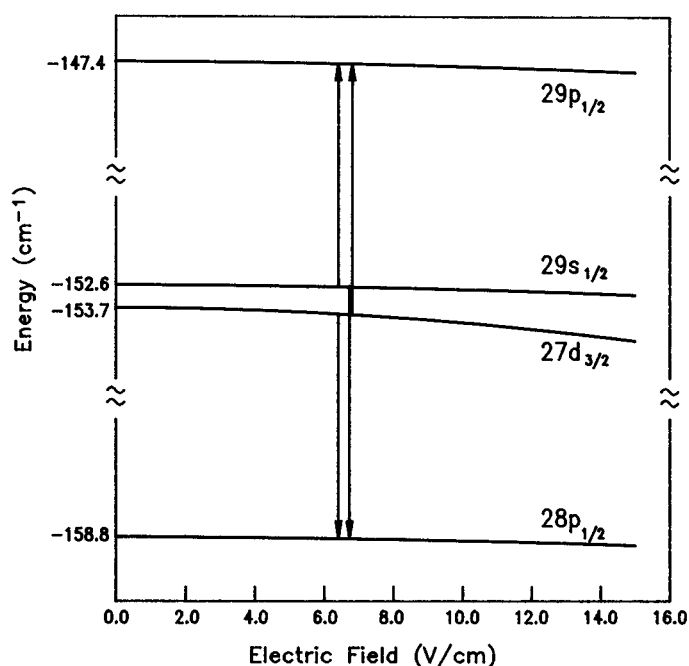


Fig. 14.15 Energy level diagram for the Rydberg states of K involved in the resonant energy transfer collisions being studied. The two transitions shown have been separated for clarity; they are degenerate in electric field (from ref. 21).

populated, and the fact that a resonant collision has occurred is ascertained by detecting the population in the higher lying 29p state by selective field ionization.

As shown by Fig. 14.15, the resonances occur near zero field, and it is easy to calculate the small Stark shifts with an accuracy greater than the linewidths of the collisional resonances. As a result it is straightforward to use the locations of the collisional resonances to determine the zero field energies of the *p* states relative to the energies of the *s* and *d* states. Since the energies of the *ns* and *nd* states have been measured by Doppler free, two photon spectroscopy,²² these resonant collision measurements for *n* = 27, 28, and 29 allow the same precision to be transferred to the *np* states. If we write the quantum defect δ_p of the K *np* states as

$$\delta_p = \delta_p^0 + \delta_p^1(n^*)^{-2} + \delta_p^2(n^*)^{-4}, \quad (14.31)$$

these measurements have allowed a new determination of δ_p^0 with a factor of 5 smaller uncertainty than the previous value.²³ Explicitly, they yield $\delta_p^0 = 1.711925(3)$, in which the major uncertainty is from the Doppler free laser spectroscopy, not from the resonant collision measurements.¹⁷

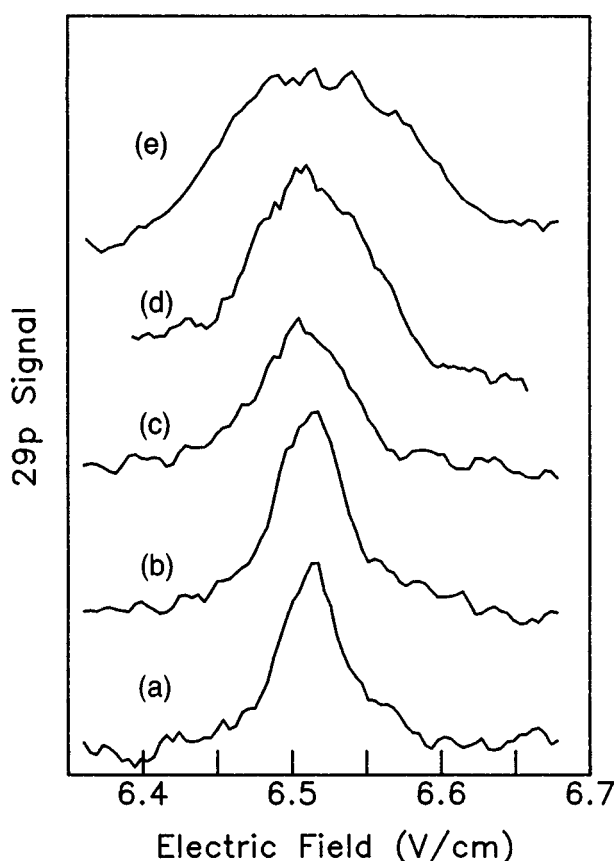


Fig. 14.16 Observed 29s resonance in K for several values of maximum-allowed collision time: (a) $\tau = 3.0 \mu\text{s}$, FWHM = 1.4 MHz, (b), $\tau = 2.0 \mu\text{s}$, FWHM = 1.8 MHz, (c) $\tau = 1.0 \mu\text{s}$, FWHM = 2.2 MHz, (d), $\tau = 0.4 \mu\text{s}$, FWHM = 3.1 MHz, and (e) $\tau = 0.2 \mu\text{s}$, FWHM = 5.2 MHz (from ref. 21).

Transform limited collisions

While collisional resonances 5 MHz wide are interesting for spectroscopic purposes, what makes them most interesting is that the 5 MHz linewidth implies that the collision lasts at least 200 ns, a time not much less than the $1 \mu\text{s}$ period allowed for the collisions to occur. If the collision linewidths can be reduced to the inverse of the time allowed for the collisions to occur, the collisional resonances become transform limited, and we know when each collision begins and ends.

Thomson *et al.* reached the transform limit by incorporating two improvements.²¹ First, Helmholtz coils were placed around the interaction region to cancel the earth's magnetic field. Second, the time interval allowed for the collisions had previously been defined by the laser pulse and the field ionization

pulse, which rises slowly at its onset. In these experiments the interval was terminated by the application of a rapidly rising, ~ 50 ns, low voltage detuning pulse which switches the field away from the field value of the collisional resonance at a well defined time.²⁴ These improvements allowed the observation of collisional resonances as narrow as 1 MHz. These narrow resonances were then transform broadened by reducing the time the atoms are allowed to collide to below 1 μ s. In Fig. 14.16 we show collisional resonances observed as the time between the laser pulse and the detuning pulse is shortened from 1 μ s to 0.4 μ s, showing clearly the transform broadening of the collisional resonances as the time interval is shortened. If the collisional resonances are transform limited, we know when individual collisions begin and end, and we can perturb the colliding atoms at well defined times during the collision, for example, when the atoms are at their point of closest approach.

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Radiative collisions

A radiative collision is a resonant energy transfer collision in which two atoms absorb or emit photons during the collision.¹ Alternatively, a radiative collision is the emission or absorption of a photon from a transient molecule, and, as shown by Gallagher and Holstein,² radiative collisions can also be described in terms of line broadening. In a line broadening experiment there are typically many atoms and weak radiation fields, and in a radiative collision experiment there are few atoms and intense radiation fields. The only real difference is whether there are many atoms or many photons. Due to the short collision times, $\sim 10^{-12}$ s, simply observing radiative collisions between low lying states requires high optical powers, and entering the regime where the optical field is no longer a minor perturbation seems unlikely. Due to their long collision times and large dipole moments, Rydberg atoms provide the ideal system in which to study radiative collisions in a quantitative fashion. As we shall see, it is straightforward to enter the strong field regime in which the radiation field, a microwave or rf field to be precise, is no longer a minor perturbation. Ironically, while the experiments are radiative collision experiments, with few atoms and many photons, the description of the strong field regime is given in terms of dressed molecular states, which is more similar to a line broadening description.³

Using a simple, three level model we can develop a feeling for the microwave powers required to observe radiatively assisted collisional energy transfer between Rydberg atoms.³ Consider the dipole–dipole atomic system shown in Fig. 15.1(a). In the Na $ns + ns \rightarrow np + (n-1)p$ resonant collisions described in the previous chapter the ns state corresponds to both s and s' of Fig. 15.1(a) and the $n-1$ and np states correspond to p and p' of Fig. 15.1(a), respectively. The collisions occurs via the interaction

$$V = \frac{\mu_1 \mu_2}{r^3}. \quad (15.1)$$

In Fig. 15.1(b) we show the level system of a collision radiatively assisted by the absorption of one photon of frequency ω . In this case the interaction leading to the production of the p and d states is given by

$$V = \frac{\mu_1 \mu_2 \mu_3 E}{r^3 \Delta}, \quad (15.2)$$

where Δ is the detuning between the real and virtual intermediate p' states and E is

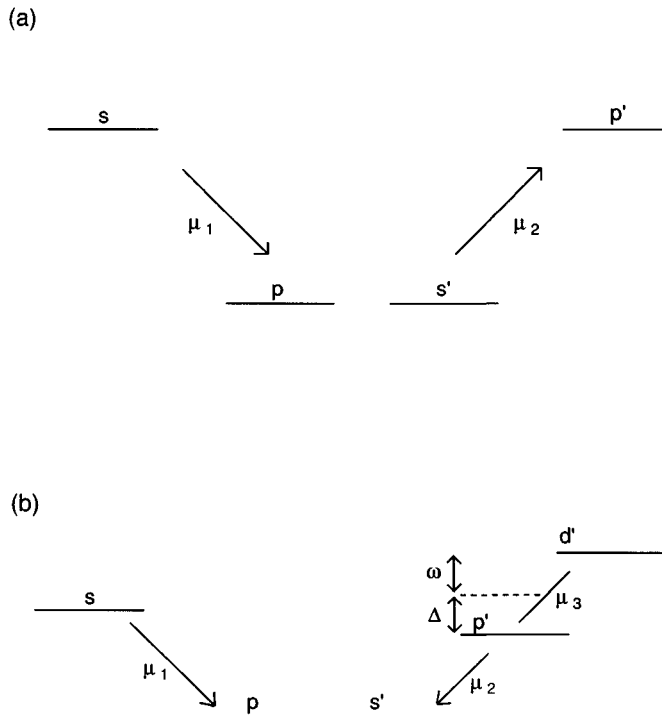


Fig. 15.1 (a) Energy levels and dipole matrix elements for the resonant collision of two atoms in the s and s' states resulting in the production of two atoms in the p and p' states. (b) Energy levels and matrix elements for the radiative collision in which an s and an s' atom collide to produce atoms in the p and d' states. The production of the d' state is via the virtual p' state which is detuned from the real p' state by an energy Δ .

the amplitude of the microwave field in which the collision occurs. The expressions of Eqs. (15.1) and (15.2) differ by the factor $\mu_3 E/\Delta$. For the radiatively assisted collision shown in Fig. 15.1(b) to be $\sim 10\%$ as probable as the resonant collision of Fig. 15.1a, we simply require that

$$\frac{\mu_3 E}{\Delta} = \frac{1}{3}. \quad (15.3)$$

If we assume typical values of $\mu_3 \approx n^2$ and $\Delta \sim 0.1n^{-3}$, the required field is given by

$$E = \frac{1}{30n^5}. \quad (15.4)$$

Evaluating Eq. (15.4) at $n = 20$ leads to a field of 50 V/cm, or a power of ~ 6 W/cm², a power roughly six orders of magnitude lower than those used in optical radiative collision experiments with low lying atomic states.⁴⁻⁷ These powers are not only readily obtained, but are easily exceeded by orders of magnitude, so that

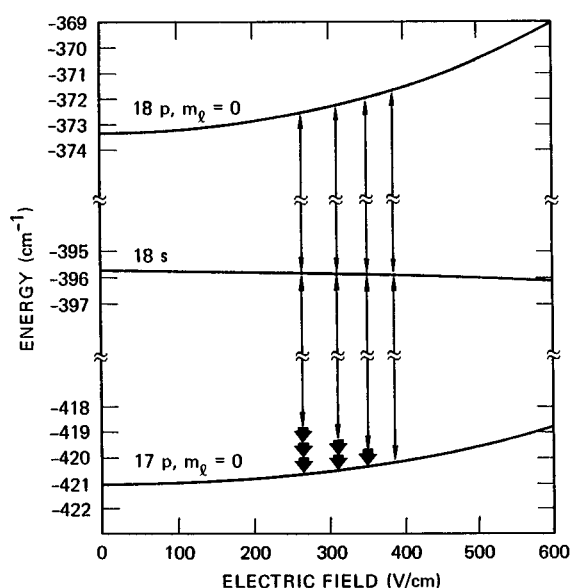
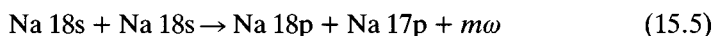


Fig. 15.2 Stark energy level diagram of the Na $|m_l| = 0$ states, relevant to the multiphoton-assisted collisions. The vertical lines indicate the collisional transfer and are drawn at the fields where they occur. The thick arrows correspond to the emitted photons (from ref. 8).

it is a straightforward matter to reach the strong field regime, in which the effect of the field is no longer a minor perturbation.⁸

The Na $ns + Na ns \rightarrow Na np + Na (n-1)p$ collisions in combined static and microwave fields were the first Rydberg atom radiative collisions studied. Specifically, the process



has been examined in detail. Here m is the number of photons emitted in the collision. When $m > 0$ there is emission, and when $m < 0$ there is absorption. The energy levels of the Na 18s, 17p, and 18p states are shown as a function of static field in Fig. 15.2.⁸ For simplicity we have only shown the $m_l = 0$ levels in Fig. 15.2. The (0,0) resonant collision, with no microwaves, occurs at a field of 390 V/cm, and the radiative collisions accompanied by the stimulated emission of one, two, and three photons occur at lower static fields, as shown. We follow the convention of labelling the collisional resonances by the m_l values of the lower and upper final p states of the collision, i.e. by (m_l, m_u) , where m_l and m_u are the m_l values of the lower and upper p states. The number of photons emitted is shown by the number of short bold arrows. In Fig. 15.2 the field displacement of a radiative collision resonance from the normal collisional resonance is proportional to the microwave frequency and inversely proportional to the sum of the differential Stark shifts of the p states. As shown in Fig. 15.2, the 18s state has a very small Stark shift, which we ignore. In principle, it is equally possible to observe collisions in which there is absorption, as opposed to emission, of several microwave photons. However, in

the $\text{Na } ns + \text{Na } ns \rightarrow \text{Na } np + \text{Na } (n-1)p$ system these radiatively assisted collisions occur at higher static fields, where there are many collisional resonances due to the Stark states which are not shown in Fig. 15.2.⁹ The $n = 17$ and $n=16$ Stark manifolds lie just below the $18p$ and $17p$ states, and at fields $\sim 10\%$ higher than the fields at which the resonant collisions of Fig. 15.2 occur, resonant collisions involving the Stark states appear. The presence of these collisional resonances precludes the unambiguous observation of $ns + ns \rightarrow np + (n-1)p$ collisional resonances in which microwave photons are absorbed.

Initial experimental study of radiative collisions

The initial experimental study of radiative collisions between Rydberg atoms was done by simply introducing a microwave field into the region between the two field plates of an apparatus such as the one shown in Fig. 14.7.¹⁰ The microwave field was introduced using a horn outside the plates, and several microwave frequencies between 12 and 18 GHz which propagated well between the plates were used. The maximum intensities were estimated to be $\sim 3 \text{ W/cm}^2$. The data are accumulated in the same fashion as in the resonant collision experiment, the only difference being the presence of a microwave field. The atoms are excited by two pulsed lasers, and collisions are allowed to occur for a $1 \mu\text{s}$ period, after which a high voltage pulse is applied to the lower field plate to field ionize atoms in the np state. The signal from the np state atoms is recorded as the static field is scanned. On each scan the microwave field amplitude is held constant. The dependence of the radiative collision cross section on the microwave power is determined by repeating the static field scan at different microwave powers. In these experiments single photon radiative collisions were observed, at static fields slightly below the fields at which the resonant collisions were observed, as expected from Fig. 15.2. A typical example is shown in Fig. 15.3. The field displacement of the radiative collision signals from the resonant collision signal is equal to the field required to produce a Stark shift equal to the microwave frequency. Two points about this experiment are worth noting. First, the radiatively assisted collision cross sections are within an order of magnitude of the resonant collision cross sections for incident microwave intensities of a few W/cm^2 , confirming the estimate of Eq. (15.4). Second, it is interesting to observe that at low microwave power the radiatively assisted collision signal is linear in the microwave power, but at higher powers the radiatively assisted collision signal rises less rapidly, suggesting the entrance into a nonperturbative regime.

Replacing the field plates of Fig. 14.7 with the resonant microwave cavity shown in Fig. 15.4 allowed an increase in the circulating microwave power by Q , the quality factor, of the cavity.⁸ The cavity is a piece of WR-90 (X band) waveguide 20 cm long which is closed at both ends. The inside dimensions of the cavity are

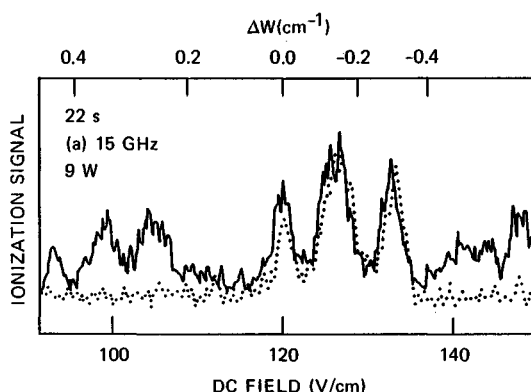


Fig. 15.3 The observed Na 22p ion signal after the population of 22s state vs the dc field in the presence of a 15 GHz microwave field (solid trace). The dotted trace was observed with no microwave field, and the sharp resonances in the center (for both solid and dotted traces) result from the resonant collisions, while the displaced resonances on the side (solid trace) are due to microwave assisted collisions. The scale on the top of the figure shows the detuning from the (0,0) resonance (from ref. 10).

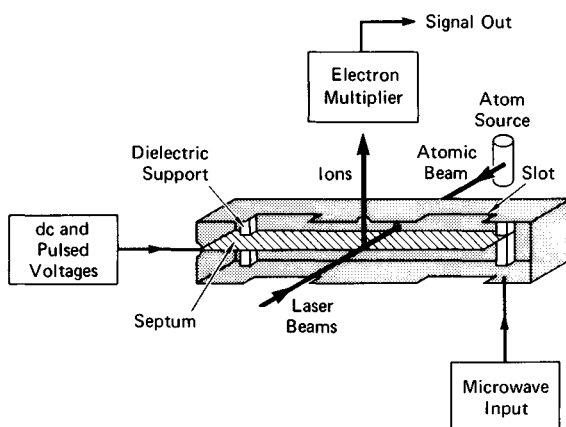


Fig. 15.4 Main features of the atomic beam apparatus, the atomic source, the microwave cavity, and the electron multiplier. The microwave cavity is shown sliced in half. The Cu septum bisects the height of the cavity. Two holes of diameter 1.3 mm are drilled in the sidewalls to admit the collinear laser and Na atomic beams, and a 1 mm diameter hole in the top of the cavity allows Na^+ resulting from field ionization of Na to be extracted. Note the slots for pumping (from ref. 8).

20.32 cm long, 1.02 cm high, and 2.28 cm wide. The cavity contains a septum, as shown in Fig. 15.4, which allows the application of a static tuning field and a pulsed field in the vertical direction. The presence of the septum does not affect the TE_{10n} cavity modes since they have only vertical electric fields.

The cavity is operated on modes of odd n , which have an electric field antinode at the center of the cavity. Specifically the $n = 15, 17$, and 19 modes, with resonant

frequencies of 12.8, 14.4, and 15.5 GHz have been used. These modes have $Q \sim 1300$, and when excited with a power of 20 W, the circulating microwave intensity is $\sim 10 \text{ kW/cm}^2$, and the microwave field is $\sim 500 \text{ V/cm}$. As shown in Fig. 15.4, the atomic and laser beams enter the cavity through opposing holes in the sidewalls of the cavity $\sim 1 \text{ mm}$ above the septum. Thus a cylindrical volume of Rydberg atoms is produced. There is a small, $\sim 1 \text{ mm}$ diameter, hole in the center of the top of the cavity through which ions produced by field ionization of atoms directly beneath the hole escape from the cavity en route to the detector. Since the atoms only travel $\sim 1 \text{ mm}$ during the time between laser excitation and field ionization, the location of the hole in the top ensures that the signals observed are due to atoms at an antinode of the microwave electric field.

Data are accumulated in the same way as used to acquire those shown in Fig. 15.3. Not surprisingly, the radiative collisions which had previously been observed with 10 W of microwave power can be observed with milliwatts of power, due to the high Q of the cavity. The most interesting aspect of the data, though, is what happens in strong microwave fields. In Fig. 15.5 we show the development of the resonances observed for the radiative collision process of Eq. (15.5) as the 15.5 GHz microwave power incident on the cavity is increased.⁸ With no microwave power only the four collisional resonances corresponding to the different $|m|$ values of the 17p and 18p states are observed. When the microwave power is raised this pattern of four resonances is repeated three times at lower static tuning fields, corresponding to the stimulated emission of one, two, and three microwave photons during the collision. A straightforward way of labeling the resonances is simply to extend our previous notation to account for the number of photons emitted. Explicitly, we label the collisional resonances as $(m_l, m_u)^m$ where m_l and m_u are the m_ℓ values of the upper and lower p states and m is the number of photons emitted. In Fig. 15.5 the $(0,0)^m$ resonances are indicated by m with an arrow.

It is hardly surprising that, as the microwave power is raised, higher order multiphoton processes are observed. On the other hand, it may be surprising that for each $m \neq 0$ the cross sections first increase then decrease with microwave power. For example, the $m = 1$ cross section is clearly zero in the lowest trace. Similarly, the $m = 0$ cross section vanishes in the trace one above the lowest but reappears in the lowest trace. Such behavior, typical of the strong field regime, is not predicted by perturbation theory. Close inspection of Fig. 15.5 reveals that the positions of the collisional resonances shift to lower static field as the microwave power is raised. Finally, in contrast to the usual observation of broadening with increased power, the $(0,0)^m$ resonances, which are well isolated from other resonances, develop from broad asymmetric resonances to narrow symmetric ones as the microwave power is raised.

That the observed resonances for $m \neq 0$ become narrower and more symmetric as the power is increased can be understood in the following way, using the one-photon process as an example. The coupling matrix element of Eq. (15.2) has two

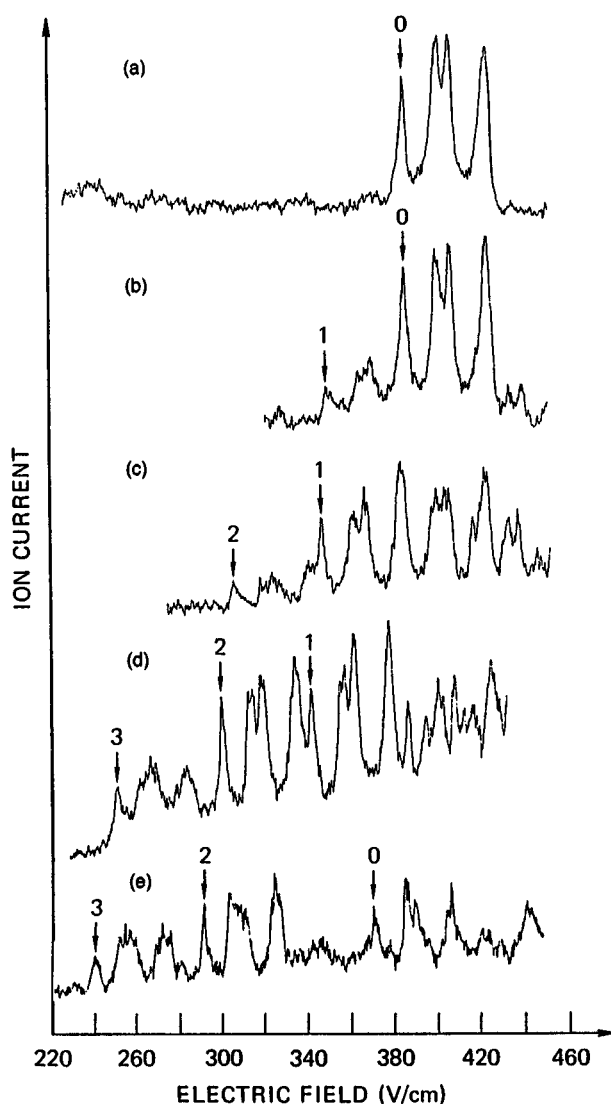


Fig. 15.5 Observed Na 18p ion signal after the population of the 18s level vs the static field with a 15.4 GHz microwave field. Trace (a) corresponds to no microwave power input to the cavity and shows the set of four zero-photon collisional resonances. Traces (b), (c), (d), and (e) correspond, respectively, to 13.5, 50, 105, and 165 V/cm microwave field amplitudes inside the cavity and show additional sets of four collisional resonances corresponding to one, two, and three-photon radiatively assisted collisions. The peaks labelled 0, 1, 2, and 3 correspond to the lowest field member of the set of four resonances corresponding to zero-, one-, two-, and three-photon assisted collisions, $(0,0)^0$, $(0,0)^1$, $(0,0)^2$, $(0,0)^3$ (from ref. 8).

factors, a dipole–dipole term, and a dipole–field term. For the collision to be observable, the time integral of the matrix element over the collision must exceed some minimum value ~ 1 . Thus, if the microwave field is smaller, the radiative collision must occur with a smaller impact parameter to compensate. A small impact parameter has two effects. First, it implies a short collision time and a wide resonance. Second, as the atoms come closer together the molecular energy levels deviate from their $r = \infty$ values, producing the asymmetry. As the microwave field is raised, the impact parameter can become larger, and the resonances become narrower and more symmetric.

Theoretical description

The experimental data shown in Figs. 15.3 and 15.5 were obtained with a microwave frequency $\omega/2\pi > 1/\tau$ where τ is the time or duration of the collision and $1/\tau$ is the linewidth. In this case the resonances corresponding to the absorption or emission of different numbers of photons are resolved. Here we describe radiative collisions starting from the high frequency regime, $\omega/2\pi > 1/\tau$ and progressing to the low frequency regime, $\omega/2\pi < 1/\tau$.

The strong field, high frequency regime

To describe the shifts and intensities of the m -photon assisted collisional resonances with the microwave field Pillet *et al.* developed a picture based on dressed molecular states,³ and we follow that development here. As in the previous chapter, we break the Hamiltonian into an unperturbed Hamiltonian H_0 , and a perturbation V . The difference from our previous treatment of resonant collisions is that now H_0 describes the isolated, noninteracting, atoms in both static and microwave fields. Each of the two atoms is described by a dressed atomic state, and we construct the dressed molecular state as a direct product of the two atomic states. The dipole–dipole interaction V is still given by Eq. (14.12), and using it we can calculate the transition probabilities and cross sections for the radiatively assisted collisions.

We begin by developing the wavefunction for one atom in combined static and microwave fields, both of which are in the same direction. If the nominal $n\ell$ state has an energy $W_{n\ell}(E)$ in a static field E , we can describe the energy in the vicinity of the static field E_S by

$$W_{n\ell}(E_S + \Delta E) = W'_{n\ell} + k_{n\ell}\Delta E + \frac{1}{2}a_{n\ell}(\Delta E)^2 \quad (15.6)$$

where

$$W'_{n\ell} = W_{n\ell}(E_S), \quad k_{n\ell} = \left. \frac{dW_{n\ell}}{dE} \right|_{E_S}, \quad \text{and} \quad a_{n\ell} = \left. \frac{d^2W_{n\ell}}{dE^2} \right|_{E_S}$$

If we now imagine that ΔE varies in time, but slowly, its only effect is to cause a time variation of the energy of the $n\ell$ state. We assume that the spatial wavefunction is unaffected by ΔE and that no transitions occur. This approximation is the adiabatic approximation of Autler and Townes.¹¹ Now let us consider the time variation of ΔE of particular interest to us, $E_{mw} \cos \omega t$. If the assumptions stated above are valid, we can use the energy of Eq. (15.6) as the unperturbed Hamiltonian H_0 in the Schrodinger equation. Explicitly,

$$W(t) \psi_{n\ell}(t) = (W'_{n\ell} + k_{n\ell} E_{mw} \cos \omega t + \frac{a_{n\ell}}{2} (E_{mw} \cos \omega t)^2) \psi_{n\ell}(t) = \frac{i\partial \psi_{n\ell}(t)}{\partial t}, \quad (15.7)$$

which has the solution

$$\psi_{n\ell}(t) = \psi_{n\ell} e^{-i \int_0^t W(t') dt'}. \quad (15.8)$$

Carrying out the integrations and discarding the phase factors from t_0 yields

$$\psi_{n\ell}(t) = \psi_{n\ell} e^{-i(W' + \frac{a_{n\ell}}{4} E_{mw}^2)t} e^{-i \frac{k_{n\ell} E_{mw}}{\omega} \sin \omega t} e^{-i \frac{a_{n\ell} E_{mw}^2}{8\omega} \sin 2\omega t}. \quad (15.9)$$

The exponentials with sinusoidal arguments may be expanded as Bessel functions using¹²

$$e^{ix \sin \omega t} = \sum_{-\infty}^{\infty} J_k(x) e^{ik\omega t}. \quad (15.10)$$

With the Bessel function expansion of Eq. (15.10), Eq. (15.9) becomes

$$\psi_{n\ell}(t) = \psi_{n\ell} e^{-i(W'_{n\ell} + \frac{a_{n\ell} E_{mw}^2}{4})t} \left(\sum_{-\infty}^{\infty} J_k \left(\frac{k_{n\ell} E_{mw}}{\omega} \right) e^{-ik\omega t} \right) \left(\sum_{-\infty}^{\infty} J_{k'} \left(\frac{a_{n\ell} E_{mw}^2}{8\omega} \right) e^{-i2k'\omega t} \right). \quad (15.11)$$

If $a_{n\ell} E_{mw}^2 \ll \omega$, as is usually the case for the experiments described here, in the last expansion of Eq. (15.11) only the $k' = 0$ term contributes and the summation is equal to one. In this case the wavefunction is written as

$$\psi_{n\ell}(t) = \psi_{n\ell} e^{-i(W'_{n\ell} + \frac{a_{n\ell} E_{mw}^2}{4})t} \sum_{-\infty}^{\infty} J_k \left(\frac{k_{n\ell} E_{mw}}{\omega} \right) e^{-ik\omega t}. \quad (15.12)$$

We note that for collisions involving states of good parity, $k_{n\ell} = 0$, and the first summation of Eq. (15.11) collapses to 1. In this case we may not disregard the second summation of Eq. (15.11).

As discussed in Chapter 10, we can think of the effect of the microwave field on the $n\ell$ state as modulating its energy. Just as modulating the frequency of a radio wave produces sidebands, the microwave field can be thought of as modulating the energy of the $n\ell$ state and breaking it into carrier and sideband states.

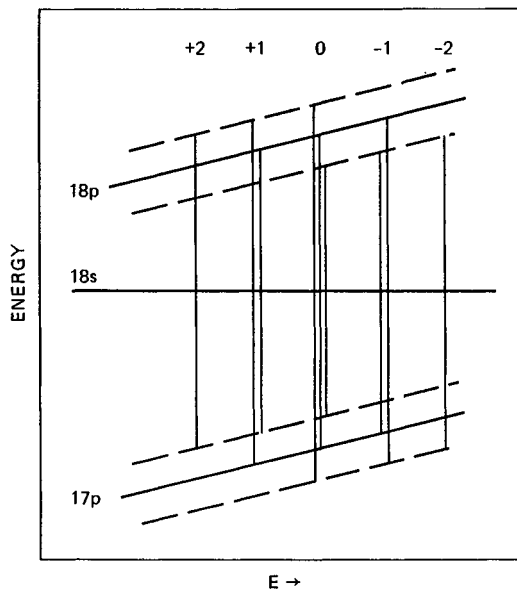


Fig. 15.6 Energy levels of the Na 17p, 18s, and 18p states showing the first upper and lower sideband states of the p states. The numbers +2, . . . , -2 refer to the net number of photons emitted in a radiative collision at these tunings of the field. Note that there are several processes which lead to the net emission of, for example, zero photons (from ref. 3).

In Eq. (15.12) $\psi_{n\ell}$ is the spatial wavefunction of the $n\ell$ state at the static field E_S , $W'_{n\ell}$ is the energy in the static field E_S , $\alpha_{n\ell}E_{mw}^2/4$ is the ac Stark shift produced by the microwave field, and $W'_{n\ell} + \alpha_{n\ell}E_{mw}^2/4$ is the carrier energy. The Bessel function $J_k(kE_{mw}/\omega)$ gives the amplitude in the k th sideband state, which is displaced in energy from the carrier state by $k\omega$, k microwave photons. For the np states k_{np} is large, but for the ns states k_{ns} is very small, and the amplitudes of the s sidebands are negligible.

A graphic presentation of the effect of the microwave fields is shown in Fig. 15.6, a plot of the energies of the 18s, 17p and 18p states showing the carrier energies of these states as well as the energies of the ± 1 sideband states of the p states.³ As mentioned above, and shown explicitly in Eq. (15.12), the amplitude of each sideband depends on the strength of the microwave field, and while this point is not made explicitly in Fig. 15.6, it is implicit in that no sidebands of the s state are shown and second and higher sidebands of the p states are not shown. Thus Fig. 15.6 corresponds to a microwave field too weak to have observable amplitudes for the second sidebands of the p states. With no microwaves all sidebands would have zero amplitude and the carriers amplitude one. A further interesting point to observe in Fig. 15.6 is that there are two processes corresponding to the net emission of ± 1 photon, and three corresponding to the net emission of zero photons. These processes correspond to the transitions between the

energetically allowed combinations of sideband states, raising the question of how to account for these different processes, all of which occur at the same static field and are therefore experimentally indistinguishable.

Using the dressed atomic states we construct the dressed molecular states as direct products. Explicitly, we construct the two states $\psi_A(t)$ and $\psi_B(t)$ given by³

$$\psi_A(t) = \psi_{1ns}(t) \otimes \psi_{2ns}(t) \quad (15.13a)$$

and

$$\psi_B(t) = \psi_{1np}(t) \otimes \psi_{2(n-1)p}(t) \quad (15.13b)$$

in which each of the individual wavefunctions is given by Eq. (15.12). If we write out the expression for ψ_B explicitly we find

$$\psi_B(t) = \psi_{1np}\psi_{2(n-1)p} e^{-iW_B t} \sum_{kk'} J_k \left(\frac{k_{np} E_{mw}}{\omega} \right) J_{k'} \left(\frac{k_{(n-1)p} E_{mw}}{\omega} \right) e^{-i(k+k')\omega t}, \quad (15.14)$$

where $W_B = W'_{np} + W'_{(n-1)p} + (\alpha_{np} + \alpha_{(n-1)p})E_{mw}^2/4$, the sum of the carrier energies of the np and $(n-1)p$ states. With the Bessel function identity,¹³

$$J_N(x \pm y) = \sum_{-\infty}^{\infty} J_{N \mp k}(x) J_k(y), \quad (15.15)$$

the double summation of Eq. (15.14) may be condensed to a single sum, yielding

$$\psi_B(t) = \psi_{1np}\psi_{2(n-1)p} e^{-iW_B t} \sum_k J_k \left(\frac{k_B E_{mw}}{\omega} \right) e^{-ik\omega t}, \quad (15.16)$$

where $k_B = k_{np} + k_{(n-1)p}$. Similarly we can write the wavefunction for $\psi_A(t)$ as

$$\psi_A(t) = \psi_{1ns}\psi_{2ns} e^{-iW_A t} \sum_{k'} J_{k'} \left(\frac{k_A E_{mw}}{\omega} \right) e^{-ik'\omega t}, \quad (15.17)$$

where $W_A = 2W'_{ns}$ and $k_A = 2k_{ns}$

Both $\psi_A(t)$ and $\psi_B(t)$ are solutions to H_0 , and the total wavefunction is, in general, a linear combination of the two. Explicitly, it is given by

$$\psi(t) = C_A(t)\psi_A(t) + C_B(t)\psi_B(t), \quad (15.18)$$

where $C_A(t)$ and $C_B(t)$ are time dependent coefficients analogous to those in Eq. (14.15). If we insert the wavefunction of Eq. (15.18) into the time dependent Schrodinger equation,

$$H\Psi(t) = i\partial\Psi/\partial t, \quad (15.19)$$

and use the fact that $\psi_A(t)$ and $\psi_B(t)$ are solutions of the unperturbed Schrodinger equation, we find the equation

$$VC_A(t)\psi_A(t) + VC_B(t)\psi_B(t) = i\dot{C}_A(t)\psi_A(t) + i\dot{C}_B(t)\psi_B(t). \quad (15.20)$$

Assuming that the diagonal matrix elements of V vanish, multiplying Eq. (15.20) in turn by $\psi_A^*(t)$ and $\psi_B^*(t)$ and carrying out the spatial integrations yields

$$\langle\psi_B(t)|V|\psi_A(t)\rangle C_A(t) = i\dot{C}_B(t) \quad (15.21a)$$

and

$$\langle \psi_A(t) | V | \psi_B(t) \rangle C_B(t) = i \dot{C}_A(t). \quad (15.21b)$$

We note that Eqs. (15.21) are similar to Eqs. (14.16) but for the time dependence of $\psi_A(t)$ and $\psi_B(t)$. We recall that V is always time dependent since it depends on the internuclear separation of the atoms. However, this time dependence is slow compared to those of $\psi_A(t)$ and $\psi_B(t)$.

Our interest is in computing the transition probability at resonance, a condition which is met when $W_A = W_B + m\omega$ where m is an integer. First, we explicitly write the matrix element $\langle \psi_B(t) | V | \psi_A(t) \rangle$ using the wavefunctions of Eqs. (15.16) and (15.17) and the Bessel function relation of Eq. (15.15). It is given by

$$\langle \psi_B(t) | V | \psi_A(t) \rangle = \langle \psi_B | V | \psi_A \rangle \sum_k J_k \left(\frac{(k_A - k_B) E_{mw}}{\omega} \right) e^{-i[W_A - W_B + k\omega]t}. \quad (15.22)$$

In this form it is apparent that the matrix element $\langle \psi_B(t) | V | \psi_A(t) \rangle$ is a product of the spatial matrix element, which varies slowly with time as the atoms move past each other, and the Bessel function sum, which contains an enormous range of frequencies. At resonance one term of the sum is constant in time and the others oscillate at multiples of ω . The oscillating terms are ineffective in causing transitions in collisions of duration much longer than the microwave period, since they average to zero. Thus at the resonance corresponding to the radiative collision in which m photons are emitted, i.e. when $W_A = W_B + m\omega$, the constant and significant part of the matrix element of Eq. (15.22) is given by

$$\langle \psi_B(t) | V | \psi_A(t) \rangle = \langle \psi_B | V | \psi_A \rangle J_m \left(\frac{K E_{mw}}{\omega} \right) \quad (15.23)$$

where $K = k_A - k_B$. The matrix element of Eq. (15.23) is identical to the matrix element for a normal resonant collision multiplied by a constant factor, the Bessel function, and as a result the transition probability can be computed using the same approaches as in Chapter 14.

All radiative collision experiments with Rydberg atoms have been done with the collision velocity perpendicular to the static and microwave fields. Thus the results obtained are an average over spatial orientations, and it makes little sense to use a detailed model of the interaction. Accordingly, we assume that

$$|\langle \psi_B | V | \psi_A \rangle| = \begin{cases} \frac{\chi}{b^3} & r < \frac{\sqrt{5}b}{2} \\ 0 & r > \frac{\sqrt{5}b}{2} \end{cases} \quad (15.24)$$

where b is the impact parameter and $\chi = \mu_1 \mu_2$. With this assumption, the atoms interact for a time b/v . Using matrix elements of the form of Eq. (15.24) leads to

sinusoidally oscillating solutions of $C_A(t)$ and $C_B(t)$ during the collision. If we choose as initial conditions $C_A(-\infty) = 1$ and $C_B(-\infty) = 0$, then after the collision,

$$C_A^2(t) = \cos^2 \left[\frac{\chi}{b^2 v} J_m \left(\frac{KE_{mw}}{\omega} \right) \right], \quad (15.25a)$$

and the transition probability at impact parameter b , $P_m(b)$, is given by $C_B^2(t)$. Explicitly,

$$P_m(b) = C_B^2(t) = \sin^2 \left[\frac{\chi}{b^2 v} J_m \left(\frac{KE_{mw}}{\omega} \right) \right]. \quad (15.25b)$$

The transition probability of Eq. (15.25b) is of the same form as the transition probability of Eq. (20.25), with $D = \chi |J_m(KE_{mw}/\omega)|$. Accordingly, if we define an impact parameter $b_m(E_{mw})$ such that

$$\frac{\chi}{b_m^2(E_{mw})v} \left| J_m \left(\frac{KE_{mw}}{\omega} \right) \right| = \frac{\pi}{2}, \quad (15.26)$$

$P_m(b)$ oscillates between 0 and 1 as b increases from zero to $b_m(E_{mw})$, at which point $P_m(b) = 1$ for the last time. As b increases from $b_m(E_{mw})$ to infinity $P_m(b)$ decreases monotonically from 1 to 0.

To obtain the cross section for the m -photon assisted collision in the presence of a microwave field E_{mw} we integrate the transition probability over impact parameter. Explicitly

$$\sigma_m(E_{mw}) = \int_0^\infty 2\pi b \sin^2 \frac{\chi}{b^2 v} \left| J_m \left(\frac{KE_{mw}}{\omega} \right) \right| db. \quad (15.27)$$

Carrying out the integral explicitly yields

$$\sigma_m(E_{mw}) = \frac{\pi^3}{4} b_m^2(E_{mw}) = \frac{\pi^2}{2} \frac{\chi}{v} \left| J_m \left(\frac{KE_{mw}}{\omega} \right) \right|. \quad (15.28)$$

A radiative collision with the emission of no photons in the absence of a microwave field is simply a resonant collision. Since $J_0(0) = 1$, we can express the radiative collision cross section of Eq. (15.28) as

$$\sigma_m(E_{mw}) = \sigma_R \left| J_m \left(\frac{KE_{mw}}{\omega} \right) \right|, \quad (15.29)$$

i.e. as the product of the resonant collision cross section σ_R and the magnitude of the $J_m(KE_{mw}/\omega)$ Bessel function.

Eq. (15.29) is only valid in the strong field regime. It is clear that if $J_m(KE_{mw}/\omega)$ is small, using Eq. (15.26) we compute a small value of $b_m(E_{mw})$. When this value approaches the size of the atom, Eq. (15.29) is no longer valid, for the integral of Eq. (15.27) is dominated by contributions for $b \sim b_m(E_{mw})$, where neither the assumption of the energies being independent of r nor the dipole approximation

are valid. Under such circumstances a collision with a small impact parameter does not lead to as large a transition probability as indicated by Eq. (15.25b).

Connection to the weak field regime

In the experiments with Rydberg atoms it is very difficult to observe radiatively assisted collisions with cross sections more than a factor of 10 smaller than the resonant collision cross sections, so the deviations from Eq. (15.29) are not apparent. However, in other contexts, such as laser assisted collisions, this limitation does not apply, and it is interesting to consider how the above description passes over into the weak field regime, in which $J_m(KE_{\text{mw}}/\omega)$ is small. If we restrict the integration in Eq. (15.27) to the large r region of space, in which the approximations we have used are valid, we can rewrite Eq. (15.27) as

$$\begin{aligned}\sigma_m &= \int_B \frac{\chi^2}{b^4 v^2} J_m^2\left(\frac{KE_{\text{mw}}}{\omega}\right) 2\pi b \, db \\ &= \frac{\pi \chi^2}{B^2 v^2} J_m^2\left(\frac{KE_{\text{mw}}}{\omega}\right),\end{aligned}\quad (15.30)$$

in which an obvious lower limit for B is the geometrical radius of the atom. Accordingly, $B = n^2 a_0$ for a Rydberg atom of principal quantum number n . An important fact to note about Eq. (15.30) is that $\sigma_m(E_{\text{mw}})$ is proportional to $J_m^2(KE_{\text{mw}}/\omega)$. Using the fact that for small arguments $J_m(x) = (x/2)^m$,¹² at low microwave fields

$$\begin{aligned}\sigma_m &\propto E_{\text{mw}}^{2m} \\ &\propto I_{\text{mw}}^m,\end{aligned}\quad (15.31)$$

i.e. the cross section for an m -photon assisted collision is proportional to the microwave intensity to the m th power, as expected on the basis of time dependent perturbation theory.

The low frequency regime

In the previous discussion of radiatively assisted collisions we have assumed that there are many cycles of the radiation field during the time of one collision, i.e. $\omega/2\pi \gg 1/\tau$. It is interesting to consider the opposite extreme, $\omega/2\pi \ll 1/\tau$. We consider collisions in the total field

$$E = E_S + E_{\text{rf}} \cos(\omega t + \phi), \quad (15.32)$$

and we assume that the collisional resonances are observed by scanning the static field. If, in the absence of the rf field, the resonance is found at $E_S = E_R$, it is found at

$$E_S = E_R - E_{\text{rf}} \cos(\omega t + \phi) \quad (15.33)$$

with the added rf field of Eq. (15.32). If we can control the phase of the rf field at which the collision occurs the resonance is simply shifted as shown by Eq. (15.33). If we cannot control the phase, either because the exciting laser is not synchronized with the rf field, or because collisions can occur randomly over several rf cycles, then the probability of observing the resonant collision signal at any given static field is proportional to the likelihood of the total field's being equal to E_R at that static field. If the resonant collision cross sections is given as a function of field by $\sigma_R(E)$, then in the field $E_S + E_{\text{rf}} \cos \omega t$ the cross section observed at E_S is given by the convolution of $\sigma_R(E)$ with the function

$$f(E) = \begin{cases} \frac{1}{\pi E_{\text{rf}} [1 - (E - E_S/E_{\text{rf}})^2]^{1/2}} & |E - E_S| \leq E_{\text{rf}} \\ 0 & |E - E_S| > E_{\text{rf}} \end{cases} \quad (15.34)$$

In other words $\sigma(E_S)$ is given by the time weighted average of the cross section in the static and oscillating fields. Since the oscillating field spends most of its time at its turning points, the observed cross section can be expected to exhibit the behavior shown in Fig. 15.7.³

The intermediate frequency regime

We have considered the high frequency regime, which is independent of the rf phase, and the low frequency regime which is critically dependent on the rf phase. We now consider the intermediate frequency regime, $\omega/2\pi \sim 1/\tau$, approaching it from the low frequency side. To date the only way of calculating the lineshape of the collisional resonances has been explicit numerical integration. As we shall see, when comparisons with experimental results are made, the calculations predict multiple peaks in the cross section. The separation between peaks does not, however, correspond to the rf frequency. We can use a simple picture to understand the origin of the peaks and their locations, although the picture does not give lineshapes.¹⁴

For concreteness we assume that the collision occurs over the time interval $-T/2 < t < T/2$ in the presence of a rf field of well defined phase so that the total field is given by Eq. (15.32). We consider the transitions between the two molecular states A and B which are degenerate (at $r = \infty$) at the static field E_R . We may assume with no loss of generality that, in the vicinity of E_R , B has no Stark shift and A has a linear Stark shift. Ignoring the dipole-dipole coupling, the energy,

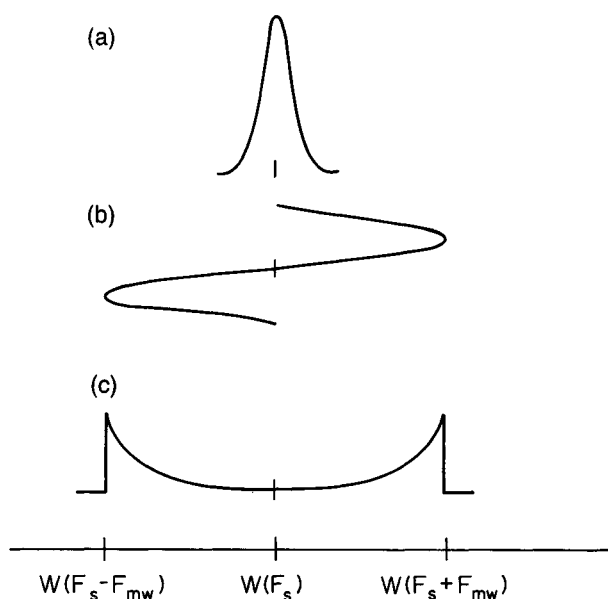


Fig. 15.7 Collisions in the presence of a low frequency microwave field: (a) the resonant collision profile without the microwaves; (b) one period of microwave field showing both the amplitude and the time weighting of the extreme field values; (c) typical time weighted radiative collision lineshape when the microwave period becomes long compared to the duration of one collision (from ref. 3).

W_B , of molecular state B is a constant, and the energy of molecular state A is given by

$$W_A = W_B + k_A(E - E_R), \quad (15.35)$$

where E is given by Eq. (15.32). Inspecting Eq. (15.35) we see that W_A follows the rf field. Imagine that we choose the rf phase $\phi = 0$ so that W_A reaches a maximum in the center of the allowed collision time. In Figs. 15.8(a) and (b), we show the energies W_A and W_B as functions of time for two choices of the phase of a 0.75 MHz rf field, $\phi = 0$ and $\pi/2$, respectively. The allowed collision time $T = 0.7 \mu\text{s}$. The dipole-dipole interaction lifts the degeneracy at resonance, leading to the avoided level crossings shown by the insets of Fig. 15.8. In Fig. 15.8(a) the energies are shown for two values of the static field E_S . For $E_S < E_R - E_{rf}$, the two levels never come into resonance. When $E_S = E_R - E_{rf}$ the two levels are resonant at $t = 0$, as shown by the broken line, and a broad resonance can be expected. For larger E_S the two levels come into resonance twice, as shown by the solid line, and the transition amplitude is the coherent sum of the amplitudes acquired at the two times the levels become resonant, at $t = \pm t'$.

As shown by Fig. 15.8(a), if the system is initially in state A it traverses two avoided level crossings to reach the final state B, and there are two possible paths to go from A to B. Whether the transition amplitudes for these two paths add

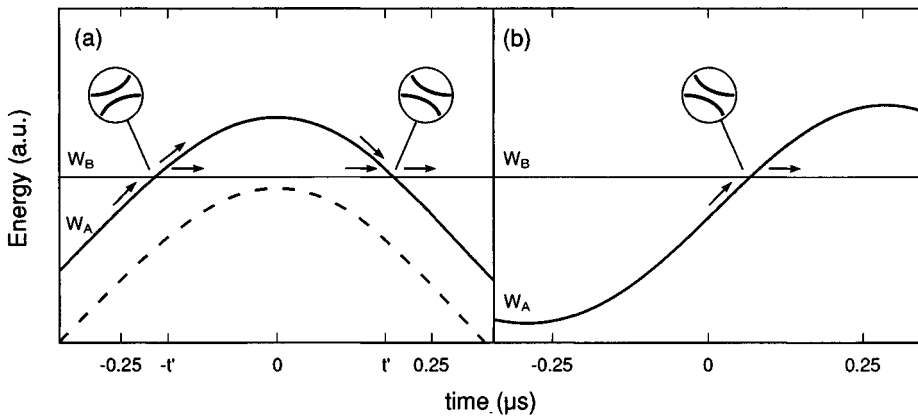


Fig. 15.8 The initial and final state energies W_A and W_B in the total electric field $E = E_S + E_{rf} \cos(\omega t + \phi)$ where the frequency $\omega/2\pi = 0.75$ MHz and the allowed collision time $T = 0.7 \mu s$. W_B is field independent and W_A varies linearly with the field. (a) The phase $\phi = 0$. No resonance occurs for $E_S < E_R - E_{rf}$. As the static field is increased, when $E_S = E_R - E_{rf}$ (---) the two levels come into resonance at $t = 0$. When $E_S > E_R - E_{rf}$ (—) the two levels come into resonance twice, and there are two resonant interaction periods at $t = \pm t'$. The levels do not actually cross but have avoided crossings as shown by the insets. There are two paths by which atoms initially in state A can reach state B, and they interfere constructively when Φ_B , the phase difference accumulated between $-t'$ and t' , is $2\pi N$. In (b) where $\phi = \pi/2$ and $E_S \approx E_R$, only one interaction period occurs for all values of E_S between $E_R - E_{rf}$ and $E_R + E_{rf}$ (after ref. 14).

constructively or destructively depends on the phase difference Φ between the two paths. Φ is simply the area between the two curves in Fig. 15.8(a), i.e.,

$$\Phi = \int_{-t'}^{t'} (W_A - W_B) dt. \quad (15.36)$$

If $\Phi = 2\pi N$, where N is an integer the interference is constructive, whereas if $\Phi = 2\pi(N + 1/2)$ it is destructive. As E_S is raised Φ passes through even and odd multiples of π leading to constructive and destructive interference in the cross section. Inspecting Fig. 15.8(a) we can see that the fringes should become closer together as E_S is increased. This description of the origin of the oscillations in the cross section makes it apparent that they are a form of Stückelberg oscillations observed in differential scattering.¹⁵

The two resonant interactions at $t = \pm t'$ can also be thought of as being analogous to the two separated oscillating field regions in a Ramsey magnetic resonance experiment, in which case the oscillations in the cross section can be thought of as Ramsey fringes.¹⁶

For $\phi = 0$, as shown in Fig. 15.8(a) the oscillations in the cross sections should extend up to the static field

$$E_S = E_R - E_{rf} \cos(\omega T/2). \quad (15.37)$$

With the rf phase $\phi = \pi$, the collisional resonance is reversed with respect to E_R . With $\phi = \pi/2$, as shown in Fig. 15.8(b) for $E_S \approx E_R$, the result is qualitatively different. The two levels come into resonance at most once for any value of E_S , and as a result there is one broad collisional resonance extending from $E_S = E_R - E_{\text{rf}} \sin(\omega T/2)$ to $E_S = E_R + E_{\text{rf}} \sin(\omega T/2)$.

The above description is obviously an extension of the low frequency description, but it can also be connected to the high frequency limit. Consider the case in which $\omega \gg 1/T$, i.e., there are many cycles of the field, and $|E_S - E_R| \ll E_{\text{rf}}$, the static tuning field is near resonance compared to the rf field amplitude. For a large transition probability the transition amplitudes of successive rf cycles must add constructively. This only happens when the phase difference Φ over an rf cycle satisfies $\Phi = 2\pi N$ with N an integer. Using the energies of Eq. (15.35),

$$\begin{aligned}\Phi &= \int_{-\pi/\omega}^{\pi/\omega} k_A(E_S + E_{\text{rf}} \cos \omega t - E_R) dt \\ &= \frac{2\pi k_A(E_S - E_R)}{\omega}.\end{aligned}\quad (15.38)$$

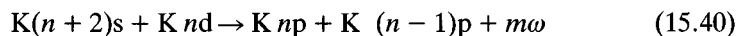
Equivalently,

$$k_A(E_S - E_R) = N\omega. \quad (15.39)$$

In other words the resonances are detuned from E_R by the static field shift equivalent to N photons, the same result as obtained in the dressed state picture.

Eq. (15.39) tells us where the collisional resonances occur, but it does not tell us how strong they are. The strength is determined by the phase contributions of the two parts of the rf cycle, when $W_A < W_B$ and $W_A > W_B$. When the phases accumulated in these two half cycles are both integral multiples of 2π the collisional resonances are strong. This requirement leads to an oscillation in the intensity of the N photon assisted collisional resonance proportional to $\cos(k_A E_{\text{rf}}/\omega + \gamma)$, where γ is a small constant, which has the same period as the Bessel function expression of Eq. (15.29).

While the experiments with Na atoms have validated many aspects of the theory of radiatively assisted collisions in the strong field, high frequency regime, experiments with velocity selected K atoms have extended the measurements both to higher order processes and through the regime in which the collision time matches the frequency of the oscillating field.^{14,17,18} Specifically, studying the process



with velocity selected atoms has several attractive features. First, since the collisional resonances are so narrow, it is straightforward to satisfy the condition $\omega/2\pi \gg 1/\tau$ yet still observe collisions radiatively assisted by many photons without exceeding the available tuning range. Second, since the collision times are long, it is possible to observe the predicted phase dependence of the collisions in the transition region $\tau \sim 2\pi/\omega$.

The techniques employed are the same as those used to study the K resonant collisions described in the previous chapter. The apparatus shown in Fig. 14.13 is used, the only difference from the previous experiment being the application of a rf voltage to the top plate of the interaction region. The rf field, at frequencies from 0.5 MHz to 4.0 MHz is either left free running or phase locked to the laser excitation.

Most of the data are acquired in the same way as described above. K atoms are excited to the 29s and 27d states by the laser excitation, $4s \rightarrow 4p \rightarrow 29s, 27d$. The atoms are allowed to collide for $1 \mu s$, after which a rapidly rising detuning pulse is applied, followed by the more slowly rising field ionization pulse. Atoms which have made the transition to the 29p state are selectively ionized by the field ionization pulse and detected. This signal is monitored as the small static tuning field is scanned. The amplitude and phase of the rf field are changed as parameters.

Using such low collision frequencies it is possible to study radiative collision processes of high order, a point made explicitly by Fig. 15.9. In Fig. 15.9 we show the development of the radiative collision signal for collisions in the presence of 4 MHz rf fields of amplitudes from 0 to 0.76 V/cm.¹⁷ Unlike the Na case described above, there is only a single collisional resonance when there is no rf field, as shown by Fig. 15.9(a). As the rf field amplitude is raised, progressively higher order resonant collision processes are observed, out to the seven photon assisted collisions. It is interesting to note that both $m > 0$ and $m < 0$ radiatively assisted collisions are observed, approximately symmetrically about the position of the collisional resonance at 6.44 V/cm. A slight asymmetry is expected at the extreme static tuning fields due to the second order Stark shift.

Close inspection of Fig. 15.9 suggests that the strength of a particular m -photon assisted cross section oscillates with the rf field as implied by the Bessel function dependence of Eq. (15.29). To demonstrate this point explicitly, the static field was fixed at the fields corresponding to several m photon assisted resonances and the amplitude of the rf field swept. Following this procedure for $m = 0, +1, -2$, and $+3$ leads to the results shown in Fig. 15.10, which shows quite clearly several zeroes in each of the σ_m cross sections. Furthermore, the experimental results agree almost perfectly with the expression of Eq. (15.28) and the dependence obtained by simply numerically integrating Eq. (14.16) using time dependent energies W_A and W_B . Close examination of Fig. 15.10 reveals that the observed cross sections at the highest rf field amplitudes are smaller than the calculated cross sections. This discrepancy is attributed to the fact that the static fields at which the collisional resonances occur shift slightly due to the second order ac Stark shift as the rf field is increased.

A final aspect of these radiation collisions, shown experimentally by Thomson *et al.*,¹⁷ is that the cross section integrated over all the collisional resonances increases with the rf field. Since the m th collisional resonance has a cross section of $\sigma = \sigma_R |J_m(KE_{mw}/\omega)|^2$ and $\sum_m J_m^2(x) = 1$, in general, $\sum \sigma_m > \sigma_R$.¹⁷

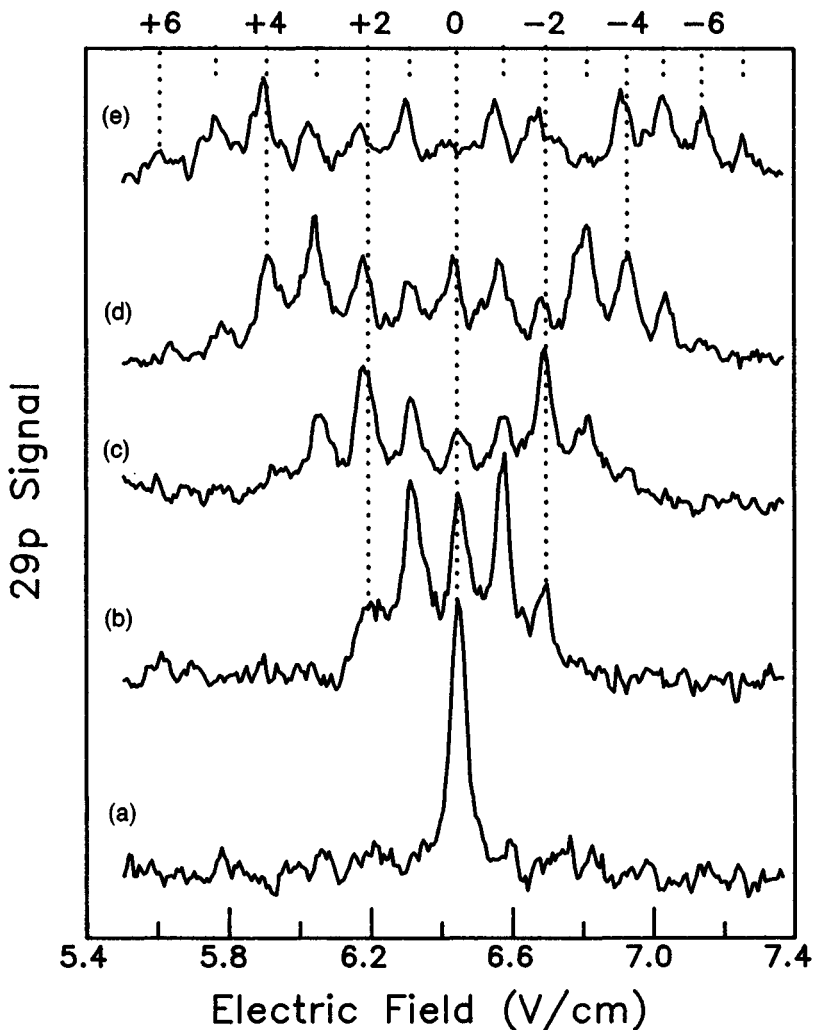


Fig. 15.9 The $n = 29$ resonance of K in the presence of a 4 MHz rf field. The rf field amplitudes for the five sets of data are: (a) 0 V/cm, (b) 0.19 V/cm, (c) 0.38 V/cm, (d) 0.57 V/cm, and (e) 0.76 V/cm. The top axis labels the sideband resonances according to the number of rf photons emitted by the atomic system (from ref. 17).

As we pointed out above, an attractive feature of the velocity selected K collisions is that we can examine the region $\omega/2\pi < 1/\tau$. We consider first the case in which there is no control over the phase of the field at which the collisions occur. This case is exemplified by Figs. 15.11(b)–(d), which show the effect of adding progressively larger amplitude 1 MHz rf fields of uncontrolled phase. Since the laser fires at an uncontrolled phase of the rf field, the observed cross section can be calculated using Eq. (15.34). As shown by Fig. 15.11 the cross section is broadened in approximately the manner shown in Fig. 15.7. As shown by a

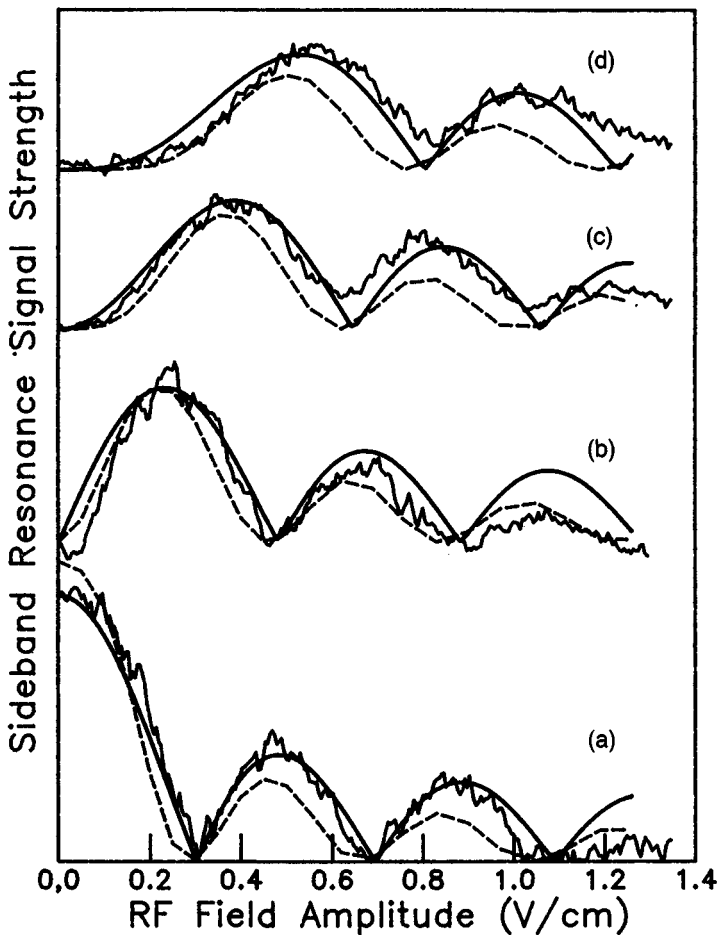


Fig. 15.10 Cross sections as a function of rf field strength for the first four orders of sideband resonances of the K 29s + K 27d radiative collisions in a 4 MHz rf field. (a) The zero-photon resonant collision cross section, (b) the +1 sideband resonance, (c) the -2 sideband resonance, and (d) the +3 sideband resonance. The solid line shows the experimental data, the bold line indicates the prediction the Floquet theory, and the dashed line is the result of numerical integration of the transition probability (from ref. 17).

comparison of Figs. 15.11(d) and 15.11(e), obtained with 1 and 0.5 MHz 0.2 V/cm rf fields, respectively, the broadening of the collisional resonance is independent of the frequency of the applied field, depending only on its amplitude, as predicted by Eq. (15.34).¹⁸

When the phase of a low frequency rf field is controlled the observed collisional resonances change dramatically. With the field of Eq. (15.32), when the atoms are allowed to collide during the interval $-0.5 \mu\text{s} < t < 0.5 \mu\text{s}$ with $\phi = 0$ or π , we observe the collisional resonances shown in Figs. 15.12(b) and (c) respectively, in agreement with Eq. (15.33). With no rf field the resonance occurs at $E_S = 6.44$

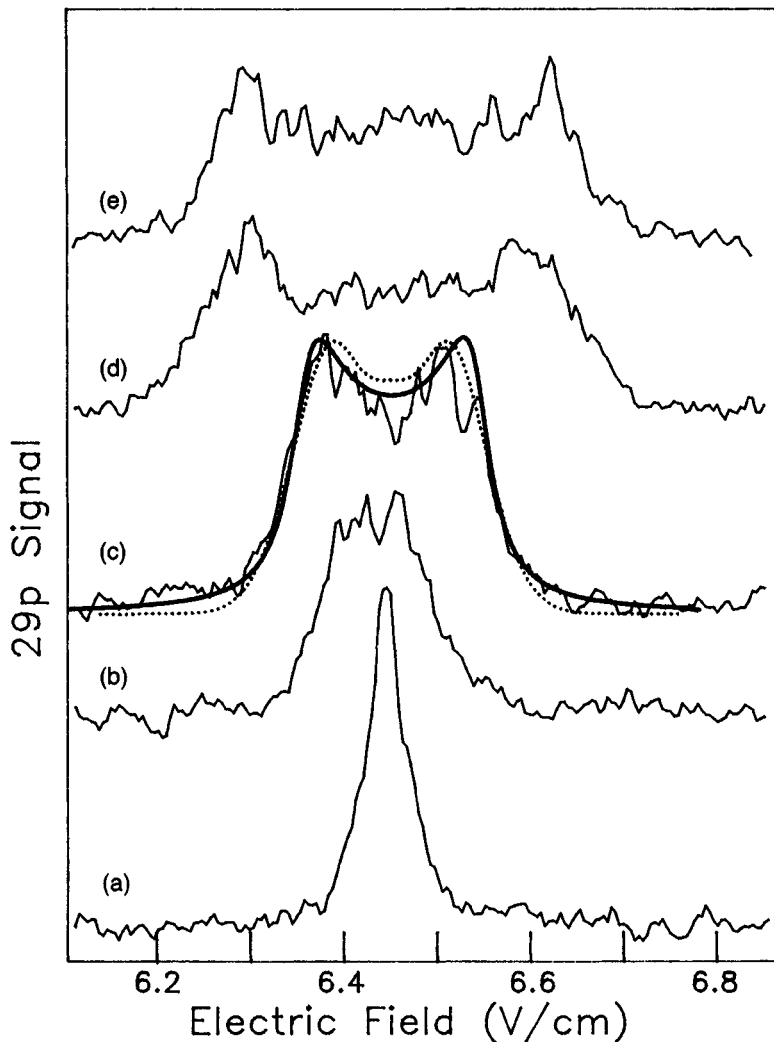


Fig. 15.11 The K 29s + 27d resonance in the presence of a low frequency rf field. In zero rf field (a), the FWHM is 1.6 MHz. In (b)–(d), a 1.0 MHz field of strength 0.05 V/cm, 0.1 V/cm, and 0.2 V/cm respectively is present. The solid line in (b) is a numerical integration of the transition probability, and the bold line is the convolution of a Lorentzian lineshape with a sinusoidal shift from resonance. In (e), the rf frequency is 0.5 MHz and its strength is 0.2 V/cm. For these low frequencies, the features are no longer frequency dependent but rather are field strength dependent (from ref. 18).

V/cm, as shown by Fig. 15.12(a). When $\phi=0$ the collision occurs at the maximum of the rf field, and the resonance occurs at a lower static field, as shown in Fig. 15.12(b). Similarly when the collision occurs at the minimum of the rf field the resonance occurs at a higher static field, as shown in Fig. 15.12(c).

Finally we consider the case in which $\omega/2\pi \approx 1/\tau$. In Fig. 15.13 we show the resonances obtained with an allowed collision time $T = 0.8 \mu\text{s}$ (the collisions are

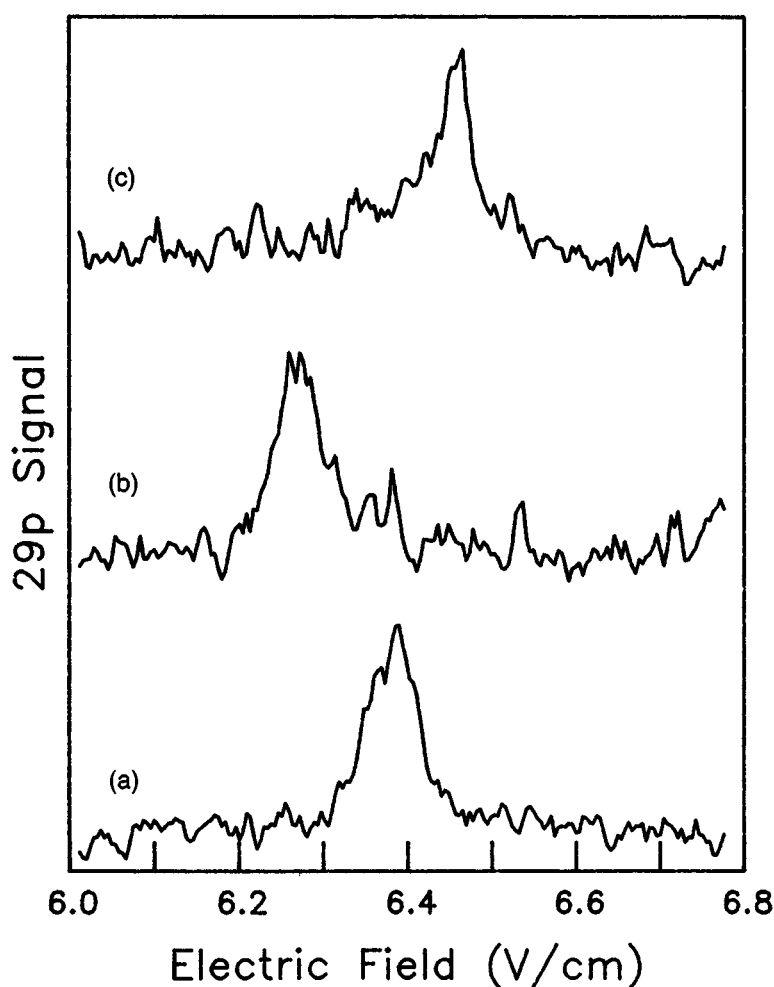


Fig. 15.12 (a) The K 29s + K 27d resonance with a FWHM of 2.0 MHz. In (b) and (c), a 0.5 MHz, 0.1 V/cm rf field is present with a phase of 0 and π respectively. A phase of 0 means the cosine wave is at maximum during the temporal center of the collision (from ref. 18).

allowed to occur for $-0.4\mu\text{s} < t < 0.4\mu\text{s}$ and two choices of the phase of an rf field, $E_{\text{rf}} \cos(\omega t + \phi)$, of amplitude $E_{\text{rf}} = 0.21$ V/cm and frequency 0.75 MHz.¹⁴ In Fig. 15.13(a) we show the resonance with no rf field at 6.44 V/cm. In Fig. 15.13(b), in which $\phi = 0$, the experimental curve exhibits a broad peak at $E_S = 6.27$ V/cm, a subsidiary peak at 6.35 V/cm, and a possible third subsidiary peak at 6.40 V/cm. The dotted lines are numerical calculations and the locations of the peaks for $N = 0-2$ from Eq. (15.36) are shown as well. As can be seen, the locations of the experimental peaks agree well with both the numerical calculation and the interference picture of Fig. 15.8(a) which leads to Eq. (15.36). When $\phi = \pi/2$, as shown in Fig. 15.13(c), a single broad resonance is observed, with no apparent

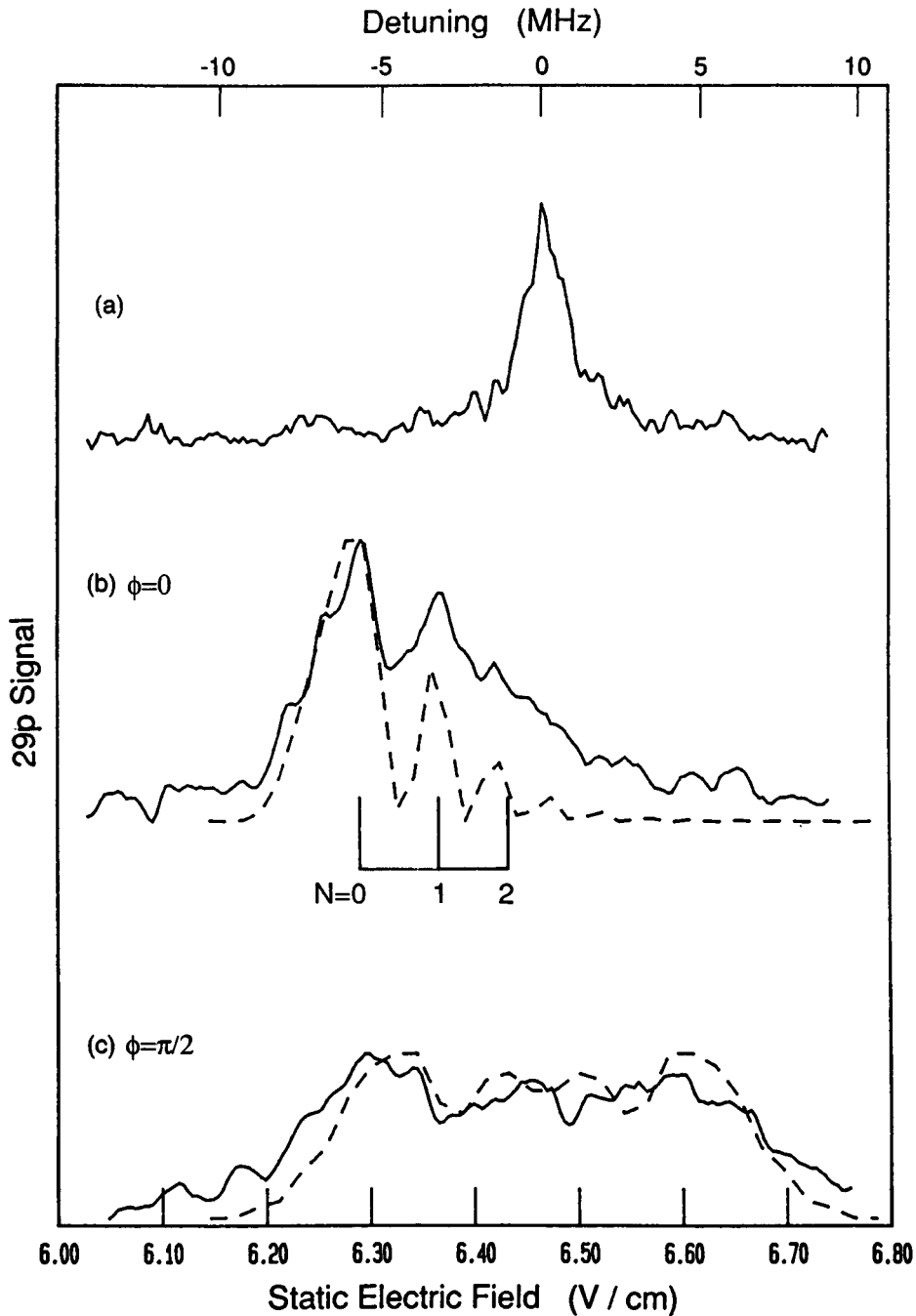


Fig. 15.13 The experimental (—) and calculated (---) lineshapes of the $n=29$ radiatively assisted resonance of K as functions of the static field E_S . (a) The unperturbed resonance with no rf field present, FWHM = 1.4 MHz. (b), (c) The resonance in the presence of a 0.75 MHz, 0.21 V/cm rf electric field and a phase of (b) $\phi = 0$ and (c) $\phi = \pi/2$. Indicated below the spectra are the calculated positions of the peaks using the $\Phi = 2\pi N$ condition (from ref. 14).

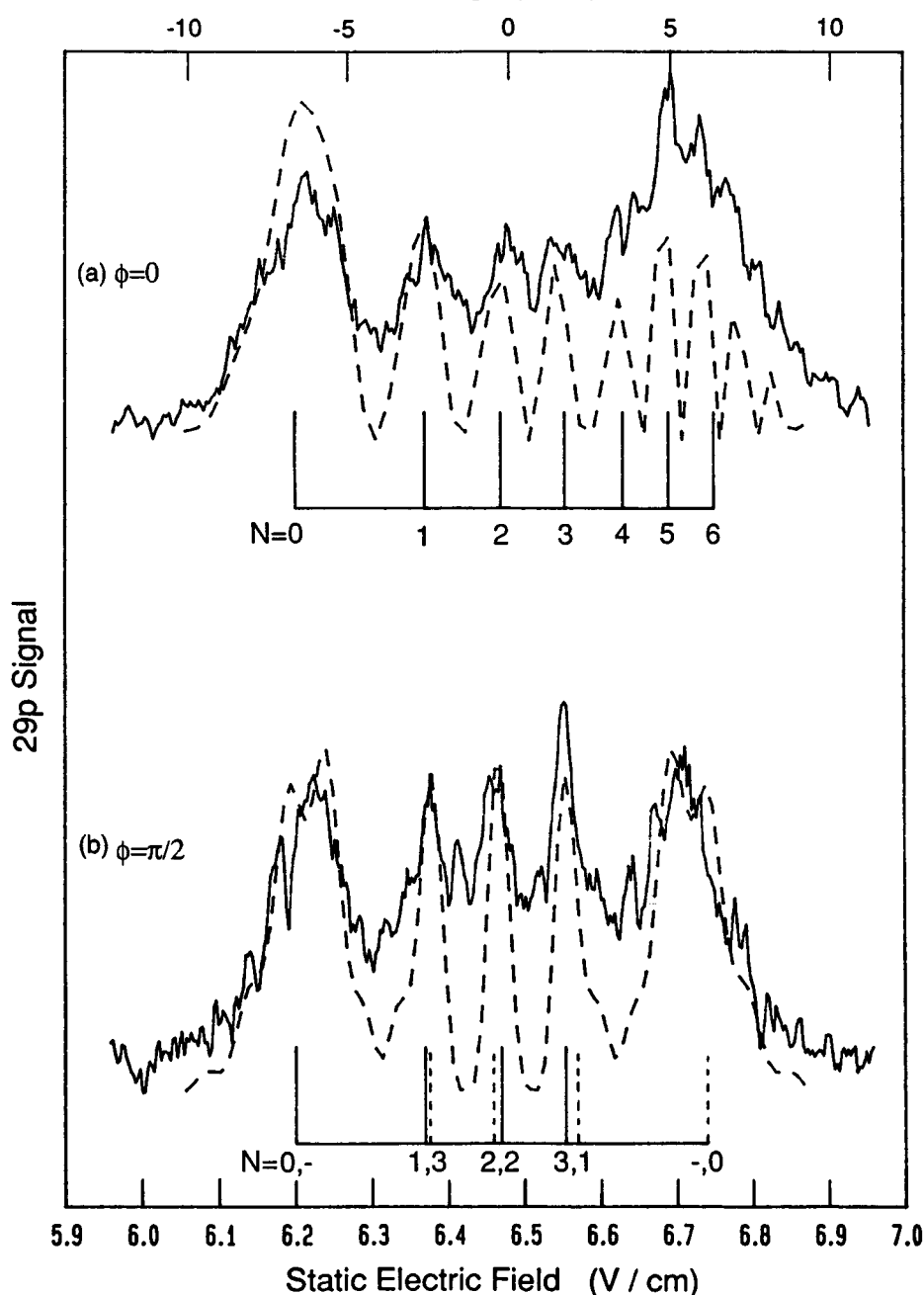


Fig. 15.14 The experimental (—) and calculated (---) lineshapes as functions of the static field E_S in a 0.31 V/cm, 1.48 MHz, rf field of phases (a) $\phi = 0$ and (b) $\phi = \pi/2$. The allowed collision time is 0.8 μ s. The calculated fringe positions using $\Phi = 2\pi N$ are indicated below the spectra. In (b) the fringe positions due to interference between the first and second (solid vertical marker) and the second and third (dashed marker) interactions are shown (from ref. 14).

subsidiary peaks. As shown, the numerical calculations agree with the experimental result. Since the levels are only degenerate once, there is no possibility of interference, as shown by Fig. 15.8(b).

As the rf frequency is raised, interference becomes possible for all phases, a point made by Figs. 15.14(a) and (b), showing the results for a 1.48 MHz field of amplitude 0.31 V/cm and phases $\phi = 0$ and $\pi/2$, respectively. The result for $\phi = \pi$ is the mirror image of the $\phi = 0$ result. As shown, the experimental data match the numerical calculation and the prediction of Eq. (15.36).

In addition to the fact that the $\phi = \pi/2$ trace shows interference peaks, we note that they occur at approximately the same static field as the $\phi = 0$ peaks, although they are narrower. As the rf frequency is raised, the phase dependence slowly disappears, until at 4 MHz it is undetectable, and we are in the high frequency regime described by the dressed state picture.

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Double Rydberg states

The autoionizing two electron states we have considered so far are those which can be represented sensibly by an independent electron picture. For example, an autoionizing Ba 6pnd state is predominantly 6pnd with only small admixtures of other states, and the departures from the independent electron picture can usually be described using perturbation theory or with a small number of interacting channels. In all these cases one of the electrons spends most of its time far from the core, in a coulomb potential, and the deviation of the potential from a coulomb potential occurs only within a small zone around the origin.

In contrast, in highly correlated states the noncoulomb potential seen by the outer electron is not confined to a small region. In most of its orbit the electron does not experience a coulomb potential, and an independent electron description based on $n\ell n'\ell'$ states becomes nearly useless. There are two ways in which this situation can arise. The first, and most obvious, is that the inner electron's wavefunction becomes nearly as large as that of the outer electron. If we assign the two electrons the quantum numbers $n_i\ell_i$ and $n_o\ell_o$, this requirement is met when n_i approaches n_o , which leads to what might be called radial correlation. The sizes of the two electron's orbits are related. The second way the potential seen by the outer electron can have a long range noncoulomb part is if the presence of the outer electron polarizes the inner electron states. This polarization occurs most easily if the inner electron ℓ states of the same n are degenerate, for then even a very weak field from the outer electron, equal to $1/r_o^2$, converts them to Stark states, which have permanent dipole moments. The anisotropic potential from the induced dipole produces an angular correlation in the motion of the two electrons, even if the orbit of the outer electron is many times larger than the orbit of the inner electron. Angular correlation may be found in all the doubly excited states of He since He^+ is hydrogenic. On the other hand, radial correlation is presumably only found when n_i approaches n_o . Angular correlation may also be observed in the doubly excited states converging to the nearly degenerate high ℓ states of a singly charged ion, Sr^+ for example, even when the outer electron is in a substantially larger orbit than the inner electron. On the other hand, the doubly excited states converging to low ℓ states of Sr^+ , which have large quantum defects, do not exhibit angular correlation before they exhibit radial correlation. To see remarkable correlation in these states, the requirement that n_i approach n_o must be met.

Doubly excited states of He

Although the existence of autoionizing states has been known since the 1920s, it was the observation of the doubly excited states of He by Madden and Codling¹ which aroused interest in highly correlated states. Using synchrotron radiation, they recorded the absorption spectrum of He using an 83 cm path length of He at a pressure of 0.3 Torr. One might naively expect to see two series, $2snp$ and $2pns$, both converging to the $\text{He}^+ n = 2$ limit. However, they observed one strong series, the + series, and a very weak series, the – series.² In addition to being weaker, the – series lies below the + series and has smaller autoionization widths.

More recently the measurements have been repeated and extended by Domke *et al.*,³ who also used a synchrotron as a light source, but detected the photoions produced instead of the absorption of the synchrotron radiation. Their spectra are shown in Fig. 23.1. As shown by Fig. 23.1 the strong $n = 2 +$ peaks appear as classic Beutler–Fano profiles on top of a small background $1s\epsilon p$ photoionization signal. They are roughly 100 times as intense as the – resonances. In addition, the widths of the – peaks are about 10% of the widths of the + peaks. If we examine the spectra converging to the higher He^+ limits, shown in Fig. 23.2, several features are apparent. First, we can see that these spectra contain only one series, the + series. In addition, in the spectra converging to higher n states of the ion the resonances become smaller relative to the background photoionization, which is hardly surprising, as there are more available continua. Stated another way, the magnitude of the Fano q parameter decreases with increasing n , although it also changes sign, which is not as simply explained. In Fig. 23.2(c) we can see that the series converging to the $\text{He}^+ n = 5$ and 6 and $n = 6$ and 7 states overlap, producing the same sort of pattern seen in Fig. 19.2.

The initial experiment of Madden and Codling, the observation of the states converging to the $\text{He}^+ n = 2$ state, was first explained by Cooper *et al.*² in the following way. Rather than use the $2snp$ and $2pns$ states as the atomic basis, they proposed using the combinations²

$$\Psi_{\pm} = \frac{1}{\sqrt{2}} [u_{2snp} \pm u_{2pns}] \quad (23.1)$$

where $u(2snp)$ is a product of a $2s$ and an np wavefunction. The + and – states have characteristics which are different from those of angular momentum states. In the + states the radial oscillations of the two electrons are in phase, and in the – states they are out of phase. In addition, by considering the properties of the + and – states near the origin we can explain the $n = 2$ spectra, shown in Fig. 23.1. Both u_{2snp} and u_{2pns} are products of one s wavefunction and one p wavefunction. Thus at the origin they are functionally the same. Since the normalization factors of wavefunctions of principal quantum number 2 and n are given by $2^{-3/2}$ and $n^{-3/2}$ respectively, the overall normalization factors of u_{2snp} and u_{2pns} are nearly the same. Near the origin $u_{2snp} \approx u_{2pns}$, and Ψ_- nearly vanishes. As a result, the – states are only weakly excited from the ground state, and they have lower autoionization rates than the + states, as observed.

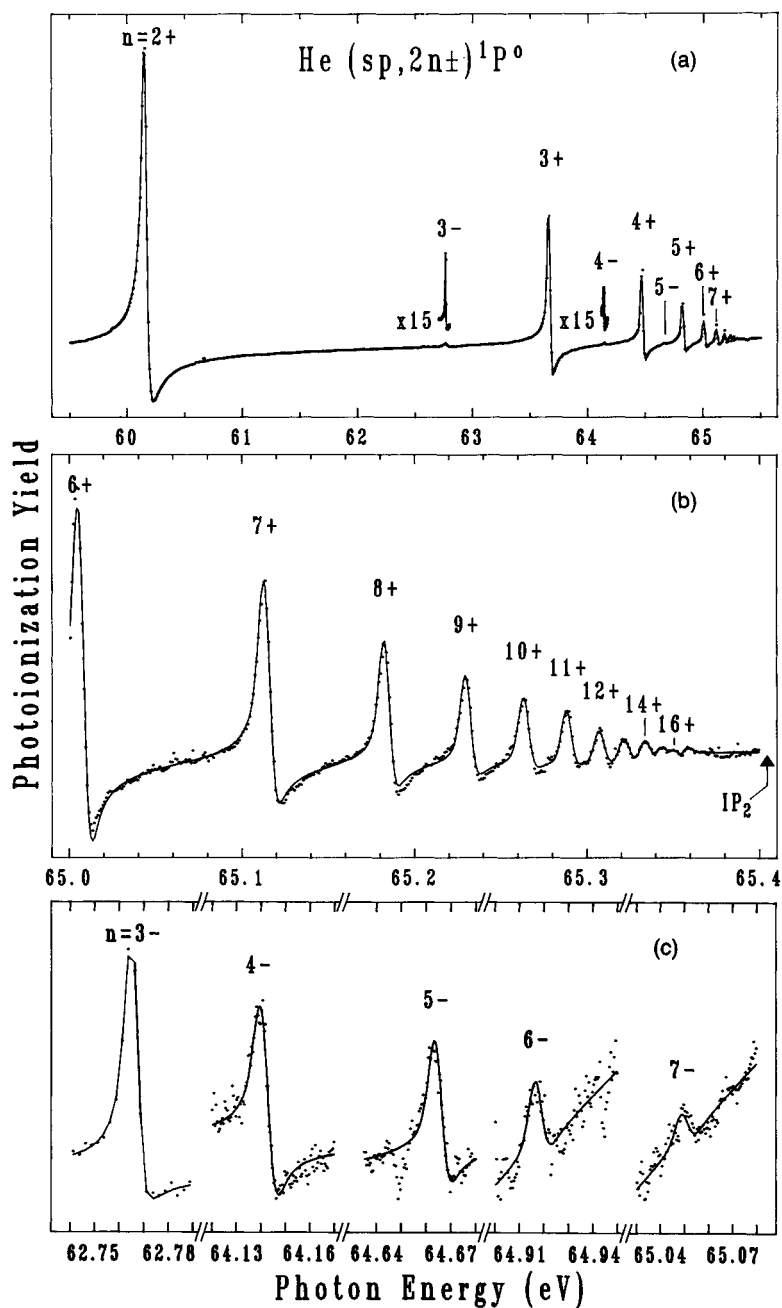


Fig. 23.1 Autoionizing states of doubly excited He below the $\text{He}^+ n=2$ threshold: (a) overview, (b) magnification of the $n \geq 6$ region, and (c) $2n-$ states (from ref. 3).

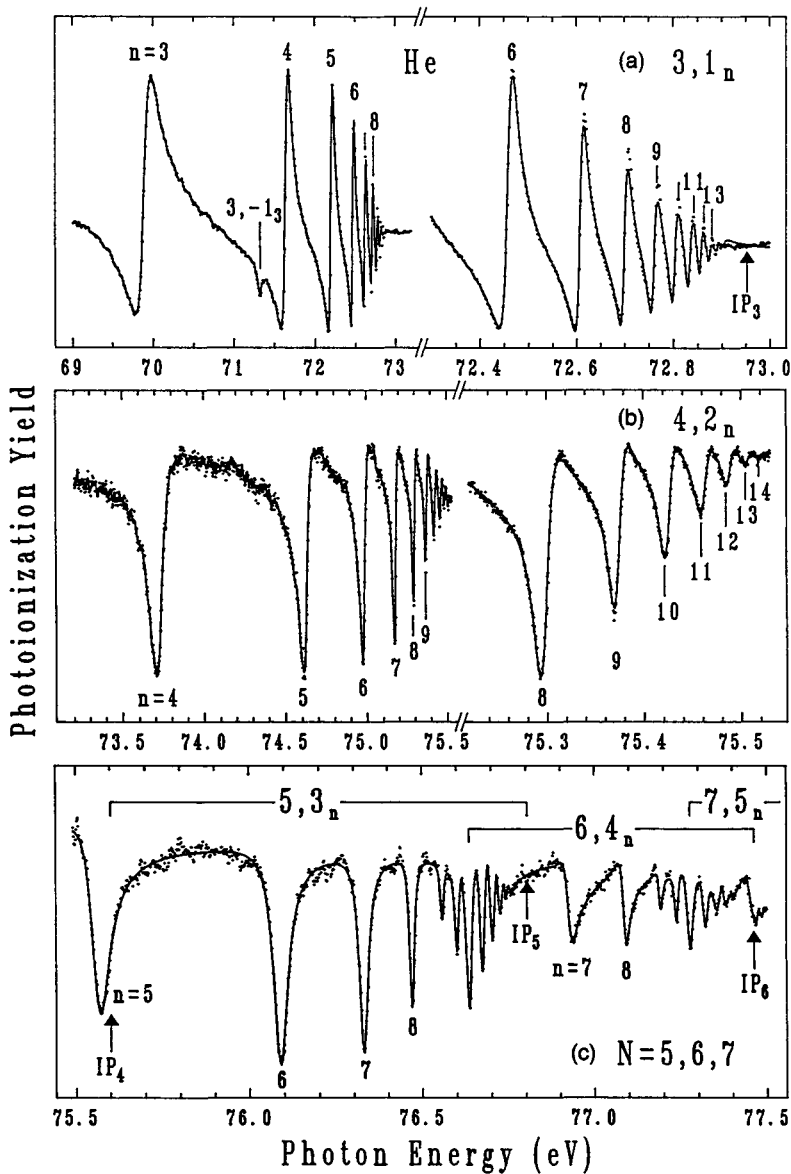


Fig. 23.2 Autoionizing states of He: (a) below the $\text{He}^+ n = 3$ threshold, (b) below the $\text{He}^+ n = 4$ threshold, and (c) below the $\text{He}^+ n = 5$ and 6 thresholds. The high n regions are shown magnified on the right-hand sides in (a) and (b). Note the overlapping of series in (c) (from ref. 3).

Theoretical descriptions

The work of Cooper *et al.*² was the beginning of a steady stream of theoretical work. Shortly after the experiments of Madden and Codling more detailed

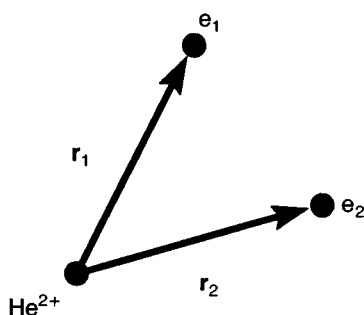


Fig. 23.3 The He atom.

calculations of the doubly excited He levels were carried out by a variety of techniques.⁴⁻⁷ While these calculations showed that the description of Cooper *et al.*² omitted aspects of the problem, such as the contribution of the 2pnd states,⁷ the basic feature of their explanation, the creation of superposition states, somewhat similar to Stark states, was found in all these treatments.⁴⁻⁷

While the detailed calculations gave energies and widths in reasonable agreement with the observed experimental results, they did not provide a simple physical picture of why the He spectrum was the way it was. To address this issue Macek described the He atom using hyperspherical coordinates. In this approach, the two electrons, each described by three coordinates, are replaced by an equivalent single particle in six dimensions.^{8,9} In Fig. 23.3 we show the He atom in which $\mathbf{r}_1$ and $\mathbf{r}_2$ are the vectors from He²⁺ to each of the two electrons. The hyperradius R_h is defined by

$$R_h = \sqrt{r_1^2 + r_2^2}, \quad (23.2)$$

and the ratio of r_1 to r_2 defines the hyperangle α , i.e.

$$\tan \alpha = r_2/r_1. \quad (23.3)$$

At $\alpha = 0$ and $\pi/2$ electrons 1 and 2, respectively, are infinitely far from the He²⁺, while at $\alpha = \pi/4$ they are equidistant from the He²⁺. Following the approach of Macek⁸ and Fano,⁹ we replace the usual wavefunction ψ by Ψ , which is related to ψ by

$$\Psi(r, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = \sin \alpha \cos \alpha R_h^{5/2} \psi(R_h, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2), \quad (23.4)$$

where the unit vector $\hat{\mathbf{r}}_i = \mathbf{r}_i/r_i$. This substitution is roughly analogous to replacing the radial function $R(r)$ by $\rho(r)/r$ in the H atom. Using Ψ the Schroedinger equation in hyperspherical coordinates is given by⁹

$$\left[\frac{\hbar^2}{2m} \left(\frac{-\partial^2}{\partial R_h^2} + \frac{\Lambda^2}{R_h^2} \right) + V(R_h, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) - E \right] \Psi(R_h, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = 0 \quad (23.5)$$

where Λ is the grand angular momentum of Smith¹⁰ defined by

$$\Lambda^2 = \frac{-\partial^2}{\partial \alpha^2} - \frac{1}{4} + \frac{\ell_1^2}{\cos^2 \alpha} + \frac{\ell_2^2}{\sin^2 \alpha}, \quad (23.6)$$

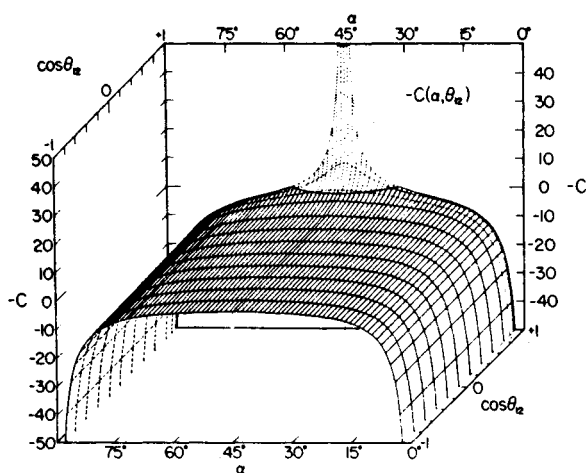


Fig. 23.4 Three dimensional plot of $-C(\alpha, \theta_{12})$ with $Z = 1$ in hyperspherical coordinates; the ordinates represent a potential surface in Rydberg units at $R_h = 1a_0$ (from ref. 11).

where ℓ_1 and ℓ_2 are the angular momenta of the two electrons. From Eqs. (23.5) and (23.6) it is apparent that Λ^2 plays the role of ℓ^2 in the H atom.

A key point of the hyperspherical approach is that the potential V can be factored;

$$V(R_h, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = \frac{e^2}{R_h} C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2), \quad (23.7)$$

where

$$C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = \frac{-Z}{\cos \alpha} - \frac{Z}{\sin \alpha} + \frac{1}{\sqrt{1 - \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2 \sin 2\alpha}}. \quad (23.8)$$

The scaled potential $C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2)$ is shown in Fig. 23.4 using $\cos \theta_{12} = \hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2$.¹¹ Inspecting it we can see that C is flat at $\alpha = 45^\circ$, corresponding to $r_1 = r_2$, for all angles θ_{12} except $\theta_{12} = 0$. The potential valleys, at $\alpha = 0$ and $\pi/2$ correspond to one of the two electrons being far from the He^{2+} core and the other being close to the core. Inspecting Fig. 23.4 we can see that there is a saddle point in $C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2)$ at $\theta_{12} = \pi$ and $\alpha = \pi/4$.

If we define the operator

$$H_R = \frac{\hbar^2 \Lambda^2}{2mR} + \frac{e^2 C}{R_h}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) \quad (23.9)$$

and solve the eigenvalue equation

$$H_R \phi_\mu(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = U_\mu(R_h) \phi_\mu(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) \quad (23.10)$$

at constant R_h , we find the eigenvalue $U_\mu(R_h)$ corresponding to the solution ϕ_μ with the set of quantum numbers μ corresponding to the α , $\hat{\mathbf{r}}_1$, and $\hat{\mathbf{r}}_2$ coordinates.

An alternative approach is to use the well known eigenfunctions of Λ^2 and cope with off diagonal matrix elements due to $C(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2)$.

If we now assume that

$$\Psi(R_h, \alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) = \sum_{\mu} F_{\mu}(R_h) \phi_{\mu}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2), \quad (23.11)$$

where $F_{\mu}(R_h)$ is a wavefunction analogous to the vibrational wavefunction in a diatomic molecule, Eq. (23.5) becomes

$$\left[\frac{\hbar^2}{2M} \frac{\partial^2}{\partial R_h^2} + H_R \right] F_{\mu}(R_h) \phi_{\mu} R = W F_{\mu}(R_h) \phi_{\mu}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2). \quad (23.12)$$

If we replace $H_R \phi_{\mu}$ by $V_{\mu}(R_h) \phi_{\mu}$, use the orthogonality of the ϕ_{μ} functions, and make the adiabatic assumption that $\partial \phi_{\mu}(\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2) / \partial R_h$ is negligible, we can remove the angular variation from Eq. (23.12), leaving the radial equation

$$\left[\frac{\hbar^2}{2M} \frac{\partial^2}{\partial R_h^2} + U_{\mu}(R_h) - W \right] F_{\mu}(R_h) = 0 \quad (23.13)$$

in which $U_{\mu}(R_h)$ is the effective radial potential, which plays the same role as the internuclear potential curve in a diatomic molecule. In fact, the adiabatic approximation is in essence the same approximation as the Born–Oppenheimer approximation used to treat diatomic molecules. In the descriptions of a diatomic molecule the fact that the nuclei are more massive is usually given as the reason why the separation works. However, all that is really necessary is that the motion in the $\alpha, \hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2$ coordinates not change rapidly with R_h .

In Fig. 23.5 we show the set of potential curves calculated by Sadeghpour and Greene for H^- , which differs from He in having shallower wells.¹² Each curve corresponds to a set of μ quantum numbers. In Fig. 23.5(a) we can see that there is a family of curves which connects to H levels of different n at $R_h \rightarrow \infty$. We can also see that as R_h is reduced from infinity that the potential curves from each n split into a fan of curves, with larger spacing at higher n . This splitting is a manifestation of the Stark effect from the field of the well removed electron on the remaining H atom. It is also apparent that only the lowest lying curves have wells at small R_h allowing the formation of quasi-stable states of H^- . Each of the potential curves $U_{\mu}(R_h)$ of Fig. 23.5(a) supports a series of $F_{\mu}(R_h)$ vibrational solutions. In Fig. 23.5(b) we show the energies of the lowest lying potential wells of Fig. 23.5(a). These are the states which are accessible from the ground state of H^- , and the energies agree well with the experimental observations.

In arriving at the radial equation of Eq. (23.13) we assumed $\partial \phi / \partial R_h$ to be negligible. The neglected terms couple states on different potential curves, allowing the autoionization of He or autodetachment of H^- doubly excited states. In other words the adiabatic approximation gives excellent energies, but it does not give the decay rates of doubly excited states.

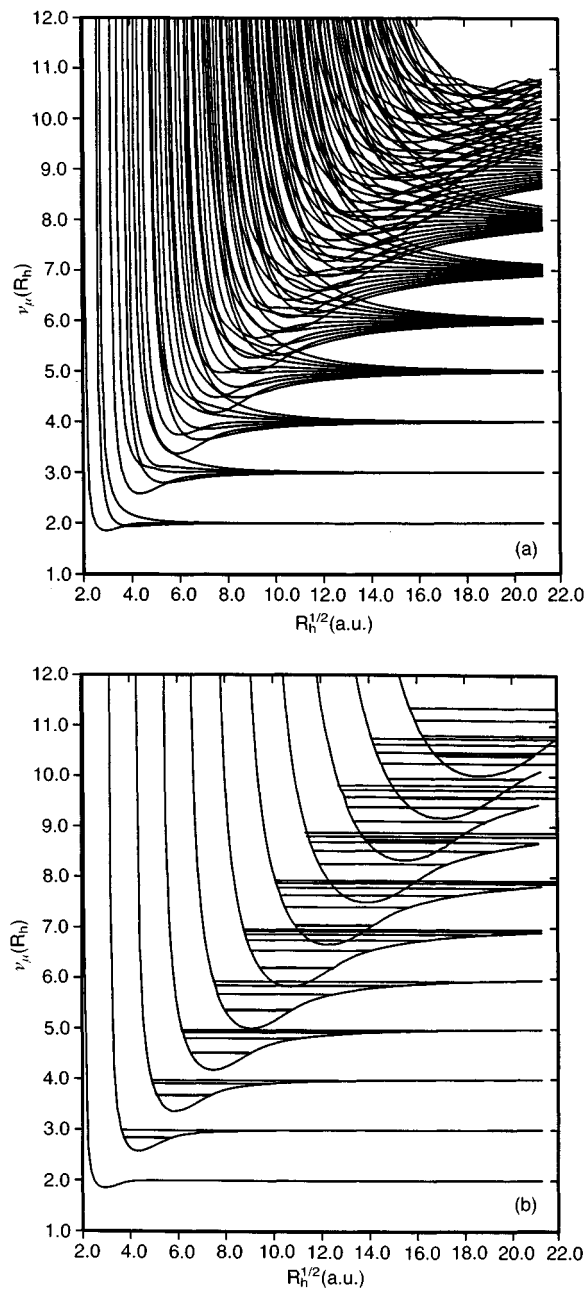


Fig. 23.5 (a) Adiabatic potential curves for $1P^\circ H^-$ shown as effective quantum numbers vs $\sqrt{R_h}$. One should look edgewise along the the Wannier ridge line, at $v_\mu = 18^{-1/4}\sqrt{R_h}$, to see the lowest + channels. In (b), only the lowest + channels within each n manifold are plotted along with the level positions in each potential. The Wannier ridge as an imaginary straight line through the avoided crossings is clearly evident in this figure (from ref. 12).

Rau has pointed out that the doubly excited states in which both electrons are at the same distance from the He^{2+} core have $\alpha = \pi/4$ and are likely to be at the saddle point of the potential of Fig. 23.4.¹³ Furthermore, if the potential V of Eq. (23.7) is expanded in the vicinity of $\theta_{12} = \pi$ and $\alpha = \pi/4$ it can be approximated by its leading term, i.e.¹³

$$V = \frac{-Z_0(Z)}{R_h}, \quad (23.14)$$

where $Z_0(Z) = 2\sqrt{2}(Z - 1/4)$. For this potential the energies relative to the double ionization limit are given by

$$W = \frac{-Z_0^2(Z)}{2(n + 5/2)^2}, \quad (23.15)$$

an expression appropriate for the six dimensional coulomb problem. To account for the higher order terms neglected in Eq. (23.14) a screening constant σ and a quantum defect δ are introduced into the expression for the energy. Explicitly,¹³

$$W = -\frac{Z_0^2(Z - \sigma)}{2(n + 5/2 - \delta)^2} \quad (23.16)$$

A similar formula was given by Read,¹⁴ who has, in addition, summarized much of the work on the development of Rydberg formulae for the energies of the double Rydberg states.¹⁵ The formula of Rau, given by the expression of Eq. (23.16) gives an excellent description of the $\text{He}^- 1s(ns)^2$ states observed by Buckman *et al.*¹⁶ if $\sigma = 1/2\sqrt{2}$ and $\delta = 1.67$. An interesting aspect of this approach is that all the nodes in the wavefunction are in the R_h direction, whereas in the hyperspherical approach the ns^2 states have no nodes in the R_h direction, but are the lowest states in R_h potential curves corresponding to different values of the set of quantum numbers μ .

The hyperspherical approach has enjoyed enormous success in the calculations of energies. It does, however, have the drawback that the physical interpretation of the set of internal μ quantum numbers is not transparent. Had we used the eigenfunctions of Λ^2 , it is not clear that the situation would be much improved since the individual angular momenta of the two electrons ℓ_1 and ℓ_2 appear as quantum numbers, and the $\pm$ description of Eq. (23.1) and the Stark like splittings of Fig. 23.5 suggest that ℓ_1 and ℓ_2 are not good quantum numbers. To address this issue methods of classifying states using group theory were explored by Sinanoğlu and Herrick,¹⁷ and they made substantial progress in classifying the states.¹⁸ One of the more interesting aspects of this approach is that the notion that the doubly excited states are related to Stark states arises naturally from the original group theory work of Park¹⁹ relating Stark and angular momentum states in H. Molmer and Taulbjerg²⁰ make the point that some of the states described by Herrick are simple linear combinations of products of Stark states.

An approach which joins these two disparate points of view is the molecular orbital approach of Feagin and Briggs,^{21,22} who treat the three body coulomb

problem in which two of the particles have the same charge, $Z_1 = Z_2$, and mass $m_1 = m_2$, and the third has a different charge Z_3 and mass m_3 . The approach is similar to the hyperspherical approach in using an adiabatic approximation. The difference lies in the choice of coordinates. If $\mathbf{r}_1$ and $\mathbf{r}_2$ are the vectors defining the locations of the two electrons relative to the He^{2+} , as shown in Fig. 23.1, the two vectors

$$\mathbf{R} = \mathbf{r}_2 - \mathbf{r}_1 \quad (23.17a)$$

and

$$\mathbf{r} = \mathbf{r}_1 + \mathbf{r}_2 \quad (23.17b)$$

are introduced. R plays a role comparable to the role of R_h in the hyperspherical approach. The magnitude R is the distance between the two like particles 1 and 2, the two electrons in He or H^- , and r is the distance from the center of mass of particles 1 and 2 to the third particle, He^{2+} or H^+ for He and H^- . These coordinates are precisely the coordinates commonly used to describe H_2^+ . The Schrodinger equation for the three body coulomb system takes the form²²

$$\left(\frac{-\nabla_R^2}{2\mu_{12}} - \frac{\nabla_r^2}{2\mu_{12,3}} + \frac{Z_1 Z_2}{R} + \frac{Z_1 Z_3}{r_1} - \frac{Z_2 Z_3}{r_2} \right) \psi(\mathbf{r}, \mathbf{R}) = W\psi(\mathbf{r}, \mathbf{R}), \quad (23.18)$$

where μ_{12} is the reduced mass of the two like particles, $\mu_{12} = m_1 m_2 / (m_1 + m_2) = 1/2$ for He or H^- , and $\mu_{12,3}$ is the reduced mass of the center of mass of the two like particles and the third particle; $\mu_{12,3} = (m_1 + m_2)m_3 / (m_1 + m_2 + m_3) \cong 1/2$ for He or H^- .

The eigenfunctions of the atom must be eigenfunctions of the total angular momentum and its projection on a space fixed z axis. If we ignore the electron spins for simplicity, the total angular momentum L and its projection M on the space fixed z axis are conserved. The spatial wavefunctions $\Psi_{LM}(\mathbf{r}, \mathbf{R})$ are related to the body fixed wavefunctions by the Euler transformation

$$\Psi_{LM}(\mathbf{r}, \mathbf{R}) = D_{MK}^L(\psi, \theta, 0) \frac{1}{R} f_{iK}(R) \phi_{iK}^L(\mathbf{r}, R), \quad (23.19)$$

where ψ, θ , and $\phi = 0$ are the Euler angles required to rotate the space fixed z axis into the direction of the vector $\mathbf{R}$, the body fixed frame, D_{MK}^L is a rigid top wavefunction of total angular momentum L and projections M and K on the space fixed and body fixed (along $\mathbf{R}$) axes.²² D_{MK}^L describes the overall rotation of the atom.

$\phi_{iK}(\mathbf{r}, R)$ is the molecular orbital having the projection of its angular momentum on the $\mathbf{R}$ direction equal to number K and internal quantum numbers represented by i . At a fixed separation R , ϕ_{iK} describes the motion of He^+ relative to the center of mass of the two electrons. $f_{iK}(R)$ describes the effect of separating the two electrons, which is analogous to the vibrational motion of the nuclei in a diatomic molecule. Note that the third Euler angle is zero in $D_{MK}^L(\psi, \theta, 0)$ since it coincides with the azimuthal motion of $\mathbf{r}$ about $\mathbf{R}$.

We can write the Hamiltonian of Eq. (23.18) as

$$H = \frac{-\nabla_R^2}{2\mu_{12}} + h, \quad (23.20)$$

where

$$h = \frac{-\nabla_r^2}{2\mu_{12,3}} + \frac{Z_1 Z_2}{R} + \frac{Z_1 Z_3}{r_1} + \frac{Z_2 Z_3}{r_2} \quad (23.21)$$

is the Hamiltonian for the motion of the third particle, He^{2+} or H^+ , in the body fixed frame.

One of the aspects of the molecular orbital approach which is most interesting is that the eigenfunctions of the operator h are apparently not too different from the final atomic eigenfunctions. We shall discuss this point further, but for the moment we simply note that the eigenfunctions $\phi_{iK}(\mathbf{r}, R)$ satisfy the eigenvalue equation

$$h\phi_{iK}(\mathbf{r}, R) = \varepsilon_{iK}(R)\phi_{iK}(\mathbf{r}, R). \quad (23.22)$$

The ∇_R^2 term of the Hamiltonian represents the kinetic energy of the relative motion of the two electrons. It can be broken into radial and angular parts using

$$\frac{-\nabla_R^2}{2\mu_{12}} = -\frac{\partial^2/\partial R^2}{2\mu_{12}} + \frac{L_R^2}{2\mu_{12}}, \quad (23.23)$$

where $\mathbf{L}_R = -i\mathbf{R} \times \nabla_R$.

If we now ignore the off diagonal couplings between the $\phi_{iK}(\mathbf{r}, R)$ states introduced by L_R^2 and the derivatives of ϕ_{iK} with respect to R , the Schroedinger equation, Eq. (23.15), can be reduced to the ordinary differential equation

$$\left[\frac{\partial^2/\partial R^2}{2\mu_{12}} + U_{iK}^L(R) - W \right] f_{iK}(R) = 0 \quad (23.24)$$

where

$$U_{iK}^L(R) = \left\langle \phi_{iK} \left| h + \frac{L_R^2}{2\mu_{12}} \right| \phi_{iK} \right\rangle = \varepsilon_{iK}(R) + \left\langle \phi_{iK} \left| \frac{L_R^2}{2\mu_{12}} \right| \phi_{iK} \right\rangle. \quad (23.25)$$

Eq. (23.24) leads to a series of vibrational functions in the potential of $U_{iK}^L(R)$ described by Eq. (23.25).

A useful feature of the molecular orbital approach is that the eigenvalue equation of Eq. (23.22) can be separated in confocal elliptic coordinates,²³ and, equally important, these eigenfunctions are apparently somewhat similar to the final atomic eigenfunctions.²² The coordinates are given by²²

$$\lambda = (r_1 + r_2)/R \quad (23.26a)$$

and

$$\mu = (r_1 - r_2)/R. \quad (23.26b)$$

The surfaces of constant μ and λ are surfaces of revolution about the interelectron axis. The eigenfunction $\phi_{iK}(\mathbf{r}, R)$ of Eq. (23.22) has the form

$$\phi_{iK}(\mathbf{r}, R) = \phi_\lambda(\lambda) \phi_\mu(\mu) e^{iK\phi} \quad (23.27)$$

and has the quantum numbers K , n_λ , and n_μ , the latter two specifying the number of nodes in the λ and μ directions. Since these quantum numbers are good irrespective of R , they allow the connection between the case $R = 0$ and $R = \infty$. For $R \rightarrow 0$, analogous to the united atom limit in a diatomic molecule, the problem reduces to one with spherical symmetry identical to the H atom, and $n_\lambda = n - \ell - 1$, the number of radial nodes, and $n_\mu = \ell - |m|$, the number of polar nodes. These evolutions are easily envisioned by considering Eqs. (23.26). As the two electrons are moved together the lines of constant λ become circles, and the lines of constant μ radial lines. As $R \rightarrow \infty$ n_λ and n_μ become directly related to the parabolic quantum numbers, n_1 and n_2 . Explicitly, $n_\lambda = n_1$ and $n_\mu = 2n_2$ if n_μ is even and $2n_2 + 1$ if n_μ is odd.²² In a real atom $R \neq 0$ so the actual atomic states have some relation to the Stark states, as mentioned in the discussion of the initial approach of Cooper *et al.*,² and as can be seen in Fig. 23.5(a).

An attractive feature of the molecular orbital approach is that the nodal lines of the atomic wavefunctions fall on surfaces of constant μ and λ . This fact seems to be true whether the wavefunctions are obtained by a diagonalization technique, a hyperspherical coordinate approach, or the molecular orbital approach. To illustrate this point we show in Fig. 23.6²⁴ a density plot of a wavefunction obtained in hyperspherical coordinates for $R = 80a_0$ by Sadeghpour and Greene.¹² Although the figure is drawn using the hyperspherical co-ordinates θ_{12} and α , the lines of constant λ and μ are shown as well, and it is apparent that $n_\lambda = 6$ and $n_\mu = 0$. As shown by Rost *et al.*,²⁴ in most cases the nodal lines of the wavefunctions fall on surfaces of constant μ or λ , implying that the μ and λ motions are almost always separable. However, the μ and R motions are in general found to be coupled.²⁵

In the discussion of both the hyperspherical method and the molecular orbital method we assumed that the motion in the R_n or R direction was slow. While in the Born–Oppenheimer approximation the slowness is generally thought to arise from the much higher mass of the nuclei, as pointed out by Feagin and Briggs,²² it is probably the repulsive nature of the internuclear potential which allows the separability. In other words, the success of these methods in treating He is due to the interelectronic repulsion of the $1/r_{12}$ potential.

Rost and Briggs adopted the opposite approach, ignoring the electron electron interaction to generate the diabatic potential $U(R)$.²⁶ Specifically they approximated the Hamiltonian as²⁶

$$H = \frac{-\nabla_{r_1}^2}{2} - \frac{\nabla_{r_2}^2}{2} - \frac{Z}{r_1} - \frac{Z}{r_2}, \quad (23.28)$$

ignoring the $1/r_{12}$ electron electron interaction.

They used trial wavefunctions of the form²⁶

$$\psi = \psi_n(\mathbf{r}_1) \psi_m(\mathbf{r}_2), \quad (23.29)$$

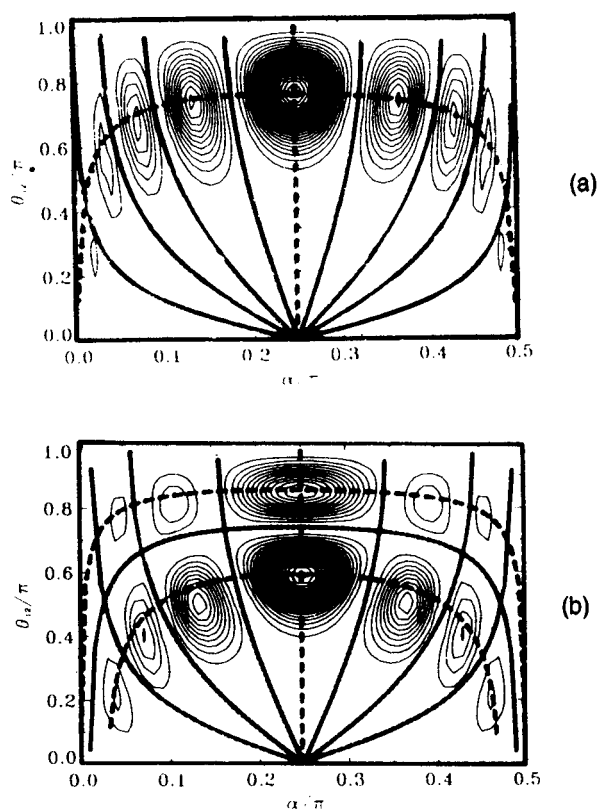


Fig. 23.6 Contour plot of adiabatic, two electron densities taken from ref. 12. The nodal (solid) and antinodal (dashed) lines of constant λ and μ are overdrawn for (a) the state with $(n_\lambda n_\mu) = (0,8)$ and (b) the state with $(n_\lambda n_\mu) = (1,6)$ (from ref. 24).

products of one electron wavefunctions with Z equal to a variational parameter α . The wavefunction of Eq. (23.29) can also be expressed in molecular orbital fashion as

$$\psi = \frac{1}{R} f(R) \phi(\mathbf{r}, R), \quad (23.30)$$

and the function $U(R)$ is calculated using

$$U(R) = \left\langle \phi \left| -\nabla_R^2 - \frac{\nabla_r^2}{4} - \frac{Z}{|\mathbf{r} - \mathbf{R}/2|} - \frac{Z}{|\mathbf{r} + \mathbf{R}/2|} \right| \phi \right\rangle, \quad (23.31)$$

i.e. by integrating over all coordinates save R . Since $U(R)$ depends on α , both through Z in Eq. (23.31), and through ϕ , α can be varied to find the minimum energy of any state. Using nodeless $1s$ trial wavefunctions for ψ_n and ψ_m they found that $\alpha = 1.815$ gives the minimum He $1s^2$ energy and leads to the He potential curve of Fig. 23.7. It passes approximately through the avoided crossings of the adiabatic curves analogous to those shown in Fig. 23.5, and gives a series of

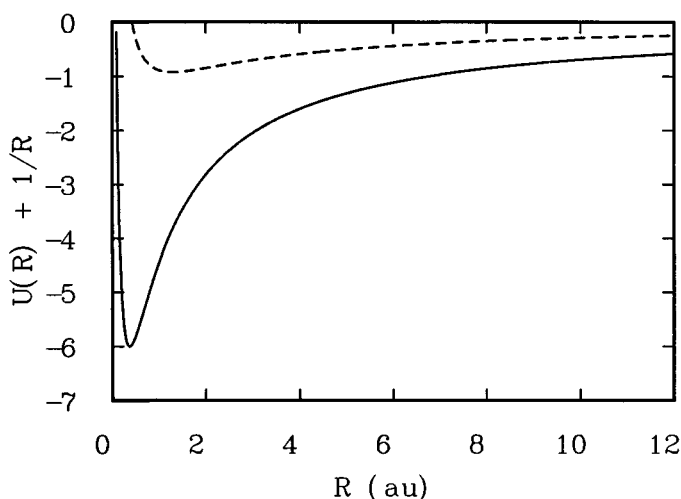


Fig. 23.7 Electronic vibrational potentials $U(R)$ for He (full curve, $\alpha = 1.8315$) and H^- (broken curve, $\alpha = 0.828$) (from ref. 26).

energies corresponding to increasing vibrational motion of R . The energies of the wavefunctions with even numbers of nodes correspond to the He ns^2 states and are in excellent agreement with energies calculated in other ways. It is interesting that in these wavefunctions all the nodes are in $f(R)$, as in the approach of Rau,¹³ but in the molecular orbital and hyperspherical wavefunctions most often calculated, $f(R)$ is nodeless and the nodes appear in other directions. However, in either case the wavefunction has the largest amplitude at large R , so the presence or absence of an oscillating wavefunction at small R does not have much bearing on the energy, although it would make a difference in the excitation probability from an initially small atomic state. The situation is approximately the same as the $\text{H } n\ell$ states. All are degenerate, all have the same total number of nodes, and all have wavefunctions most likely to be found at similar orbital radii.

All of the above approaches are based on separating, approximately, the motion of the electrons in some coordinate system. A radically different approach is the one used by Richter and Wintgen,²⁷ who have rewritten the Hamiltonian using coordinate systems which are not separable but lead to equations simple in form.

One of the inherent problems in all the quantum mechanical approaches is the high density of states near a double ionization limit. As pointed out by Percival²⁸ and Leopold and Percival,²⁹ classical techniques offer an attractive alternative, and recent calculations have demonstrated the usefulness of this approach. For example, Ezra *et al.*³⁰ have studied the doubly excited He states of total angular momentum $L = 0$. In these states the motion of the electrons is confined to a plane. The rather surprising result is that the stable orbits correspond to an “asymmetric stretch” motion of the two electrons as shown in Fig. 23.8. The

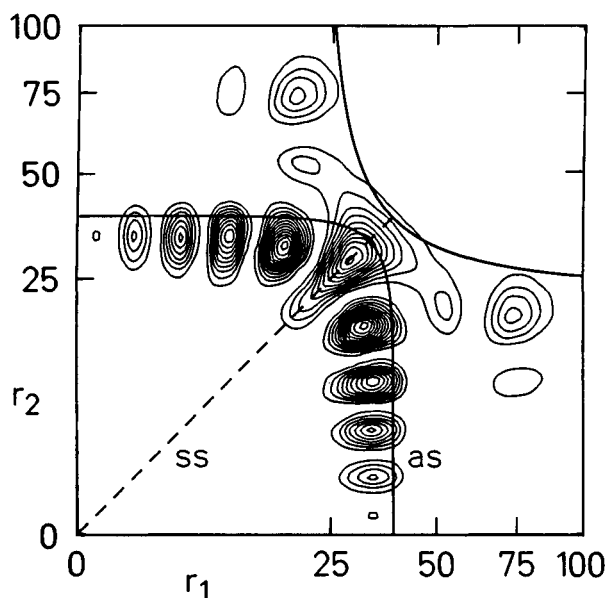


Fig. 23.8 Probability distribution $|\Psi_{Nn}(x, y, z)|^2$ for the intrashell wavefunction $N = n = 6$ in the $x = 0$ plane corresponding to the collinear arrangement $r_{12} = r_1 + r_2$. The axes have a quadratic scale to account for the wave propagation in coulombic systems, where nodal distances increase quadratically. The fundamental orbit (—) (as) as well as the symmetric stretch motion (---) (ss) along the Wannier ridge are overlaid on the figure (from ref. 30).

motions of the two electrons are not symmetric about the He^{2+} , i.e. they are not mirror images of each other. In other words the motion is not along the Wannier ridge but perpendicular to it. This conclusion is not really different from those drawn from quantum mechanical calculations as shown by Fig. 23.8, in which it is apparent that the asymmetric stretch classical orbit falls almost precisely on the antinodes of the quantum mechanical wavefunction.

Laser excitation

In addition to the synchrotron experiments on He, laser experiments have been done using alkaline earth atoms. The attractions of the laser are two. First, it is in principle possible to obtain better resolution with a laser than with a synchrotron, and, second, it is a straightforward matter to start from bound states other than the ground state, affording access to a wider range of doubly excited states. The drawback to laser excitation is that it is not presently possible to produce tunable radiation at $200 \text{ \AA}$, as used in the He experiments. In fact, $2000 \text{ \AA}$, equivalent to a photon energy of 6 eV , is a more realistic lower wavelength limit. The wavelength

restriction of available lasers makes alkaline earth atoms, with their low double ionization limits, very attractive. In Ba, for example, the double ionization limit lies only 10 eV above the first limit and can be reached from a bound Rydberg state by the absorption of two 2500 Å photons, a technically manageable proposition.

The typical experimental approach is to use a thermal beam of alkaline earth atoms, which is first excited to a bound Rydberg state by one or two pulsed dye lasers. Typically, the bound Rydberg state is a high n state, although sometimes it is a high ℓ state as well. The bound Rydberg atom is excited to a high lying doubly excited state in one of two ways. First it can be excited by a two-photon ICE, for example a Ba $6snd$ state can be excited to the $9dnd$ state. The most straightforward approach is to use a single laser operating at half the $6snd \rightarrow 9dnd$ transition frequency.³¹ This approach has the substantial attraction of simplicity, but a significant drawback as well. Absorption of a single photon leads to photoionization to the $6ped$ continua. Since the virtual intermediate state of the two photon $6snd \rightarrow 9dnd$ transition is far off resonance, driving it requires a relatively high laser intensity, and the undesired single photon ionization is not a negligible problem.

An alternative two photon excitation scheme has been used to excite the Ba $6snd \rightarrow 7snd$ transitions.³² Two frequencies are used, the first coincident with an overlap integral zero in the $6snd \rightarrow 6pnd$ transitions. This approach has the attraction that the first photon produces no single photon ionization, yet the virtual intermediate state is insignificantly detuned from the real intermediate state. It has the advantage of requiring low laser powers and producing little single photon ionization. However it does have two significant drawbacks. First, it requires one more laser than does a single color, two-photon excitation, and second, it fixes the energy of the virtual state at 2.5 eV above the first ionization limit, which is rather low.

The second approach to exciting high lying doubly excited states is to make a sequence of single photon ICE transitions, for example from Ba $6snd \rightarrow 6pnd \rightarrow 6dnd$. Eichmann *et al.*³³ have used this method to excite Ba $n_i g n_o d$ states by the route $6s n_o d \rightarrow 6p n_o d \rightarrow 6d n_o d \rightarrow 6f n_o d \rightarrow n_i g n_o d$, using four dye lasers. All four transitions are strong single photon transitions which are easily driven by lasers with 100 μ J pulse energies. The two problems are the multiple lasers and the fact that each of the several intermediate autoionizing states decays, diminishing the signal from the final doubly excited state and adding to the background ion signal.

As is by now apparent, in all these schemes for exciting high lying autoionizing states there is a substantial number of ions and electrons created having nothing to do with the final doubly excited state. As a result, simply detecting ionization is rarely adequate, although it is when studying relatively low lying states such as the Ba $7snd$ and $6dnd$ states. Detection schemes which are only sensitive to the final $n_i \ell_i n_o \ell_o$ state must be used. A relatively straightforward approach is to detect only the high energy electrons which the $n_i \ell_i n_o \ell_o$ state ejects in autoionization to the

ground state of the ion. For example, in the two photon excitation of the Ba $6dnd$ states from the $6snd$ states such electrons have energies of ~ 4.6 eV, while those from single photon ionization have energies of at most 2.3 eV. The success of this approach depends upon there being a favorable branching ratio for the autoionization to the continua of the ground state of the ion. As we consider doubly excited states converging to progressively higher energy states of the ion, autoionization is more likely to occur to high lying states of the ion, and detecting the high energy electrons associated with autoionization to the ground state becomes impractical. As a result, this approach has only been used on relatively low lying doubly excited states.

As mentioned above, if we excite a Ba $n_i\ell_i n_o\ell_o$ double Rydberg state, it is likely to autoionize to an excited $n_i\ell_i$ state of Ba^+ , which is energetically nearby. The other two techniques for selectively detecting the Ba $n_i\ell_i n_o\ell_o$ states depend upon this fact. A laser, often one of the ones used to produce the Ba $n_i\ell_i n_o\ell_o$ doubly excited state, can be used to photoionize the $Ba^+ n_i\ell_i$ state to produce Ba^{2+} , which is easily separated from the copious Ba^+ signal to provide a clear signature of the production of the Ba $n_i\ell_i n_o\ell_o$ state. This approach assumes that the doubly excited state autoionizes and that it does not give up very much energy to the ejected electron when it does so. The photon energy of the laser photoionizing the Ba^+ must be high enough to ionize the ions resulting from the final doubly excited state, but low enough not to ionize background ions produced along the way.

When doubly excited states converging to Rydberg states of the ion are produced, the product of autoionization is often an ionic Rydberg state, which is easily ionized by microwave ionization³⁴ or pulsed field ionization,^{33,35} although for field ionization to be useful, risetimes of $\sim 1 \mu s$ are required. Since either of these methods is only sensitive to Rydberg states of the ion, they provide an excellent means of detecting the high lying doubly excited states.

Experimental observations of electron correlation

The strategy of the experimental study has been to start at the autoionizing states converging to low lying ionic states, which are understood using an independent electron picture, and proceed to the doubly excited states converging to higher lying ionic states, all the while looking for clear manifestations of correlation of the motion of the two electrons.

Most of the experiments to date have been carried out on doubly excited states converging to isolated low angular momentum states of the ion. In general the spectra are well characterized as two photon ICE spectra in which the observed spectrum is characterized by the product of an overlap integral and the spectral density of the doubly excited autoionizing states. As an example we show in Fig. 23.9 the Ba $6s19d \rightarrow 9dn'd$ spectra observed by Camus *et al.*³¹ As shown in Fig.

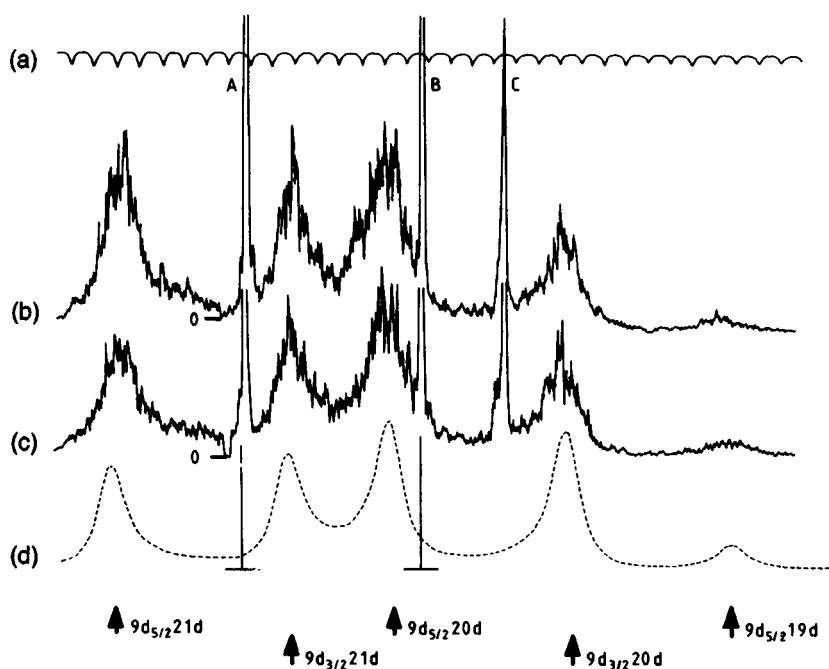


Fig. 23.9 Observed excitation spectra $6s19d \rightarrow 9dn'd$ in neutral Ba: (b) circular, (c) linear polarization of laser beams and (d) computed spectra. The sharp resonances A, B and C correspond respectively to $6s_{1/2} \rightarrow 9d_{5/2}$, $6s_{1/2} \rightarrow 9d_{3/2}$ and $5d \rightarrow 10g$ ionic transitions used as standard calibrations with the Fabry-Perot fringes (a) (from ref. 31).

23.9, there are four peaks in the spectra, corresponding to the ionic $Ba^+ 6s_{1/2} \rightarrow 9d_{3/2}$ and $6s_{1/2} \rightarrow 9d_{5/2}$ transitions for $n = 20$ and 21 , in agreement with the calculated spectrum.

In spite of the fact that the spectra can be understood as two-photon ICE spectra, implying that there is no serious departure from the independent electron picture, the quantum defects and autoionization widths of these $n_i \ell_i n_o \ell_o$ states display an interesting dependence on n_i . As shown by Bloomfield *et al.*,³⁶ the quantum defects of both the $n_i s n_o s$ and $n_i s n_o d$ states increase linearly with n_i , as shown by Fig. 23.10. If the quantum defect is due mostly to core penetration, as it must be when the quantum defects are so large, evaluating the n_i dependence of the expectation value of the penetration energy for an outer $n_o s$ or $n_o d$ electron yields a quantum defect which increases with n_i as shown in Fig. 23.10.³⁶

Not only the quantum defects of the $n_i s n_o s$ and $n_i s n_o d$ states, but also the widths change. The scaled widths $n_o^3 \Gamma$ of the $n_i s n_o s$ and $n_i s n_o d$ states are plotted in Fig. 23.11.³⁶ If we first consider the scaled widths of the $n_i s n_o d$ states, we can see that they rise sharply with n_i , a dependence which is not surprising if we assume that most of the autoionization rate comes from dipole autoionization to a nearby state of the ion. If the outer $n_o d$ electron remains outside the inner $n_i s$ electron, the autoionization rate Γ is given by

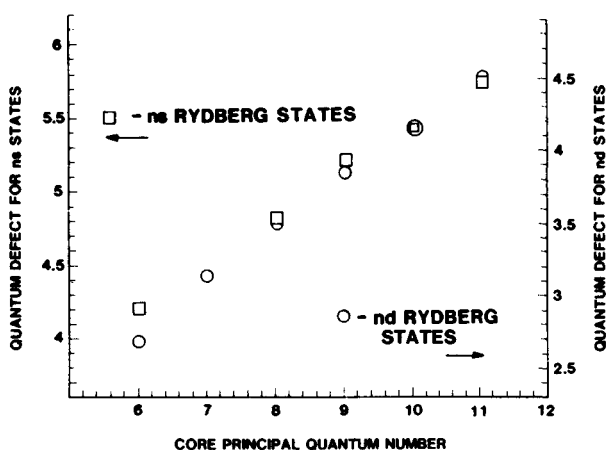


Fig. 23.10 Quantum defects of the Ba *msns* and *msnd* states as a function of the core principal quantum number *m* (from ref. 36).

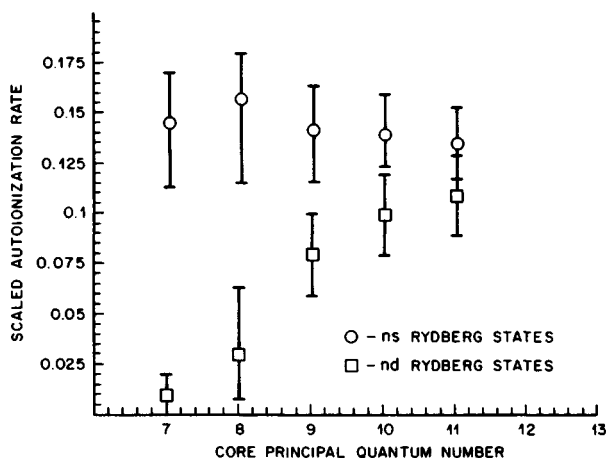


Fig. 23.11. Core scaling of autoionization rates of the Ba *msns* and *msnd* states as a function of the principal quantum number of the core. The *nd* states show increasing autoionization width with increasing core size, while the *ns* autoionization rates are independent of the core principal quantum number (from ref. 36).

$$\Gamma \approx 2\pi |\langle n_i s | \mu | n_i p \rangle|^2 |\langle n_0 d | 1/r^3 | \epsilon f \rangle|^2, \quad (23.32)$$

where the predominant dependence on n_i is through the dipole moment $\langle n_i s | \mu | n_i p \rangle$ which scales approximately as n_i^{-2} . As a result $\Gamma \propto n_i^4$ and the rapid rise of $n_0^3 \Gamma$ with n_i shown in Fig. 23.11 is not unexpected. Once the inner electron $n_i s$ orbit

becomes large enough that the outer electron's orbit passes through most of it, so that core penetration contributes significantly to the autoionization rate, increasing the size of the inner electron's orbit does not affect the autoionization rate, because making the $n_i s$ orbit larger also makes it more dilute.

One of the ways of defining correlation is that the wavefunctions are not well represented by the designation $n_i \ell_i n_o \ell_o$ but rather as the + and - states of Eq. (23.1) or some other linear combination of $n_i \ell_i n_o \ell_o$ states. Excluding the configuration mixing of series converging to the two fine structure levels of an ionic state, several such cases have been observed. In any atom perturbing levels converging to higher limits which lie below the first limit often satisfy this criterion. The Ba 5d7d states are excellent examples. There have also been observations in more highly excited states. In the Ca $4s7s \rightarrow 7d7s$ transition Morita and Suzuki³⁷ observed clear structure due to interaction with the $5gn\ell$ states. Similarly, Jones *et al.*³⁸ observed structure in the Ba $5d_{3/2}5g J = 2 \rightarrow 4f_{5/2}7g J = 3$ transition due to the degenerate $6d_{5/2}nf$ and $6d_{5/2}nh$ states. In all of these cases there is evidently angular momentum exchange between the two electrons, as in Eq. (23.1); however, in both cases the Rydberg electron spends much of its time far from the ionic core where it is in a purely coulomb potential. In this sense these states are not highly correlated states.

When the series converging to high lying isolated states of Ba^+ are excited it is possible to observe clear evidence of correlation. A good example occurs in the work of Camus *et al.*,³⁴ who observed the two photon spectra from the bound Ba $6sn_p$ states to the $n_i sn_p$ and $n_i dn_p$ states. In Fig. 23.12 we show the two photon spectrum from the Ba 6s45p state to the doubly excited states lying near the $n = 30$ states of Ba^+ . The spectrum of Fig. 23.12 corresponds to changing the inner electron from 28s to 35s. The $6s45p \rightarrow n_i d45p$ and $6s45p \rightarrow n_i s45p$ transitions for $n_i < 28$ appear as simple two photon ICE transitions, which are for all intents and purposes identical to the spectrum of Fig. 23.9. As shown by Fig. 23.12, between the 27d45p resonance and the 29s45p resonance there are weak, but easily observed resonances which correspond to $28pn_o\ell$ and $25fn_o\ell$ states. As n_i is increased these p and f resonances become more pronounced, to the point that the $31pn_o\ell$ resonance is almost as large as the 30d45p resonance. At energies just above the $30pn_o\ell$ and $31pn_o\ell$ states there is a signal which is clearly nonzero, and by the time the energy of the $33pn_o\ell$ state is reached the spectrum is essentially continuous, i.e. doubly excited states are excited at any wavelength. By the time we have reached the 32d45p state the outer electron is not in a pure 45p state; however, we have used the label for simplicity. There are two important aspects of these observations; first, the appearance of the $n_i pn_o\ell$ and $n_i fn_o\ell$ resonances, and, second, which is not apparent in Fig. 23.12, the center of gravity of the resonances shifts from the ionic transition frequency as the inner electron quantum number n_i is increased. We shall return to this point shortly. Eichmann *et al.*³³ and Jones and Gallagher³⁵ have made observations which are similar to those shown in Fig. 23.12, so they may be regarded as representative.

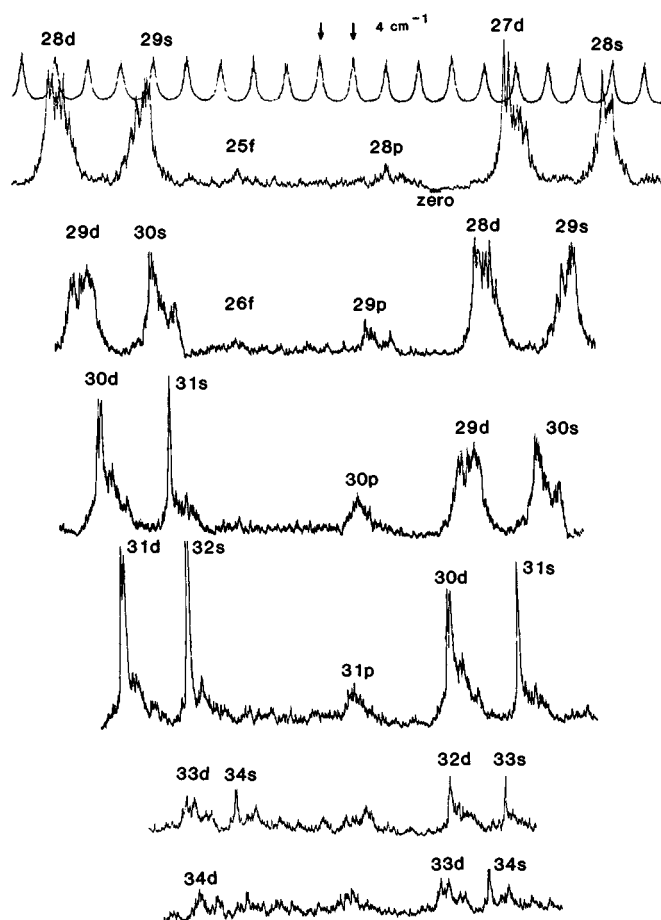


Fig. 23.12 Ba two-photon absorption spectra $6s45p \rightarrow n\ell n'\ell'$. Narrow resonance lines correspond to ionic resonances of Ba^+ due to coincidences between atomic (Ba) and ionic (Ba^+) resonances. They are indicated by an arrow. Note the evolution to a continuous spectrum at high photon energies (from ref. 34).

Camus *et al.*³⁴ explained their observations by a picture which has sometimes been called the frozen planet model. Qualitatively, the relatively slowly moving outer electron produces a quasi-static field at the inner electron given by $1/r_o^2$, and this field leads to the Stark effect in Ba^+ . The field allows the transitions to the $n_i p n_o \ell$ and $n_i f n_o \ell$ states and leads to shifts of the ionic energies. The presence of the $n_i p n_o \ell$ and $n_i f n_o \ell$ resonances in the spectrum of Fig. 23.12 is quite evident. Camus *et al.* compared the shifts to those calculated in a fashion similar to a Born–Oppenheimer calculation. With the outer electron frozen in place at r_o they calculated the Ba^+ energies, $W_i(r_o)$, and wavefunctions. They then added the energy $W_o(r_o)$ to the normal screened coulomb potential seen by the outer electron. This procedure leads to a phase shift in the outer electron wavefunction

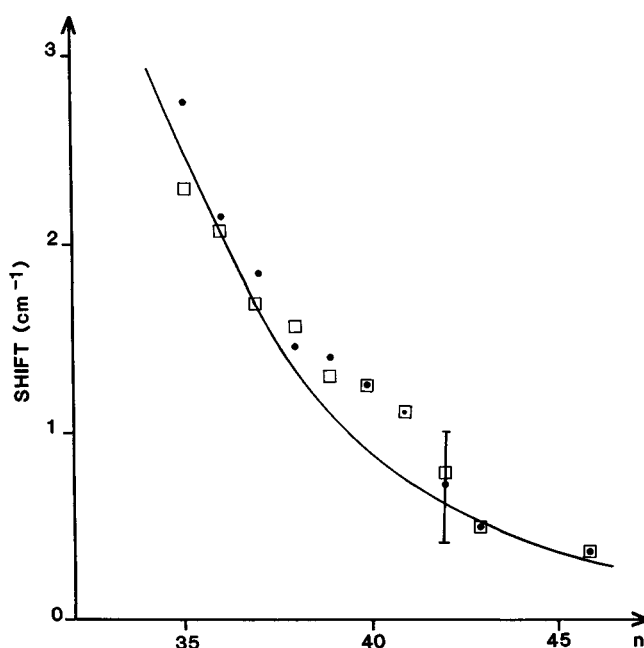


Fig. 23.13 Shifts of the centers of gravity compared to the ionic parent line positions for the observed structures for double Rydberg resonances $6snp \rightarrow 26dn'p$ ($\square$) and $6snp \rightarrow 27sn'p$ ($\bullet$) vs n^* ; $n = 39$ – 47 and 50 . The average quantum defect is 4.1 for the $6snp$ series. The full line is the theoretical curve (from ref. 34).

which varies with r_o . Using this approach they were able to compare the shifts from the ionic lines of the centers of gravity of the observed resonances.

The cross section for a two photon ICE transition is given by

$$\sigma \propto A^2(\nu) \langle \nu_b \ell^B | \nu \ell \rangle^2, \quad (23.33)$$

where we have omitted all terms which are either slowly varying or constant, and ν_b is the effective quantum number of the outer electron in the initial bound state. Camus *et al.*³⁴ observed that the resonances shown in Fig. 23.12 are too wide to come from a single final outer electron state. Rather, each resonance comes from a band of possible final states, in which case A^2 of Eq. (23.33) is a constant, and the cross section is determined by only the squared overlap integral. The overlap integral is centered on the ionic transition frequency. Thus by finding the center of gravity of the observed resonances we can find the transition frequency of the Ba^+ ion with the outer electron present, and we can compare the shifts from the bare ion frequencies to the values calculated using the procedure outlined in the previous paragraph. In Fig. 23.13 we show the excellent agreement between the measured and calculated shifts.

The frozen planet model is simple and physically appealing. In addition, it is clearly related to the treatment based on $+$ and $-$ states originally given by Cooper *et al.*² The most convincing demonstrations of the legitimacy of the frozen

planet model is one which appears, in retrospect, obvious. Eichmann *et al.*³³ simply replaced the outer electron by a static field to check that the spectrum is the same. Using six dye lasers they excited Ba atoms first to the bound $6s78d$ states and then to final $n_i\ell_i78d$ states of $n_i > 30$ via the route $6s78d \rightarrow 6p78d \rightarrow 6d78d \rightarrow 6f78d \rightarrow n_i\ell_i78d$, and field ionized the Ba^+ ions resulting from autoionization of the final state. Their spectra are shown in Fig. 23.14. Fig. 23.14(a) shows the spectrum obtained by scanning the last laser, driving the $6f78d \rightarrow n_i\ell_i78d$ transition. It shows strong background peaks at the location of the ionic transitions to the n_id and n_ig states, mostly due to excitation of Ba ions produced along the way. Fig. 23.14(c) is the spectrum obtained when Ba is photoionized by the first two lasers, instead of being excited to the $6s78d$ state, and then excited to produce the nd and ng Rydberg states of Ba^+ . Fig. 23.14(d) is obtained by the same procedure as Fig. 23.14(c) except that a field of 60 V/cm is added. This field corresponds to $\langle r \rangle^{-2}$ of the $78d$ electron. Fig. 23.14(b) is composed of the normalized superposition of the zero field ion spectrum of Fig. 23.14(c) and the 60 V/cm ion spectrum of Fig. 23.14(d). These two components should correspond to the background ion signal and the doubly excited state signals in Fig. 23.14(a). The normalization of Figs. 23.14(c) and (d) to produce Fig. 23.14(b) effectively allows the amount of background ion signal in the synthetic spectrum of Fig. 23.14(b) to be adjusted. Finally, Fig. 23.14(e) is the computed Ba^+ spectrum in the 60 V/cm field. As shown, the $34g$ state has disappeared, being replaced by the $n = 34$ Stark manifold which is more likely to be excited on its red end than its blue end.

The good match between the actual spectrum of Fig. 23.14(a) and the spectrum of Fig. 23.14(b), synthesized from ion spectra, validates the frozen planet model. Several other aspects of this experiment are different from the other double Rydberg experiments. First, the effect of the outer electron is quite noticeable at a much lower value of n_i/n_o than in the experiments of Camus *et al.*³⁴ and Jones and Gallagher.³⁵ The reason is that the $Ba^+ ng$ states are essentially degenerate with the higher ℓ states so that much smaller fields from the outer electron convert the ℓ states to Stark states. The situation is similar to He, and is a good example of angular correlation between the two electrons. Eichmann *et al.* noted that in the calculated spectrum of Fig. 23.14(e) there is almost no nd character in the Stark manifold of states and that the pronounced angular correlation shown in Fig. 23.14(a) could not have been observed if only the n_id78d states had been excited, in agreement with other results.

The frozen planet model also agrees with classical calculations of Richter and Wintgen,³⁹ who find a classically stable state in which both electrons are on the same side of the atom in orbits which exhibit pronounced angular correlation, even though the orbital radii of the two electrons are very different.

One of the recurring manifestations of correlation between the two valence electrons is the conversion of ℓ states to Stark states. Eichmann *et al.*⁴⁰ have studied doubly excited Sr states in which both electrons have $\ell > 2$, so that both inner and outer electrons are in approximately hydrogenic, easily polarized states

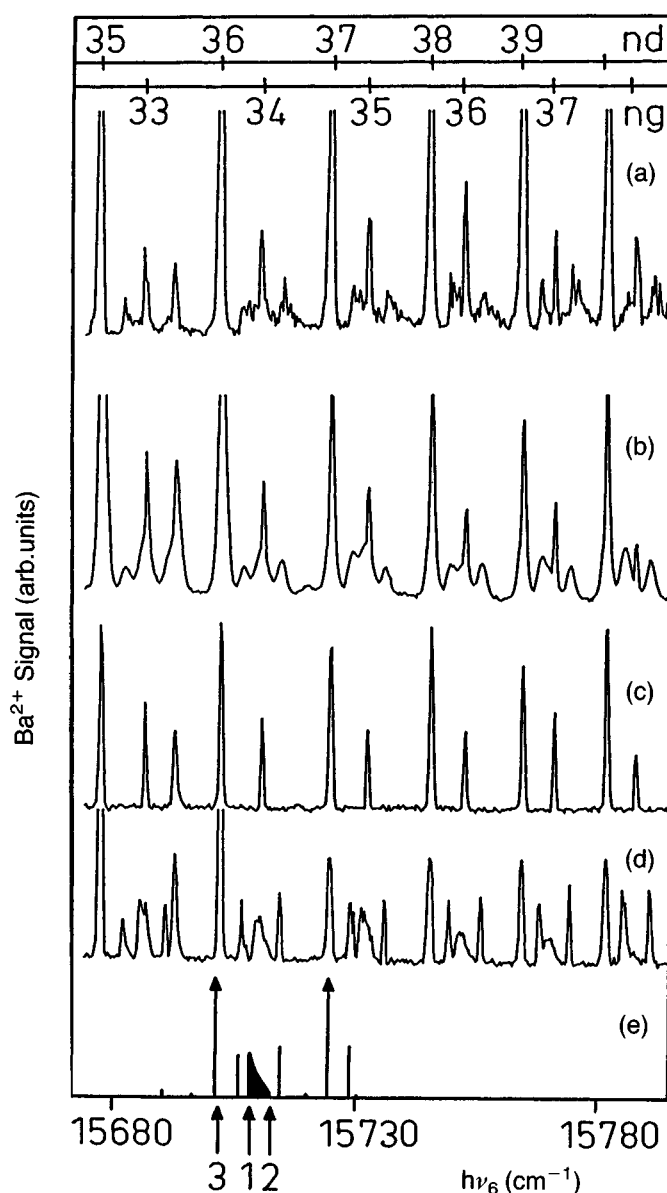


Fig. 23.14 (a) Six laser absorption spectrum of $Ba\ n\ell 78d$ planetary states, recorded as Ba^{2+} signal vs. the photon energy, $h\nu_6$, of the laser driving the last transition, from the $6f_{5/2}$ state to the $nd78d$ and $ng78d$ states. (b) Synthetic spectrum as in (a), composed from a normalized superposition of (c) and (d), including artificial autoionization broadening of (d). (c) Structure of the purely ionic component in (a), obtained by six laser excitation of ionic $Ba^+ nd$ and $Ba^+ ng$ states. (The extra peak between $33g$ and $36d$ appears through accidental excitation and photoionization of a $8p_{3/2}$ state.) (d) Structure of the planetary component in (a), obtained by repeating the spectrum (c) in the presence of an external field of 60 V/cm. (e) Theoretical bar spectrum equivalent to (d), calculated around $n = 34$ (from ref. 33).

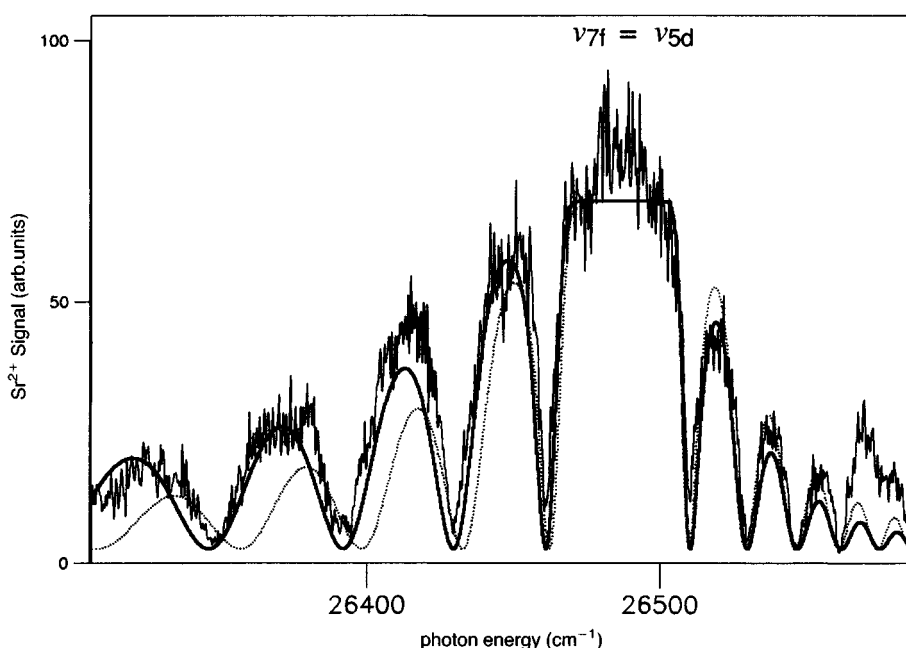


Fig. 23.15 Observed and calculated squared overlap integral $|\langle \nu_{5d} | \nu_{7f} \rangle|^2$ for the transition from the $5d20\ell$ state of $\ell > 9$ to the $7fn\ell$ states as a function of the photon energy of the laser driving the transition. The experimental spectrum (noisy trace) was taken at high laser power and shows some saturation. The smooth solid trace is the squared overlap integral calculated using as the final planetary state the wavefunction $\Psi(\nu_{7f})$, calculated using the potential of Eq. (23.34) which contains an induced dipole. The polarizability α of the $\text{Sr}^+ 7f$ state is independently calculated to be $5 \times 10^5 a_0^3$. The dotted curve is the squared overlap integral obtained with a final state calculated using a coulomb potential ($\alpha = 0$ in Eq. (23.34)). Note the characteristic phase shift and height asymmetry of the side maxima of the overlap integral calculated with the induced dipole potential for the final state when compared to the overlap integral calculated with a pure coulomb potential for the final state (from ref. 40).

with small quantum defects. Specifically, they produced bound $\text{Sr } 5sn_o\ell_o$ states of $n_o \sim 20$ and $\ell_o \approx 9$ using the Stark switching technique. They then further excited these states via the route $5sn_o\ell_o \rightarrow 5pn_o\ell_o \rightarrow 5dn_o\ell_o \rightarrow n_i fn_o\ell_o$. The final $n_i fn_o\ell_o$ state, which is a nominal description only, decays to an excited state of Sr^+ which is photoionized to produce Sr^{2+} . Sr^{2+} is detected as the wavelength of the laser driving the $5dn_o\ell_o \rightarrow n_i fn_o\ell_o$ transition is scanned.

They observed two clear signatures of the Stark, or dipole, structure of the doubly excited states; first in the quantum defects and second, in the overlap integrals. They observed several Rydberg series converging to excited Sr^+ states. From the $5dn_o = 17 \ell_o = 9$ state they observed a series with a quantum defect of $0.70(\text{mod } 1)$ converging to the $6g$ state of Sr^+ . While the observation of a series converging to this limit alone is indicative of correlation, what is most interesting is the quantum defect. It is simply impossible for a Sr coulomb state of $\ell = 9$ to have a quantum defect of 0.70 . On the other hand if the outer electron is not in a

purely coulomb potential, it is possible for it to have such a quantum defect. Following Watanabe and Greene⁴¹ they assume that the outer electron is in the potential

$$V(r_o) = \frac{-1}{r_o} + \frac{\bar{\ell}_o(\bar{\ell}_o + 1)}{2r_o^2} + \frac{\tilde{\alpha}}{r_o^4}, \quad (23.34)$$

where $\bar{\ell}_o$ and $\tilde{\alpha}$ depend on r_o and are equal to the orbital angular momentum of the outer electron, ℓ_o , and the polarizability of the ion, α , as $r_o \rightarrow \infty$. Using this potential they calculate a quantum defect of 6.68 which is in excellent, perhaps fortuitous agreement with the observations.

The second manifestation of the dipole nature of the outer electron states is found in the overlap integral observed in the final ICE transition. For transitions between coulomb states of effective quantum numbers ν_1 and ν_2 the overlap integral has a $\sin(\nu_1 - \nu_2)/(\nu_1 - \nu_2)$ form. However, for the dipole states it does not, as shown by Fig. 23.15, a recording of the Sr $5dn_o\ell \rightarrow 7fn'_o\ell'_o$ transition. As shown by Fig. 23.15, the observed spectrum, taken at high laser power, is not symmetric about the ionic $Sr^+ 5d \rightarrow 7f$ transition. Rather, it is more intense on the low frequency side. While the observed spectrum clearly disagrees with the assumption of the outer electron's being in a coulomb potential, it is in quite good agreement with the overlap integral assuming that in the final state the outer electron is in the potential of Eq. (23.34).

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